

- [E16] Knuth, D. E., *The Art of Computer Programming*. 3 vols. 3rd ed. Reading, MA: Addison-Wesley, 1997–2009.
- [E17] Kreyszig, E., *Introductory Functional Analysis with Applications*. New York: Wiley, 1989.
- [E18] Kreyszig, E., On methods of Fourier analysis in multigrid theory. *Lecture Notes in Pure and Applied Mathematics* 157. New York: Dekker, 1994, pp. 225–242.
- [E19] Kreyszig, E., Basic ideas in modern numerical analysis and their origins. *Proceedings of the Annual Conference of the Canadian Society for the History and Philosophy of Mathematics*. 1997, pp. 34–45.
- [E20] Kreyszig, E., and J. Todd, *QR* in two dimensions. *Elemente der Mathematik* 31 (1976), pp. 109–114.
- [E21] Mortensen, M. E., *Geometric Modeling*. 2nd ed. New York: Wiley, 1997.
- [E22] Morton, K. W., and D. F. Mayers, *Numerical Solution of Partial Differential Equations: An Introduction*. New York: Cambridge University Press, 1994.
- [E23] Ortega, J. M., *Introduction to Parallel and Vector Solution of Linear Systems*. New York: Plenum Press, 1988.
- [E24] Overton, M. L., *Numerical Computing with IEEE Floating Point Arithmetic*. Philadelphia: SIAM, 2004.
- [E25] Press, W. H. et al., *Numerical Recipes in C: The Art of Scientific Computing*. 2nd ed. New York: Cambridge University Press, 1992.
- [E26] Shampine, L. F., *Numerical Solutions of Ordinary Differential Equations*. New York: Chapman and Hall, 1994.
- [E27] Varga, R. S., *Matrix Iterative Analysis*. 2nd ed. New York: Springer, 2000.
- [E28] Varga, R. S., *Geršgorin and His Circles*. New York: Springer, 2004.
- [E29] Wilkinson, J. H., *The Algebraic Eigenvalue Problem*. Oxford: Oxford University Press, 1988.
- [F8] Harary, F., *Graph Theory*. Reprinted. Reading, MA: Addison-Wesley, 2000.
- [F9] Merris, R., *Graph Theory*. Hoboken, NJ: Wiley-Interscience, 2000.
- [F10] Ralston, A., and P. Rabinowitz, *A First Course in Numerical Analysis*. 2nd ed. Mineola, NY: Dover, 2001.
- [F11] Thulasiraman, K., and M. N. S. Swamy, *Graph Theory and Algorithms*. New York: Wiley-Interscience, 1992.
- [F12] Tucker, A., *Applied Combinatorics*. 5th ed. Hoboken, NJ: Wiley, 2007.

Part G. Probability and Statistics (Chaps. 24–25)

- [G1] American Society for Testing Materials, *Manual on Presentation of Data and Control Chart Analysis*. 7th ed. Philadelphia: ASTM, 2002.
- [G2] Anderson, T. W., *An Introduction to Multivariate Statistical Analysis*. 3rd ed. Hoboken, NJ: Wiley, 2003.
- [G3] Cramér, H., *Mathematical Methods of Statistics*. Reprinted. Princeton, NJ: Princeton University Press, 1999.
- [G4] Dodge, Y., *The Oxford Dictionary of Statistical Terms*. 6th ed. Oxford: Oxford University Press, 2006.
- [G5] Gibbons, J. D. and S. Chakraborti, *Nonparametric Statistical Inference*. 4th ed. New York: Dekker, 2003.
- [G6] Grant, E. L. and R. S. Leavenworth, *Statistical Quality Control*. 7th ed. New York: McGraw-Hill, 1996.
- [G7] IMSL, *Fortran Numerical Library*. Houston, TX: Visual Numerics, 2002.
- [G8] Kreyszig, E., *Introductory Mathematical Statistics. Principles and Methods*. New York: Wiley, 1970.
- [G9] O'Hagan, T. et al., *Kendall's Advanced Theory of Statistics 3-Volume Set*. Kent, U.K.: Hodder Arnold, 2004.
- [G10] Rohatgi, V. K. and A. K. MD. E. Saleh, *An Introduction to Probability and Statistics*. 2nd ed. Hoboken, NJ: Wiley-Interscience, 2001.

Web References

- [W1] upgraded version of [GenRef1] online at <http://dlmf.nist.gov/>. Hardcopy and CD-Rom: Oliver, W. J. et al. (eds.), *NIST Handbook of Mathematical Functions*. Cambridge; New York: Cambridge University Press, 2010.
- [W2] O'Connor, J. and E. Robertson, MacTutor History of Mathematics Archive. St. Andrews, Scotland: University of St. Andrews, School of Mathematics and Statistics. Online at <http://www-history.mcs.st-andrews.ac.uk>. (Biographies of mathematicians, etc.).

Part F. Optimization, Graphs (Chaps. 22–23)

- [F1] Bondy, J. A. and U.S.R. Murty, *Graph Theory with Applications*. Hoboken, NJ: Wiley-Interscience, 1991.
- [F2] Cook, W. J. et al., *Combinatorial Optimization*. New York: Wiley, 1997.
- [F3] Diestel, R., *Graph Theory*. 4th ed. New York: Springer, 2006.
- [F4] Diwekar, U. M., *Introduction to Applied Optimization*. 2nd ed. New York: Springer, 2008.
- [F5] Gass, S. L., *Linear Programming. Method and Applications*. 3rd ed. New York: McGraw-Hill, 1969.
- [F6] Gross, J. T. and J. Yellen (eds.), *Handbook of Graph Theory and Applications*. 2nd ed. Boca Raton, FL: CRC Press, 2006.
- [F7] Goodrich, M. T., and R. Tamassia, *Algorithm Design: Foundations, Analysis, and Internet Examples*. Hoboken, NJ: Wiley, 2002.



APPENDIX 2

Answers to Odd-Numbered Problems

Problem Set 1.1, page 8

1. $y = \frac{1}{\pi} \cos 2\pi x + c$
3. $y = ce^x$
5. $y = 2e^{-x}(\sin x - \cos x) + c$
7. $y = \frac{1}{5.13} \sinh 5.13x + c$
9. $y = 1.65e^{-4x} + 0.35$
11. $y = (x + \frac{1}{2})e^x$
13. $y = 1/(1 + 3e^{-x})$
15. $y = 0$ and $y = 1$ because $y' = 0$ for these y
17. $\exp(-1.4 \cdot 10^{-11}t) = \frac{1}{2}$, $t = 10^{11}(\ln 2)/1.4$ [sec]
19. Integrate $y'' = g$ twice, $y'(t) = gt + v_0$, $y'(0) = v_0 = 0$ (start from rest), then $y(t) = \frac{1}{2}gt^2 + y_0$, where $y(0) = y_0 = 0$

Problem Set 1.2, page 11

11. Straight lines parallel to the x -axis
13. $y = x$
15. $mv' = mg - bv^2$, $v' = 9.8 - v^2$, $v(0) = 10$, $v' = 0$ gives the limit $9.8 = 3.1$ [meter/sec]
17. Errors of steps 1, 5, 10: 0.0052, 0.0382, 0.1245, approximately
19. $x_5 = 0.0286$ (error 0.0093), $x_{10} = 0.2196$ (error 0.0189)

Problem Set 1.3, page 18

1. If you add a constant later, you may not get a solution.
Example: $y' = y$, $\ln |y| = x + c$, $y = e^{x+c} = \tilde{c}e^x$ but not $e^x + c$ (with $c \neq 0$)
3. $\cos^2 y \, dy = dx$, $\frac{1}{2}y + \frac{1}{4} \sin 2y + c = x$
5. $y^2 + 36x^2 = c$, ellipses
7. $y = x \arctan(x^2 + c)$
9. $y = x/(c - x)$
11. $y = 24/x$, hyperbola
13. $dy/\sin^2 y = dx/\cosh^2 x$, $-\cot y = \tanh x + c$, $c = 0$, $y = -\operatorname{arccot}(\tanh x)$
15. $y^2 + 4x^2 = c = 25$
17. $y = x \arctan(x^3 - 1)$
19. $y_0 e^{kt} = 2y_0$, $e^k = 2$ (1 week), $e^{2k} = 2^2$ (2 weeks), $e^{4k} = 2^4$
21. 69.6% of y_0
23. $PV = c = \text{const}$
25. $T = 22 - 17e^{-0.5306t} = 21.9$ [$^{\circ}\text{C}$] when $t = 9.68$ min
27. $e^{-k \cdot 10} = \frac{1}{2}$, $k = \frac{1}{10}$, $\ln \frac{1}{2}$, $e^{-kt_0} = 0.01$, $t = (\ln 100)/k = 66$ [min]
29. No. Use Newton's law of cooling.
31. $y = ax$, $y' = g(y/x) = a = \text{const}$, independent of the point (x, y)
33. $\Delta S = 0.15S\Delta\phi$, $dS/d\phi = 0.15S$, $S = S_0 e^{0.15\phi} = 1000S_0$,
 $\phi = (1/0.15) \ln 1000 = 7.3 \cdot 2\pi$. Eight times.

Problem Set 1.4, page 26

1. Exact, $2x = 2x$, $x^2y = c$, $y = c/x^2$
3. Exact, $y = \arccos(c/\cos x)$
5. Not exact, $y = \sqrt{x^2 + cx}$
7. $F = e^{x^2}$, $e^{x^2} \tan y = c$
9. Exact, $u = e^{2x} \cos y + k(y)$, $u_y = -e^{2x} \sin y + k'$, $k' = 0$. Ans. $e^{2x} \cos y = 1$
11. $F = \sinh x$, $\sinh^2 x \cos y = c$
13. $u = e^x + k(y)$, $u_y = k' = -1 + e^y$, $k = -y + e^y$. Ans. $e^x - y + e^y = c$
15. $b = k$, $ax^2 + 2kxy + ly^2 = c$

Problem Set 1.5, page 34

3. $y = ce^x - 5.2$
5. $y = (x + c)e^{-kx}$
7. $y = x^2(c + e^x)$
9. $y = (x - 2.5/e)e^{\cos x}$
11. $y = 2 + c \sin x$
13. Separate. $y - 2.5 = c \cosh^4 1.5x$
15. $(y_1 + y_2)' + p(y_1 + y_2) = (y_1' + py_1) + (y_2' + py_2) = 0 + 0 = 0$
17. $(y_1 + y_2)' + p(y_1 + y_2) = (y_1' + py_1) + (y_2' + py_2) = r + 0 = r$
19. Solution of $cy_1' + pcy_1 = c(y_1' + py_1) = cr$
21. $y = uy^*$, $y' + py = u'y^* + uy^{*'} + puy^* = u'y^* + u(y^{*' + py^*}) = u'y^* + u \cdot 0 = r$, $u' = r/y^* = re^{\int p dx}$, $u = \int e^{\int p dx} r dx + c$. Thus, $y = uy_h$ gives (4). We shall see that this method extends to higher-order ODEs (Secs. 2.10 and 3.3).
23. $y^2 = 1 + 8e^{-x^2}$
25. $y = 1/u$, $u = ce^{-3.2x} + 10/3.2$
27. $dx/dy = 6e^y - 2x$, $x = ce^{-2y} + 2e^y$
31. $T = 240e^{kt} + 60$, $T(10) = 200$, $k = -0.0539$, $t = 102$ min
33. $y' = A - ky$, $y(0) = 0$, $y = A(1 - e^{-kt})/k$
35. $y' = 175(0.0001 - y/450)$, $y(0) = 450 \cdot 0.0004 = 0.18$,
 $y = 0.135e^{-0.3889t} + 0.045 = 0.18/2$,
 $e^{-0.3889t} = (0.09 - 0.045)/0.135 = 1/3$,
 $t = (\ln 3)/0.3889 = 2.82$. Ans. About 3 years
37. $y' = y - y^2 - 0.2y$, $y = 1/(1.25 - 0.75e^{-0.8t})$, limit 0.8, limit 1
39. $y' = By^2 - Ay = By(y - A/B)$, $A > 0$, $B > 0$. Constant solutions $y = 0$,
 $y = A/B$, $y' > 0$ if $y > A/B$ (unlimited growth), $y' < 0$ if $0 < y < A/B$
(extinction). $y = A/(ce^{At} + B)$, $y(0) > A/B$ if $c < 0$, $y(0) < A/B$ if $c > 0$.

Problem Set 1.6, page 38

1. $x^2/(c^2 + 9) + y^2/c^2 - 1 = 0$
3. $y - \cosh(x - c) - c = 0$
5. $y/x = c$, $y'/x = y/x^2$, $y' = y/x$, $\tilde{y}' = -x/\tilde{y}$, $\tilde{y}^2 + x^2 = \tilde{c}$, circles
7. $2\tilde{y}^2 - x^2 = \tilde{c}$
9. $y' = -2xy$, $\tilde{y}' = 1/(2x\tilde{y})$, $x = \tilde{c}e^{\tilde{y}^2}$
11. $\tilde{y} = \tilde{c}x$
13. $y' = -4x/9y$. Trajectories $\tilde{y}' = 9\tilde{y}/4x$, $\tilde{y} = \tilde{c}x^{9/4}$ ($\tilde{c} > 0$).
Sketch or graph these curves.
15. $u = c$, $u_x dx + u_y dy = 0$, $y' = -u_x/u_y$. Trajectories $\tilde{y}' = u_{\tilde{y}}/u_x$. Now
 $v = \tilde{c}$, $v_x dx + v_y dy = 0$, $y' = -v_x/v_y$. This agrees with the trajectory ODE
in u if $u_x = v_y$ (equal denominators) and $u_y = -v_x$ (equal numerators). But these
are just the Cauchy–Riemann equations.

Problem Set 1.7, page 42

1. $y' = f(x, y) = r(x) - p(x)y$; hence $\partial f/\partial y = -p(x)$ is continuous and is thus bounded in the closed interval $|x - x_0| \leq a$.
3. In $|x - x_0| < a$; just take b in $\alpha = b/K$ large, namely, $b = \alpha K$.
5. R has sides $2a$ and $2b$ and center $(1, 1)$ since $y(1) = 1$. In R ,
 $f = 2y^2 \leq 2(b+1)^2 = K$, $\alpha = b/K = b/(2(b+1)^2)$, $d\alpha/db = 0$ gives $b = 1$,
 and $\alpha_{\text{opt}} = b/K = \frac{1}{8}$. Solution by $dy/y^2 = 2 dx$, etc., $y = 1/(3 - 2x)$.
7. $|1 + y^2| \leq K = 1 + b^2$, $\alpha = b/K$, $d\alpha/db = 0$, $b = 1$, $\alpha = \frac{1}{2}$.
9. No. At a common point (x_1, y_1) they would both satisfy the “initial condition” $y(x_1) = y_1$, violating uniqueness.

Chapter 1 Review Questions and Problems, page 43

11. $y = ce^{-2x}$
13. $y = 1/(ce^{-4x} + 4)$
15. $y = ce^{-x} + 0.01 \cos 10x + 0.1 \sin 10x$
17. $y = ce^{-2.5x} + 0.640x - 0.256$
19. $25y^2 - 4x^2 = c$
21. $F = x, x^3 e^y + x^2 y = c$
23. $y = \sin(x + \frac{1}{4}\pi)$
25. $3 \sin x + \frac{1}{3} \sin y = 0$
27. $e^k = 1.25$, $(\ln 2)/\ln 1.25 = 3.1$, $(\ln 3)/\ln 1.25 = 4.9$ [days]
29. $e^k = 0.9$, 6.6 days. 43.7 days from $e^{kt} = 0.5$, $e^{kt} = 0.01$

Problem Set 2.1, page 53

1. $F(x, z, z') = 0$
3. $y = c_1 e^{-x} + c_2$
5. $y = (c_1 x + c_2)^{-1/2}$
7. $(dz/dy)z = -z^3 \sin y$, $-1/z = -dx/dy = \cos y + \tilde{c}_1$, $x = -\sin y + c_1 y + c_2$
9. $y_2 = x^3 \ln x$
11. $y = c_1 e^{2x} + c_2$
13. $y(t) = c_1 e^{-t} + kt + c_2$
15. $y = 3 \cos 2.5x - \sin 2.5x$
17. $y = -0.75x^{3/2} - 2.25x^{-1/2}$
19. $y = 15e^{-x} - \sin x$

Problem Set 2.2, page 59

1. $y = c_1 e^{-2.5x} + c_2 e^{2.5x}$
3. $y = c_1 e^{-2.8x} + c_2 e^{-3.2x}$
5. $y = (c_1 + c_2 x)e^{-\pi x}$
7. $y = c_1 + c_2 e^{-4.5x}$
9. $y = c_1 e^{-2.6x} + c_2 e^{0.8x}$
11. $y = c_1 e^{-x/2} + c_2 e^{3x/2}$
13. $y = (c_1 + c_2 x)e^{5x/3}$
15. $y = e^{-0.27x} (A \cos(\sqrt{\pi}x) + B \sin(\sqrt{\pi}x))$
17. $y'' + 2\sqrt{5}y' + 5y = 0$
19. $y'' + 4y' + 5y = 0$
21. $y = 4.6 \cos 5x - 0.24 \sin 5x$
23. $y = 6e^{2x} + 4e^{-3x}$
25. $y = 2e^{-x}$
27. $y = (4.5 - x)e^{-\pi x}$
29. $y = \frac{1}{\sqrt{\pi}} e^{-0.27x} \sin(\sqrt{\pi}x)$
31. Independent
33. $c_1 x^2 + c_2 x^2 \ln x = 0$ with $x = 1$ gives $c_1 = 0$; then $c_2 = 0$ for $x = 2$, say.
Hence independent
35. Dependent since $\sin 2x = 2 \sin x \cos x$
37. $y_1 = e^{-x}$, $y_2 = 0.001e^x + e^{-x}$

Problem Set 2.3, page 61

1. $4e^{2x}$, $-e^{-x} + 8e^{2x}$, $-\cos x - 2 \sin x$
3. 0, 0, $(D - 2I)(-4e^{-2x}) = 8e^{-2x} + 8e^{-2x}$
5. 0, $5e^{2x}$, 0
7. $(2D - I)(2D + I)$, $y = c_1e^{0.5x} + c_2e^{-0.5x}$
9. $(D - 2.1I)^2$, $y = (c_1 + c_2x)e^{2.1x}$
11. $(D - 1.6I)(D - 2.4I)$, $y = c_1e^{1.6x} + c_2e^{2.4x}$
15. Combine the two conditions to get $L(cy + kw) = L(cy) + L(kw) = cLy + kLw$.
The converse is simple.

Problem Set 2.4, page 69

1. $y' = y_0 \cos \omega_0 t + (v_0/\omega_0) \sin \omega_0 t$. At integer t (if $\omega_0 = \pi$), because of periodicity.
3. (i) Lower by a factor $\sqrt{2}$, (ii) higher by $\sqrt{2}$
5. 0.3183, 0.4775, $\sqrt{(k_1 + k_2)/m}/(2\pi) = 0.5738$
7. $mL\theta'' = -mg \sin \theta \approx -mg\theta$ (tangential component of $W = mg$),
 $\theta'' + \omega_0^2\theta = 0$, $\omega_0/(2\pi) = \sqrt{g/L}/(2\pi)$
9. $my'' = -\tilde{a}\gamma y$, where $m = 1$ kg, $ay = \pi \cdot 0.01^2 \cdot 2y$ meter³ is the volume of the water that causes the restoring force $a\gamma y$ with $\gamma = 9800$ nt (= weight/meter³).
 $y'' + \omega_0^2 y = 0$, $\omega_0^2 = a\gamma/m = a\gamma = 0.000628\gamma$. Frequency $\omega_0/2\pi = 0.4$ [sec⁻¹].
13. $y = [y_0 + (v_0 + \alpha y_0)t]e^{-\alpha t}$, $y = [1 + (v_0 + 1)t]e^{-t}$;
(ii) $v_0 = -2, -\frac{3}{2}, -\frac{4}{3}, -\frac{5}{4}, -\frac{6}{5}$
15. $\omega^* = [\omega_0^2 - c^2/(4m^2)]^{1/2} = \omega_0[1 - c^2/(4mk)]^{1/2} \approx \omega_0(1 - c^2/8mk) = 2.9583$
17. The positive solutions of $\tan t = 1$, that is, $\pi/4$ (max), $5\pi/4$ (min). etc
19. $0.0231 = (\ln 2)/30$ [kg/sec] from $\exp(-10 \cdot 3c/2m) = \frac{1}{2}$.

Problem Set 2.5, page 73

3. $y = (c_1 + c_2 \ln x)x^{-1.8}$
5. $\sqrt{x}(c_1 \cos(\ln x) + c_2 \sin(\ln x))$
7. $y = c_1x^2 + c_2x^3$
9. $y = (c_1 + c_2 \ln x)x^{0.6}$
11. $y = x^2(c_1 \cos(\sqrt{6} \ln x) + c_2 \sin(\sqrt{6} \ln x))$
13. $y = x^{-3/2}$
15. $y = (3.6 + 4.0 \ln x)/x$
17. $y = \cos(\ln x) + \sin(\ln x)$
19. $y = -0.525x^5 + 0.625x^{-3}$

Problem Set 2.6, page 79

3. $W = -2.2e^{-3x}$
5. $W = -x^4$
7. $W = a$
9. $y'' + 25y = 0$, $W = 5$, $y = 3 \cos 5x - \sin 5x$
11. $y'' + 5y + 6.34 = 0$, $W = 0.3e^{-5x}$, $3e^{-2.5} \cos 0.3x$
13. $y'' + 2y' = 0$, $W = -2e^{-2x}$, $y = 0.5(1 + e^{-2x})$
15. $y'' - 3.24y = 0$, $W = 1.8$, $y = 14.2 \cosh 1.8x + 9.1 \sinh 1.8x$

Problem Set 2.7, page 84

1. $y = c_1e^{-x} + c_2e^{-4x} - 5e^{-3x}$
3. $y = c_1e^{-2x} + c_2e^{-x} + 6x^2 - 18x + 21$
5. $y = (c_1 + c_2x)e^{-2x} + \frac{1}{2}e^{-x} \sin x$
7. $y = c_1e^{-x/2} + c_2e^{-3x/2} + \frac{4}{5}e^x + 6x - 16$
9. $y = c_1e^{4x} + c_2e^{-4x} + 1.2xe^{4x} - 2e^x$
11. $y = \cos(\sqrt{3}x) + 6x^2 - 4$

13. $y = e^{x/4} - 2e^{x/2} + \frac{1}{5}e^{-x} + e^x$ 15. $y = \ln x$
 17. $y = e^{-0.1x}(1.5 \cos 0.5x - \sin 0.5x) + 2e^{0.5x}$

Problem Set 2.8, page 91

3. $y_p = 1.0625 \cos 2t + 3.1875 \sin 2t$
 5. $y_p = -1.28 \cos 4.5t + 0.36 \sin 4.5t$
 7. $y_p = 25 + \frac{4}{3} \cos 3t + \sin 3t$
 9. $y = e^{-1.5t}(A \cos t + B \sin t) + 0.8 \cos t + 0.4 \sin t$
 11. $y = A \cos \sqrt{2}t + B \sin \sqrt{2}t + t(\sin \sqrt{2}t - \cos \sqrt{2}t)/(2\sqrt{2})$
 13. $y = A \cos t + B \sin t - (\cos \omega t)/(\omega^2 - 1)$
 15. $y = e^{-2t}(A \cos 2t + B \sin 2t) + \frac{1}{4} \sin 2t$
 17. $y = \frac{1}{3} \sin t - \frac{1}{15} \sin 3t - \frac{1}{105} \sin 5t$
 19. $y = e^{-t}(0.4 \cos t + 0.8 \sin t) + e^{-t/2}(-0.4 \cos \frac{1}{2}t + 0.8 \sin \frac{1}{2}t)$
 25. **CAS Experiment.** The choice of ω needs experimentation, inspection of the curves obtained, and then changes on a trial-and-error basis. It is interesting to see how in the case of beats the period gets increasingly longer and the maximum amplitude gets increasingly larger as $\omega/(2\pi)$ approaches the resonance frequency.

Problem Set 2.9, page 98

1. $RI' + I/C = 0$, $I = ce^{-t/(RC)}$
 3. $LI' + RI = E$, $I = (E/R) + ce^{-Rt/L} = 4.8 + ce^{-40t}$
 5. $I = 2(\cos t - \cos 20t)/399$
 7. I_0 is maximum when $S = 0$; thus, $C = 1/(\omega^2 L)$.
 9. $I = 0$ 11. $I = 5.5 \cos 10t + 16.5 \sin 10t$ A
 13. $I = e^{-5t}(A \cos 10t + B \sin 10t) - 400 \cos 25t + 200 \sin 25t$ A
 15. $R > R_{\text{crit}} = 2\sqrt{L/C}$ is Case I, etc.
 17. $E(0) = 600$, $I'(0) = 600$, $I = e^{-3t}(-100 \cos 4t + 75 \sin 4t) + 100 \cos t$
 19. $R = 2 \Omega$, $L = 1$ H, $C = \frac{1}{12}$ F, $E = 4.4 \sin 10t$ V

Problem Set 2.10, page 102

1. $y = A \cos 3x + B \sin 3x + \frac{1}{9}(\cos 3x) \ln |\cos 3x| + \frac{1}{3}x \sin 3x$
 3. $y = c_1 x + c_2 x^2 - x \sin x$ 5. $y = A \cos x + B \sin x + \frac{1}{2}x(\cos x + \sin x)$
 7. $y = (c_1 + c_2 x)e^{2x} + x^{-2}e^{2x}$ 9. $y = (c_1 + c_2 x)e^x + 4x^{7/2}e^x$
 11. $y = c_1 x^2 + c_2 x^3 + 1/(2x^4)$ 13. $y = c_1 x^{-3} + c_2 x^3 + 3x^5$

Chapter 2 Review Questions and Problems, page 102

7. $y = c_1 e^{-4.5x} + c_2 e^{-3.5x}$ 9. $y = e^{-3x}(A \cos 5x + B \sin 5x)$
 11. $y = (c_1 + c_2 x)e^{0.8x}$ 13. $y = c_1 x^{-4} + c_2 x^3$
 15. $y = c_1 e^{2x} + c_2 e^{-x/2} - 3x + x^2$ 17. $y = (c_1 + c_2 x)e^{1.5x} + 0.25x^2 e^{1.5x}$
 19. $y = 5 \cos 4x - \frac{3}{4} \sin 4x + e^x$ 21. $y = -4x + 2x^3 + 1/x$
 23. $I = -0.01093 \cos 415t + 0.05273 \sin 415t$ A

25. $I = \frac{1}{73}(50 \sin 4t - 110 \cos 4t) \text{ A}$
 27. RLC -circuit with $R = 20 \Omega$, $L = 4 \text{ H}$, $C = 0.1 \text{ F}$, $E = -25 \cos 4t \text{ V}$
 29. $\omega = 3.1$ is close to $\omega_0 = \sqrt{k/m} = 3$, $y = 25(\cos 3t - \cos 3.1t)$.

Problem Set 3.1, page 111

9. Linearly independent
 11. Linearly independent
 13. Linearly independent
 15. Linearly dependent

Problem Set 3.2, page 116

1. $y = c_1 + c_2 \cos 5x + c_3 \sin 5x$
 3. $y = c_1 + c_2 x + c_3 \cos 2x + c_4 \sin 2x$
 5. $y = A_1 \cos x + B_1 \sin x + A_2 \cos 3x + B_2 \sin 3x$
 7. $y = 2.398 + e^{-1.6x}(1.002 \cos 1.5x - 1.998 \sin 1.5x)$
 9. $y = 4e^{-x} + 5e^{-x/2} \cos 3x$
 11. $y = \cosh 5x - \cos 4x$
 13. $y = e^{0.25x} + 4.3e^{-0.7x} + 12.1 \cos 0.1x - 0.6 \sin 0.1x$

Problem Set 3.3, page 122

1. $y = (c_1 + c_2 x + c_3 x^2)e^{-x} + \frac{1}{8}e^x - x + 2$
 3. $y = c_1 \cos x + c_2 \sin x + c_3 \cos 3x + c_4 \sin 3x + 0.1 \sinh 2x$
 5. $y = c_1 x^2 + c_2 x + c_3 x^{-1} - \frac{1}{12}x^{-2}$
 7. $y = (c_1 + c_2 x + c_3 x^2)e^{3x} - \frac{1}{4}(\cos 3x - \sin 3x)$
 9. $y = \cos x + \frac{1}{2} \sin 4x$
 11. $y = e^{-3x}(-1.4 \cos x - \sin x)$
 13. $y = 2 - 2 \sin x + \cos x$

Chapter 3 Review Questions and Problems, page 122

7. $y = c_1 + e^{-2x}(A \cos 3x + B \sin 3x)$
 9. $y = c_1 \cosh 2x + c_2 \sinh 2x + c_3 \cos 2x + c_4 \sin 2x + \cosh x$
 11. $y = (c_1 + c_2 x + c_3 x^2)e^{-1.5x}$
 13. $y = (c_1 + c_2 x + c_3 x^2)e^{-2x} + x^2 - 3x + 3$
 15. $y = c_1 x + c_2 x^{1/2} + c_3 x^{3/2} - \frac{10}{3}$
 17. $y = 2e^{-2x} \cos 4x + 0.05x - 0.06$
 19. $y = 4e^{-4x} + 5e^{-5x}$

Problem Set 4.1, page 136

1. Yes
 5. $y_1' = 0.02(-y_1 + y_2)$, $y_2' = 0.02(y_1 - 2y_2 + y_3)$, $y_3' = 0.02(y_2 - y_3)$
 7. $c_1 = 1$, $c_2 = -5$
 9. $c_1 = 10$, $c_2 = 5$
 11. $y_1' = y_2$, $y_2' = y_1 + \frac{15}{4}y_2$, $\mathbf{y} = c_1 \begin{bmatrix} 1 & 4 \end{bmatrix}^T e^{4t} + c_2 \begin{bmatrix} 1 & -\frac{1}{4} \end{bmatrix}^T e^{-t/4}$
 13. $y_1' = y_2$, $y_2' = 24y_1 - 2y_2$, $y_1 = c_1 e^{4t} + c_2 e^{-6t} = y$, $y_2 = y'$
 15. (a) For example, $C = 1000$ gives -2.39993 , -0.000167 . (b) -2.4 , 0 .
 (d) $a_{22} = -4 + 2\sqrt{6.4} = 1.05964$ gives the critical case. C about 0.18506 .

Problem Set 4.3, page 147

1. $y_1 = c_1 e^{-2t} + c_2 e^{2t}$, $y_2 = -3c_1 e^{-2t} + c_2 e^{2t}$
3. $y_1 = 2c_1 e^{2t} + 2c_2$, $y_2 = c_1 e^{2t} - c_2$
5. $y_1 = 5c_1 + 2c_2 e^{14.5t}$
 $y_2 = -2c_1 + 5c_2 e^{14.5t}$
7. $y_1 = -c_2 \cos \sqrt{2}t + c_3 \sin \sqrt{2}t + c_1$
 $y_2 = c_2 \sqrt{2} \sin \sqrt{2}t + c_3 \sqrt{2} \cos \sqrt{2}t$
 $y_3 = c_2 \cos \sqrt{2}t - c_3 \sin \sqrt{2}t + c_1$
9. $y_1 = \frac{1}{2}c_1 e^{-18t} + 2c_2 e^{9t} - c_3 e^{18t}$
 $y_2 = c_1 e^{-18t} + c_2 e^{9t} + c_3 e^{18t}$
 $y_3 = c_1 e^{-18t} - 2c_2 e^{9t} - \frac{1}{2}c_3 e^{18t}$
11. $y_1 = -20e^t + 8e^{-t/2}$
 $y_2 = 4e^t - 4e^{-t/2}$
13. $y_1 = 2 \sinh t$, $y_2 = 2 \cosh t$
15. $y_1 = \frac{1}{2}e^t$
 $y_2 = \frac{1}{2}e^t$
17. $y_2 = y_1' + y_1$, $y_2' = y_1'' + y_1' = -y_1 - y_2 = -y_1 - (y_1' + y_1)$,
 $y_1'' + 2y_1' + 2y_1 = 0$, $y_1 = e^{-t}(A \cos t + B \sin t)$,
 $y_2 = y_1' + y_1 = e^{-t}(B \cos t - A \sin t)$. Note that $r^2 = y_1^2 + y_2^2 = e^{-2t}(A^2 + B^2)$.
19. $I_1 = c_1 e^{-t} + 3c_2 e^{-3t}$, $I_2 = -3c_1 e^{-t} - c_2 e^{-3t}$

Problem Set 4.4, page 151

1. Unstable improper node, $y_1 = c_1 e^t$, $y_2 = c_2 e^{2t}$
3. Center, always stable, $y_1 = A \cos 3t + B \sin 3t$, $y_2 = 3B \cos 3t - 3A \sin 3t$
5. Stable spiral, $y_1 = e^{-2t}(A \cos 2t + B \sin 2t)$, $y_2 = e^{-2t}(B \cos 2t - A \sin 2t)$
7. Saddle point, always unstable, $y_1 = c_1 e^{-t} + c_2 e^{3t}$, $y_2 = -c_1 e^{-t} + c_2 e^{3t}$
9. Unstable node, $y_1 = c_1 e^{6t} + c_2 e^{2t}$, $y_2 = 2c_1 e^{6t} - 2c_2 e^{2t}$
11. $y = e^{-t}(A \cos t + B \sin t)$. Stable and attractive spirals
15. $p = 0.2 \neq 0$ (was 0), $\Delta < 0$, spiral point, unstable.
17. For instance, (a) -2 , (b) -1 , (c) $-\frac{1}{2}$, (d) $=1$, (e) 4 .

Problem Set 4.5, page 159

5. Center at $(0, 0)$. At $(2, 0)$ set $y_1 = 2 + \tilde{y}_1$. Then $\tilde{y}_2' = \tilde{y}_1$. Saddle point at $(2, 0)$.
7. $(0, 0)$, $y_1' = -y_1 + y_2$, $y_2' = -y_1 - y_2$, stable and attractive spiral point; $(-2, 2)$,
 $y_1 = -2 + \tilde{y}_1$, $y_2 = 2 + \tilde{y}_2$, $\tilde{y}_1' = -\tilde{y}_1 - 3\tilde{y}_2$, $\tilde{y}_2' = -\tilde{y}_1 - \tilde{y}_2$, saddle point
9. $(0, 0)$ saddle point, $(-3, 0)$ and $(3, 0)$ centers
11. $(\frac{1}{2}\pi \pm 2n\pi, 0)$ saddle points; $(-\frac{1}{2}\pi \pm 2n\pi, 0)$ centers.
Use $-\cos(\pm\frac{1}{2}\pi + \tilde{y}_1) = \sin(\pm\tilde{y}_1) \approx \pm\tilde{y}_1$.
13. $(\pm 2n\pi, 0)$ centers; $y_1 = (2n + 1)\pi + \tilde{y}_1$, $(\pi \pm 2n\pi, 0)$ saddle points
15. By multiplication, $y_2 y_2' = (4y_1 - y_1^3)y_1'$. By integration,
 $y_2^2 = 4y_1^2 - \frac{1}{2}y_1^4 + c^* = \frac{1}{2}(c + 4 - y_1^2)(c - 4 + y_1^2)$, where $c^* = \frac{1}{2}c^2 - 8$.

Problem Set 4.6, page 163

3. $y_1 = c_1 e^{-t} + c_2 e^t$, $y_2 = -c_1 e^{-t} + c_2 e^t - e^{3t}$
5. $y_1 = c_1 e^{5t} + c_2 e^{2t} - 0.43t - 0.24$, $y_2 = c_1 e^{5t} - 2c_2 e^{2t} + 1.12t + 0.53$

7. $y_1 = c_1 e^t + 4c_2 e^{2t} - 3t - 4 - 2e^{-t}$, $y_2 = -c_1 e^t - 5c_2 e^{2t} + 5t + 7.5 + e^{-t}$
9. The formula for $\mathbf{v}$ shows that these various choices differ by multiples of the eigenvector for $\lambda = -2$, which can be absorbed into, or taken out of, c_1 in the general solution $\mathbf{y}^{(h)}$.
11. $y_1 = -\frac{8}{3} \cosh t - \frac{4}{3} \sinh t + \frac{11}{3} e^{2t}$, $y_2 = -\frac{8}{3} \sinh t - \frac{4}{3} \cosh t + \frac{4}{3} e^{2t}$
13. $y_1 = \cos 2t + \sin 2t + 4 \cos t$, $y_2 = 2 \cos 2t - 2 \sin 2t + \sin t$
15. $y_1 = 4e^{-t} - 4e^t + e^{2t}$, $y_2 = -4e^{-t} + t$
17. $I_1 = 2c_1 e^{\lambda_1 t} + 2c_2 e^{\lambda_2 t} + 100$,
 $I_2 = (1.1 + \sqrt{0.41})c_1 e^{\lambda_1 t} + (1.1 - \sqrt{0.41})c_2 e^{\lambda_2 t}$,
 $\lambda_1 = -0.9 + \sqrt{0.41}$, $\lambda_2 = -0.9 - \sqrt{0.41}$
19. $c_1 = 17.948$, $c_2 = -67.948$

Chapter 4 Review Questions and Problems, page 164

11. $y_1 = c_1 e^{4t} + c_2 e^{-4t}$, $y_2 = 2c_1 e^{4t} - 2c_2 e^{-4t}$. Saddle point
13. $y_1 = e^{-4t}(A \cos t + B \sin t)$, $y_2 = \frac{1}{5} e^{-4t}[(B - 2A) \cos t - (A + 2B) \sin t]$; asymptotically stable spiral point
15. $y_1 = c_1 e^{-5t} + c_2 e^{-t}$, $y_2 = c_1 e^{-5t} - c_2 e^{-t}$. Stable node
17. $y_1 = e^{-t}(A \cos 2t + B \sin 2t)$, $y_2 = e^{-t}(B \cos 2t - A \sin 2t)$. Stable and attractive spiral point
19. Unstable spiral point
21. $y_1 = c_1 e^{-4t} + c_2 e^{4t} - 1 - 8t^2$, $y_2 = -c_1 e^{-4t} + c_2 e^{4t} - 4t$
23. $y_1 = 2c_1 e^{-t} + 2c_2 e^{3t} + \cos t - \sin t$, $y_2 = -c_1 e^{-t} + c_2 e^{3t}$
25. $I_1' + 2.5(I_1 - I_2) = 169 \sin t$, $2.5(I_2' - I_1') + 25I_2 = 0$,
 $I_1 = (19 + 32.5t)e^{-5t} - 19 \cos t + 62.5 \sin t$,
 $I_2 = (-6 - 32.5t)e^{-5t} + 6 \cos t + 2.5 \sin t$
27. $(0, 0)$ saddle point; $(-1, 0)$, $(1, 0)$ centers
29. $(n\pi, 0)$ center when n is even and saddle point when n is odd

Problem Set 5.1, page 174

3. $\sqrt{|k|}$
5. $\sqrt{3/2}$
7. $y = a_0(1 - x^2 + x^4/2! - x^6/3! + \dots) = a_0 e^{-x^2}$
9. $y = a_0 + a_1 x - \frac{1}{2} a_0 x^2 - \frac{1}{6} a_1 x^3 + \dots = a_0 \cos x + a_1 \sin x$
11. $a_0(1 - \frac{1}{12}x^4 - \frac{1}{60}x^5 - \dots) + a_1(x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4 - \frac{1}{24}x^5 - \dots)$
13. $a_0(1 - \frac{1}{2}x^2 - \frac{1}{24}x^4 + \frac{13}{720}x^6 + \dots) + a_1(x - \frac{1}{6}x^3 - \frac{1}{24}x^5 + \frac{5}{1008}x^7 + \dots)$
15. $\sum_{m=1}^{\infty} \frac{(m+1)(m+2)}{(m+1)^2 + 1} x^m$, $\sum_{m=5}^{\infty} \frac{(m-4)^2}{(m-3)!} x^m$
17. $s = 1 + x - x^2 - \frac{5}{6}x^3 + \frac{2}{3}x^4 + \frac{11}{24}x^5$, $s(\frac{1}{2}) = \frac{923}{768}$
19. $s = 4 - x^2 - \frac{1}{3}x^3 + \frac{1}{30}x^5$, $s(2) = -\frac{8}{5}$; but $x = 2$ is too large to give good values. Exact: $y = (x - 2)^2 e^x$

Problem Set 5.2, page 179

5. $P_6(x) = \frac{1}{16}(231x^6 - 315x^4 + 105x^2 - 5)$,
 $P_7(x) = \frac{1}{16}(429x^7 - 693x^5 + 315x^3 - 35x)$

11. Set $x = az$. $y = c_1 P_n(x/a) + c_2 Q_n(x/a)$
 15. $P_1^1 = \sqrt{1-x^2}$, $P_2^1 = 3x\sqrt{1-x^2}$, $P_2^2 = 3(1-x^2)$,
 $P_4^2 = (1-x^2)(105x^2-15)/2$

Problem Set 5.3, page 186

3. $y_1 = 1 - \frac{x^2}{3!} + \frac{x^4}{5!} - + \cdots = \frac{\sin x}{x}$, $y_2 = \frac{1}{x} - \frac{x}{2!} + \frac{x^3}{4!} - + \cdots = \frac{\cos x}{x}$
 5. $b_0 = 1$, $c_0 = 0$, $r^2 = 0$, $y_1 = e^{-x}$, $y_2 = e^{-x} \ln x$
 7. $y_1 = 1 + \frac{1}{2}x^2 - \frac{1}{6}x^3 + \frac{1}{24}x^4 - \frac{1}{30}x^5 + \frac{1}{144}x^6 - \cdots$,
 $y_2 = x + \frac{1}{6}x^3 - \frac{1}{12}x^4 + \frac{1}{120}x^5 - \frac{1}{120}x^6 + \cdots$
 9. $y_1 \sqrt{x}$, $y_2 = 1 + x$
 11. $y_1 = e^x$, $y_2 = e^x/x$
 13. $y_1 = e^x$, $y_2 = e^x \ln x$
 15. $y = AF(1, 1, -\frac{1}{2}; x) + Bx^{3/2}F(\frac{5}{2}, \frac{5}{2}, \frac{5}{2}; x)$
 17. $y = A(1 - 8x + \frac{32}{5}x^2) + Bx^{3/4}F(\frac{7}{4}, -\frac{5}{4}, \frac{7}{4}; x)$
 19. $y = c_1 F(2, -2, -\frac{1}{2}; t-2) + c_2 (t-2)^{3/2} F(\frac{7}{2}, -\frac{1}{2}, \frac{5}{2}; t-2)$

Problem Set 5.4, page 195

3. $c_1 J_0(\sqrt{x})$
 5. $c_1 J_\nu(\lambda x) + c_2 J_{-\nu}(\lambda x)$, $\nu \neq 0, \pm 1, \pm 2, \cdots$
 7. $c_1 J_{1/2}(\frac{1}{2}x) + c_2 J_{-1/2}(\frac{1}{2}x) = x^{-1/2}(\tilde{c}_1 \sin \frac{1}{2}x + \tilde{c}_2 \cos \frac{1}{2}x)$
 9. $x^{-\nu}(c_1 J_\nu(x) + c_2 J_{-\nu}(x))$, $\nu \neq 0, \pm 1, \pm 2, \cdots$
 13. $J_n(x_1) = J_n(x_2) = 0$ implies $x_1^{-n} J_n(x_1) = x_2^{-n} J_n(x_2) = 0$ and $[x^{-n} J_n(x)]' = 0$ somewhere between x_1 and x_2 by Rolle's theorem.
 Now use (21b) to get $J_{n+1}(x) = 0$ there. Conversely, $J_{n+1}(x_3) = J_{n+1}(x_4) = 0$, thus $x_3^{n+1} J_{n+1}(x_3) = x_4^{n+1} J_{n+1}(x_4) = 0$ implies $J_n(x) = 0$ in between by Rolle's theorem and (21a) with $\nu = n+1$.
 15. By Rolle, $J'_0 = 0$ at least once between two zeros of J_0 . Use $J'_0 = -J_1$ by (21b) with $\nu = 0$. Together $J_1 = 0$ at least once between two zeros of J_0 . Also use $(xJ_1)' = xJ_0$ by (21a) with $\nu = 1$ and Rolle.
 19. Use (21b) with $\nu = 0$, (21a) with $\nu = 1$, (21d) with $\nu = 2$, respectively.
 21. Integrate (21a).
 23. Use (21a) with $\nu = 1$, partial integration, (21b) with $\nu = 0$, partial integration.
 25. Use (21d) to get

$$\begin{aligned} \int J_5(x) dx &= -2J_4(x) + \int J_3(x) dx = -2J_4(x) - 2J_2(x) + \int J_1(x) dx \\ &= -2J_4(x) - 2J_2(x) - J_0(x) + c. \end{aligned}$$

Problem Set 5.5, page 200

1. $c_1 J_4(x) + c_2 Y_4(x)$
 3. $c_1 J_{2/3}(x^2) + c_2 Y_{2/3}(x^2)$
 5. $c_1 J_0(\sqrt{x}) + c_2 Y_0(\sqrt{x})$

$$7. \sqrt{x} (c_1 J_{1/4}(\tfrac{1}{2} kx^2) + c_2 Y_{1/4}(\tfrac{1}{2} kx^2))$$

$$9. x^3 (c_1 J_3(x) + c_2 Y_3(x))$$

$$11. \text{Set } H^{(1)} = kH^{(2)} \text{ and use (10).}$$

$$13. \text{Use (20) in Sec. 5.4.}$$

Chapter 5 Review Questions and Problems, page 200

$$11. \cos 2x, \sin 2x$$

$$13. (x-1)^{-5}, (x-1)^7; \text{ Euler-Cauchy with } x-1 \text{ instead of } x$$

$$15. J_{\sqrt{x}}(x), J_{-\sqrt{x}}(x)$$

$$17. e^x, 1+x$$

$$19. \sqrt{x} J_1(\sqrt{x}), \sqrt{x} Y_1(\sqrt{x})$$

Problem Set 6.1, page 210

$$1. 3/s^2 + 12/s$$

$$5. 1/((s-2)^2 - 1)$$

$$9. \frac{1}{s} + \frac{e^{-s} - 1}{s^2}$$

$$13. \frac{(1 - e^{-s})^2}{s}$$

$$19. \text{Use } e^{at} = \cosh at + \sinh at.$$

$$23. \text{Set } ct = p. \text{ Then } \mathcal{L}(f(ct)) = \int_0^\infty e^{-st} f(ct) dt = \int_0^\infty e^{-(s/c)p} f(p) dp/c = F(s/c)/c.$$

$$25. 0.2 \cos 1.8t + \sin 1.8t$$

$$29. 2t^3 - 1.9t^5$$

$$33. \frac{2}{(s+3)^3}$$

$$37. \pi t e^{-\pi t}$$

$$41. e^{-5\pi t} \sinh \pi t$$

$$45. (k_0 + k_1 t) e^{-at}$$

$$3. s/(s^2 + \pi^2)$$

$$7. (\omega \cos \theta + s \sin \theta)/(s^2 + \omega^2)$$

$$11. \frac{1 - e^{-bs}}{s^2} - \frac{be^{-bs}}{s}$$

$$15. \frac{e^{-s} - 1}{2s^2} - \frac{e^{-s}}{2s} + \frac{1}{s}$$

$$27. \frac{1}{L^2} \cos \frac{n\pi t}{L}$$

$$31. \mathcal{L}^{-1} \left(\frac{4}{s-2} - \frac{3}{s+1} \right) = 4e^{2t} - 3e^{-t}$$

$$35. \frac{0.5 \cdot 2\pi}{(s+4.5)^2 + 4\pi^2}$$

$$39. \frac{7}{2} t^3 e^{-t\sqrt{2}}$$

$$43. e^{3t} (2 \cos 3t + \frac{5}{3} \sin 3t)$$

Problem Set 6.2, page 216

$$1. y = 1.25e^{-5.2t} - 1.25 \cos 2t + 3.25 \sin 2t$$

$$3. (s-3)(s+2) = 11s + 28 - 11 = 11s + 17, \quad Y = 10/(s-3) + 1/(s+2), \\ y = 10e^{3t} + e^{-2t}$$

$$5. (s^2 - \frac{1}{4})Y = 12s, \quad y = 12 \cosh \frac{1}{2}t$$

$$7. y = \frac{1}{2}e^{3t} + \frac{5}{2}e^{-4t} + \frac{1}{2}e^{-3t} \quad 9. y = e^t - e^{3t} + 2t$$

$$11. (s+1.5)^2 Y = s + 31.5 + 3 + 54/s^4 + 64/s, \\ Y = 1/(s+1.5) + 1/(s+1.5)^2 + 24/s^4 - 32/s^3 + 32/s^2, \\ y = (1+t)e^{-1.5t} + 4t^3 - 16t^2 + 32t$$

$$13. t = \tilde{t} - 1, \quad \tilde{Y} = 4/(s-6), \quad \tilde{y} = 4e^{6t}, \quad y = 4e^{6(t+1)}$$

15. $t = \tilde{t} + 1.5$, $(s - 1)(s + 4)\tilde{Y} = 4s + 17 + 6/(s - 2)$, $y = 3e^{t-1.5} + e^{2(t-1.5)}$
17. $\frac{1}{(s + a)^2}$ 19. $\frac{2\omega^2}{s(s^2 + 4\omega^2)}$
21. $\mathcal{L}(f') = \mathcal{L}(\sinh 2t) = s\mathcal{L}(f) - 1$. Answer: $(s^2 - 2)/(s^3 - 4s)$
23. $12(1 - e^{-t/4})$ 25. $(1 - \cos \omega t)/\omega^2$
27. $\frac{1}{9}(1 + t - \cos 3t - \frac{1}{3}\sin 3t)$ 29. $\frac{1}{a^2}(e^{-at} - 1) + \frac{t}{a}$

Problem Set 6.3, page 223

3. $\mathcal{L}((t - 2)u(t - 2)) = e^{-2s}/s^2$
5. $\left(e^t \left(1 - u \left(t - \frac{1}{2}\pi \right) \right) \right) = \frac{1}{s - 1} (1 - e^{-\pi s/2 + \pi/2})$
7. $\frac{1}{s + \pi} (e^{-2(s + \pi)} - e^{-4(s + \pi)})$
9. $e^{-3s/2} \left(\frac{2}{s^3} + \frac{3}{s^2} + \frac{9}{s} \right)$
11. $(se^{-\pi s/2} + e^{-\pi s})/(s^2 + 1)$ 13. $2[1 + u(t - \pi)] \sin 3t$
15. $(t - 3)^3 u(t - 3)/6$ 17. $e^{-t} \cos t$ ($0 < t < 2\pi$)
19. $\frac{1}{3}(e^t - 1)^3 e^{-5t}$ 21. $\sin 3t + \sin t$ ($0 < t < \pi$); $\frac{4}{3} \sin 3t$ ($t > \pi$)
23. $e^t - \sin t$ ($0 < t < 2\pi$), $e^t - \frac{1}{2} \sin 2t$ ($t > 2\pi$)
25. $t - \sin t$ ($0 < t < 1$), $\cos(t - 1) + \sin(t - 1) - \sin t$ ($t > 1$)
27. $t = 1 + \tilde{t}$, $\tilde{y}'' + 4\tilde{y} = 8(1 + \tilde{t})^2(1 - u(\tilde{t} - 4))$, $\cos 2t + 2t^2 - 1$ if $t < 5$,
 $\cos 2t + 49 \cos(2t - 10) + 10 \sin(2t - 10)$ if $t > 5$
29. $0.1i' + 25i = 490e^{-5t}[1 - u(t - 1)]$,
 $i = 20(e^{-5t} - e^{-250t}) + 20u(t - 1)[-e^{-5t} + e^{-250t + 245}]$
31. $Rq' + q/C = 0$, $Q = \mathcal{L}(q)$, $q(0) = CV_0$, $i = q'(t)$,
 $R(sQ - CV_0) + Q/C = 0$, $q = CV_0 e^{-t/(RC)}$
33. $10I + \frac{100}{s}I = \frac{100}{s^2}e^{-2s}$, $I = e^{-2s} \left(\frac{1}{s} - \frac{1}{s + 10} \right)$, $i = 0$ if $t < 2$ and
 $1 - e^{-10(t-2)}$ if $t > 2$
35. $i = (10 \sin 10t + 100 \sin t)(u(t - \pi) - u(t - 3\pi))$
37. $(0.5s^2 + 20)I = 78s(1 + e^{-\pi s})/(s^2 + 1)$,
 $i = 4 \cos t - 4 \cos \sqrt{40}t - 4u(t - \pi)[\cos t + \cos(\sqrt{40}(t - \pi))]$
39. $i' + 2i + 2 \int_0^t i(\tau) d\tau = 1000(1 - u(t - 2))$, $I = 1000(1 - e^{-2s})/(s^2 + 2s + 2)$,
 $i = 1000e^{-t} \sin t - 1000u(t - 2)e^{-t+2} \sin(t - 2)$

Problem Set 6.4, page 230

3. $y = 8 \cos 2t + \frac{1}{2}u(t - \pi) \sin 2t$
5. $\sin t$ ($0 < t < \pi$); 0 ($\pi < t < 2\pi$); $-\sin t$ ($t > 2\pi$)
7. $y = e^{-t} + 4e^{-3t} \sin \frac{1}{2}t + \frac{1}{2}u(t - \frac{1}{2})e^{-3(t-1/2)} \sin(\frac{1}{2}t - \frac{1}{4})$
9. $y = 0.1[e^t + e^{-2t}(-\cos t + 7 \sin t)] + 0.1u(t - 10)[-e^{-t} + e^{-2t+30}(\cos(t - 10) - 7 \sin(t - 10))]$

11. $y = -e^{-3t} + e^{-2t} + \frac{1}{6}u(t-1)(1 - 3e^{-2(t-1)} + 2e^{-3(t-1)}) + u(t-2)(e^{-2(t-2)} - e^{-3(t-2)})$

15. $ke^{-ps}/(s - se^{-ps})$ ($s > 0$)

Problem Set 6.5, page 237

1. *t*

3. $(e^t - e^{-t})/2 = \sinh t$

5. $\frac{1}{2} t \sin \omega t$

7. $e^t - t - 1$

9. $y - 1 * y = 1, \quad y = e^t$

11. $y = \cos t$

13. $y(t) + 2 \int_0^t e^{t-\tau} y(\tau) d\tau = te^t, \quad y = \sinh t$

17. $e^{4t} - e^{-1.5t}$

19. $t \sin \pi t$

21. $(\omega t - \sin \omega t)/\omega^2$

23. $4.5(\cosh 3t - 1)$

25. $1.5t \sin 6t$

Problem Set 6.6, page 241

$$3. \frac{\frac{1}{2}}{(s+3)^2}$$

5. $\frac{s^2 - \omega^2}{(s^2 + \omega^2)^2}$

7. $\frac{2s^3 + 24s}{(s^2 - 4)^3}$

9. $\frac{\pi(3s^2 - \pi^2)}{(s^2 + \pi^2)^3}$

11. $\frac{4s^2 - \pi^2}{(s^2 + \frac{1}{4}\pi^2)^2}$

15. $F(s) = -\frac{1}{2}\left(\frac{1}{s^2 - 9}\right)', \quad f(t) = \frac{1}{6}t \sinh 3t$

17. $\ln s - \ln (s - 1); \quad (-1 + e^t)/t$

19. $[\ln(s^2 + 1) - 2 \ln(s - 1)]' = 2s/(s^2 + 1) - 2/(s - 1); \quad 2(-\cos t + e^t)/t$

Problem Set 6.7, page 246

3. $y_1 = -e^{-5t} + 4e^{2t}$, $y_2 = e^{-5t} + 3e^{2t}$

5. $y_1 = -\cos t + \sin t + 1 + u(t-1)[-1 + \cos(t-1) - \sin(t-1)]$
 $y_2 = \cos t + \sin t - 1 + u(t-1)[1 - \cos(t-1) - \sin(t-1)]$

7. $y_1 = -e^{-2t} + 4e^t + \frac{1}{3}u(t-1)(-e^{3-2t} + e^t)$,
 $y_2 = -e^{-2t} + e^t + \frac{1}{3}u(t-1)(-e^{3-2t} + e^t)$

9. $y_1 = (3 + 4t)e^{3t}$, $y_2 = (1 - 4t)e^{3t}$

11. $y_1 = e^t + e^{2t}, \quad y_2 = e^{2t}$

13. $y_1 = -4e^t + \sin 10t + 4 \cos t$, $y_2 = 4e^t - \sin 10t + 4 \cos t$

15. $y_1 = e^t, \quad y_2 = e^{-t}, \quad y_3 = e^t - e^{-t}$

19. $4i_1 + 8(i_1 - i_2) + 2i_1' = 390 \cos t, \quad 8i_2 + 8(i_2 - i_1) + 4i_2' = 0,$
 $i_1 = -26e^{-2t} - 16e^{-8t} + 42 \cos t + 15 \sin t,$
 $i_2 = -26e^{-2t} + 8e^{-8t} + 18 \cos t + 12 \sin t$

Chapter 6 Review Questions and Problems, page 251

11. $\frac{5s}{s^2 - 4} - \frac{3}{s^2 - 1}$

13. $\frac{1}{2}(1 - \cos \pi t)$, $\pi^2/(2s^3 + 2\pi^2s)$

15. $e^{-3s+3/2}/(s - \frac{1}{2})$

17. Sec. 6.6; $2s^2/(s^2 + 1)^2$

19. $12/(s^2(s+3))$
 23. $\sin(\omega t + \theta)$
 27. $e^{-2t}(3 \cos t - 2 \sin t)$
 31. $e^{-t} + u(t - \pi)[1.2 \cos t - 3.6 \sin t + 2e^{-t+\pi} - 0.8e^{2t-2\pi}]$
 33. 0 ($0 \leq t \leq 2$), $1 - 2e^{-(t-2)} + e^{-2(t-2)}$ ($t > 2$)
 35. $y_1 = 4e^t - e^{-2t}$, $y_2 = e^t - e^{-2t}$
 37. $y_1 = \cos t - u(t - \pi) \sin t + 2u(t - 2\pi) \sin^2 \frac{1}{2}t$,
 $y_2 = -\sin t - 2u(t - \pi) \cos^2 \frac{1}{2}t + u(t - 2\pi) \sin t$
 39. $y_1 = (1/\sqrt{10}) \sin \sqrt{10}t$, $y_2 = -(1/\sqrt{10}) \sin \sqrt{10}t$
 41. $1 - e^{-t}$ ($0 < t < 4$), $(e^4 - 1)e^{-t}$ ($t > 4$)
 43. $i(t) = e^{-4t}(\frac{3}{26} \cos 3t - \frac{10}{39} \sin 3t) - \frac{3}{26} \cos 10t + \frac{8}{65} \sin 10t$
 45. $5i_1' + 20(i_1 - i_2) = 60$, $30i_2' + 20(i_2' - i_1') + 20i_2 = 0$,
 $i_1 = -8e^{-2t} + 5e^{-0.8t} + 3$, $i_2 = -4e^{-2t} + 4e^{-0.8t}$

Problem Set 7.1, page 261

3. 3×3 , 3×4 , 3×6 , 2×2 , 2×3 , 3×2
 5. $\mathbf{B} = \frac{1}{5}\mathbf{A}$, $\frac{1}{10}\mathbf{A}$
 7. No, no, yes, no, no
 9. $\begin{bmatrix} 0 & 6 & 12 \\ 18 & 15 & 15 \\ 3 & 0 & -9 \end{bmatrix}$, $\begin{bmatrix} 0 & 2.5 & 1 \\ 2.5 & 1.5 & 2 \\ -1 & 2 & -1 \end{bmatrix}$, $\begin{bmatrix} 0 & 8.5 & 13 \\ 20.5 & 16.5 & 17 \\ 2 & 2 & -10 \end{bmatrix}$, undefined
 11. $\begin{bmatrix} 0 & 26 \\ 34 & 32 \\ 28 & -10 \end{bmatrix}$, same, $\begin{bmatrix} 5.4 & 0.6 \\ -4.2 & 2.4 \\ -0.6 & 0.6 \end{bmatrix}$, same
 13. $\begin{bmatrix} 70 & 28 \\ -28 & 56 \\ 14 & 0 \end{bmatrix}$, same, $-\mathbf{D}$, undefined
 15. $\begin{bmatrix} 5.5 \\ 33.0 \\ -11.0 \end{bmatrix}$, same, undefined, undefined
 17. $\begin{bmatrix} -4.5 \\ -27.0 \\ 9.0 \end{bmatrix}$

Problem Set 7.2, page 270

5. $10, n(n+1)/2$
 7. $\mathbf{0}$, $\mathbf{I}$, $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$

$$11. \begin{bmatrix} 10 & -14 & -6 \\ -5 & 7 & -12 \\ -5 & -1 & -4 \end{bmatrix}, \text{ same, } \begin{bmatrix} 10 & -5 & -15 \\ -14 & 7 & -33 \\ -2 & -4 & -4 \end{bmatrix}, \text{ same}$$

$$13. \begin{bmatrix} 1 & 2 & 0 \\ 2 & 13 & -6 \\ 0 & -6 & 4 \end{bmatrix}, \begin{bmatrix} -9 & -5 \\ 3 & -1 \\ 4 & 0 \end{bmatrix}, \text{ undefined, } \begin{bmatrix} -9 & 3 & 4 \\ -5 & -1 & 0 \end{bmatrix}$$

$$15. \text{ Undefined, } \begin{bmatrix} 8 \\ -4 \\ -3 \end{bmatrix}, [7 \quad -1 \quad 3], \text{ same}$$

$$17. \begin{bmatrix} -30 & -18 \\ 45 & 9 \\ 5 & -7 \end{bmatrix}, \text{ undefined, } \begin{bmatrix} 22 \\ 4 \\ -12 \end{bmatrix}, \text{ undefined}$$

$$19. \text{ Undefined, } \begin{bmatrix} 10.5 \\ 0 \\ -3 \end{bmatrix}, \begin{bmatrix} 7 \\ -3 \\ 1 \end{bmatrix}, \text{ same}$$

$$25. (d) \mathbf{AB} = (\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T = \mathbf{BA}; \text{ etc.}$$

$$(e) \text{ Answer. If } \mathbf{AB} = -\mathbf{BA}.$$

$$29. \mathbf{p} = [85 \quad 62 \quad 30]^T, \quad \mathbf{v} = [44,920 \quad 30,940]^T$$

Problem Set 7.3, page 280

$$1. x = -2, \quad y = 0.5$$

$$3. x = 1, \quad y = 3, \quad z = -5$$

$$5. x = 6, \quad y = -7$$

$$7. x = -3t, \quad y = t \text{ arb.}, \quad z = 2t$$

$$9. x = 3t - 1, \quad y = -t + 4, \quad z = t \text{ arb.}$$

$$11. w = 1, \quad x = t_1 \text{ arb.}, \quad y = 2t_2 - t_1, \quad z = t_2 \text{ arb.}$$

$$13. w = 4, \quad x = 0, \quad y = 2, \quad z = 6 \quad 17. I_1 = 2, \quad I_2 = 6, \quad I_3 = 8$$

$$19. I_1 = (R_1 + R_2)E_0/(R_1R_2) \text{ A}, \quad I_2 = E_0/R_1 \text{ A}, \quad I_3 = E_0/R_2 \text{ A}$$

$$21. x_2 = 1600 - x_1, \quad x_3 = 600 + x_1, \quad x_4 = 1000 - x_1. \text{ No}$$

$$23. \text{ C: } 3x_1 - x_3 = 0, \quad \text{H: } 8x_1 - 2x_4 = 0, \quad \text{O: } 2x_2 - 2x_3 - x_4 = 0, \quad \text{thus} \\ \text{C}_3\text{H}_8 + 5\text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$$

Problem Set 7.4, page 287

$$1. 1; [2 \quad -1 \quad 3]; [2 \quad -1]^T \quad 3. 3; \{[3 \quad 5 \quad 0], [0 \quad 3 \quad 5], [0 \quad 0 \quad 1]\}$$

$$5. 3; \{[2 \quad -1 \quad 4], [0 \quad 1 \quad -46], [0 \quad 0 \quad 1]\}; \{[2 \quad 0 \quad 1], [0 \quad 3 \quad 23], [0 \quad 0 \quad 1]\}$$

7. 2; $[8 \ 0 \ 4 \ 0]$, $[0 \ 2 \ 0 \ 4]$; $[8 \ 0 \ 4]$, $[0 \ 2 \ 0]$
 9. 3; $[9 \ 0 \ 1 \ 0]$, $[0 \ 9 \ 8 \ 9]$, $[0 \ 0 \ 1 \ 0]$
 11. (c) 1
 19. Yes
 23. Yes
 27. 2, $[-2 \ 0 \ 1]$, $[0 \ 2 \ 1]$
 29. No
 33. 1, solution of the given system $c[1 \ \frac{10}{3} \ 3]$, basis $[1 \ \frac{10}{3} \ 3]$
 35. 1, $[4 \ 2 \ \frac{4}{3} \ 1]$

17. No

21. No

25. Yes

31. No

Problem Set 7.7, page 300

7. $\cos(\alpha + \beta)$
 11. 40
 15. -64
 19. 2
 23. $x = 0$, $y = 4$, $z = -1$
9. 1
 13. 289
 17. 2
 21. $x = 3.5$, $y = -1.0$
 25. $w = 3$, $x = 0$, $y = 2$, $z = -2$

Problem Set 7.8, page 308

1. $\begin{bmatrix} 1.20 & 4.64 \\ 0.50 & 3.60 \end{bmatrix}$
3. $\begin{bmatrix} 54 & 0.9 & -3.4 \\ 2 & 0.2 & -0.2 \\ -30 & -0.5 & 2 \end{bmatrix}$
5. $\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & -4 & 1 \end{bmatrix}$
7. $\mathbf{A}^{-1} = \mathbf{A}$
9. $\begin{bmatrix} 0 & 0 & \frac{1}{2} \\ \frac{1}{8} & 0 & 0 \\ 0 & \frac{1}{4} & 0 \end{bmatrix}$
11. $(\mathbf{A}^2)^{-1} = (\mathbf{A}^{-1})^2 = \begin{bmatrix} 3.760 & 22.272 \\ 2.400 & 15.280 \end{bmatrix}$

15. $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$, $(\mathbf{A}\mathbf{A}^{-1})^{-1} = (\mathbf{A}^{-1})^{-1}\mathbf{A}^{-1} = \mathbf{I}$. Multiply by $\mathbf{A}$ from the right.

Problem Set 7.9, page 318

1. $[1 \ 0]^T$, $[0 \ 1]^T$; $[1 \ 0]^T$, $[0 \ -1]^T$; $[1 \ 1]^T$, $[-1 \ 1]^T$
 3. 1, $[1 \ 11 \ -7]^T$
 5. No
 7. Dimension 2, basis xe^{-x} , e^{-x}
 9. 3; basis $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$
 11. $x_1 = 5y_1 - y_2$, $x_2 = 3y_1 - y_2$
 13. $x_1 = 2y_1 - 3y_2$, $x_2 = -10y_1 + 16y_2 + y_3$, $x_3 = -7y_1 + 11y_2 + y_3$

15. $\sqrt{26}$

19. 1

23. $\mathbf{a} = \begin{bmatrix} 3 & 1 & -4 \end{bmatrix}^T$, $\mathbf{b} = \begin{bmatrix} -4 & 8 & -1 \end{bmatrix}^T$, $\|\mathbf{a} + \mathbf{b}\| = \sqrt{107} \leq 5.099 + 9$

25. $\mathbf{a} = \begin{bmatrix} 5 & 3 & 2 \end{bmatrix}^T$, $\mathbf{b} = \begin{bmatrix} 3 & 2 & -1 \end{bmatrix}^T$, $90 + 14 = 2(38 + 14)$

17. $\sqrt{5}$

21. $k = -20$

Chapter 7 Review Questions and Problems, page 318

11. $\begin{bmatrix} -1 & 6 & 1 \\ -18 & 8 & -7 \\ -13 & -2 & -7 \end{bmatrix}$, $\begin{bmatrix} 1 & 18 & 13 \\ -6 & -8 & 2 \\ -1 & 7 & 7 \end{bmatrix}$

13. $\begin{bmatrix} 21 & -8 & -31 \end{bmatrix}^T$, $\begin{bmatrix} 21 & -8 & 31 \end{bmatrix}$

15. 197, 0

17. -5, $\det \mathbf{A}^2 = (\det \mathbf{A})^2 = 25$, 0

19. $\begin{bmatrix} -2 & -12 & -12 \\ -12 & 16 & -9 \\ -12 & -9 & -14 \end{bmatrix}$

21. $x = 4$, $y = -2$, $z = 8$

23. $x = 6$, $y = 2t + 2$, $z = t$ arb.

25. $x = 0.4$, $y = -1.3$, $z = 1.7$

27. $x = 10$, $y = -2$

29. Ranks 2, 2, ∞

31. Ranks 2, 2, 1

33. $I_1 = 16.5$ A, $I_2 = 11$ A, $I_3 = 5.5$ A

35. $I_1 = 4$ A, $I_2 = 5$ A, $I_3 = 1$ A

Problem Set 8.1, page 329

1. $3, \begin{bmatrix} 1 & 0 \end{bmatrix}^T$; $-0.6, \begin{bmatrix} 0 & 1 \end{bmatrix}^T$ 3. $-4, \begin{bmatrix} 2 & 9 \end{bmatrix}^T$; $3, \begin{bmatrix} 1 & 1 \end{bmatrix}^T$

5. $-3i, \begin{bmatrix} 1 & -i \end{bmatrix}$; $3i, \begin{bmatrix} 1 & i \end{bmatrix}$, $i = \sqrt{-1}$

7. $\lambda^2 = 0$, $\begin{bmatrix} 1 & 0 \end{bmatrix}^T$

9. $0.8 + 0.6i, \begin{bmatrix} 1 & -i \end{bmatrix}^T$; $0.8 - 0.6i, \begin{bmatrix} 1 & i \end{bmatrix}^T$

11. $-(\lambda^3 - 18\lambda^2 + 99\lambda - 162)/(\lambda - 3) = -(\lambda^2 - 15\lambda + 54)$; $3, \begin{bmatrix} 2 & -2 & 1 \end{bmatrix}^T$; $6, \begin{bmatrix} 1 & 2 & 2 \end{bmatrix}^T$; $9, \begin{bmatrix} 2 & 1 & -2 \end{bmatrix}^T$

13. $-(\lambda - 9)^3$; $9, \begin{bmatrix} 2 & -2 & 1 \end{bmatrix}^T$, defect 2

15. $(\lambda + 1)^2(\lambda^2 + 2\lambda - 15)$; $-1, \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}^T, \begin{bmatrix} 0 & 1 & 0 & 0 \end{bmatrix}^T$; $-5, \begin{bmatrix} -3 & -3 & 1 & 1 \end{bmatrix}^T, 3, \begin{bmatrix} 3 & -3 & 1 & -1 \end{bmatrix}^T$

17. $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Eigenvalues $i, -i$. Corresponding eigenvectors are complex,

indicating that no direction is preserved under a rotation.

19. $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$; $1, \begin{bmatrix} 0 \\ 1 \end{bmatrix}$; $0, \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. A point onto the x_2 -axis goes onto itself,

a point on the x_1 -axis onto the origin.

23. Use that real entries imply real coefficients of the characteristic polynomial.

Problem Set 8.2, page 333

1. $1.5, [1 \ -1]^T, -45^\circ$; $4.5, [1 \ 1]^T, 45^\circ$
3. $1, [-1/\sqrt{6} \ 1]^T, 112.2^\circ$; $8, [1 \ 1/\sqrt{6}]^T, 22.2^\circ$
5. $0.5, [1 \ -1]^T$; $1.5, [1 \ 1]^T$; directions -45° and 45°
7. $[5 \ 8]^T$
9. $[11 \ 12 \ 16]^T$
11. 1.8
13. $c[10 \ 18 \ 25]^T$
15. $\mathbf{x} = (\mathbf{I} - \mathbf{A})^{-1}\mathbf{y} = [0.6747 \ 0.7128 \ 0.7543]^T$
17. $\mathbf{A}\mathbf{x}_j = \lambda_j\mathbf{x}_j$ ($\mathbf{x}_j \neq \mathbf{0}$), $(\mathbf{A} - k\mathbf{I})\mathbf{x}_j = \lambda_j\mathbf{x}_j - k\mathbf{x}_j = (\lambda_j - k)\mathbf{x}_j$.
19. From $\mathbf{A}\mathbf{x}_j = \lambda_j\mathbf{x}_j$ ($\mathbf{x}_j \neq \mathbf{0}$) and Prob. 18 follows $k_p\mathbf{A}^p\mathbf{x}_j = k_p\lambda_j^p\mathbf{x}_j$ and $k_q\mathbf{A}^q\mathbf{x}_j = k_q\lambda_j^q\mathbf{x}_j$ ($p \geq 0, q \geq 0$, integer). Adding on both sides, we see that $k_p\mathbf{A}^p + k_q\mathbf{A}^q$ has the eigenvalue $k_p\lambda_j^p + k_q\lambda_j^q$. From this the statement follows.

Problem Set 8.3, page 338

1. $0.8 \pm 0.6i, [1 \ \pm i]^T$; orthogonal
3. $2 \pm 0.8i, [1 \ \pm i]$. Not skew-symmetric!
5. $1, [0 \ 2 \ 1]^T$; $6, [1 \ 0 \ 0]^T, [0 \ 1 \ -2]^T$; symmetric
7. $0, \pm 25i$, skew-symmetric
9. $1, [0 \ 1 \ 0]^T$; $i, [1 \ 0 \ i]^T$; $-i, [1 \ 0 \ -i]^T$, orthogonal
15. No
17. $\mathbf{A}^{-1} = (-\mathbf{A}^T)^{-1} = -(\mathbf{A}^{-1})^T$
19. No since $\det \mathbf{A} = \det (\mathbf{A}^T) = \det (-\mathbf{A}) = (-1)^3 \det (\mathbf{A}) = -\det (\mathbf{A}) = 0$.

Problem Set 8.4, page 345

1. $\begin{bmatrix} -25 & 12 \\ -50 & 25 \end{bmatrix}, -5, \begin{bmatrix} 3 \\ 5 \end{bmatrix}; 5, \begin{bmatrix} 2 \\ 5 \end{bmatrix}; \mathbf{x} = \begin{bmatrix} -2 \\ 4 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix}$
3. $\begin{bmatrix} 3.008 & -0.544 \\ 5.456 & 6.992 \end{bmatrix}, 4, \begin{bmatrix} -17 \\ 31 \end{bmatrix}; 6, \begin{bmatrix} -2 \\ 11 \end{bmatrix}; \mathbf{x} = \begin{bmatrix} 25 \\ 25 \end{bmatrix}, \begin{bmatrix} 10 \\ 5 \end{bmatrix}$
5. $\begin{bmatrix} 4 & 3 & -9 \\ 0 & -5 & 15 \\ 0 & -5 & 15 \end{bmatrix}, 0, \begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix}; 4, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}; 10, \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}; \mathbf{x} = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$
9. $\begin{bmatrix} \frac{1}{5} & \frac{2}{5} \\ -\frac{2}{5} & \frac{1}{5} \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ 0 & 0 \end{bmatrix}$
11. $\begin{bmatrix} -2 & 1 \\ 3 & -1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & -5 \end{bmatrix}$

$$13. \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$15. \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ -\frac{1}{3} & \frac{1}{6} & \frac{1}{6} \\ 0 & -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & -2 & 0 \\ 1 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 10 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$17. \mathbf{C} = \begin{bmatrix} 7 & 3 \\ 3 & 7 \end{bmatrix}, \quad 4y_1^2 + 10y_2^2 = 200, \quad \mathbf{x} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \mathbf{y}, \quad \text{ellipse}$$

$$19. \mathbf{C} = \begin{bmatrix} 3 & 11 \\ 11 & 3 \end{bmatrix}, \quad 14y_1^2 - 8y_2^2 = 0, \quad \mathbf{x} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \mathbf{y}; \quad \text{pair of straight lines}$$

$$21. \mathbf{C} = \begin{bmatrix} 1 & -6 \\ -6 & 1 \end{bmatrix}, \quad 7y_1^2 - 5y_2^2 = 70, \quad \mathbf{x} = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \mathbf{y}, \quad \text{hyperbola}$$

$$23. \mathbf{C} = \begin{bmatrix} -11 & 42 \\ 42 & 24 \end{bmatrix}, \quad 52y_1^2 - 39y_2^2 = 156, \quad \mathbf{x} = \frac{1}{\sqrt{13}} \begin{bmatrix} 2 & 3 \\ 3 & -2 \end{bmatrix} \mathbf{y}, \quad \text{hyperbola}$$

Problem Set 8.5, page 351

1. Hermitian, 5, $[-i \ 1]^T$, 7, $[i \ 1]^T$

3. Unitary, $(1 - i\sqrt{3})/2$, $[-1 \ 1]^T$; $(1 + i\sqrt{3})/2$, $[1 \ 1]^T$

5. Skew-Hermitian, unitary, $-i$, $[0 \ -1 \ 1]^T$, i , $[1 \ 0 \ 0]^T$, $[0 \ 1 \ 1]^T$

7. Eigenvalues -1 , 1 ; eigenvectors $[1 \ -1]^T$, $[1 \ 1]^T$; $[1 \ -i]^T$, $[1 \ i]^T$; $[0 \ 1]^T$, $[1 \ 0]^T$, resp.

9. Hermitian, 16

11. Skew-Hermitian, $-6i$

$$13. (\mathbf{ABC})^T = \overline{\mathbf{C}}^T \overline{\mathbf{B}}^T \overline{\mathbf{A}}^T = \mathbf{C}^{-1}(-\mathbf{B})\mathbf{A}$$

$$15. \mathbf{A} = \mathbf{H} + \mathbf{S}, \quad \mathbf{H} = \frac{1}{2}(\mathbf{A} + \overline{\mathbf{A}}^T), \quad \mathbf{S} = \frac{1}{2}(\mathbf{A} - \overline{\mathbf{A}}^T) \quad (\mathbf{H} \text{ Hermitian, } \mathbf{S} \text{ skew-Hermitian})$$

$$19. \mathbf{A}\overline{\mathbf{A}}^T - \overline{\mathbf{A}}^T\mathbf{A} = (\mathbf{H} + \mathbf{S})(\mathbf{H} - \mathbf{S}) - (\mathbf{H} - \mathbf{S})(\mathbf{H} + \mathbf{S}) = 2(-\mathbf{HS} + \mathbf{SH}) = \mathbf{0}$$

if and only if $\mathbf{HS} = \mathbf{SH}$.

Chapter 8 Review Questions and Problems, page 352

$$11. 3, [1 \ 1]^T; \quad 2, [1 \ -1]^T$$

$$13. 3, [1 \ 5]^T; \quad 7, [1 \ 1]^T$$

$$15. 0, [2 \ -2 \ 1]^T; \quad 9i, [-1 + 3i \ 1 + 3i \ 4]^T; \quad -9i, [-1 - 3i \ 1 - 3i \ 4]^T$$

$$17. -1, 1; \quad \mathbf{A} = \frac{1}{16} \begin{bmatrix} 5 & -3 \\ -3 & 5 \end{bmatrix} \begin{bmatrix} 23 & 2 \\ 39 & 1 \end{bmatrix} = \frac{1}{8} \begin{bmatrix} -1 & 1 \\ 63 & 1 \end{bmatrix}$$

$$19. \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \mathbf{A} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} -0.9 & 0 \\ 0 & 0.6 \end{bmatrix}$$

$$21. \frac{1}{3} \begin{bmatrix} 1 & 1 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & 2 & 1 \\ 1 & -1 & 1 \\ -1 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 0 & 0 \\ 0 & -20 & 0 \\ 0 & 0 & 22 \end{bmatrix}$$

$$23. \mathbf{C} = \begin{bmatrix} 4 & 12 \\ 12 & -14 \end{bmatrix}, \quad 10y_1^2 - 20y_2^2 = 20, \quad \mathbf{x} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix} \mathbf{y}, \quad \text{hyperbola}$$

$$25. \mathbf{C} = \begin{bmatrix} 3.7 & 1.6 \\ 1.6 & 1.3 \end{bmatrix}, \quad 4.5y_1^2 + 0.5y_2^2 = 4.5, \quad \mathbf{x} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix} \mathbf{y}, \quad \text{ellipse}$$

Problem Set 9.1, page 360

1. 5, 1, 0; $\sqrt{26}$; $[5/\sqrt{26}, 1/\sqrt{26}, 0]$
3. 8.5, -4.0, 1.7; $\sqrt{91.14}$, $[0.890, -0.419, 0.178]$
5. 2, 1, -2; $\mathbf{u} = [\frac{2}{3}, \frac{1}{3}, -\frac{2}{3}]$, position vector of Q
7. $Q: (4, 0, \frac{1}{2})$, $|\mathbf{v}| = \sqrt{16.25}$
9. $Q: (0, 0, -8)$, $|\mathbf{v}| = 8$
11. $[6, 4, 0]$, $[\frac{3}{2}, 1, 0]$, $[-3, -2, 0]$
13. $[1, 5, 8]$
15. $7[9, -7, 8] = [63, -49, 56]$
17. $[12, 8, 0]$
21. $[4, 9, -3]$, $\sqrt{106}$
23. $[0, 0, 5]$, 5
25. $[6, 2, -14] = 2\mathbf{u}$, $\sqrt{236}$
27. $\mathbf{p} = [0, 0, -5]$
29. $\mathbf{v} = [v_1, v_2, 3]$, v_1, v_2 arbitrary
31. $k = 10$
33. $|\mathbf{p} + \mathbf{q} + \mathbf{u}| \leq 18$. Nothing
35. $v_B - v_A = [-19, 0] - [22/\sqrt{2}, 22/\sqrt{2}] = [-19 - 22/\sqrt{2}, -22/\sqrt{2}]$
37. $\mathbf{u} + \mathbf{v} + \mathbf{p} = [-k, 0] + [l, l] + [0, -1000] = \mathbf{0}$, $-k + l + 0 = 0$,
 $0 + l - 1000 = 0$, $l = 1000$, $k = 1000$

Problem Set 9.2, page 367

1. 44, 44, 0
3. $\sqrt{35}$, $\sqrt{320}$, $\sqrt{86}$
5. $|[2, 9, 9]| = \sqrt{166} = 12.88 < \sqrt{80} + \sqrt{86} = 18.22$
7. $|-24| = 24$, $|a||c| = \sqrt{35}\sqrt{86} = \sqrt{3010} = 54.86$; cf. (6)
9. 300; cf. (5a) and (5b)
13. Use (1) and $|\cos \gamma| \leq 1$.
15. $|\mathbf{a} + \mathbf{b}|^2 + |\mathbf{a} - \mathbf{b}|^2 = \mathbf{a} \cdot \mathbf{a} + 2\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{b} + (\mathbf{a} \cdot \mathbf{a} - 2\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{b})$
 $= 2|\mathbf{a}|^2 + 2|\mathbf{b}|^2$
17. $[2, 5, 0] \cdot [2, 2, 2] = 14$
19. $[0, 4, 3] \cdot [-3, -2, 1] = -5$ is negative! Why?
21. Yes, because $W = (\mathbf{p} + \mathbf{q}) \cdot \mathbf{d} = \mathbf{p} \cdot \mathbf{d} + \mathbf{q} \cdot \mathbf{d}$.
23. $\arccos 0.5976 = 53.3^\circ$
27. $\beta - \alpha$ is the angle between the unit vectors $\mathbf{a}$ and $\mathbf{b}$. Use (2).
29. $\gamma = \arccos(12/(6\sqrt{13})) = 0.9828 = 56.3^\circ$ and 123.7°
31. $a_1 = -\frac{28}{3}$
33. $\pm[\frac{3}{5}, -\frac{4}{5}]$
35. $(\mathbf{a} + \mathbf{b}) \cdot (\mathbf{a} - \mathbf{b}) = |\mathbf{a}|^2 - |\mathbf{b}|^2 = 0$, $|\mathbf{a}| = |\mathbf{b}|$. A square.
37. 0. Why?
39. If $|\mathbf{a}| = |\mathbf{b}|$ or if $\mathbf{a}$ and $\mathbf{b}$ are orthogonal.

Problem Set 9.3, page 374

5. $-\mathbf{m}$ instead of $\mathbf{m}$, tendency to rotate in the opposite sense.
 7. $|\mathbf{v}| = |[0, 20, 0] \times [8, 6, 0]| = |[0, 0, -160]| = 160$
 9. Zero volume in Fig. 191, which can happen in several ways.
 11. $[0, 0, 7], [0, 0, -7], -4$ 13. $[6, 2, 7], [-6, -2, -7]$
 15. $\mathbf{0}$ 17. $[-32, -58, 34], [-42, -63, 19]$
 19. $1, -1$
 21. $[-48, -72, -168], 12\sqrt{248} = 189.0, 189.0$
 23. $0, 0, 13$
 25. $\mathbf{m} = [-2, -2, 0] \times [2, 3, 0] = [0, 0, -10], m = 10$ clockwise
 27. $[6, 2, 0] \times [1, 2, 0] = [0, 0, 10]$ 29. $\frac{1}{2} |[-12, 2, 6]| = \sqrt{46}$
 31. $3x + 2y - z = 5$ 33. $474/6 = 79$

Problem Set 9.4, page 380

1. Hyperbolas
 3. Parallel straight lines (planes in space) $y = \frac{3}{4}x + c$
 5. Circles, centers on the y -axis
 7. Ellipses 9. Parallel planes
 11. Elliptic cylinders 13. Paraboloids

Problem Set 9.5, page 390

1. Circle, center $(3, 0)$, radius 2 3. Cubic parabola $x = 0, z = y^3$
 5. Ellipse 7. Helix
 9. A "Lissajous curve"
 11. $\mathbf{r} = [3 + \sqrt{13} \cos t, 2 + \sqrt{13} \sin t, 1]$
 13. $\mathbf{r} = [2 + t, 1 + 2t, 3]$ 15. $\mathbf{r} = [t, 4t - 1, 5t]$
 17. $\mathbf{r} = [\sqrt{2} \cos t, \sin t, \sin t]$ 19. $\mathbf{r} = [\cosh t, (\sqrt{3}/2) \sinh t, -2]$
 21. Use $\sin(-\alpha) = -\sin \alpha$.
 25. $\mathbf{u} = [-\sin t, 0, \cos t]$. At P , $\mathbf{r}' = [-8, 0, 6]$. $\mathbf{q}(w) = [6 - 8w, i, 8 + 6w]$.
 27. $\mathbf{q}(w) = [2 + w, \frac{1}{2} - \frac{1}{4}w, 0]$ 29. $\sqrt{\mathbf{r}' \cdot \mathbf{r}'} = \cosh t, l = \sinh t = 1.175$
 31. $\sqrt{\mathbf{r}' \cdot \mathbf{r}'} = a, l = a\pi/2$ 33. Start from $\mathbf{r}(t) = [t, f(t)]$.
 35. $\mathbf{v} = \mathbf{r}' = [1, 2t, 0], |\mathbf{v}| = \sqrt{1 + 4t^2}, \mathbf{a} = [0, 2, 0]$
 37. $\mathbf{v}(0) = (\omega + 1)R\mathbf{i}, \mathbf{a}(0) = -\omega^2 R\mathbf{j}$
 39. $\mathbf{v} = [-\sin t - 2 \sin 2t, \cos t - 2 \cos 2t], |\mathbf{v}|^2 = 5 - 4 \cos 3t,$
 $\mathbf{a} = [-\cos t - 4 \cos 2t, -\sin t + 4 \sin 2t],$ and $\mathbf{a}_{\tan} = \frac{6 \sin 3t}{5 - 4 \cos 3t} \mathbf{v}.$
 41. $\mathbf{v} = [-\sin t, 2 \cos 2t, -2 \sin 2t], |\mathbf{v}|^2 = 4 + \sin^2 t,$
 $\mathbf{a} = [-\cos t, -4 \sin 2t, -4 \cos 2t],$ and $\mathbf{a}_{\tan} = \frac{\frac{1}{2} \sin 2t}{4 + \sin^2 t} \mathbf{v}.$
 43. 1 year $= 365 \cdot 86,400$ sec, $R = 30 \cdot 365 \cdot 86,400/2\pi = 151 \cdot 10^6$ [km],
 $|\mathbf{a}| = \omega^2 R = |\mathbf{v}|^2/R = 5.98 \cdot 10^{-6}$ [km/sec²]
 45. $R = \frac{3960 + 80}{10^8} \text{ mi} = 2.133 \cdot 10^7$ ft, $g = |\mathbf{a}| = \omega^2 R = |\mathbf{v}|^2/R, |\mathbf{v}| = \sqrt{gR} =$
 $\sqrt{6.61 \cdot 10^8} = 25,700$ [ft/sec] $= 17,500$ [mph]
 49. $\mathbf{r}(t) = [t, y(t), 0], \mathbf{r}' = [1, y', 0] \mathbf{r} \cdot \mathbf{r}' = 1 + y'^2$, etc.

$$51. \frac{d\mathbf{r}}{ds} = \frac{d\mathbf{r}}{dt} \bigg/ \frac{ds}{dt}, \quad \frac{d^2\mathbf{r}}{ds^2} = \frac{d^2\mathbf{r}}{dt^2} \bigg/ \left(\frac{ds}{dt}\right)^2 + \dots, \quad \frac{d^3\mathbf{r}}{ds^3} = \frac{d^3\mathbf{r}}{dt^3} \bigg/ \left(\frac{ds}{dt}\right)^3 + \dots$$

$$53. 3/(1 + 9t^2 + 9t^4)$$

Problem Set 9.7, page 402

1. $[2y - 1, 2x + 2]$
3. $[-y/x^2, 1/x]$
5. $[4x^3, 4y^3]$
7. Use the chain rule.
9. Apply the quotient rule to each component and collect terms.
11. $[y, x], [5, -4]$
13. $[2x/(x^2 + y^2), 2y/(x^2 + y^2)], [0.16, 0.12]$
15. $[8x, 18y, 2z], [40, -18, -22]$
17. For P on the x - and y -axes.
19. $[-1.25, 0]$
21. $[0, -e]$
23. Points with $y = 0, \pm\pi, \pm2\pi, \dots$
25. $-\nabla T(P) = [0, 4, -1]$
31. $\nabla f = [32x, -2y], \nabla f(P) = [160, -2]$
33. $[12x, 4y, 2z], [60, 20, 10]$
35. $[-2x, -2y, 1], [-6, -8, 1]$
37. $[2, 1] \cdot [1, -1]/\sqrt{5} = 1/\sqrt{5}$
39. $[1, 1, 1] \cdot [-3/125, 0, -4/125]/\sqrt{3} = -7/(125\sqrt{3})$
41. $\sqrt{8/3}$
43. $f = xyz$
45. $f = \int v_1 dx + \int v_2 dy + \int v_3 dz$

Problem Set 9.8, page 405

1. $2x + 8y + 18z; 7$
3. 0, after simplification; solenoidal
5. $9x^2y^2z^2; 1296$
7. $-2e^x(\cos y)z$
9. (b) $(fv_1)_x + (fv_2)_y + (fv_3)_z = f[(v_1)_x + (v_2)_y + (v_3)_z] + f_xv_1 + f_yv_2 + f_zv_3$, etc.
11. $[v_1, v_2, v_3] = \mathbf{r}' = [x', y', z'] = [y, 0, 0], z' = 0, z = c_3, y' = 0, y = c_2$, and $x' = y = c_2, x = c_2t + c_1$. Hence as t increases from 0 to 1, this “shear flow” transforms the cube into a parallelepiped of volume 1.
13. $\text{div}(\mathbf{w} \times \mathbf{r}) = 0$ because v_1, v_2, v_3 do not depend on x, y, z , respectively.
15. $-2 \cos 2x + 2 \cos 2y$
17. 0
19. $2/(x^2 + y^2 + z^2)^2$

Problem Set 9.9, page 408

3. Use the definitions and direct calculation.
5. $[x(z^2 - y^2), y(x^2 - z^2), z(y^2 - x^2)]$
7. $e^{-x}[\cos y, \sin y, 0]$
9. $\text{curl } \mathbf{v} = [-6z, 0, 0]$ incompressible, $\mathbf{v} = \mathbf{r}' = [x', y', z'] = [0, 3z^2, 0], x = c_1, z = c_3, y' = 3z^2 = 3c_3^2, y = 3c_3^2t + c_2$
11. $\text{curl } \mathbf{v} = [0, 0, -3]$, incompressible, $x' = y, y' = -2x, 2xx' + yy' = 0, x^2 + \frac{1}{2}y^2 = c, z = c_3$
13. $\text{curl } \mathbf{v} = 0$, irrotational, $\text{div } \mathbf{v} = 1$, compressible, $\mathbf{r} = [c_1e^t, c_2e^t, c_3e^{-t}]$. Sketch it.
15. $[-1, -1, -1]$, same (why?)
17. $-yz - zx - xy, 0$ (why?), $-y - z - x$
19. $[-2z - y, -2x - z, -2y - x]$, same (why?)

Chapter 9 Review Questions and Problems, page 409

11. $-10, 1080, 1080, 65$
 13. $[-10, -30, 0], [10, 30, 0], 0, 40$
 15. $[-1260, -1830, -300], [-210, 120, -540], \text{undefined}$
 17. $-125, 125, -125$
 19. $[70, -40, -50], 0, \sqrt{35^2 + 20^2 + 25^2} = \sqrt{2250}$
 21. $[-2, -6, -13]$
 23. $\gamma_1 = \arccos(-10/\sqrt{65 \cdot 40}) = 1.7682 = -101.3^\circ, \gamma_2 = 23.7^\circ$
 25. $[5, 2, 0] \cdot [4 - 1, 3 - 1, 0] = 19$ 27. $\mathbf{v} \cdot \mathbf{w}/|\mathbf{w}| = 22/\sqrt{8} = 7.78$
 29. $[0, 0, -14], \text{tendency of clockwise rotation}$ 31. 4
 33. 1, $-2y$
 35. 0, same (why?), $2(y^2 + x^2 - xz)$
 37. $[0, -2, 0]$ 39. $9/\sqrt{225} = \frac{3}{5}$

Problem Set 10.1, page 418

3. 4
 5. $\mathbf{r} = [2 \cos t, 2 \sin t], 0 \leq t \leq \pi/2; \frac{8}{5}$
 7. "Exponential helix," $(e^{6\pi} - 1)/3$ 9. 23.5, 0
 11. $2e^{-t} + 2te^{-t^2}, -2e^{-2} - e^{-4} + 3$ 15. $18\pi, \frac{4}{3}(4\pi)^3, 18\pi$
 17. $[4 \cos t, + \sin t, \sin t, 4 \cos t], [2, 2, 0]$ 19. $144t^4, 1843.2$

Problem Set 10.2, page 425

3. $\sin \frac{1}{2}x \cos 2y, 1 - 1/\sqrt{2} = 0.293$ 5. $e^{xy} \sin z, e - 0$
 7. $\cosh 1 - 2 = -0.457$
 9. $e^x \cosh y + e^z \sinh y, e - (\cosh 1 + \sinh 1) = 0$
 13. $e^{a^2} \cos 2b$ 15. Dependent, $x^2 \neq -4y^2$, etc.
 17. Dependent, $4 \neq 0$, etc. 19. $\sin(a^2 + 2b^2 + c^2)$

Problem Set 10.3, page 432

3. $8y^3/3, 54$ 5. $\int_0^1 [x - x^3 - (x^2 - x^5)] dx = \frac{1}{12}$
 7. $\cosh 2x - \cosh x, \frac{1}{2} \sinh 4 - \sinh 2$ 9. $36 + 27y^2, 144$
 11. $z = 1 - r^2, dx dy = r dr d\theta, \text{Answer: } \pi/2$
 13. $\bar{x} = 2b/3, \bar{y} = h/3$ 15. $\bar{x} = 0, \bar{y} = 4r/3\pi$
 17. $I_x = bh^3/12, I_y = b^3h/4$
 19. $I_x = (a + b)h^3/24, I_y = h(a^4 - b^4)/(48(a - b))$

Problem Set 10.4, page 438

1. $(-1 - 1) \cdot \pi/4 = -\pi/2$ 3. $9(e^2 - 1) - \frac{8}{3}(e^3 - 1)$
 5. $2x - 2y, 2x(1 - x^2) - (2 - x^2)^2 + 1, x = -1 \cdots 1, -\frac{56}{15}$
 7. 0. Why? 9. $\frac{16}{5}$
 13. $\nabla^2 w = \cosh x, y = x/2 \cdots 2, \frac{1}{2} \cosh 4 - \frac{1}{2}$

15. $\nabla^2 w = 6xy$, $3x(10 - x^2)^2 - 3x$, 486 17. $\nabla^2 w = 6x - 6y$, -38.4
 19. $|\text{grad } w|^2 = e^{2x}$, $\frac{5}{2}(e^4 - 1)$

Problem Set 10.5, page 442

1. Straight lines, $\mathbf{k}$
 3. $z = c\sqrt{x^2 + y^2}$, circles, straight lines, $[-cu \cos v, -cu \sin v, u]$
 5. $z = x^2 + y^2$, circles, parabolas, $[-2u^2 \cos v, -2u^2 \sin v, u]$
 7. $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$, $[bc \cos^2 v \cos u, ac \cos^2 v \sin u, ab \sin v \cos v]$, ellipses
 11. $[\tilde{u}, \tilde{v}, \tilde{u}^2 + \tilde{v}^2]$, $\tilde{\mathbf{N}} = [-2\tilde{u}, -2\tilde{v}, 1]$
 13. Set $x = u$ and $y = v$.
 15. $[2 + 5 \cos u, -1 + 5 \sin u, v]$, $[5 \cos u, 5 \sin u, 0]$
 17. $[a \cos v \cos u, -2.8 + a \cos v \sin u, 3.2 + a \sin v]$, $a = 1.5$;
 $[a^2 \cos^2 v \cos u, a^2 \cos^2 v \sin u, a^2 \cos v \sin v]$
 19. $[\cosh u, \sinh u, v]$, $[\cosh u, -\sinh u, 0]$

Problem Set 10.6, page 450

1. $\mathbf{F}(\mathbf{r}) \cdot \mathbf{N} = [-u^2, v^2, 0] \cdot [-3, 2, 1] = 3u^2 + 2v^2$, 29.5
 3. $\mathbf{F}(\mathbf{r}) \cdot \mathbf{N} = \cos^3 v \cos u \sin u$ from (3), Sec. 10.5. Answer: $\frac{1}{3}$
 5. $\mathbf{F}(\mathbf{r}) \cdot \mathbf{N} = -u^3$, -128π
 7. $\mathbf{F} \cdot \mathbf{N} = [0, \sin u, \cos v] \cdot [1, -2u, 0]$, $4 + (-2 + \pi^2/16 - \pi/2)\sqrt{2} = -0.1775$
 9. $\mathbf{r} = [2 \cos u, 2 \sin u, v]$, $0 \leq u \leq \pi/4$, $0 \leq v \leq 5$. Integrate $2 \sinh v \sin u$ to get $2(1 - 1/\sqrt{2})(\cosh 5 - 1) = 42.885$.
 13. $7\pi^3/\sqrt{6} = 88.6$
 15. $G(\mathbf{r}) = (1 + 9u^4)^{3/2}$, $|\mathbf{N}| = (1 + 9u^4)^{1/2}$. Answer: 54.4
 21. $I_{x=y} = \iint_S [\frac{1}{2}(x - y)^2 + z^2] \sigma \, dA$
 23. $[u \cos v, u \sin v, u]$, $\int_0^{2\pi} \int_0^h u^2 \cdot u\sqrt{2} \, du \, dv = \frac{\pi}{\sqrt{2}} h^4$
 25. $[\cos u \cos v, \cos u \sin v, \sin u]$, $dA = (\cos u) \, du \, dv$, B the z -axis, $I_B = 8\pi/3$,
 $I_K = I_B + 1^2 \cdot 4\pi = 20.9$.

Problem Set 10.7, page 457

1. 224
 3. $-e^{-1-z} + e^{-y-z}$, $-2e^{-1-z} + e^{-z}$, $2e^{-3} - e^{-2} - 2e^{-1} + 1$
 5. $\frac{1}{2}(\sin 2x)(1 - \cos 2x)$, $\frac{1}{8}$, $\frac{3}{4}$
 7. $[r \cos u \cos v, \cos u \sin v, r \sin u]$, $dV = r^2 \cos u \, dr \, du \, dv$, $\sigma = v$, $2\pi^2 a^3/3$
 9. $\text{div } \mathbf{F} = 2x + 2z$, 48 11. $12(e - 1/e) = 24 \sinh 1$
 13. $\text{div } \mathbf{F} = -\sin z$, 0 15. $1/\pi + \frac{5}{24} = 0.5266$
 17. $h^4 \pi/2$ 19. $8abc(b^2 + c^2)/3$
 21. $(a^4/4) \cdot 2\pi \cdot h = ha^4 \pi/2$ 23. $h^5 \pi/10$
 25. Do Prob. 20 as the last one.

Problem Set 10.8, page 462

1. $x = 0, y = 0, z = 0$, no contributions. $x = a$: $\partial f / \partial n = \partial f / \partial x = -2x = -2a$, etc.
Integrals $x = a$: $(-2a)bc$, $y = b$: $(-2b)ac$, $z = c$: $(4c)ab$. Sum 0
3. The volume integral of $8y^2 + [0, 8y] \cdot [2x, 0] = 8y^2$ is $8y^3/3 = \frac{8}{3}$. The surface integral of $f \partial g / \partial n = f \cdot 2x = 2f = 8y^2$ over $x = 1$ is $8y^3/3 = \frac{8}{3}$. Others 0.
5. The volume integral of $6y^2 \cdot 4 - 2x^2 \cdot 12$ is 0; $8(x = 1)$, $-8(y = 1)$, others 0.
7. $\mathbf{F} = [x, 0, 0]$, $\operatorname{div} \mathbf{F} = 1$, use (2*), Sec. 10.7, etc.
9. $z = 0$ and $z = \sqrt{a^2 - x^2 - y^2} = \sqrt{a^2 - r^2}$, $dx dy = r dr d\theta$,
 $-2\pi \cdot \frac{1}{2}(a^2 - r^2)^{3/2} \cdot \frac{2}{3} \Big|_0^a = \frac{2}{3}\pi a^3$
11. $r = a$, $\phi = 0$, $\cos \phi = 1$, $v = \frac{1}{3}a \cdot (4\pi a^2)$

Problem Set 10.9, page 468

1. $S: z = y$ ($0 \leq x \leq 1, 0 \leq y \leq 4$), $[0, 2z, -2z] \cdot [0, -1, 1]$, ± 20
3. $[2e^{-z} \cos y, -e^{-z}, 0] \cdot [0, -y, 1] = ye^{-z}$, $\pm(2 - 2/\sqrt{e})$
5. $[0, 2z, \frac{3}{2}] \cdot [0, 0, 1] = \frac{3}{2}$, $\pm \frac{3}{2}a^2$
7. $[-e^z, -e^x, -e^y] \cdot [-2x, 0, 1]$, $\pm(e^4 - 2e + 1)$
9. The sides contribute $a, 3a^2/2, -a, 0$.
11. -2π ; $\operatorname{curl} \mathbf{F} = \mathbf{0}$
13. $5\mathbf{k}, 80\pi$
15. $[0, -1, 2x - 2y] \cdot [0, 0, 1]$, $\frac{1}{3}$
17. $\mathbf{r} = [\cos u, \sin u, v]$, $[-3v^2, 0, 0] \cdot [\cos u, \sin u, 0]$, -1
19. $\mathbf{r} = [u \cos v, u \sin v, u]$, $0 \leq u \leq 1, 0 \leq v \leq \pi/2$,
 $[-e^z, 1, 0] \cdot [-u \cos v, -u \sin v, u]$. Answer: $1/2$

Chapter 10 Review Questions and Problems, page 469

11. $\mathbf{r} = [4 - 10t, 2 + 8t]$, $\mathbf{F}(\mathbf{r}) \cdot d\mathbf{r} = [2(4 - 10t)^2, -4(2t + 8t)^2] \cdot [-10, 8] dt$;
 $-4528/3$. Or using exactness.
13. Not exact, $\operatorname{curl} \mathbf{F} = (5 \cos x)\mathbf{k}$, ± 10
15. 0 since $\operatorname{curl} \mathbf{F} = \mathbf{0}$
17. By Stokes, $\pm 18\pi$
19. $\mathbf{F} = \operatorname{grad}(y^2 + xz)$, 2π
21. $M = 8$, $\bar{x} = \frac{8}{5}$, $\bar{y} = \frac{16}{5}$
23. $M = \frac{63}{20}$, $\bar{x} = \frac{8}{7} = 1.14$, $\bar{y} = \frac{118}{49} = 2.41$
25. $M = 4k/15$, $\bar{x} = \frac{5}{16}$, $\bar{y} = \frac{4}{7}$
27. $288(a + b + c)\pi$
29. $\operatorname{div} \mathbf{F} = 20 + 6z^2$. Answer: 21
31. $24 \sinh 1 = 28.205$
33. Direct integration, $\frac{224}{3}$
35. 72π

Problem Set 11.1, page 482

1. $2\pi, 2\pi, \pi, \pi, 1, 1, \frac{1}{2}, \frac{1}{2}$
5. There is no *smallest* $p > 0$.
13. $\frac{4}{\pi}(\cos x + \frac{1}{9} \cos 3x + \frac{1}{25} \cos 5x + \cdots) + 2(\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x + \cdots)$
15. $\frac{4}{3}\pi^2 + 4(\cos x + \frac{1}{4} \cos 2x + \frac{1}{9} \cos 3x + \cdots) - 4\pi(\sin x + \frac{1}{2} \sin 2x + \frac{1}{3} \sin 3x + \cdots)$
17. $\frac{\pi}{2} + \frac{4}{\pi} \left(\cos x + \frac{1}{9} \cos 3x + \frac{1}{25} \cos 5x + \cdots \right)$

19. $\frac{\pi}{4} - \frac{2}{\pi} \left(\cos x + \frac{1}{9} \cos 3x + \frac{1}{25} \cos 5x + \cdots \right) + \sin x - \frac{1}{2} \sin 2x + \frac{1}{3} \sin 3x - \cdots$
21. $2(\sin x + \frac{1}{2} \sin 2x + \frac{1}{3} \sin 3x + \frac{1}{4} \sin 4x + \frac{1}{5} \sin 5x + \cdots)$

Problem Set 11.2, page 490

1. Neither, even, odd, odd, neither 3. Even 5. Even
9. Odd, $L = 2$, $\frac{4}{\pi} \left(\sin \frac{\pi x}{2} + \frac{1}{3} \sin \frac{3\pi x}{2} + \frac{1}{5} \sin \frac{5\pi x}{2} + \cdots \right)$
11. Even, $L = 1$, $\frac{1}{3} - \frac{4}{\pi^2} \left(\cos \pi x - \frac{1}{4} \cos 2\pi x + \frac{1}{9} \cos 3\pi x - \cdots \right)$
13. Rectifier, $L = \frac{1}{2}$, $\frac{1}{8} - \frac{1}{\pi^2} \left(\cos 2\pi x + \frac{1}{9} \cos 6\pi x + \frac{1}{25} \cos 10\pi x + \cdots \right) + \frac{1}{\pi} \left(\frac{1}{2} \sin 2\pi x - \frac{1}{4} \sin 4\pi x + \frac{1}{6} \sin 6\pi x - \frac{1}{8} \sin 8\pi x + \cdots \right)$
15. Odd, $L = \pi$, $\frac{4}{\pi} \left(\sin x - \frac{1}{9} \sin 3x + \frac{1}{25} \sin 5x - \cdots \right)$
17. Even, $L = 1$, $\frac{1}{2} + \frac{4}{\pi^2} \left(\cos \pi x + \frac{1}{9} \cos 3\pi x + \frac{1}{25} \cos 5\pi x + \cdots \right)$
19. $\frac{3}{8} + \frac{1}{2} \cos 2x + \frac{1}{8} \cos 4x$
23. $L = 4$, (a) 1, (b) $\frac{4}{\pi} \left(\sin \frac{\pi x}{4} + \frac{1}{3} \sin \frac{3\pi x}{4} + \frac{1}{5} \sin \frac{5\pi x}{4} + \cdots \right)$
25. $L = \pi$, (a) $\frac{\pi}{2} + \frac{4}{\pi} \left(\cos x + \frac{1}{9} \cos 3x + \frac{1}{25} \cos 5x + \cdots \right)$,
 (b) $2(\sin x + \frac{1}{2} \sin 2x + \frac{1}{3} \sin 3x + \frac{1}{4} \sin 4x + \cdots)$
27. $L = \pi$, (a) $\frac{3\pi}{8} + \frac{2}{\pi} \left(\cos x - \frac{1}{2} \cos 2x + \frac{1}{9} \cos 3x + \frac{1}{25} \cos 5x - \frac{1}{18} \cos 6x + \frac{1}{49} \cos 7x + \frac{1}{81} \cos 9x - \frac{1}{50} \cos 10x + \frac{1}{121} \cos 11x + \cdots \right)$
 (b) $\left(1 + \frac{2}{\pi} \right) \sin x + \frac{1}{2} \sin 2x + \left(\frac{1}{3} - \frac{2}{9\pi} \right) \sin 3x + \frac{1}{4} \sin 4x + \left(\frac{1}{5} + \frac{2}{25\pi} \right) \sin 5x + \frac{1}{6} \sin 6x + \cdots$
29. Rectifier, $L = \pi$,
 (a) $\frac{2}{\pi} - \frac{4}{\pi} \left(\frac{1}{1 \cdot 3} \cos x + \frac{1}{3 \cdot 5} \cos 3x + \frac{1}{5 \cdot 7} \cos 5x + \cdots \right)$, (b) $\sin x$

Problem Set 11.3, page 494

3. The output becomes a pure cosine series.
5. For A_n this is similar to Fig. 54 in Sec. 2.8, whereas for the phase shift B_n the sense is the same for all n .

7. $y = C_1 \cos \omega t + C_2 \sin \omega t + a(\omega) \sin t$, $a(\omega) = 1/(\omega^2 - 1) = -1.33, -5.26, 4.76, 0.8, 0.01$. Note the change of sign.
11. $y = C_1 \cos \omega t + C_2 \sin \omega t + \frac{4}{\pi} \left(\frac{1}{\omega^2 - 9} \sin t + \frac{1}{\omega^2 - 49} \sin 3t + \frac{1}{\omega^2 - 121} \sin 5t + \cdots \right)$
13. $y = \sum_{n=1}^N (A_n \cos nt + B_n \sin nt)$, $A_n = [(1 - n^2)a_n - nb_n c]/D_n$,
 $B_n = [(1 - n^2)b_n + nca_n]/D_n$, $D_n = (1 - n^2)^2 + n^2 c^2$
15. $b_n = (-1)^{n+1} \cdot 12/n^3$ (n odd), $y = \sum_{n=1}^{\infty} (A_n \cos nt + B_n \sin nt)$,
 $A_n = (-1)^n \cdot 12nc/n^3 D_n$, $B_n = (-1)^{n+1} \cdot 12(1 - n^2)/(n^3 D_n)$ with D_n as in Prob. 13.
17. $I = 50 + A_1 \cos t + B_1 \sin t + A_3 \cos 3t + B_3 \sin 3t + \cdots$, $A_n = (10 - n^2)a_n/D_n$,
 $B_n = 10na_n/D_n$, $a_n = -400/(n^2 \pi)$, $D_n = (n^2 - 10)^2 + 100n^2$
19. $I(t) = \sum_{n=1}^{\infty} (A_n \cos nt + B_n \sin nt)$, $A_n = (-1)^{n+1} \frac{2400(10 - n^2)}{n^2 D_n}$,
 $B_n = (-1)^{n+1} \frac{24,000}{n D_n}$, $D_n = (10 - n^2)^2 + 100n^2$

Section 11.4, page 498

3. $F = \frac{\pi}{2} - \frac{4}{\pi} \left(\cos x + \frac{1}{9} \cos 3x + \frac{1}{25} \cos 5x + \cdots \right)$, $E^* = 0.0748, 0.0748, 0.0119, 0.0119, 0.0037$
5. $F = \frac{4}{\pi} \left(\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x + \cdots \right)$, $E^* = 1.1902, 1.1902, 0.6243, 0.6243, 0.4206$ (0.1272 when $N = 20$)
7. $F = 2[(\pi^2 - 6) \sin x - \frac{1}{8}(4\pi^2 - 6) \sin 2x + \frac{1}{27}(9\pi^2 - 6) \sin 3x - \cdots]$;
 $E^* = 674.8, 454.7, 336.4, 265.6, 219.0$. Why is E^* so large?

Section 11.5, page 503

3. Set $x = ct + k$. 5. $x = \cos \theta$, $dx = -\sin \theta d\theta$, etc.
7. $\lambda_m = (m\pi/10)^2$, $m = 1, 2, \dots$; $y_m = \sin(m\pi x/10)$
9. $\lambda = [(2m + 1)\pi/(2L)]^2$, $m = 0, 1, \dots$, $y_m = \sin((2m + 1)\pi x/(2L))$
11. $\lambda_m = m^2$, $m = 1, 2, \dots$, $y_m = x \sin(m \ln |x|)$
13. $p = e^{8x}$, $q = 0$, $r = e^{8x}$, $\lambda_m = m^2$, $y_m = e^{-4x} \sin mx$, $m = 1, 2, \dots$

Section 11.6, page 509

1. $8(P_1(x) - P_3(x) + P_5(x))$
3. $\frac{4}{5}P_0(x) - \frac{4}{7}P_2(x) - \frac{8}{35}P_4(x)$
9. $-0.4775P_1(x) - 0.6908P_3(x) + 1.844P_5(x) - 0.8236P_7(x) + 0.1658P_9(x) + \cdots$,
 $m_0 = 9$. **Rounding** seems to have considerable influence in Probs. 8–13.

11. $0.7854P_0(x) - 0.3540P_2(x) + 0.0830P_4(x) - \cdots, m_0 = 4$
 13. $0.1212P_0(x) - 0.7955P_2(x) + 0.9600P_4(x) - 0.3360P_6(x) + \cdots, m_0 = 8$
 15. (c) $a_m = (2/J_1^2(\alpha_{0,m}))(J_1(\alpha_{0,m})/\alpha_{0,m}) = 2/(\alpha_{0,m}J_1(\alpha_{0,m}))$

Section 11.7, page 517

1. $f(x) = \pi e^{-x} (x > 0)$ gives $A = \int_0^\infty e^{-v} \cos wv \, dv = \frac{1}{1+w^2}, B = \frac{w}{1+w^2}$
 (see Example 3), etc.
 3. Use (11); $B = \frac{2}{\pi} \int_0^\infty \frac{\pi}{2} \sin wv \, dv = \frac{1 - \cos \pi w}{w}$
 5. $B(w) = \frac{2}{\pi} \int_0^1 \frac{1}{2} \pi v \sin wv \, dv = \frac{\sin w - w \cos w}{w^2}$
 7. $\frac{2}{\pi} \int_0^\infty \frac{\sin w \cos xw}{w} \, dw$
 9. $A(w) = \frac{2}{\pi} \int_0^\infty \frac{\cos wv}{1+v^2} \, dv = e^{-w} (w > 0)$
 11. $\frac{2}{\pi} \int_0^\infty \frac{\cos \pi w + 1}{1-w^2} \cos xw \, dw$
 15. For $n = 1, 2, 11, 12, 31, 32, 49, 50$ the value of $\text{Si}(n\pi) - \pi/2$ equals 0.28, -0.15 , 0.029, -0.026 , 0.0103, -0.0099 , 0.0065, -0.0064 (rounded).
 17. $\frac{2}{\pi} \int_0^\infty \frac{1 - \cos w}{w} \sin xw \, dw$
 19. $\frac{2}{\pi} \int_0^\infty \frac{w - e(w \cos w - \sin w)}{1+w^2} \sin xw \, dw$

Section 11.8, page 522

1. $\hat{f}_c(w) = \sqrt{(2/\pi)} (2 \sin w - \sin 2w)/w$
 3. $\hat{f}_c(w) = \sqrt{(2/\pi)} (\cos 2w + 2w \sin 2w - 1)/w^2$
 5. $\hat{f}_c(w) = \sqrt{\frac{2}{\pi}} \frac{(w^2 - 2) \sin w + 2w \cos w}{w^3}$
 7. Yes. No
 9. $\sqrt{2/\pi} w/(a^2 + w^2)$
 11. $\sqrt{2/\pi} ((2 - w^2) \cos w + 2w \sin w - 2)/w^3$
 13. $\mathcal{F}_s(e^{-x}) = \frac{1}{w} \left(-\mathcal{F}_c(e^{-x}) + \sqrt{\frac{2}{\pi}} \cdot 1 \right) = \frac{1}{w} \left(\sqrt{\frac{2}{\pi}} \cdot \frac{1}{w^2 + 1} + \sqrt{\frac{2}{\pi}} \right) = \sqrt{\frac{2}{\pi}} \frac{w}{w^2 + 1}$

Problem Set 11.9, page 533

3. $i(e^{-ibw} - e^{-iaw})/(w\sqrt{2\pi})$ if $a < b$; 0 otherwise
 5. $[e^{(1-iw)a} - e^{-(1-iw)a}]/(\sqrt{2\pi}(1-iw))$
 7. $(e^{-iaw}(1+iaw) - 1)/(\sqrt{2\pi}w^2)$ 9. $\sqrt{2/\pi}(\cos w + w \sin w - 1)/w^2$
 11. $i\sqrt{2/\pi}(\cos w - 1)/w$ 13. $e^{-w^2/2}$ by formula 9

17. No, the assumptions in Theorem 3 are not satisfied.

19. $[f_1 + f_2 + f_3 + f_4, f_1 - if_2 - f_3 + if_4, f_1 - f_2 + f_3 - f_4, f_1 + if_2 - f_3 - if_4]$

$$21. \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \end{bmatrix} = \begin{bmatrix} f_1 + f_2 \\ f_1 - f_2 \end{bmatrix}$$

Chapter 11 Review Questions and Problems, page 537

$$11. 1 + \frac{4}{\pi} \left(\sin \frac{\pi x}{2} + \frac{1}{3} \sin \frac{3\pi x}{2} + \frac{1}{5} \sin \frac{5\pi x}{2} + \cdots \right)$$

$$13. \frac{1}{4} - \frac{2}{\pi^2} \left(\cos \pi x + \frac{1}{9} \cos 3\pi x + \frac{1}{25} \cos 5\pi x + \cdots \right) + \frac{1}{\pi} \left(\sin \pi x - \frac{1}{2} \sin 2\pi x + \frac{1}{3} \sin 3\pi x - + \cdots \right)$$

15. $\cosh x, \sinh x$ ($-5 < x < 5$), respectively

17. Cf. Sec. 11.1.

$$19. \frac{1}{2} - \frac{4}{\pi^2} \left(\cos \pi x + \frac{1}{9} \cos 3\pi x + \cdots \right), \quad \frac{2}{\pi} \left(\sin \pi x - \frac{1}{2} \sin 2\pi x + - \cdots \right)$$

$$21. y = C_1 \cos \omega t + C_2 \sin \omega t + \frac{\pi^2}{\omega^2} - 12 \left(\frac{\cos t}{\omega^2 - 1} - \frac{1}{4} \cdot \frac{\cos 2t}{\omega^2 - 4} + \frac{1}{9} \cdot \frac{\cos 3t}{\omega^2 - 9} - \frac{1}{16} \cdot \frac{\cos 4t}{\omega^2 - 16} + - \cdots \right)$$

23. 0.82, 0.50, 0.36, 0.28, 0.23

25. 0.0076, 0.0076, 0.0012, 0.0012, 0.0004

$$27. \frac{1}{\pi} \int_0^\infty \frac{(\cos w + w \sin w - 1) \cos wx + (\sin w - w \cos w) \sin wx}{w^2} dw$$

$$29. \sqrt{2/\pi} (\cos aw - \cos w + aw \sin aw - w \sin w)/w^2$$

Problem Set 12.1, page 542

$$1. L(c_1 u_1 + c_2 u_2) = c_1 L(u_1) + c_2 L(u_2) = c_1 \cdot 0 + c_2 \cdot 0 = 0$$

$$3. c = 2$$

$$5. c = a/b$$

7. Any c and ω

$$9. c = \pi/25$$

$$15. u = 110 - (110/\ln 100) \ln(x^2 + y^2)$$

$$17. u = a(y) \cos 4\pi x + b(y) \sin 4\pi x$$

$$19. u = c(x) e^{-y^{3/3}}$$

$$21. u = e^{-3y}(a(x) \cos 2y + b(x) \sin 2y) + 0.1e^{3y}$$

$$23. u = c_1(y)x + c_2(y)/x^2 \text{ (Euler–Cauchy)}$$

$$25. u(x, y) = axy + bx + cy + k; a, b, c, k \text{ arbitrary constants}$$

Problem Set 12.3, page 551

$$5. k \cos 3\pi t \sin 3\pi x$$

$$7. \frac{8k}{\pi^3} \left(\cos \pi t \sin \pi x + \frac{1}{27} \cos 3\pi t \sin 3\pi x + \frac{1}{125} \cos 5\pi t \sin 5\pi x + \cdots \right)$$

$$9. \frac{0.8}{\pi^2} \left(\cos \pi t \sin \pi x - \frac{1}{9} \cos 3\pi t \sin 3\pi x + \frac{1}{25} \cos 5\pi t \sin 5\pi x - + \cdots \right)$$

11. $\frac{2}{\pi^2} \left((2 - \sqrt{2}) \cos \pi t \sin \pi x - \frac{1}{9} (2 + \sqrt{2}) \cos 3\pi t \sin 3\pi x \right.$
 $\left. + \frac{1}{25} (2 + \sqrt{2}) \cos 5\pi t \sin 5\pi x - + \cdots \right)$
13. $\frac{4}{\pi^3} \left((4 - \pi) \cos \pi t \sin \pi x + \cos 2\pi t \sin 2\pi x + \frac{4 + 3\pi}{27} \cos 3\pi t \sin 3\pi x \right.$
 $\left. + \frac{4 - 5\pi}{125} \cos 5\pi t \sin 5\pi x + \cdots \right)$. No terms with $n = 4, 8, 12, \dots$.
17. $u = \frac{8L^2}{\pi^3} \left(\cos \left[c \left(\frac{\pi}{L} \right)^2 t \right] \sin \frac{\pi x}{L} + \frac{1}{3^3} \cos \left[c \left(\frac{3\pi}{L} \right)^2 t \right] \sin \frac{3\pi x}{L} + \cdots \right)$
19. (a) $u(0, t) = 0$, (b) $u(L, t) = 0$, (c) $u_x(0, t) = 0$, (d) $u_x(L, t) = 0$. $C = -A$, $D = -B$ from (a), (c). Insert this. The coefficient determinant resulting from (b), (d) must be zero to have a nontrivial solution. This gives (22).

Problem Set 12.4, page 556

3. $c^2 = 300/[0.9/(2 \cdot 9.80)] = 80.83^2 \text{ [m}^2/\text{sec}^2]$
9. Elliptic, $u = f_1(y + 2ix) + f_2(y - 2ix)$
11. Parabolic, $u = xf_1(x - y) + f_2(x - y)$
13. Hyperbolic, $u = f_1(y - 4x) + f_2(y - x)$
15. Hyperbolic, $xy'^2 + yy' = 0$, $y = v$, $xy = w$, $u_w = z$, $u = \frac{1}{y} f_1(xy) + f_2(y)$
17. Elliptic, $u = f_1(y - (2 - i)x) + f_2(y - (2 + i)x)$. Real or imaginary parts of any function u of this form are solutions. Why?

Problem Set 12.6, page 566

3. $u_1 = \sin x e^{-t}$, $u_2 = \sin 2x e^{-4t}$, $u_3 = \sin 3x e^{-9t}$ differ in rapidity of decay.
5. $u = \sin 0.1\pi x e^{-1.752\pi^2 t/100}$
7. $u = \frac{800}{\pi^3} \left(\sin 0.1\pi x e^{-0.01752\pi^2 t} + \frac{1}{3^3} \sin 0.3\pi x e^{-0.01752(3\pi)^2 t} + \cdots \right)$
9. $u = u_I + u_{II}$, where $u_{II} = u - u_I$ satisfies the boundary conditions of the text,
 so that $u_{II} = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{L} e^{-(cn\pi/L)^2 t}$, $B_n = \frac{2}{L} \int_0^L [f(x) - u_I(x)] \sin \frac{n\pi x}{L} dx$.
11. $F = A \cos px + B \sin px$, $F'(0) = Bp = 0$, $B = 0$, $F'(L) = -Ap \sin pL = 0$,
 $p = n\pi/L$, etc.
13. $u = 1$
15. $\frac{1}{2} + \frac{4}{\pi^2} \left(\cos x e^{-t} + \frac{1}{9} \cos 3x e^{-9t} + \frac{1}{25} \cos 5x e^{-25t} + \cdots \right)$
17. $-\frac{K\pi}{L} \sum_{n=1}^{\infty} n B_n e^{-\lambda_n^2 t}$
19. $u = 1000 (\sin \frac{1}{2} \pi x \sinh \frac{1}{2} \pi y) / \sinh \pi$
21. $u = \frac{100}{\pi} \sum_{n=1}^{\infty} \frac{1}{(2n-1) \sinh (2n-1)\pi} \sin \frac{(2n-1)\pi x}{24} \sinh \frac{(2n-1)\pi y}{24}$

$$23. u = A_0 x + \sum_{n=1}^{\infty} A_n \frac{\sinh(n\pi x/24)}{\sinh n\pi} \cos \frac{n\pi y}{24},$$

$$A_0 = \frac{1}{24^2} \int_0^{24} f(y) dy, \quad A_n = \frac{1}{12} \int_0^{24} f(y) \cos \frac{n\pi y}{24} dy$$

$$25. \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{a} \sinh \frac{n\pi(b-y)}{a}, \quad A_n = \frac{2}{a \sinh(n\pi b/a)} \int_0^a f(x) \sin \frac{n\pi x}{a} dx$$

Problem Set 12.7, page 574

$$3. A = \frac{2}{\pi} \int_0^{\infty} \frac{\cos pv}{1+v^2} dv = \frac{2}{\pi} \cdot \frac{\pi}{2} e^{-p}, \quad u = \int_0^{\infty} e^{-p-c^2 p^2 t} \cos px dp$$

$$5. A = \frac{2}{\pi} \int_0^1 v \cos pv dv = \frac{2}{\pi} \cdot \frac{\cos p + p \sin p - 1}{p^2}, \text{ etc.}$$

$$7. A = \frac{2}{\pi} \int_0^{\infty} \frac{\sin v}{v} \cos pv dv = \frac{2}{\pi} \cdot \frac{\pi}{2} = 1 \text{ if } 0 < p < 1 \text{ and } 0 \text{ if } p > 1,$$

$$u = \int_0^1 \cos px e^{-c^2 p^2 t} dp$$

9. Set $w = -v$ in (21) to get $\operatorname{erf}(-x) = -\operatorname{erf} x$.

13. In (12) the argument $x + 2cz\sqrt{t}$ is 0 (the point where f jumps) when $z = -x/(2c\sqrt{t})$. This gives the lower limit of integration.

15. Set $w = s/\sqrt{2}$ in (21).

Problem Set 12.9, page 584

1. (a), (b) It is multiplied by $\sqrt{2}$. (c) Half

5. $B_{mn} = (-1)^{n+1} 8/(mn\pi^2)$ if m odd, 0 if m even

7. $B_{mn} = (-1)^{m+n} 4ab/(mn\pi^2)$

11. $u = 0.1 \cos \sqrt{20}t \sin 2x \sin 4y$

$$13. \frac{6.4}{\pi^2} \sum_{m=1}^{\infty} \sum_{\substack{n=1 \\ m,n \text{ odd}}}^{\infty} \frac{1}{m^3 n^3} \cos(t\sqrt{m^2 + n^2}) \sin mx \sin ny$$

17. $c\pi\sqrt{260}$ (corresponding eigenfunctions $F_{4,16}$ and $F_{16,14}$), etc.

$$19. \cos\left(\pi t \sqrt{\frac{36}{a^2} + \frac{4}{b^2}}\right) \sin \frac{6\pi x}{a} \sin \frac{4\pi y}{b}$$

Problem Set 12.10, page 591

$$5. 110 + \frac{440}{\pi} (r \cos \theta - \frac{1}{3} r^3 \cos 3\theta + \frac{1}{5} r^5 \cos 5\theta - \dots)$$

$$7. 55\pi - \frac{440}{\pi} (r \cos \theta + \frac{1}{9} r^3 \cos 3\theta + \frac{1}{25} r^5 \cos 5\theta + \dots)$$

11. Solve the problem in the disk $r < a$ subject to u_0 (given) on the upper semicircle and $-u_0$ on the lower semicircle.

$$u = \frac{4u_0}{\pi} \left(\frac{r}{a} \sin \theta + \frac{1}{3a^3} r^3 \sin 3\theta + \frac{1}{5a^5} r^5 \sin 5\theta + \cdots \right)$$

13. Increase by a factor $\sqrt{2}$

15. $T = 6.826\rho R^2 f_1^2$

17. No

25. $\alpha_{11}/(2\pi) = 0.6098$; See Table A1 in App. 5.

Problem Set 12.11, page 598

5. $A_4 = A_6 = A_8 = A_{10} = 0$, $A_5 = 605/16$, $A_7 = -4125/128$, $A_9 = 7315/256$

9. $\nabla^2 u = u'' + 2u'/r = 0$, $u''/u' = -2/r$, $\ln |u'| = -2 \ln |r| + c_1$,

$u' = \tilde{c}/r^2$, $u = c/r + k$

13. $u = 320/r + 60$ is smaller than the potential in Prob. 12 for $2 < r < 4$.

17. $u = 1$

19. $\cos 2\phi = 2 \cos^2 \phi - 1$, $2w^2 - 1 = \frac{4}{3}P_2(w) - \frac{1}{3}$, $u = \frac{4}{3}r^2 P_2(\cos \phi) - \frac{1}{3}$

25. Set $1/r = \rho$. Then $u(\rho, \theta, \phi) = rv(r, \theta, \phi)$, $u_\rho = (v + rv_r)(-1/\rho^2)$,

$u_{\rho\rho} = (2v_r + rv_{rr})(1/\rho^4) + (v + rv_r)(2/\rho^3)$, $u_{\rho\rho} + (2/\rho)u_\rho = r^5(v_{rr} + (2/r)v_r)$.

Substitute this and $u_{\phi\phi} = rv_{\phi\phi}$ etc. into (7) [written in terms of ρ] and divide by r^5 .

Problem Set 12.12, page 602

5. $W = \frac{c(s)}{x^s} + \frac{x}{s^2(s+1)}$, $W(0, s) = 0$, $c(s) = 0$, $w(x, t) = x(t - 1 + e^{-t})$

7. $w = f(x)g(t)$, $xf'g + f_2^2 = xt$, take $f(x) = x$ to get $g = ce^{-t} + t - 1$ and $c = 1$ from $w(x, 0) = x(c - 1) = 0$.

11. Set $x^2/(4c^2\tau) = z^2$. Use z as a new variable of integration. Use $\text{erf}(\infty) = 1$.

Chapter 12 Review Questions and Problems, page 603

17. $u = c_1(x)e^{-3y} + c_2(x)e^{2y} - 3$

19. Hyperbolic, $f_1(x) + f_2(y + x)$

21. Hyperbolic, $f_1(y + 2x) + f_2(y - 2x)$

23. $\frac{3}{4} \cos 2t \sin x - \frac{1}{4} \cos 6t \sin 3x$

25. $\sin 0.01\pi x e^{-0.001143t}$

27. $\frac{3}{4} \sin 0.01\pi x e^{-0.001143t} - \frac{1}{4} \sin 0.03\pi x e^{-0.01029t}$

29. $100 \cos 2x e^{-4t}$

39. $u = (u_1 - u_0)(\ln r)/\ln(r_1/r_0) + (u_0 \ln r_1 - u_1 \ln r_0)/\ln(r_1/r_0)$

Problem Set 13.1, page 612

1. $1/i = i/i^2 = -i$, $1/i^3 = i/i^4 = i$

3. $4.8 - 1.4i$

5. $x - iy = -(x + iy)$, $x = 0$

9. $-117, 4$

11. $-8 - 6i$

13. $-120 - 40i$

15. $3 - i$

17. $-4x^2y^2$

19. $(x^2 - y^2)/(x^2 + y^2)$, $2xy/(x^2 + y^2)$

Problem Set 13.2, page 618

1. $\sqrt{2}(\cos \frac{1}{4}\pi + i \sin \frac{1}{4}\pi)$

3. $2(\cos \frac{1}{2}\pi + i \sin \frac{1}{2}\pi)$, $2(\cos \frac{1}{2}\pi - i \sin \frac{1}{2}\pi)$

5. $\frac{1}{2}(\cos \pi + i \sin \pi)$
 9. $3\pi/4$
 13. -1024 . Answer: π
 17. $2 + 2i$
 23. $6, -3 \pm 3\sqrt{3}i$
 25. $\cos(\frac{1}{8}\pi + \frac{1}{2}k\pi) + i \sin(\frac{1}{8}\pi + \frac{1}{2}k\pi), k = 0, 1, 2, 3$
 27. $\cos \frac{1}{5}\pi \pm i \sin \frac{1}{5}\pi, \cos \frac{3}{5}\pi \pm i \sin \frac{3}{5}\pi, -1$
 29. $i, -1 - i$
 31. $\pm(1 - i), \pm(2 + 2i)$
 33. $|z_1 + z_2|^2 = (z_1 + z_2)(\overline{z_1 + z_2}) = (z_1 + z_2)(\overline{z_1} + \overline{z_2})$. Multiply out and use $\operatorname{Re} z_1 \overline{z_2} \leq |z_1 \overline{z_2}|$ (Prob. 34).
 $z_1 \overline{z_1} + z_1 \overline{z_2} + z_2 \overline{z_1} + z_2 \overline{z_2} = |z_1|^2 + 2 \operatorname{Re} z_1 \overline{z_2} + |z_2|^2 \leq |z_1|^2 + 2|z_1||z_2| + |z_2|^2 = (|z_1| + |z_2|)^2$. Hence $|z_1 + z_2|^2 \leq (|z_1| + |z_2|)^2$. Taking square roots gives (6).
 35. $[(x_1 + x_2)^2 + (y_1 + y_2)^2] + [(x_1 - x_2)^2 + (y_1 - y_2)^2] = 2(x_1^2 + y_1^2 + x_2^2 + y_2^2)$

Problem Set 13.3, page 624

1. Closed disk, center $-1 + 5i$, radius $\frac{3}{2}$
 3. Annulus (circular ring), center $4 - 2i$, radii π and 3π
 5. Domain between the bisecting straight lines of the first quadrant and the fourth quadrant.
 7. Half-plane extending from the vertical straight line $x = -1$ to the right.
 11. $u(x, y) = (1 - x)/((1 - x)^2 + y^2), u(1, -1) = 0,$
 $v(x, y) = y/((1 - x)^2 + y^2), v(1, -1) = -1$
 15. Yes, since $\operatorname{Im}(|z|^2/z) = \operatorname{Im}(|z|^2 \overline{z}/(z\overline{z})) = \operatorname{Im} \overline{z} = -r \sin \theta \rightarrow 0$.
 17. Yes, because $\operatorname{Re} z = r \cos \theta \rightarrow 0$ and $1 - |z| \rightarrow 1$ as $r \rightarrow 0$.
 19. $f'(z) = 8(z - 4i)^7$. Now $z - 4i = 3$, hence $f'(3 + 4i) = 8 \cdot 3^7 = 17,496$.
 21. $n(1 - z)^{-n-1}i, ni$
 23. $3iz^2/(z + i)^4, -3i/16$

Problem Set 13.4, page 629

1. $r_x = x/r = \cos \theta, r_y = \sin \theta, \theta_x = -(\sin \theta)/r, \theta_y = (\cos \theta)/r$
 (a) $0 = u_x - v_y = u_r \cos \theta + u_\theta(-\sin \theta)/r - v_r \sin \theta - v_\theta(\cos \theta)/r$
 (b) $0 = u_y + v_x = u_r \sin \theta + u_\theta(\cos \theta)/r + v_r \cos \theta + v_\theta(-\sin \theta)/r$
 Multiply (a) by $\cos \theta$, (b) by $\sin \theta$, and add. Etc. _____
 3. Yes
 5. No, $f(z) = \overline{(z^2)}$
 7. Yes, when $z \neq 0$. Use (7).
 9. Yes, when $z \neq 0, -2\pi i, 2\pi i$
 11. Yes
 13. $f(z) = -\frac{1}{2}i(z^2 + c), c$ real
 15. $f(z) = 1/z + c$ (c real)
 17. $f(z) = z^2 + z + c$ (c real)
 19. No
 21. $a = \pi, v = e^{\pi x} \sin \pi y$
 23. $a = 0, v = \frac{1}{2}b(y^2 - x^2) + c$
 27. $f = u + iv$ implies $if = -v + iu$.
 29. Use (4), (5), and (1).

Problem Set 13.5, page 632

3. $e^{2\pi i} e^{-2\pi} = e^{-2\pi} = 0.001867$
 5. $e^2(-1) = -7.389$
 7. $e^{\sqrt{2}i} = 4.113i$
 9. $5e^{i \arctan(3/4)} = 5e^{0.644i}$
 11. $6.3e^{\pi i}$
 13. $\sqrt{2}e^{\pi i/4}$

15. $\exp(x^2 - y^2) \cos 2xy, \exp(x^2 - y^2) \sin 2xy$
 17. $\operatorname{Re}(\exp(z^3)) = \exp(x^3 - 3xy^2) \cos(3x^2y - y^3)$
 19. $z = 2n\pi i, \quad n = 0, 1, \dots$

Problem Set 13.6, page 636

1. Use (11), then (5) for e^{iy} , and simplify. 7. $\cosh 1 = 1.543, i \sinh 1 = 1.175i$
 9. Both $-0.642 - 1.069i$. Why? 11. $i \sinh \pi = 11.55i$, both
 15. Insert the definitions on the left, multiply out, and simplify.
 17. $z = \pm(2n + 1)i/2$ 19. $z = \pm n\pi i$

Problem Set 13.7, page 640

5. $\ln 11 + \pi i$ 7. $\frac{1}{2} \ln 32 - \pi i/4 = 1.733 - 0.785i$
 9. $i \arctan(0.8/0.6) = 0.927i$ 11. $\ln e + \pi i/2 = 1 + \pi i/2$
 13. $\pm 2n\pi i, \quad n = 0, 1, \dots$
 15. $\ln |e^i| + i \arctan \frac{\sin 1}{\cos 1} \pm 2n\pi i = 0 + i + 2n\pi i, \quad n = 0, 1, \dots$
 17. $\ln(i^2) = \ln(-1) = (1 \pm 2n)\pi i, \quad 2 \ln i = (1 \pm 4n)\pi i, \quad n = 0, 1, \dots$
 19. $e^{4-3i} = e^4(\cos 3 - i \sin 3) = -54.05 - 7.70i$
 21. $e^{0.6} e^{0.4i} = e^{0.6}(\cos 0.4 + i \sin 0.4) = 1.678 + 0.710i$
 23. $e^{(1-i) \operatorname{Ln}(1+i)} = e^{\ln \sqrt{2} + \pi i/4 - i \ln \sqrt{2} + \pi/4} = 2.8079 + 1.3179i$
 25. $e^{(3-i) \operatorname{Ln}(3+i)} = 27e^\pi(\cos(3\pi - \ln 3) + i \sin(3\pi - \ln 3)) = -284.2 + 556.4i$
 27. $e^{(2-i) \operatorname{Ln}(-1)} = e^{(2-i)\pi i} = e^\pi = 23.14$

Chapter 13 Review Questions and Problems, page 641

1. $2 - 3i$ 3. $27.46e^{0.9929i}, 7.616e^{1.976i}$
 11. $-5 + 12i$ 13. $0.16 - 0.12i$
 15. i 17. $4\sqrt{2}e^{-3\pi i/4}$
 19. $15e^{-\pi i/2}$ 21. $\pm 3, \pm 3i$
 23. $(\pm 1 \pm i)/\sqrt{2}$ 25. $f(z) = -iz^2/2$
 27. $f(z) = e^{-2z}$ 29. $f(z) = e^{-z^2/2}$
 31. $\cos 3 \cosh 1 + i \sin 3 \sinh 1 = -1.528 + 0.166i$
 33. $i \tanh 1 = 0.7616i$
 35. $\cosh \pi \cos \pi + i \sinh \pi \sin \pi = -11.592$

Problem Set 14.1, page 651

1. Straight segment from (2, 1) to (5, 2.5).
 3. Parabola $y = x^2$ from (1, 2) to (2, 8).
 5. Circle through (0, 0), center (3, -1), radius $\sqrt{10}$, oriented clockwise.
 7. Semicircle, center 2, radius 4.
 9. Cubic parabola $y = x^3 \quad (-2 \leq x \leq 2)$
 11. $z(t) = t + (2 + t)i \quad (-1 \leq t \leq 1)$
 13. $z(t) = 2 - i + 2e^{it} \quad (0 \leq t \leq \pi)$

15. $z(t) = 2 \cosh t + i \sinh t$ ($-\infty < t < \infty$)
 17. Circle $z(t) = -a - ib + re^{-it}$ ($0 \leq t \leq 2\pi$)
 19. $z(t) = t + (1 - \frac{1}{4}t^2)i$ ($-2 \leq t \leq 2$)
 21. $z(t) = (1 + i)t$ ($1 \leq t \leq 3$), $\operatorname{Re} z = t$, $z'(t) = 1 + i$. Answer: $4 + 4i$
 23. $e^{2\pi i} - e^{\pi i} = 1 - (-1) = 2$
 25. $\frac{1}{2} \exp z^2 \Big|_1^i = \frac{1}{2}(e^{-1} - e^1) = -\sinh 1$
 27. $\tan \frac{1}{4}\pi i - \tan \frac{1}{4} = i \tanh \frac{1}{4} - 1$
 29. $\operatorname{Im} z^2 = 2xy = 0$ on the axes. $z = 1 + (-1 + i)t$ ($0 \leq t \leq 1$),
 $(\operatorname{Im} z^2) \dot{z} = 2(1 - t)y(-1 + i)$ integrated: $(-1 + i)/3$.
 35. $|\operatorname{Re} z| = |x| \leq 3 = M$ on C , $L = \sqrt{8}$

Problem Set 14.2, page 659

1. Use (12), Sec. 14.1, with $m = 2$. 3. Yes 5. 5
 7. (a) Yes. (b) No, we would have to move the contour across $\pm 2i$.
 9. 0, yes 11. πi , no
 13. 0, yes 15. $-\pi$, no
 17. 0, no 19. 0, yes
 21. $2\pi i$ 23. $1/z + 1/(z - 1)$, hence $2\pi i + 2\pi i = 4\pi i$.
 25. 0 (Why?) 27. 0 (Why?)
 29. 0

Problem Set 14.3, page 663

1. $2\pi i z^2/(z - 1) \Big|_{z=-1} = -\pi i$ 3. 0
 5. $2\pi i (\cos 3z)/6 \Big|_{z=0} = \pi i/3$ 7. $2\pi i (i/2)^3/2 = \pi/8$
 11. $2\pi i \cdot \frac{1}{z + 2i} \Big|_{z=2i} = \frac{\pi}{2}$ 13. $2\pi i (z + 2) \Big|_{z=2} = 8\pi i$
 15. $2\pi i \cosh(-\pi^2 - \pi i) = -2\pi i \cosh \pi^2 = -60,739i$ since $\cosh \pi i = \cos \pi = -1$
 and $\sinh \pi i = i \sin \pi = 0$.
 17. $2\pi i \frac{\operatorname{Ln}(z + 1)}{z + i} \Big|_{z=i} = 2\pi i \frac{\operatorname{Ln}(1 + i)}{2i} = \pi(\ln \sqrt{2} + i\pi/4) = 1.089 + 2.467i$
 19. $2\pi i e^{2i}/(2i) = \pi e^{2i}$

Problem Set 14.4, page 667

1. $(2\pi i/3!)(-\cos 0) = -\pi i/3$ 3. $(2\pi i/(n - 1)!)e^0$
 5. $\frac{2\pi i}{3!}(\cosh 2z)''' = \frac{\pi i}{3} \cdot 8 \sinh 1 = 9.845i$
 7. $(2\pi i/(2n)!) (\cos z)^{(2n)} \Big|_{z=0} = (2\pi i/(2n)!)(-1)^n \cos 0 = (-1)^n 2\pi i/(2n)!$
 9. $-2\pi i (\tan \pi z)' \Big|_{z=0} = \frac{-2\pi i \cdot \pi}{\cos^2 \pi z} \Big|_{z=0} = -2\pi^2 i$
 11. $\frac{2\pi i}{4}((1 + z)\sin z)' \Big|_{z=1/2} = \frac{1}{2}\pi i(\sin z + (1 + z)\cos z) \Big|_{z=1/2}$
 $= \frac{1}{2}\pi i(\sin \frac{1}{2} + \frac{3}{2}\cos \frac{1}{2})$
 $= 2.821i$

13. $2\pi i \cdot \frac{1}{z} \Big|_{z=2} = \pi i$ 15. 0. Why?
17. 0 by Cauchy's integral theorem for a doubly connected domain; see (6) in Sec. 14.2.
19. $(2\pi i/2!)4^{-3}(e^{3z})''|_{z=\pi i/4} = -9\pi(1+i)/(64\sqrt{2})$

Chapter 14 Review Questions and Problems, page 668

21. $\frac{1}{2} \cosh(-\frac{1}{4}\pi^2) - \frac{1}{2} = 2.469$
23. $2\pi i(e^z)^{(4)}|_{z=0} = ie^z/12|_{z=0} = \pi i/12$ by Cauchy's integral formula.
25. $-2\pi i(\tan \pi z)'|_{z=1} = -2\pi^2 i/\cos^2 \pi z|_{z=1} = -2\pi^2 i$
27. 0 since $z^2 + \bar{z} - 2 = 2(x^2 - y^2)$ and $y = x$
29. $-4\pi i$

Problem Set 15.1, page 679

1. $z_n = (2i/2)^n$; bounded, divergent, $\pm 1, \pm i$
3. $z_n = -\frac{1}{2}\pi i/(1 + 2/(ni))$ by algebra; convergent to $-\pi i/2$
5. Bounded, divergent, $\pm 1 + 10i$
7. Unbounded, hence divergent
9. Convergent to 0, hence bounded
17. Divergent; use $1/\ln n > 1/n$. 19. Convergent; use $\Sigma 1/n^2$.
21. Convergent 23. Convergent
25. Divergent
29. By absolute convergence and Cauchy's convergence principle, for given $\epsilon > 0$ we have for every $n > N(\epsilon)$ and $p = 1, 2, \dots$

$$|z_{n+1}| + \dots + |z_{n+p}| < \epsilon,$$

hence $|z_{n+1} + \dots + z_{n+p}| < \epsilon$ by (6*), Sec. 13.2, hence convergence by Cauchy's principle.

Problem Set 15.2, page 684

1. No! Nonnegative integer powers of z (or $z - z_0$) only!
3. At the center, in a disk, in the whole plane
5. $\Sigma a_n z^{2n} = \Sigma a_n (z^2)^n$, $|z^2| < R = \lim |a_n/a_{n+1}|$; hence $|z| < \sqrt{R}$.
7. $\pi/2, \infty$ 9. $i, \sqrt{3}$ 11. $0, \sqrt{\frac{26}{5}}$
13. $-i, \frac{1}{2}$ 15. $2i, 1$ 17. $1/\sqrt{2}$

Problem Set 15.3, page 689

3. $f = \sqrt[n]{n}$. Apply l'Hôpital's rule to $\ln f = (\ln n)/n$.
5. 2 7. $\sqrt{3}$ 9. $1/\sqrt{2}$
11. $\sqrt{\frac{7}{3}}$ 13. 1 15. $\frac{3}{4}$

Problem Set 15.4, page 697

$$3. 2z^2 - \frac{(2z^2)^3}{3!} + \dots = 2z^2 - \frac{4}{3}z^6 + \frac{4}{15}z^{10} - \dots, \quad R = \infty$$

$$5. \frac{1}{2} - \frac{1}{4}z^4 + \frac{1}{8}z^8 - \frac{1}{16}z^{12} + \frac{1}{32}z^{16} - + \dots, \quad R = \sqrt[4]{2}$$

$$7. \frac{1}{2} + \frac{1}{2} \cos z = 1 - \frac{1}{2 \cdot 2!} z^2 + \frac{1}{2 \cdot 4!} z^4 - \frac{1}{2 \cdot 6!} z^6 + - \dots, \quad R = \infty$$

$$9. \int_0^z \left(1 - \frac{1}{2}t^2 + \frac{1}{8}t^4 - + \dots \right) dt = z - \frac{1}{6}z^3 + \frac{1}{40}z^5 - + \dots, \quad R = \infty$$

$$11. z^3/(1!3) - z^7/(3!7) + z^{11}/(5!11) - + \dots, \quad R = \infty$$

$$13. (2/\sqrt{\pi})(z - z^3/3 + z^5/(2!5) - z^7/(3!7) + \dots), \quad R = \infty$$

$$17. \text{Team Project. (a)} (\ln(1+z))' = 1 - z + z^2 - + \dots = 1/(1+z).$$

(c) Use that the terms of $(\sin iy)/(iy)$ are all positive, so that the sum cannot be zero.

$$19. \frac{1}{2} + \frac{1}{2}i + \frac{1}{2}i(z-i) + (-\frac{1}{4} + \frac{1}{4}i)(z-i)^2 - \frac{1}{4}(z-i)^3 + \dots, \quad R = \sqrt{2}$$

$$21. 1 - \frac{1}{2!} \left(z - \frac{1}{2}\pi \right)^2 + \frac{1}{4!} \left(z - \frac{1}{2}\pi \right)^4 - \frac{1}{6!} \left(z - \frac{1}{2}\pi \right)^6 + - \dots, \quad R = \infty$$

$$23. -\frac{1}{4} - \frac{2}{8}i(z-i) + \frac{3}{16}(z-i)^2 + \frac{4}{32}i(z-i)^3 - \frac{5}{64}(z-i)^4 + \dots, \quad R = 2$$

$$25. 2 \left(z - \frac{1}{2}i \right) + \frac{2^3}{3!} \left(z - \frac{1}{2}i \right)^3 + \frac{2^5}{5!} \left(z - \frac{1}{2}i \right)^5 + \dots, \quad R = \infty$$

Problem Set 15.5, page 704

$$3. |z+i| \leq \sqrt{3} - \delta, \quad \delta > 0$$

$$5. |z + \frac{1}{2}i| \leq \frac{1}{4} - \delta, \quad \delta > 0$$

7. Nowhere

$$9. |z - 2i| \leq 2 - \delta, \quad \delta > 0$$

$$11. |z^n| \leq 1 \text{ and } \sum 1/n^2 \text{ converges. Use Theorem 5.}$$

$$13. |\sin^n z| \leq 1 \text{ for all } z, \text{ and } \sum 1/n^2 \text{ converges. Use Theorem 5.}$$

$$15. R = 4 \text{ by Theorem 2 in Sec. 15.2; use Theorem 1.}$$

$$17. R = 1/\sqrt{\pi} > 0.56; \text{ use Theorem 1.}$$

Chapter 15 Review Questions and Problems, page 706

$$11. 1$$

$$13. 3$$

$$15. \frac{1}{2}$$

$$17. \infty, \quad e^{2z}$$

$$19. \infty, \quad \cosh \sqrt{z}$$

$$21. \sum_{n=0}^{\infty} \frac{z^{4n}}{(2n+1)!}, \quad R = \infty$$

$$23. \frac{1}{2} + \frac{1}{2} \cos 2z = 1 + \frac{1}{2} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n)!} (2z)^{2n}, \quad R = \infty$$

$$25. \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n!} z^{2n-2}, \quad R = \infty$$

$$27. \cos \left[\left(z - \frac{1}{2}\pi \right) + \frac{1}{2}\pi \right] = -\left(z - \frac{1}{2}\pi \right) + \frac{1}{6} \left(z - \frac{1}{2}\pi \right)^3 - + \dots = -\sin \left(z - \frac{1}{2}\pi \right)$$

$$29. \ln 3 + \frac{1}{3}(z-3) - \frac{1}{2 \cdot 9}(z-3)^2 + \frac{1}{3 \cdot 27}(z-3)^3 - + \dots, \quad R = 3$$

Problem Set 16.1, page 714

1. $z^{-4} - \frac{1}{2}z^{-2} + \frac{1}{24} - \frac{1}{720}z^2 + \cdots, \quad 0 < |z| < \infty$
3. $z^{-3} + z^{-1} + \frac{1}{2}z + \frac{1}{6}z^3 + \frac{1}{24}z^5 + \cdots, \quad 0 < |z| < \infty$
5. $z^{-2} + z^{-1} + 1 + z + z^2 + \cdots, \quad 0 < |z| < 1$
7. $z^3 + \frac{1}{2}z + \frac{1}{24}z^{-1} + \frac{1}{720}z^3 + \cdots, \quad 0 < |z| < \infty$
9. $\exp[1 + (z - 1)](z - 1)^{-2} = e \cdot [(z - 1)^{-2} + (z - 1)^{-1} + \frac{1}{2} + \frac{1}{6}(z - 1) + \cdots],$
 $0 < |z - 1| < \infty$
11. $\frac{[\pi i + (z - \pi i)]^2}{(z - \pi i)^4} = \frac{(\pi i)^2}{(z - \pi i)^4} + \frac{2\pi i}{(z - \pi i)^3} + \frac{1}{(z - \pi i)^2}$
13. $i^{-3} \left(1 + \frac{z - i}{i}\right)^{-3} (z - i)^{-2} = \sum_{n=0}^{\infty} \binom{-3}{n} i^{-3-n} (z - i)^{n-2} = i(z - i)^{-2}$
 $-3(z - i)^{-1} - 6i + 10(z - i) + \cdots, \quad 0 < |z - i| < 1$
15. $(-\cos(z - \pi))(z - \pi)^{-2} = -(z - \pi)^{-2} + \frac{1}{2} - \frac{1}{24}(z - \pi)^2 + \cdots,$
 $0 < |z - \pi| < \infty$
19. $\sum_{n=0}^{\infty} z^{2n}, \quad |z| < 1, \quad - \sum_{n=0}^{\infty} \frac{1}{z^{2n+2}}, \quad |z| > 1$
21. $-(z + \frac{1}{2}\pi)^{-1} \cos(z + \frac{1}{2}\pi) = -(z + \frac{1}{2}\pi)^{-1} + \frac{1}{2}(z + \frac{1}{2}\pi) - \frac{1}{24}(z + \frac{1}{2}\pi)^3 + \cdots,$
 $|z + \frac{1}{2}\pi| > 0$
23. $z^8 + z^{12} + z^{16} + \cdots, \quad |z| < 1, \quad -z^4 - 1 - z^{-4} - z^{-8} - \cdots, \quad |z| > 1$
25. $\frac{i}{(z - i)^2} + \frac{1}{z - i} + i + (z - i)$

Section 16.2, page 719

1. $0 \pm 2\pi, \pm 4\pi, \cdots$, fourth order
3. $-81i$, fourth order
5. $\pm 1, \pm 2, \cdots$, second order
7. $\pm(2 + 2i), \pm i$, simple
9. $\frac{1}{2}\sin 4z, z = 0, \pm\pi/4, \pm\pi/2, \cdots$, simple
11. $f(z) = (z - z_0)^n g(z), g(z_0) \neq 0$, hence $f^2(z) = (z - z_0)^{2n} g^2(z)$.
13. Second-order poles at i and $-2i$
15. Simple pole at ∞ , essential singularity at $1 + i$
17. Fourth-order poles at $\pm n\pi i, n = 0, 1, \cdots$, essential singularity at ∞
19. $e^z(1 - e^z) = 0, e^z = 1, z = \pm 2n\pi i$ simple zeros. Answer: simple poles at $\pm 2n\pi i$,
essential singularity at ∞
21. $1, \infty$ essential singularities, $\pm 2n\pi i, n = 0, 1, \cdots$, simple poles

Section 16.3, page 725

3. $\frac{4}{15}$ at 0
5. $\pm 4i$ at $\mp i$
7. $1/\pi$ at 0, $\pm 1, \cdots$
9. -1 at $\pm 2n\pi i$
11. $(e^z)''/2!|_{z=\pi i} = -\frac{1}{2}$ at $z = \pi i$
15. Simple pole at $\frac{1}{4}$ inside C , residue $-1/(2\pi)$. Answer: $-i$
17. Simple poles at $\pi/2$, residue $e^{\pi/2}/(-\sin \pi/2)$, and at $-\pi/2$, residue $e^{-\pi/2}/\sin \pi/2 = e^{-\pi/2}$. Answer: $-4\pi i \sinh \pi/2$
19. $2\pi i (\sinh \frac{1}{2}i)/2 = -\pi \sin \frac{1}{2}$
21. $z^{-5} \cos \pi z = \cdots + \pi^4/(4!z) - \cdots$. Answer: $2\pi^5 i/24$

23. Residues $\frac{1}{2}$ at $z = \frac{1}{2}$, 2 at $z = \frac{1}{3}$. Answer: $5\pi i$
 25. Simple poles inside C at $2i, -2i, 3i, -3i$, residues $(2i \cosh 2i)/(4z^3 + 26z)|_{z=2i} = \frac{1}{10}, \frac{1}{10}, \frac{1}{10}, \frac{1}{10}$, respectively. Answer: $2\pi i \cdot \frac{4}{10}$

Problem Set 16.4, page 733

1. $2\pi/\sqrt{k^2 - 1}$ 3. $\pi/\sqrt{2}$
 5. $5\pi/12$ 7. $2a\pi/\sqrt{a^2 - 1}$
 9. 0. Why? (Make a sketch.) 11. $\pi/2$
 13. 0. Why? 15. $\pi/3$
 17. 0. Why?
 19. Simple poles at $\pm 1, i$ (and $-i$); $2\pi i \cdot \frac{1}{4}i + \pi i(-\frac{1}{4} + \frac{1}{4}) = -\frac{1}{2}\pi$
 21. Simple poles at 1 and $\pm 2\pi i$, residues i and $-i$. Answer: $\frac{\pi}{5}(\cos 1 - e^{-2})$
 23. $-\pi/2$ 25. 0
 27. Let $q(z) = (z - a_1)(z - a_2) \cdots (z - a_k)$. Use (4) in Sec. 16.3 to form the sum of the residues $1/q'(a_1) + \cdots + 1/q'(a_k)$ and show that this sum is 0; here $k > 1$.

Chapter 16 Review Questions and Problems, page 733

11. $6\pi i$ 13. $2\pi i(-10 - 10)$
 15. $2\pi i(25z^2)'|_{z=5} = 500\pi i$ 17. 0 (n even), $(-1)^{(n-1)/2}2\pi i/(n-1)!$ (n odd)
 19. $\pi/6$ 21. $\pi/60$
 23. 0. Why? 25. $\operatorname{Res}_{z=i} e^{iz}/(z^2 + 1) = 1/(2ie)$. Answer: π/e .

Problem Set 17.1, page 741

5. Only in size
 7. $x = c, w = -y + ic; y = k, w = -k + ix$
 9. Parallel displacement; each point is moved 2 to the right and 1 up.
 11. $|w| \leq \frac{1}{4}, -\pi/4 < \operatorname{Arg} w < \pi/4$ 13. $-5 \leq \operatorname{Re} z \leq -2$
 15. $u \geq 1$ 17. Annulus $\frac{1}{2} \leq |w| \leq 4$
 19. $0 < u < \ln 4, \pi/4 < v \leq 3\pi/4$
 21. $z^3 + az^2 + bz + c, z = -\frac{1}{3}(a \pm \sqrt{a^2 - 3b})$
 23. $z = (-1 \pm \sqrt{3})/2$
 25. $\sinh z = 0$ at $z = 0, \pm \pi i, \pm 2\pi i, \dots$
 29. $M = |z| = 1$ on the unit circle, $J = |z|^2$
 31. $|w'| = 1/|z|^2 = 1$ on the unit circle, $J = 1/|z|^4$
 33. $M = e^x = 1$ for $x = 0$, the y -axis, $J = e^{2x}$
 35. $M = 1/|z| = 1$ on the unit circle, $J = 1/|z|^2$

Problem Set 17.2, page 745

7. $z = \frac{w + i}{2w}$ 9. $z = \frac{4w + i}{-3iw + 1}$
 11. $z = 0, 1/(a + ib)$ 13. $z = 0, \pm \frac{1}{2}, \pm i = \pm i/2$

$$15. z = i, 2i \qquad 17. w = \frac{az}{cz + a} \qquad 19. w = \frac{az + b}{-bz + a}$$

Problem Set 17.3, page 750

3. Apply the inverse g of f on both sides of $z_1 = f(z_1)$ to get $g(z_1) = g(f(z_1)) = z_1$.
 9. $w = iz$, a rotation. Sketch to see. 11. $w = (z + i)/(z - i)$
 13. $w = 1/z$, almost by inspection 15. $w = 1/z - 1$
 17. $w = (2z - i)/(-iz - 2)$ 19. $w = (z^4 - i)/(-iz^4 + 1)$

Problem Set 17.4, page 754

1. Circle $|w| = e^c$ 3. Annulus $1/\sqrt{e} \leq |w| \leq \sqrt{e}$
 5. w -plane without $w = 0$ 7. $1 < |w| < e, v > 0$
 9. $\pm(2n + 1)\pi/2, \quad n = 0, 1, \dots$
 11. $u^2/\cosh^2 2 + v^2/\sinh^2 2 < 1, \quad u > 0, v > 0$
 13. Elliptic annulus bounded by $u^2/\cosh^2 1 + v^2/\sinh^2 1 = 1$ and $u^2/\cosh^2 3 + v^2/\sinh^2 3 = 1$
 15. $\cosh z = \cos iz = \sin(iz + \frac{1}{2}\pi)$
 17. $0 < \operatorname{Im} t < \pi$ is the image of R under $t = z^2/2$. Answer: $e^t = e^{z^2/2}$.
 19. Hyperbolas $u^2/\cos^2 c - v^2/\sin^2 c = \cosh^2 c - \sinh^2 c = 1$ when $c \neq 0, \pi$, and $u = \pm \cosh y$ (thus $|u| \geq 1$), $v = 0$ when $c = 0, \pi$.
 21. Interior of $u^2/\cosh^2 2 + v^2/\sinh^2 2 = 1$ in the fourth quadrant, or map $\pi/2 < x < \pi, 0 < y < 2$ by $w = \sin z$ (why?).
 23. $v < 0$
 25. The images of the five points in the figure can be obtained directly from the function w .

Problem Set 17.5, page 756

1. w moves once around the circle $|w| = \frac{1}{2}$.
 3. Four sheets, branch point at $z = -1$
 5. $-i/4$, three sheets
 7. z_0, n sheets
 9. $\sqrt{z(z - i)(z + i)}, 0, \pm i$, two sheets

Chapter 17 Review Questions and Problems, page 756

11. $1 < |w| < 4, |\arg w| < \pi/4$ 13. Horizontal strip $-8 < v < 8$
 15. $u = 1 - \frac{1}{4}v^2$, same (why?) 17. $|w| > 1$
 19. $\frac{1}{3} < |w| < \frac{1}{2}, \quad v < 0$ 21. $w = 1 + iv, \quad v < 0$
 23. $w = \frac{10z + 5i}{z + 2i}$ 25. Rotation $w = iz$
 27. $w = 1/z$ 29. $z = 0$
 31. $z = 2 \pm \sqrt{6}$ 33. $z = 0, \pm i, \pm 3i$
 35. $w = e^{4z}$ 37. $w = iz^2 + 1$
 39. $w = z^2/(2c)$

Problem Set 18.1, page 762

1. 2.5 mm = 0.25 cm; $\Phi = \operatorname{Re} 110(1 + (\operatorname{Ln} z)/\ln 4)$
3. $\Phi = \operatorname{Re} \left(30 - \frac{20}{\ln 10} \operatorname{Ln} z \right)$
5. $\Phi(x) = \operatorname{Re} (375 + 25z)$
7. $\Phi(r) = \operatorname{Re} (32 - z)$
13. Use Fig. 391 in Sec. 17.4 with the z - and w -planes interchanged and $\cos z = \sin(z + \frac{1}{2}\pi)$.
15. $\Phi = 220(x^3 - 3xy^2) = \operatorname{Re} (220z^3)$

Problem Set 18.2, page 766

3. $w = iz^2$ maps R onto the strip $-2 \leq u \leq 0$; and $\Phi^* = U_2 + (U_1 - U_2)(1 + \frac{1}{2}u) = U_2 + (U_1 - U_2)(1 - xy)$.
5. (a) $\frac{(x-2)(2x-1) + 2y^2}{(x-2)^2 + y^2} = c$, (b) $x^2 - y^2 = c$, $xy = c$, $e^x \cos y = c$
7. See Fig. 392 in Sec. 17.4. $\Phi = \operatorname{Re}(\sin^2 z)$, $\sin^2 x (y=0)$, $\sin^2 x \cosh^2 1 - \cos^2 x \sinh^2 1 (y=1)$, $-\sinh^2 y (x=0, \pi)$.
9. $\Phi(x, y) = \cos^2 x \cosh^2 y - \sin^2 x \sinh^2 y$; $\cosh^2 y (x=0)$, $-\sinh y (x = \frac{\pi}{2})$, $\cos^2 x (y=0)$, $\cos^2 x \cosh^2 1 - \sin^2 x \sinh^2 1 (y=1)$
13. Corresponding rays in the w -plane make equal angles, and the mapping is conformal.
15. Apply $w = z^2$.
17. $z = (2Z - i)/(-iZ - 2)$ by (3) in Sec. 17.3.
19. $\Phi = \frac{5}{\pi} \operatorname{Arg}(z - 2)$, $F = -\frac{5i}{\pi} \operatorname{Ln}(z - 2)$

Problem Set 18.3, page 769

1. $(80/d)y + 20$. Rotate through $\pi/2$.
5. $\frac{80}{\pi} \arctan \frac{y}{x} = \operatorname{Re} \left(-\frac{80i}{\pi} \operatorname{Ln} z \right)$
7. $T_1 + \frac{2}{\pi} (T_2 - T_1) \arctan \frac{y}{x} = \operatorname{Re} \left(T_1 - \frac{2i}{\pi} (T_2 - T_1) \operatorname{Ln} z \right)$
9. $\frac{T_1}{\pi} \left(\arctan \frac{y}{x-b} - \arctan \frac{y}{x-a} \right) = \operatorname{Re} \left(\frac{iT_1}{\pi} \operatorname{Ln} \frac{z-a}{z-b} \right)$
11. $\frac{100}{\pi} (\operatorname{Arg}(z-1) - \operatorname{Arg}(z+1)) = \operatorname{Re} \left(\frac{100i}{\pi} \operatorname{Ln} \frac{z+1}{z-1} \right)$
13. $\frac{100}{\pi} [\operatorname{Arg}(z^2-1) - \operatorname{Arg}(z^2+1)]$ from $w = z^2$ and Prob. 11.
15. $-20 + (320/\pi) \operatorname{Arg} z = \operatorname{Re} \left(-20 - \frac{320i}{\pi} \operatorname{Ln} z \right)$
17. $\operatorname{Re} F(z) = 100 + (200/\pi) \operatorname{Re} (\arcsin z)$

Problem Set 18.4, page 776

1. $V(z)$ continuously differentiable.
3. $|F'(iy)| = 1 + 1/y^2$, $|y| \geq 1$, is maximum at $y = \pm 1$, namely, 2.

5. Calculate or note that $\nabla^2 = \text{div grad}$ and curl grad is the zero vector; see Sec. 9.8 and Problem Set 9.7.
7. Horizontal parallel flow to the right.
9. $F(z) = z^4$
11. Uniform parallel flow upward, $V = \overline{F'} = iK$, $V_1 = 0$, $V_2 = K$
13. $F(z) = z^3$
15. $F(z) = z/r_0 + r_0/z$
17. Use that $w = \arccos z$ gives $z = \cos w$ and interchanging the roles of the z - and w -planes.
19. $y/(x^2 + y^2) = c$ or $x^2 + (y - k)^2 = k^2$

Problem Set 18.5, page 781

5. $\Phi = \frac{3}{2} r^3 \sin 3\theta$
7. $\Phi = \frac{1}{2} a + \frac{1}{2} ar^8 \cos 8\theta$
9. $\Phi = 3 - 4r^2 \cos 2\theta + r^4 \cos 4\theta$
11. $\Phi = \frac{2}{\pi} \left(r \sin \theta - \frac{1}{2} r^2 \sin 2\theta + \frac{1}{3} r^3 \sin 3\theta - + \cdots \right)$
13. $\Phi = \frac{2}{\pi} r \sin \theta + \frac{1}{2} r^2 \sin 2\theta - \frac{2}{9\pi} r^3 \sin 3\theta - \frac{1}{4} r^4 \sin 4\theta + + \cdots$
15. $\Phi = \frac{1}{2} + \frac{2}{\pi} \left(r \cos \theta - \frac{1}{3} r^3 \cos 3\theta + \frac{1}{5} r^5 \cos 5\theta - + \cdots \right)$
17. $\Phi = \frac{1}{3} - \frac{4}{\pi^2} \left(r \cos \theta - \frac{1}{4} r^2 \cos 2\theta + \frac{1}{9} r^3 \cos 3\theta - + \cdots \right)$

Problem Set 18.6, page 784

1. Use (2). $F(z_0 + e^{i\alpha}) = (\frac{7}{2} + e^{i\alpha})^3$, etc. $F(\frac{5}{2}) = \frac{343}{8}$
3. Use (2). $F(z_0 + e^{i\alpha}) = (2 + 3e^{i\alpha})^2$, etc. $F(4) = 100$
5. No, because $|z|$ is not analytic.
7. $\Phi(2, -2) = -3 = \frac{1}{\pi} \int_0^1 \int_0^{2\pi} (1 + r \cos \alpha)(-3 + r \sin \alpha)r \, dr \, d\alpha$

$$= \frac{1}{\pi} \int_0^1 \int_0^{2\pi} (-3r + \cdots) \, dr \, d\alpha = \frac{1}{\pi} \left(-\frac{3}{2} \right) \cdot 2\pi$$
9. $\Phi(1, 1) = 3 = \frac{1}{\pi} \int_0^1 \int_0^{2\pi} (3 + r \cos \alpha + r \sin \alpha + r^2 \cos \alpha \sin \alpha)r \, dr \, d\alpha$

$$= \frac{1}{\pi} \cdot \frac{3}{2} \cdot 2\pi$$
13. $|F(z)| = [\cos^2 x + \sinh^2 y]^{1/2}$, $z = \pm i$, $\text{Max} = [1 + \sinh^2 1]^{1/2} = 1.543$
15. $|F(z)|^2 = \sinh^2 2x \cos^2 2y + \cosh^2 2x \sin^2 2y = \sinh^2 2x + 1 \cdot \sin^2 2y$, $z = 1$,
 $\text{Max} = \sinh 2 = 3.627$
17. $|F(z)|^2 = 4(2 - 2 \cos 2\theta)$, $z = \pi/2, 3\pi/2$, $\text{Max} = 4$
19. No. Make up a counterexample.

Chapter 18 Review Questions and Problems, page 785

11. $\Phi = 10(1 - x + y)$, $F = 10 - 10(1 + i)z$
13. $\Phi = \operatorname{Re}(220 - 95.54 \operatorname{Ln} z) = 220 - \frac{220}{\ln 10} \ln r = 220 - 95.54 \ln r$
17. $2(1 - (2/\pi) \operatorname{Arg} z)$
19. $30(1 - (2/\pi) \operatorname{Arg}(z - 1))$
21. $\Phi = x + y = \text{const}$, $V = F'(z) = 1 - i$, parallel flow
23. $T(x, y) = x(2y + 1) = \text{const}$
25. $\overline{F'(z)} = \bar{z} + 1 = x + 1 - iy$

Problem Set 19.1, page 796

1. $0.84175 \cdot 10^2$, $-0.52868 \cdot 10^3$, $0.92414 \cdot 10^{-3}$, $-0.36201 \cdot 10^6$
3. 6.3698, 6.794, 8.15, impossible
5. Add first, then round.
7. 29.9667, 0.0335; 29.9667, 0.0333704 (6S-exact)
9. 29.97, 0.035; 29.97, 0.03337; 30, 0.0; 30, 0.033
11. $|\epsilon| = |x + y - (\tilde{x} + \tilde{y})| = |(x - \tilde{x}) + (y - \tilde{y})| = |\epsilon_x + \epsilon_y|$
 $\leq |\epsilon_x| + |\epsilon_y| = \beta_x + \beta_y$
13. $\frac{a_1}{a_2} = \frac{\tilde{a}_1 + \epsilon_1}{\tilde{a}_2 + \epsilon_2} = \frac{\tilde{a}_1 + \epsilon_1}{\tilde{a}_2} \left(1 - \frac{\epsilon_2}{\tilde{a}_2} + \frac{\epsilon_2^2}{\tilde{a}_2^2} - + \dots \right) \approx \frac{\tilde{a}_1}{\tilde{a}_2} + \frac{\epsilon_1}{\tilde{a}_2} - \frac{\epsilon_2}{\tilde{a}_2} \cdot \frac{\tilde{a}_1}{\tilde{a}_2}$,
 hence $\left| \left(\frac{a_1}{a_2} - \frac{\tilde{a}_1}{\tilde{a}_2} \right) \right| \left| \frac{a_1}{a_2} \right| \approx \left| \frac{\epsilon_1}{\tilde{a}_2} - \frac{\epsilon_2}{\tilde{a}_2} \right| \leq |\epsilon_{r1}| + |\epsilon_{r2}| \leq \beta_{r1} + \beta_{r2}$
15. (a) $1.38629 - 1.38604 = 0.00025$, (b) $\ln 1.00025 = 0.000249969$ is 6S-exact.
19. In the present case, (b) is slightly more accurate than (a) (which may produce nonsensical results; cf. Prob. 20).
21. $c_4 \cdot 2^4 + \dots + c_0 \cdot 2^0 = (1\ 0\ 1\ 1\ 1)_2$, NOT $(1\ 1\ 1\ 0\ 1)_2$
23. The algorithm in Prob. 22 repeats 0011 infinitely often.
25. $n = 26$. The beginning is 0.09375 ($n = 1$).
27. $I_{14} = 0.1812$ (0.1705 4S-exact), $I_{13} = 0.1812$ (0.1820), $I_{12} = 0.1951$ (0.1951), $I_{11} = 0.2102$ (0.2103), etc.
29. $-0.126 \cdot 10^{-2}$, $-0.402 \cdot 10^{-3}$; $-0.266 \cdot 10^{-6}$, $-0.847 \cdot 10^{-7}$

Problem Set 19.2, page 807

3. $g = 0.5 \cos x$, $x = 0.450184$ ($= x_{10}$, exact to 6S)
5. Convergence to 4.7 for all these starting values.
7. $x = x/(e^x \sin x)$; 0.5, 0.63256, $\dots$ converges to 0.58853 (5S-exact) in 14 steps.
9. $x = x^4 - 0.12$; $x_0 = 0$, $x_3 = -0.119794$ (6S-exact)
11. $g = 4/x + x^3/16 - x^5/576$; $x_0 = 2$, $x_n = 2.39165$ ($n \geq 6$), 2.405 4S-exact
13. This follows from the intermediate value theorem of calculus.
15. $x_3 = 0.450184$
17. Convergence to $x = 4.7, 4.7, 0.8, -0.5$, respectively. Reason seen easily from the graph of f .

19. 0.5, 0.375, 0.377968, 0.377964; (b) $1/\sqrt{7}$

21. 1.834243 ($= x_4$), 0.656620 ($= x_4$), -2.49086 ($= x_4$)

23. $x_0 = 4.5$, $x_4 = 4.73004$ (6S-exact)

25. (a) ALGORITHM BISECT (f, a_0, b_0, ϵ, N) Bisection Method

This algorithm computes the solution c of $f(x) = 0$ (f continuous) within the tolerance ϵ , given an initial interval $[a_0, b_0]$ such that $f(a_0)f(b_0) < 0$.

INPUT: Continuous function f , initial interval $[a_0, b_0]$, tolerance ϵ , maximum number of iterations N .

OUTPUT: A solution c (within the tolerance ϵ), or a message of failure.

For $n = 0, 1, \dots, N - 1$ do:

$$c = \frac{1}{2}(a_n + b_n)$$

If $f(c) = 0$ then OUTPUT c Stop. [Procedure completed]

Else if $f(a_n)f(b_n) < 0$ then set $a_{n+1} = a_n$ and $b_{n+1} = c$.

Else set $a_{n+1} = c$, and $b_{n+1} = b_n$.

If $|a_{n+1} - b_{n+1}| < \epsilon|c|$ then OUTPUT c . Stop. [Procedure completed]

End

OUTPUT $[a_N, b_N]$ and a message "Failure". Stop.

[Unsuccessful completion; N iterations did not give an interval of length not exceeding the tolerance.]

End BISECT

Note that $[a_N, b_N]$ gives $(a_N + b_N)/2$ as an approximation of the zero and $(b_N - a_N)/2$ as a corresponding error bound.

(b) 0.739085; (c) 1.30980, 0.429494

27. $x_2 = 1.5$, $x_3 = 1.76471, \dots$, $x_7 = 1.83424$ (6S-exact)

29. 0.904557 (6S-exact)

Problem Set 19.3, page 819

$$1. L_0(x) = -2x + 19, \quad L_1(x) = 2x - 18, \quad p_1(9.3) = L_0(9.3) \cdot f_0 + L_1(9.3) \cdot f_1 \\ = 0.1086 \cdot 9.3 + 1.230 = 2.2297$$

$$3. p_2(x) = \frac{(x - 1.02)(x - 1.04)}{(-0.02)(-0.04)} \cdot 1.0000 + \frac{(x - 1)(x - 1.04)}{0.02(-0.02)} \cdot 0.9888 \\ + \frac{(x - 1)(x - 1.02)}{0.04 \cdot 0.02} \cdot 0.9784 = x^2 - 2.580x + 2.580; \quad 0.9943, 0.9835$$

$$5. 0.8033 \text{ (error } -0.0245), 0.4872 \text{ (error } -0.0148); \text{ quadratic: } 0.7839 \text{ (} -0.0051), \\ 0.4678 \text{ (} 0.0046)$$

$$7. p_2(x) = 1.1640x - 0.3357x^2; \quad -0.5089 \text{ (error } 0.1262), 0.4053 \text{ (} -0.0226), \\ 0.9053 \text{ (} 0.0186), 0.9911 \text{ (} -0.0672)$$

$$9. p_2(x) = -0.44304x^2 + 1.30896x - 0.023220, \quad p_2(0.75) = 0.70929 \\ \text{(5S-exact } 0.71116)$$

$$11. L_0 = -\frac{1}{6}(x - 1)(x - 2)(x - 3), L_1 = \frac{1}{2}x(x - 2)(x - 3), L_2 = -\frac{1}{2}x(x - 1)(x - 3), \\ L_3 = \frac{1}{6}x(x - 1)(x - 2); \quad p_3(x) = 1 + 0.039740x - 0.335187x^2 + 0.060645x^3; \\ p_2(0.5) = 0.943654, p_3(1.5) = 0.510116, p_3(2.5) = -0.047991$$

$$13. 2x^2 - 4x + 2$$

$$15. p_3(x) = 2.1972 + (x - 9) \cdot 0.1082 + (x - 9)(x - 9.5) \cdot 0.005235$$

$$17. r = -1.5, p_2(0.3) = 0.6039 + (-1.5) \cdot 0.1755 + \frac{1}{2}(-1.5)(-0.5) \cdot (-0.0302) \\ = 0.3293$$

Problem Set 19.4, page 826

9. $[-1.39(x-5)^2 + 0.58(x-5)^3]'' = 0.004$ at $x = 5.8$ (due to roundoff; should be 0).
11. $1 - \frac{5}{4}x^2 + \frac{1}{4}x^4$
13. $1 - x^2, -2(x-1) - (x-1)^2 + 2(x-1)^3, -1 + 2(x-2) + 5(x-2)^2 - 6(x-2)^3$
15. $4 + x^2 - x^3, -8(x-2) - 5(x-2)^2 + 5(x-2)^3, 4 + 32(x-4) + 25(x-4)^2 - 11(x-4)^3$
17. Use the fact that the third derivative of a cubic polynomial is constant, so that g''' is piecewise constant, hence constant throughout under the present assumption. Now integrate three times.
19. Curvature $f''/(1 + f'^2)^{3/2} \approx f''$ if $|f'|$ is small.

Problem Set 19.5, page 839

1. 0.747131, which is larger than 0.746824. Why?
3. 0.5, 0.375, 0.34375, 0.335 (exact)
5. $\epsilon_{0.5} \approx 0.03452$ ($\epsilon_{0.5} = 0.03307$), $\epsilon_{0.25} \approx 0.00829$ ($\epsilon_{0.25} = 0.00820$)
7. 0.693254 (6S-exact 0.693147)
9. 0.073930 (6S-exact 0.073928)
11. 0.785392 (6S-exact 0.785398)
13. $(0.785398126 - 0.785392156)/15 = 0.39792 \cdot 10^{-6}$
15. (a) $M_2 = 2, |KM_2| = 2/(12n^2) = 10^{-5}/2, n = 183$. (b) $f^{\text{iv}} = 24/x^5, M_4 = 24, |CM_4| = 24/(180 \cdot (2m)^4) = 10^{-5}/2, 2m = 12.8$, hence 14.
17. 0.94614588, 0.94608693 (8S-exact 0.94608307)
19. 0.9460831 (7S-exact)
21. 0.9774586 (7S-exact 0.9774377)
23. Set $x = \frac{1}{2}(t+1)$, 0.2642411177 (10S-exact), $1 - 2/e$
25. $x = \frac{1}{2}(t+1), dx = \frac{1}{2}dt$, 0.746824127 (9S-exact 0.746824133)
27. 0.08, 0.32, 0.176, 0.256 (exact)
29. $5(0.1040 - \frac{1}{2} \cdot 0.1760 + \frac{1}{3} \cdot 0.1344 - \frac{1}{4} \cdot 0.0384) = 0.256$

Chapter 19 Review Questions and Problems, page 841

17. 4.375, 4.50, 6.0, impossible
19. $44.885 \leq s \leq 44.995$
21. The same as that of $\tilde{a}$.
23. $x = 20 \pm \sqrt{398} = 20.00 \pm 19.95, x_1 = 39.95, x_2 = 0.05, x_2 = 2/39.95 = 0.05006$ (error less than 1 unit of the last digit)
25. $x = x^4 - 0.1, -0.1, -0.999, -0.99900399$
27. 0.824
29. $-x + x^3, 2(x-1) + 3(x-1)^2 - (x-1)^3$
31. 0.26, $M_2 = 6, M_2^* = 0, -0.02 \leq \epsilon \leq 0, 0.01$
33. 0.90443, 0.90452 (5S-exact 0.90452)
35. (a) $(0.4^3 - 2 \cdot 0.2^3 + 0)/0.04 = 1.2$, (b) $(0.3^3 - 2 \cdot 0.2^3 + 0.1^3)/0.01 = 1.2$ (exact)

Problem Set 20.1, page 851

1. $x_1 = 7.3, \quad x_2 = -3.2$

3. No solution

5. $x_1 = 2, \quad x_2 = 1$

$$7. \begin{bmatrix} -3 & 6 & -9 & -46.725 \\ 0 & 9 & -13 & -51.223 \\ 0 & 0 & -2.88889 & -7.38689 \end{bmatrix}$$

$$x_1 = 3.908, \quad x_2 = -1.998, \quad x_3 = 2.557$$

$$9. \begin{bmatrix} 13 & -8 & 0 & 178.54 \\ 0 & 6 & 13 & 137.86 \\ 0 & 0 & -16 & -253.12 \end{bmatrix}$$

$$x_1 = 6.78, \quad x_2 = -11.3, \quad x_3 = 15.82$$

$$11. \begin{bmatrix} 3.4 & -6.12 & -2.72 & 0 \\ 0 & 0 & 4.32 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_1 = t_1 \text{ arbitrary}, \quad x_2 = (3.4/6.12)t_1, \quad x_3 = 0$$

$$13. \begin{bmatrix} 5 & 0 & 6 & -0.329193 \\ 0 & -4 & -3.6 & -2.143144 \\ 0 & 0 & 2.3 & -0.4 \end{bmatrix}$$

$$x_1 = 0.142856, \quad x_2 = 0.692307, \quad x_3 = -0.173912$$

$$15. \begin{bmatrix} -1 & -3.1 & 2.5 & 0 & -8.7 \\ 0 & 2.2 & 1.5 & -3.3 & -9.3 \\ 0 & 0 & -1.493182 & -0.825 & 1.03773 \\ 0 & 0 & 0 & 6.13826 & 12.2765 \end{bmatrix}$$

$$x_1 = 4.2, \quad x_2 = 0, \quad x_3 = -1.8, \quad x_4 = 2.0$$

Problem Set 20.2, page 857

$$1. \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 4 & 5 \\ 0 & -1 \end{bmatrix}, \quad x_1 = -4$$

$$x_2 = 6$$

$$3. \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 2 & 5 & 1 \end{bmatrix} \begin{bmatrix} 5 & 4 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 3 \end{bmatrix}, \quad x_1 = 0.4$$

$$x_2 = 0.8$$

$$x_3 = 1.6$$

$$5. \begin{bmatrix} 1 & 0 & 0 \\ 6 & 1 & 0 \\ 3 & 9 & 1 \end{bmatrix} \begin{bmatrix} 3 & 9 & 6 \\ 0 & -6 & 3 \\ 0 & 0 & -3 \end{bmatrix}, \quad x_1 = -\frac{1}{15}$$

$$x_2 = \frac{4}{15}$$

$$x_3 = \frac{2}{5}$$

$$\begin{aligned}
 7. & \begin{bmatrix} 3 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 1 & 3 \end{bmatrix} \begin{bmatrix} 3 & 2 & 4 \\ 0 & 3 & 1 \\ 0 & 0 & 3 \end{bmatrix}, \quad \begin{matrix} x_1 = 0.6 \\ x_2 = 1.2 \\ x_3 = 0.4 \end{matrix} \\
 9. & \begin{bmatrix} 0.1 & 0 & 0 \\ 0 & 0.4 & 0 \\ 0.3 & 0.2 & 0.1 \end{bmatrix} \begin{bmatrix} 0.1 & 0 & 0.3 \\ 0 & 0.4 & 0.2 \\ 0 & 0 & 0.1 \end{bmatrix}, \quad \begin{matrix} x_1 = 2 \\ x_2 = -11 \\ x_3 = 4 \end{matrix} \\
 11. & \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 2 & 0 & 0 \\ 3 & -1 & 3 & 0 \\ 2 & 0 & -1 & 4 \end{bmatrix} \begin{bmatrix} 1 & -1 & 3 & 2 \\ 0 & 2 & -1 & 0 \\ 0 & 0 & 3 & -1 \\ 0 & 0 & 0 & 4 \end{bmatrix}, \quad \begin{matrix} x_1 = 2 \\ x_2 = -3 \\ x_3 = 4 \\ x_4 = -1 \end{matrix} \\
 13. & \text{No, since } \mathbf{x}^T(-\mathbf{A})\mathbf{x} = -\mathbf{x}^T\mathbf{A}\mathbf{x} < 0; \quad \text{yes; yes; no} \\
 15. & \begin{bmatrix} -3.5 & 1.25 \\ 3.0 & -1.0 \end{bmatrix} \\
 17. & \frac{1}{36} \begin{bmatrix} 584 & 104 & -66 \\ 104 & 20 & -12 \\ -66 & -12 & 9 \end{bmatrix} \\
 19. & \frac{1}{16} \begin{bmatrix} 21 & -6 & -14 & 6 \\ -6 & 36 & -12 & -4 \\ -14 & -12 & 20 & -4 \\ 6 & -4 & -4 & 4 \end{bmatrix}
 \end{aligned}$$

Problem Set 20.3, page 863

5. Exact 0.5, 0.5, 0.5 7. $x_1 = 2$, $x_2 = -4$, $x_3 = 8$
 9. Exact 2, 1, 4
 11. (a) $\mathbf{x}^{(3)T} = [0.49983 \quad 0.50001 \quad 0.500017]$,
 (b) $\mathbf{x}^{(3)T} = [0.50333 \quad 0.49985 \quad 0.49968]$
 13. 8, -16, 43, 86 steps; spectral radius 0.09, 0.35, 0.72, 0.85, approximately
 15. $[1.99934 \quad 1.00043 \quad 3.99684]^T$ (Jacobi, Step 5); $[2.00004 \quad 0.998059 \quad 4.00072]^T$
 (Gauss-Seidel)
 19. $\sqrt{306} = 17.49$, 12, 12

Problem Set 20.4, page 871

1. 18, $\sqrt{110} = 10.49$, 8, $[0.125 \quad -0.375 \quad 1 \quad 0 \quad -0.75 \quad 0]$
 3. 5.9, $\sqrt{13.81} = 3.716$, 3, $\frac{1}{3}[0.2 \quad 0.6 \quad -2.1 \quad 3.0]$
 5. 5, $\sqrt{5}$, 1, $[1 \quad 1 \quad 1 \quad 1 \quad 1]$ 7. $ab + bc + ca = 0$

9. $\kappa = 5 \cdot \frac{1}{2} = 2.5$ 11. $\kappa = (5 + \sqrt{5})(1 + 1/\sqrt{5}) = 6 + 2\sqrt{5}$
 13. $\kappa = 19 \cdot 13 = 247$; ill-conditioned
 15. $\kappa = 20 \cdot 20 = 400$; ill-conditioned
 17. $167 \leq 21 \cdot 15 = 315$
 19. $[-2 \ 4]^T$, $[-144.0 \ 184.0]^T$, $\kappa = 25,921$, extremely ill-conditioned
 21. Small residual $[0.145 \ 0.120]$, but large deviation of $\tilde{\mathbf{x}}$.
 23. 27, 748, 28,375, 943,656, 29,070,279

Problem Set 20.5, page 875

1. $1.846 - 1.038x$ 3. $1.48 + 0.09x$
 5. $s = 90t - 675$, $v_{av} = 90 \text{ km/hr}$ 9. $-11.36 + 5.45x - 0.589x^2$
 11. $1.89 - 0.739x + 0.207x^2$
 13. $2.552 + 16.23x$, $-4.114 + 13.73x + 2.500x^2$, $2.730 + 1.466x - 1.778x^2 + 2.852x^3$

Problem Set 20.7, page 884

1. 5, 0, 7; radii 6, 4, 6. Spectrum $\{-1, 4, 9\}$
 3. Centers 0; radii 0.5, 0.7, 0.4. Skew-symmetric, hence $\lambda = i\mu$, $-0.7 \leq \mu \leq 0.7$.
 5. 2, 3, 8; radii $1 + \sqrt{2}$, 1, $\sqrt{2}$; actually (4S) 1.163, 3.511, 8.326
 7. $t_{11} = 100$, $t_{22} = t_{33} = 1$
 9. They lie in the intervals with endpoints $a_{jj} \pm (n-1) \cdot 10^{-5}$. Why?
 11. $\rho(\mathbf{A}) \leq \text{Row sum norm } \|\mathbf{A}\|_\infty = \max_j \sum_k |a_{jk}| = \max_j (|a_{jj}| + \text{Gerschgorin radius})$
 13. $\sqrt{122} = 11.05$
 15. $\sqrt{0.52} = 0.7211$
 17. Show that $\mathbf{A}\mathbf{A}^{\overline{\mathbf{T}}} = \overline{\mathbf{A}}^{\mathbf{T}}\mathbf{A}$.
 19. 0 lies in no Gerschgorin disk, by (3) with $>$; hence $\det \mathbf{A} = \lambda_1 \cdots \lambda_n \neq 0$.

Problem Set 20.8, page 887

1. $q = 10, 10.9908, 10.9999$; $|\epsilon| \leq 3, 0.3028, 0.0275$
 3. $q \pm \delta = 4 \pm 1.633$, 4.786 ± 0.619 , 4.917 ± 0.398
 5. Same answer as in Prob. 3, possibly except for small roundoff errors.
 7. $q = 5.5, 5.5738, 5.6018$; $|\epsilon| \leq 0.5, 0.3115, 0.1899$; eigenvalues (4S) 1.697, 3.382, 5.303, 5.618
 9. $\mathbf{y} = \mathbf{A}\mathbf{x} = \lambda\mathbf{x}$, $\mathbf{y}^{\mathbf{T}}\mathbf{x} = \lambda\mathbf{x}^{\mathbf{T}}\mathbf{x}$, $\mathbf{y}^{\mathbf{T}}\mathbf{y} = \lambda^2\mathbf{x}^{\mathbf{T}}\mathbf{x}$,
 $\epsilon^2 \leq \mathbf{y}^{\mathbf{T}}\mathbf{y}/\mathbf{x}^{\mathbf{T}}\mathbf{x} - (\mathbf{y}^{\mathbf{T}}\mathbf{x}/\mathbf{x}^{\mathbf{T}}\mathbf{x})^2 = \lambda^2 - \lambda^2 = 0$
 11. $q = 1, \dots, -2.8993$ approximates -3 (0 of the given matrix),
 $|\epsilon| \leq 1.633, \dots, 0.7024$ (Step 8)

Problem Set 20.9, page 896

1.
$$\begin{bmatrix} 0.98 & -0.4418 & 0 \\ -0.4418 & 0.8702 & 0.3718 \\ 0 & 0.3718 & 0.4898 \end{bmatrix}$$

$$3. \begin{bmatrix} 7 & -3.6056 & 0 \\ -3.6056 & 13.462 & 3.6923 \\ 0 & 3.6923 & 3.5385 \end{bmatrix}$$

$$5. \begin{bmatrix} 3 & -67.59 & 0 & 0 \\ -67.59 & 143.5 & 45.35 & 0 \\ 0 & 45.35 & 23.34 & 3.126 \\ 0 & 0 & 3.126 & -33.87 \end{bmatrix}$$

7. Eigenvalues 16, 6, 2

$$\begin{bmatrix} 11.2903 & -5.0173 & 0 \\ -5.0173 & 10.6144 & 0.7499 \\ 0 & 0.7499 & 2.0952 \end{bmatrix}, \begin{bmatrix} 14.9028 & -3.1265 & 0 \\ -3.1265 & 7.0883 & 0.1966 \\ 0 & 0.1966 & 2.0089 \end{bmatrix}, \begin{bmatrix} 15.8299 & -1.2932 & 0 \\ -1.2932 & 6.1692 & 0.0625 \\ 0 & 0.0625 & 2.0010 \end{bmatrix}$$

9. Eigenvalues (4S) 141.4, 68.64, -30.04

$$\begin{bmatrix} 141.1 & 4.926 & 0 \\ 4.926 & 68.97 & 0.8691 \\ 0 & 0.8691 & -30.03 \end{bmatrix}, \begin{bmatrix} 141.3 & 2.400 & 0 \\ 2.400 & 68.72 & 0.3797 \\ 0 & 0.3797 & -30.04 \end{bmatrix}, \begin{bmatrix} 141.4 & 1.166 & 0 \\ 1.166 & 68.66 & 0.1661 \\ 0 & 0.1661 & -30.04 \end{bmatrix}$$

Chapter 20 Review Questions and Problems, page 896

15. $[3.9 \ 4.3 \ 1.8]^T$

17. $[-2 \ 0 \ 5]^T$

19. $\begin{bmatrix} 0.28193 & -0.15904 & -0.00482 \\ -0.15904 & 0.12048 & -0.00241 \\ -0.00482 & -0.00241 & 0.01205 \end{bmatrix}$

21. $\begin{bmatrix} 5.750 \\ 3.600 \\ 0.838 \end{bmatrix}, \begin{bmatrix} 6.400 \\ 3.559 \\ 1.000 \end{bmatrix}, \begin{bmatrix} 6.390 \\ 3.600 \\ 0.997 \end{bmatrix}$

Exact: $[6.4 \ 3.6 \ 1.0]^T$

23. $\begin{bmatrix} 1.700 \\ 1.180 \\ 4.043 \end{bmatrix}, \begin{bmatrix} 1.986 \\ 0.999 \\ 4.002 \end{bmatrix}, \begin{bmatrix} 2.000 \\ 1.000 \\ 4.000 \end{bmatrix}$

Exact: $[2 \ 1 \ 4]^T$

25. 42, $\sqrt{674} = 25.96$, 21

27. 30

29. 5

31. $115 \cdot 0.4458 = 51.27$

33. $5 \cdot \frac{21}{63} = \frac{5}{3}$

35. $1.514 + 1.129x - 0.214x^2$

37. Centers 15, 35, 90; radii 30, 35, 25, respectively. Eigenvalues (3S) 2.63, 40.8, 96.6

39. Centers 0, -1, -4; radii 9, 6, 7, respectively; eigenvalues 0, 4.446, -9.446

Problem Set 21.1, page 910

1. $y = 5e^{-0.2x}$, 0.00458, 0.00830 (errors of y_5, y_{10})
3. $y = x - \tanh x$ (set $y - x = u$), 0.00929, 0.01885 (errors of y_5, y_{10})
5. $y = e^x$, 0.0013, 0.0042 (errors of y_5, y_{10})
7. $y = 1/(1 - x^2/2)$, 0.00029, 0.01187 (errors of y_5, y_{10})
9. Errors 0.03547 and 0.28715 of y_5 and y_{10} much larger
11. $y = 1/(1 - x^2/2)$; error $\cdot 10^{-8}$, $-4 \cdot 10^{-8}$, $\dots$, $-6 \cdot 10^{-7}$, $+9 \cdot 10^{-6}$;
 $\epsilon = 0.0002/15 = 1.3 \cdot 10^{-5}$ (use RK with $h = 0.2$)
13. $y = \tan x$; error $0.83 \cdot 10^{-7}$, $0.16 \cdot 10^{-6}$, $\dots$, $-0.56 \cdot 10^{-6}$, $+0.13 \cdot 10^{-5}$
15. $y = 3 \cos x - 2 \cos^2 x$; error $\cdot 10^7$: 0.18, 0.74, 1.73, 3.28, 5.59, 9.04, 14.3, 22.8, 36.8, 61.4
17. $y' = 1/(2 - x^4)$; error $\cdot 10^9$: 0.2, 3.1, 10.7, 23.2, 28.5, -32.3 , -376 , -1656 , -3489 , $+80444$
19. Errors for Euler–Cauchy 0.02002, 0.06286, 0.05074; for improved Euler–Cauchy -0.000455 , 0.012086, 0.009601; for Runge–Kutta. 0.0000011, 0.000016, 0.000536

Problem Set 21.2, page 915

1. $y = e^x$, $y_5^* = 1.648717$, $y_5 = 1.648722$, $\epsilon_5 = -3.8 \cdot 10^{-8}$,
 $y_{10}^* = 2.718276$, $y_{10} = 2.718284$, $\epsilon_{10} = -1.8 \cdot 10^{-6}$
3. $y = \tan x$, $y_4, \dots, y_{10}$ (error $\cdot 10^5$) 0.422798 (-0.49), 0.546315 (-1.2),
0.684161 (-2.4), 0.842332 (-4.4), 1.029714 (-7.5), 1.260288 (-13),
1.557626 (-22)
5. RK error smaller in absolute value, error $\cdot 10^5 = 0.4, 0.3, 0.2, 5.6$
(for $x = 0.4, 0.6, 0.8, 1.0$)
7. $y = 1/(4 + e^{-3x})$, $y_4, \dots, y_{10}$ (error $\cdot 10^5$) 0.232490 (0.34), 0.236787 (0.44),
0.240075 (0.42), 0.242570 (0.35), 0.244453 (0.25), 0.245867 (0.16), 0.246926 (0.09)
9. $y = \exp(x^3) - 1$, $y_4, \dots, y_{10}$ (error $\cdot 10^7$) 0.008032 (-4), 0.015749 (-10),
0.027370 (-17), 0.043810 (-26), 0.066096 (-39), 0.095411 (-54),
0.133156 (-74)
13. $y = \exp(x^2)$. Errors $\cdot 10^5$ from $x = 0.3$ to 0.7 : -5 , -11 , -19 , -31 , -41
15. (a) 0, 0.02, 0.0884, 0.215848, $y_4 = 0.417818$, $y_5 = 0.708887$ (poor)
(b) By 30–50%

Problem Set 21.3, page 922

1. $y_1 = -e^{-2x} + 4e^x$, $y_2 = -e^{-2x} + e^x$; errors of y_1 (of y_2) from 0.002 to 0.5
(from -0.01 to 0.1), monotone
3. $y_1' = y_2$, $y_2' = -\frac{1}{4}y_1$, $y = y_1 = 1$, 0.99, 0.97, 0.94, 0.9005, error
 -0.005 , -0.01 , -0.015 , -0.02 , -0.0229 ; exact $y = \cos \frac{1}{2}x$
5. $y_1' = y_2$, $y_2' = y_1 + x$, $y_1(0) = 1$, $y_2(0) = -2$, $y = y_1 = e^{-x} - x$, $y = 0.8$
(error 0.005), 0.61 (0.01), 0.429 (0.012), 0.2561 (0.0142), 0.0905 (0.0160)
7. By about a factor 10^5 . $\epsilon_n(y_1) \cdot 10^6 = -0.082, \dots, -0.27$,
 $\epsilon_n(y_2) \cdot 10^6 = 0.08, \dots, 0.27$
9. Errors of y_1 (of y_2) from $0.3 \cdot 10^{-5}$ to $1.3 \cdot 10^{-5}$ (from $0.3 \cdot 10^{-5}$ to $0.6 \cdot 10^{-5}$)
11. $(y_1, y_2) = (0, 1), (0.20, 0.98), (0.39, 0.92), \dots, (-0.23, -0.97), (-0.42, -0.91),$
 $(-0.59), (-0.81)$; continuation will give an “ellipse.”

Problem Set 21.4, page 930

3. $-3u_{11} + u_{12} = -200$, $u_{11} - 3u_{12} = -100$
5. 105, 155, 105, 115; Step 5: 104.94, 154.97, 104.97, 114.98
7. 0, 0, 0, 0. All equipotential lines meet at the corners (why?).
Step 5: 0.29298, 0.14649, 0.14649, 0.073245
9. 0.108253, 0.108253, 0.324760, 0.324760; Step 10: 0.108538, 0.108396, 0.324902, 0.324831
11. (a) $u_{11} = -u_{12} = -66$. (b) Reduce to 4 equations by symmetry.
 $u_{11} = u_{31} = -u_{15} = -u_{35} = -92.92$, $u_{21} = -u_{25} = -87.45$,
 $u_{12} = u_{32} = -u_{14} = -u_{34} = -64.22$, $u_{22} = -u_{24} = -53.98$,
 $u_{13} = u_{23} = u_{33} = 0$
13. $u_{12} = u_{32} = 31.25$, $u_{21} = u_{23} = 18.75$, $u_{jk} = 25$ at the others
15. $u_{21} = u_{23} = 0.25$, $u_{12} = u_{32} = -0.25$, $u_{jk} = 0$ otherwise
17. $\sqrt{3}$, $u_{11} = u_{21} = 0.0849$, $u_{12} = u_{22} = 0.3170$. (0.1083, 0.3248 are 4S-values of the solution of the linear system of the problem.)

Problem Set 21.5, page 935

5. $u_{11} = 0.766$, $u_{21} = 1.109$, $u_{12} = 1.957$, $u_{22} = 3.293$
7. A, as in Example 1, right sides $-220, -220, -220, -220$.
Solution $u_{11} = u_{21} = 125.7$, $u_{21} = u_{22} = 157.1$
13. $-4u_{11} + u_{21} + u_{12} = -3$, $u_{11} - 4u_{21} + u_{22} = -12$, $u_{11} - 4u_{12} + u_{22} = 0$,
 $2u_{21} + 2u_{12} - 12u_{22} = -14$, $u_{11} = u_{22} = 2$, $u_{21} = 4$, $u_{12} = 1$.
Here $-\frac{14}{3} = -\frac{4}{3}(1 + 2.5)$ with $\frac{4}{3}$ from the stencil.
15. $\mathbf{b} = [-200, -100, -100, 0]^T$; $u_{11} = 73.68$, $u_{21} = u_{12} = 47.37$, $u_{22} = 15.79$ (4S)

Problem Set 21.6, page 941

5. 0, 0.6625, 1.25, 1.7125, 2, 2.1, 2, 1.7125, 1.25, 0.6625, 0
7. Substantially less accurate, 0.15, 0.25 ($t = 0.04$), 0.100, 0.163 ($t = 0.08$)
9. Step 5 gives 0, 0.06279, 0.09336, 0.08364, 0.04707, 0.
11. Step 2: 0 (exact 0), 0.0453 (0.0422), 0.0672 (0.0658), 0.0671 (0.0628), 0.0394 (0.0373), 0 (0)
13. 0.3301, 0.5706, 0.4522, 0.2380 ($t = 0.04$), 0.06538, 0.10603, 0.10565, 0.6543 ($t = 0.20$)
15. 0.1018, 0.1673, 0.1673, 0.1018 ($t = 0.04$), 0.0219, 0.0355, $\dots$ ($t = 0.20$)

Problem Set 21.7, page 944

1. $u(x, 1) = 0, -0.05, -0.10, -0.15, -0.20, 0$
3. For $x = 0.2, 0.4$ we obtain 0.24, 0.40 ($t = 0.2$), 0.08, 0.16 ($t = 0.4$), $-0.08, -0.16$ ($t = 0.6$), etc.
5. 0, 0.354, 0.766, 1.271, 1.679, 1.834, $\dots$ ($t = 0.1$); 0, 0.575, 0.935, 1.135, 1.296, 1.357, $\dots$ ($t = 0.2$)
7. 0.190, 0.308, 0.308, 0.190, (3S-exact: 0.178, 0.288, 0.288, 0.178)

Chapter 21 Review Questions and Problems, page 945

17. $y = e^x$, 0.038, 0.125 (errors of y_5 and y_{10})
19. $y = \tan x$; 0 (0), 0.10050 (−0.00017), 0.20304 (−0.00033), 0.30981 (−0.00048), 0.42341 (−0.00062), 0.54702 (−0.00072), 0.68490 (−0.00076), 0.84295 (−0.00066), 1.0299 (−0.0002), 1.2593 (0.0009), 1.5538 (0.0036)
21. 0.1003346 ($0.8 \cdot 10^{-7}$), 0.2027099 ($1.6 \cdot 10^{-7}$), 0.3093360 ($2.1 \cdot 10^{-7}$), 0.4227930 ($2.3 \cdot 10^{-7}$), 0.5463023 ($1.8 \cdot 10^{-7}$)
23. $y = \sin x$, $y_{0.8} = 0.717366$, $y_{1.0} = 0.841496$ (errors $-1.0 \cdot 10^{-5}$, $-2.5 \cdot 10^{-5}$)
25. $y'_1 = y_2$, $y'_2 = x^2 y_1$, $y = y_1 = 1, 1, 1, 1.0001, 1.0006, 1.002$
27. $y'_1 = y_2$, $y'_2 = 2e^x - y_1$, $y = e^x - \cos x$, $y = y_1 = 0, 0.241, 0.571, \dots$; errors between 10^{-6} and 10^{-5}
29. 3.93, 15.71, 58.93
31. 0, 0.04, 0.08, 0.12, 0.15, 0.16, 0.15, 0.12, 0.08, 0.04, 0 ($t = 0.3$. 3 time steps)
33. $u(P_{11}) = u(P_{31}) = 270$, $u(P_{21}) = u(P_{13}) = u(P_{23}) = u(P_{33}) = 30$, $u(P_{12}) = u(P_{32}) = 90$, $u(P_{22}) = 60$
35. 0.043330, 0.077321, 0.089952, 0.058488 ($t = 0.04$), 0.010956, 0.017720, 0.017747, 0.010964 ($t = 0.20$)

Problem Set 22.1, page 953

3. $f(\mathbf{x}) = 2(x_1 - 1)^2 + (x_2 + 2)^2 - 6$; Step 3: (1.037, −1.926), value −5.992
9. Step 5: (0.11247, −0.00012), value 0.000016

Problem Set 22.2, page 957

7. No
9. x_3, x_4 is the unused time on M_1, M_2 , respectively.
11. $f(2.5, 2.5) = 100$
13. $f(-\frac{11}{3}, \frac{26}{3}) = 198 \frac{1}{3}$
15. $f(9, 6) = 360$
17. $0.5x_1 + 0.75x_2 \leq 45$ (copper), $0.5x_1 + 0.25x_2 \leq 30$, $f = 120x_1 + 100x_2$, $f_{\max} = f(45, 30) = 8400$
19. $f = x_1 + x_2$, $2x_1 + 3x_2 \leq 1200$, $4x_1 + 2x_2 \leq 1600$, $f_{\max} = f(300, 200) = 500$
21. $x_1/3 + x_2/2 \leq 100$, $x_1/3 + x_2/6 \leq 80$, $f = 150x_1 + 100x_2$, $f_{\max} = f(210, 60) = 37,500$

Problem Set 22.3, page 961

3. $f(120/11, 60/11) = 480/11$
5. Eliminate in Column 3, so that 20 goes. $f_{\min} = f(0, \frac{1}{2}) = -10$.
7. $f_{\max} = f(\frac{60}{21}, 0, \frac{1500}{105}, 0) = \frac{2200}{7}$
9. $f_{\max} = 6$ on the segment from (3, 0, 0) to (0, 0, 2)
11. We minimize! The augmented matrix is

$$\mathbf{T}_0 = \begin{bmatrix} 1 & 1.8 & 2.1 & 0 & 0 & 0 \\ 0 & 15 & 30 & 1 & 0 & 150 \\ 0 & 600 & 500 & 0 & 1 & 3900 \end{bmatrix}.$$

The pivot is 600. The calculation gives

$$\mathbf{T}_1 = \begin{bmatrix} 1 & 0 & \frac{6}{10} & 0 & -\frac{3}{1000} & -\frac{117}{10} \\ 0 & 0 & \frac{35}{2} & 1 & -\frac{1}{40} & \frac{105}{2} \\ 0 & 600 & 500 & 0 & 1 & 3900 \end{bmatrix} \quad \begin{array}{l} \text{Row 1} - \frac{1.8}{600} \text{ Row 3} \\ \text{Row 2} - \frac{15}{600} \text{ Row 3} \\ \text{Row 3} \end{array}$$

The next pivot is $\frac{35}{2}$. The calculation gives

$$\mathbf{T}_2 = \begin{bmatrix} 1 & 0 & 0 & -\frac{6}{175} & -\frac{3}{1400} & -\frac{27}{2} \\ 0 & 0 & \frac{35}{2} & 1 & -\frac{1}{40} & \frac{105}{2} \\ 0 & 600 & 0 & -\frac{200}{7} & \frac{12}{7} & 2400 \end{bmatrix} \quad \begin{array}{l} \text{Row 1} - \frac{1.2}{35} \text{ Row 2} \\ \text{Row 2} \\ \text{Row 3} - \frac{1000}{35} \text{ Row 2} \end{array}$$

Hence $-f$ has the maximum value -13.5 , so that f has the minimum value 13.5 , at the point

$$(x_1, x_2) = \left(\frac{2400}{600}, \frac{105/2}{35/2} \right) = (4, 3).$$

13. $f_{\max} = f(5, 4, 6) = 478$

Problem Set 22.4, page 968

1. $f(6, 3) = 84$
3. $f(20, 20) = 40$
5. $f(10, 5) = 5500$
7. $f(1, 1, 0) = 13$
9. $f(4, 0, \frac{1}{2}) = 9$

Chapter 22 Review Questions and Problems, page 968

9. Step 5: $[0.353 \quad -0.028]^\top$. Slower. Why?
11. Of course! Step 5: $[-1.003 \quad 1.897]^\top$
17. $f(2, 4) = 100$
19. $f(3, 6) = -54$

Problem Set 23.1, page 974

9. $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

13. $\begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

11. $\begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

15. ① — ②

③ — ④

17. If G is complete.

		Edge			
		e_1	e_2	e_3	e_4
19. Vertex	1	$\begin{bmatrix} -1 & -1 & 1 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$			
	2				
	3				
	4				

Problem Set 23.2, page 979

1. 5

3. 4

5. The idea is to go backward. There is a v_{k-1} adjacent to v_k and labeled $k-1$, etc. Now the only vertex labeled 0 is s . Hence $\lambda(v_0) = 0$ implies $v_0 = s$, so that $v_0 - v_1 - \cdots - v_{k-1} - v_k$ is a path $s \rightarrow v_k$ that has length k .

15. Delete the edge $(2, 4)$.

17. No

Problem Set 23.3, page 983

1. $(1, 2), (2, 4), (4, 3)$; $L_2 = 12, L_3 = 36, L_4 = 28$

5. $(1, 2), (2, 4), (3, 4), (3, 5)$; $L_2 = 2, L_3 = 4, L_4 = 3, L_5 = 6$

7. $(1, 2), (2, 4), (3, 4)$; $L_2 = 10, L_3 = 15, L_4 = 13$

9. $(1, 5), (2, 3), (2, 6), (3, 4), (3, 5)$; $L_2 = 9, L_3 = 7, L_4 = 8, L_5 = 4, L_6 = 14$

Problem Set 23.4, page 987

1. $\begin{array}{c} 2 \\ \diagdown \\ 4 - 3 - 5 \\ \diagup \\ 1 \end{array} \quad L = 10$

3. $5 - 3 - 6 \begin{array}{c} \diagup 1 \\ \diagdown 2 - 4 \end{array} \quad L = 17$

5. $1 \begin{array}{c} \diagup 2 \\ \diagdown 4 \end{array} \begin{array}{c} \diagup 3 \\ \diagdown 5 \end{array} \quad L = 12$

9. Yes

11. $1 - 3 - 4 \begin{array}{c} \diagup 2 \\ \diagdown 5 - 6 \end{array} \quad L = 38$

13. New York–Washington–Chicago–Dallas–Denver–Los Angeles

15. G is connected. If G were not a tree, it would have a cycle, but this cycle would provide two paths between any pair of its vertices, contradicting the uniqueness.

19. If we add an edge (u, v) to T , then since T is connected, there is a path $u \rightarrow v$ in T which, together with (u, v) , forms a cycle.

Problem Set 23.5, page 990

1. If G is a tree.
3. A shortest spanning tree of the largest connected graph that contains vertex 1.
7. $(1, 4), (1, 3), (1, 2), (2, 6), (3, 5)$; $L = 32$
9. $(1, 4), (4, 3), (4, 2), (3, 5)$; $L = 20$
11. $(1, 4), (4, 3), (4, 5), (1, 2)$; $L = 12$

Problem Set 23.6, page 997

1. $\{3, 6\}$, $11 + 3 = 14$
3. $\{4, 5, 6\}$, $10 + 5 + 13 = 28$
5. $\{3, 6, 7\}$, $8 + 4 + 4 = 16$
7. $S = \{1, 4\}$, $8 + 6 = 14$
9. One is interested in flows *from* s to t , not in the opposite direction.
13. $\Delta_{12} = 5, \Delta_{24} = 8, \Delta_{45} = 2$; $\Delta_{12} = 5, \Delta_{25} = 3$; $\Delta_{13} = 4, \Delta_{35} = 9$
 $P_1: 1 - 2 - 4 - 5, \Delta f = 2$; $P_2: 1 - 2 - 5, \Delta f = 3$; $P_3: 1 - 3 - 5, \Delta f = 4$
15. $1 - 2 - 5, \Delta f = 2$; $1 - 4 - 2 - 5, \Delta f = 2$, etc.
17. $f_{13} = f_{35} = 8, f_{14} = f_{45} = 5, f_{12} = f_{24} = f_{46} = 4, f_{56} = 13, f = 4 + 13 = 17$,
 $f = 17$ is unique.
19. For instance, $f_{12} = 10, f_{24} = f_{45} = 7, f_{13} = f_{25} = 5, f_{35} = 3, f_{32} = 2$,
 $f = 3 + 5 + 7 = 15, f = 15$ is unique.

Problem Set 23.7, page 1000

3. $(2, 3)$ and $(5, 6)$
5. By considering only edges with one labeled end and one unlabeled end
7. $1 - 2 - 5, \Delta_t = 2$; $1 - 4 - 2 - 5, \Delta_t = 1$; $f = 6 + 2 + 1 = 9$, where 6 is the given flow
9. $1 - 2 - 4 - 6, \Delta_t = 2$; $1 - 3 - 5 - 6, \Delta_t = 1$; $f = 4 + 2 + 1 = 7$, where 4 is the given flow
15. $S = \{1, 2, 4, 5\}, T = \{3, 6\}, \text{cap}(S, T) = 14$

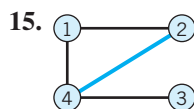
Problem Set 23.8, page 1005

1. No
3. No
5. Yes, $S = \{1, 4, 5, 8\}$
7. Yes, $S = \{1, 3, 5\}$
11. $1 - 2 - 3 - 7 - 5 - 4$
13. $1 - 2 - 3 - 7 - 5 - 4$ is augmenting and gives $1 - 2 - 3 - 7 - 5 - 4$ and $(1, 2), (3, 7), (5, 4)$ is of maximum cardinality.
15. $1 - 4 - 3 - 6 - 7 - 8$ is augmenting and gives $1 - 4 - 3 - 6 - 7 - 8$ and $(1, 4), (3, 6), (7, 8)$ is of maximum cardinality.
19. 3
21. 2
23. 3
25. K_4

Chapter 23 Review Questions and Problems, page 1006

11.
$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

13. To vertex 1 2 3 4
From vertex 1 $\begin{bmatrix} 0 & 1 & 0 & 1 \\ 2 & 1 & 0 & 1 & 0 \\ 3 & 0 & 1 & 0 & 1 \\ 4 & 1 & 0 & 1 & 0 \end{bmatrix}$



17.

Vertex	Incident Edges
1	(1, 2), (1, 4)
2	(2, 1), (2, 4)
3	(3, 4)
4	(4, 1), (4, 2), (4, 3)

19. (1, 2), (1, 4), (2, 3); $L_2 = 2, L_3 = 5, L_4 = 5$

23. (1, 6), (4, 5), (2, 3), (7, 8)

Problem Set 24.1, page 1015

1. $q_L = 19, q_M = 20, q_U = 20.5$

3. $q_L = 138, q_M = 144, q_U = 154$

5. $q_L = 199, q_M = 201, q_U = 201$

7. $q_L = 1.3, q_M = 1.4, q_U = 1.45$

9. $q_L = 89.9, q_M = 91.0, q_U = 91.8$

11. $\bar{x} = 19.875, s = 0.835, \text{IQR} = 1.5$

13. $\bar{x} = 144.67, s = 8.9735, \text{IQR} = 16$

15. $\bar{x} = 1.355, s = 0.136, \text{IQR} = 0.15$

17. 3.54, 1.29

Problem Set 24.2, page 1017

1. 2^3 outcomes: $RRR, RRL, RLR, LRR, RLL, LRL, LLR, LLL$

3. $6^2 = 36$ outcomes (1, 1), (1, 2), $\dots$, (6, 6), first number (second number) referring to the first die (second die)

5. Infinitely many outcomes $H \quad TH \quad TTH \quad TTTH \quad \dots$ ($H = \text{Head}, T = \text{Tail}$)

7. The space of ordered pairs of numbers

9. 10 outcomes: $D \quad ND \quad NND \quad \dots \quad NNNNNNNNND$

11. Yes

17. $A \cup B = B$ implies $A \subseteq B$ by the definition of union. Conversely, $A \subseteq B$ implies that $A \cup B = B$ because always $B \subseteq A \cup B$, and if $A \subseteq B$, we must have equality in the previous relation.

Problem Set 24.3, page 1024

1. $1 - 4/216 = 98.15\%$, by Theorem 1
3. (a) $0.9^3 = 72.9\%$, (b) $\frac{90}{100} \cdot \frac{89}{99} \cdot \frac{88}{98} = 72.65\%$
5. $\frac{8}{9}$
7. Small sample from a large population containing *many* items in each class we are interested in (defectives and nondefectives, etc.)
9. $\frac{498}{500} \cdot \frac{497}{499} \cdot \frac{496}{498} \cdot \frac{495}{497} \cdot \frac{494}{496} \approx 0.98008$
11. (a) $\frac{100}{200} \cdot \frac{99}{199} = 24.874\%$, (b) $\frac{100}{200} \cdot \frac{100}{199} + \frac{100}{200} \cdot \frac{100}{199} = 50.25\%$, (c) same as (a).
(a) + (b) + (c) = 1. Why?
13. $1 - 0.96^3 = 11.5\%$
15. $1 - 0.875^4 = 0.4138 < 1 - 0.75^2 = 0.4375 < 0.5$ (c < b < a)
17. $A = B \cup (A \cap B^c)$, hence $P(A) = P(B) + P(A \cap B^c) \geq P(B)$ by disjointedness of B and $A \cap B^c$

Problem Set 24.4, page 1028

1. In $10! = 3,628,800$ ways
3. $\frac{2}{6} \cdot \frac{1}{5} \cdot \frac{4}{4} \cdot \frac{3}{3} \cdot \frac{2}{2} \cdot \frac{1}{1} = \frac{4}{6} \cdot \frac{3}{5} \cdot \frac{2}{4} \cdot \frac{1}{3} \cdot \frac{2}{2} \cdot \frac{1}{1} = \frac{4!2!}{6!} = \frac{2}{6} \cdot \frac{1}{5} = \frac{1}{15}$
5. $\binom{10}{3} \binom{5}{2} \binom{6}{2} = 18,000$
7. 210, 70, 112, 28
9. In $6!/6 = 120$ ways
11. $9 \cdot 8 = 72$
13. (b) $1/(12n)$
15. $P(\text{No two people have a birthday in common}) = 365 \cdot 364 \cdots 346/365^{20} = 0.59$.
Answer: 41%, which is surprisingly large.

Problem Set 24.5, page 1034

1. $k = \frac{1}{55}$ by (6)
3. $k = \frac{1}{4}$ by (10), $P(0 \leq X \leq 2) = \frac{1}{2}$
5. No, because of (6)
7. $k = \frac{1}{100}$ because of (6) and $1 + 8 + 27 + 64 = 100$
9. $k = 5$; 50%
11. $0.5^3 = 12.5\%$
13. $F(x) = 0$ if $x < -1$, $F(x) = \frac{1}{2}(x+1)^2$ if $-1 \leq x < 0$
 $F(x) = 1 - \frac{1}{2}(x-1)^2$ if $0 \leq x < 1$, $F(x) = 1$ if $x \leq 1$
Answer: 500 cans, $P = 0.125$, 0
15. $X > b$, $X \geq b$, $X < c$, $X \leq c$, etc.

Problem Set 24.6, page 1038

1. $k = \frac{1}{2}$, $\mu = \frac{4}{3}$, $\sigma^2 = \frac{2}{9}$
3. $\mu = \pi$, $\sigma^2 = \pi^2/3$; cf. Example 2
5. $\mu = \frac{1}{4}$, $\sigma^2 = \frac{1}{16}$
7. $C = \frac{1}{2}$, $\mu = 2$, $\sigma^2 = 4$
9. 750, 1, 0.002
11. $c = 0.073$
13. \$643.50
15. $\frac{1}{2}, \frac{1}{20}, (X - \frac{1}{2})\sqrt{20}$
17. $X = \text{Product of the 2 numbers}$. $E(X) = 12.25$, 12 cents
19. $(0 + 1 \cdot 3 + 3 \cdot 8 + 1 \cdot 27)/8 = 54/8 = 6.75$

Problem Set 24.7, page 1044

3. 38%
 5. $\binom{5}{x} 0.5^5$, 0.03125, 0.15625, $1 - f(0) = 0.96875$, 0.96875
 7. 0.265
 9. $f(x) = 0.5^x e^{-0.5}/x!$, $f(0) + f(1) = e^{-0.5}(1.0 + 0.5) = 0.91$. Answer: 9%
 11. $13\frac{1}{4}\%$
 13. 42%, 47.2%, 10.5%, 0.3%
 15. $1 - e^{-0.2} = 18\%$

Problem Set 24.8, page 1050

1. 0.1587, 0.5, 0.6915, 0.6247 3. 45.065, 56.978, 2.022
 5. 15.9% 7. 31.1%, 95.4%
 9. About 58% 11. $t = 1084$ hours
 13. About 683 (Fig. 521a)

Problem Set 24.9, page 1059

1. $\frac{1}{8}, \frac{3}{16}, \frac{3}{8}$ 3. $\frac{2}{9}, \frac{1}{9}, \frac{1}{2}$
 5. $f_2(y) = 1/(\beta_2 - \alpha_2)$ if $\alpha_2 < y < \beta_2$
 7. 27.45 mm, 0.38 mm
 11. 25.26 cm, 0.0078 cm 13. 50%
 15. The distributions in Prob. 17 and Example 1
 17. No

Chapter 24 Review Questions and Problems, page 1060

11. $Q_L = 110$, $Q_M = 112$, $Q_U = 115$
 13. $\bar{x} = 111.9$, $s = 4.0125$, $s^2 = 16.1$
 21. $x_{\min} \leq x_j \leq x_{\max}$. Sum over j from 1.
 17. $\bar{x} = 6$, $s = 3.65$
 19. $f(x) = \binom{50}{x} 0.03^x 0.97^{50-x} \approx 1.5^x e^{-1.5}/x!$
 21. $f(x) = 2^{-x}$, $x = 1, 2, \dots$ 23. $1, \frac{1}{2}$
 25. 0.1587, 0.6306, 0.5, 0.4950

Problem Set 25.2, page 1067

1. In Example 1, $\mu = 0$ so $\sum_{j=1}^n x_j = 0$. $\partial \ln \ell / \partial \ell = 0$ and $\tilde{\sigma}^2$ is as before.
 3. $\ell = e^{-n\mu} \mu^{(x_1 + \dots + x_n)} / (x_1! \dots x_n!)$, $\partial \ln \ell / \partial \mu = -n + (x_1 + \dots + x_n)/\mu = 0$,
 $n\hat{\mu} = n\bar{x}$, $\hat{\mu} = \bar{x} = 15.3$
 5. $l = p^k(1-p)^{n-k}$, $\hat{p} = k/n$, k = number of successes in n trials
 7. $7/12$
 9. $l = f = p(1-p)^{x-1}$, etc., $\hat{p} = 1/x$
 11. $\hat{\theta} = n/\sum x_j = 1/\bar{x}$
 13. $\hat{\theta} = 1$
 15. Variability larger than perhaps expected

Problem Set 25.3, page 1077

3. Shorter by a factor $\sqrt{2}$ 5. 4, 16
 7. $c = 1.96$, $\bar{x} = 126$, $s^2 = 126 \cdot 674/800 = 106.155$, $k = cs/\sqrt{n} = 0.714$,
 $\text{CONF}_{0.95}\{125.3 \leq \mu \leq 126.7\}$, $\text{CONF}_{0.95}\{0.1566 \leq p \leq 0.1583\}$
 9. $\text{CONF}_{0.99}\{63.72 \leq \mu \leq 66.28\}$
 11. $n - 1 = 5$, $F(c) = 0.995$, $c = 4.03$, $\bar{x} = 9533.33$, $s^2 = 49,666.67$,
 $k = 366.66$ (Table 25.2), $\text{CONF}_{0.99}\{9166.7 \leq \mu \leq 9900\}$
 13. $\text{CONF}_{0.95}\{0.023 \leq \sigma^2 \leq 0.085\}$
 15. $n - 1 = 99$ degrees of freedom. $F(c_1) = 0.025$, $c_1 = 74.2$, $F(c_2) = 0.975$,
 $c_2 = 129.6$. Hence $k_1 = 12.41$, $k_2 = 7.10$. $\text{CONF}_{0.95}\{7.10 \leq \sigma^2 \leq 12.41\}$.
 17. $\text{CONF}_{0.95}\{0.74 \leq \sigma^2 \leq 5.19\}$
 19. $Z = X + Y$ is normal with mean 105 and variance 1.25.
Answer: $P(104 \leq Z \leq 106) = 63\%$

Problem Set 25.4, page 1086

3. $t = (0.286 - 0)/(4.31/\sqrt{7}) = 0.18 < c = 1.94$; accept the hypothesis.
 5. $c = 6090 > 6019$; do not reject the hypothesis.
 7. $\sigma^2/n = 1.8$, $c = 57.8$, accept the hypothesis.
 9. $\mu < 58.69$ or $\mu > 61.31$
 11. Alternative $\mu \neq 5000$, $t = (4990 - 5000)/(20/\sqrt{50}) = -3.54 < c = -2.01$
 (Table A9, Appendix 5). Reject the hypothesis $\mu = 5000$ g.
 13. Two-sided. $t = (0.55 - 0)/\sqrt{0.546/8} = 2.11 < c = 2.37$ (Table A9, Appendix 5),
 no difference
 15. $19 \cdot 1.0^2/0.8^2 = 29.69 < c = 30.14$ (Table A10, Appendix 5), accept the
 hypothesis
 17. By (12), $t_0 = \sqrt{16}(20.2 - 19.6)/\sqrt{0.16 + 0.36} > c = 1.70$. Assert that B is better.

Problem Set 25.5, page 1091

1. $\text{LCL} = 1 - 2.58 \cdot 0.02/2 = 0.974$, $\text{UCL} = 1.026$
 3. 27
 5. Choose 4 times the original sample size
 9. $2.58\sqrt{0.0004}/\sqrt{2} = 0.036$, $\text{LCL} = 3.464$, $\text{UCL} = 3.536$
 11. $\text{LCL} = np - 3\sqrt{np(1-p)}$, $\text{CL} = np$, $\text{UCL} = np + 3\sqrt{np(1-p)}$
 13. In about 30% (5%) of the cases
 15. $\text{LCL} = \mu - 3\sqrt{\mu}$ is negative in (b) and we set $\text{LCL} = 0$, $\text{CL} = \mu = 3.6$,
 $\text{UCL} = \mu + 3\sqrt{\mu} = 9.3$.

Problem Set 25.6, page 1095

1. 0.9825, 0.9384, 0.4060 3. 0.8187, 0.6703, 0.1353
 5. $e^{-25\theta}(1 + 25\theta)$, $P(A; 1.5) = 94.5$, $\alpha = 5.5\%$ 7. 19.5%, 14.7%
 9. $(1 - \theta)^n + n\theta(1 - \theta)^{n-1}$ 11. $(1 - \frac{1}{2})^3 + 3 \cdot \frac{1}{2}(1 - \frac{1}{2})^2 = \frac{1}{2}$
 13. $\sum_{x=0}^9 \binom{100}{x} 0.12^x 0.88^{100-x} = 22\%$ (by the normal approximation)
 15. $(1 - \theta)^5$, $[\theta(1 - \theta)^{5-1}]' = 0$, $\theta = \frac{1}{6}$, $\text{AOQL} = 6.7\%$

Problem Set 25.7, page 1099

3. $\chi_0^2 = (40 - 50)^2/50 + (60 - 50)^2/50 = 4 > c = 3.84$; no
 5. $\chi_0^2 = \frac{16}{10} > 11.07$; yes
 7. $\chi_0^2 = 10.264 < 11.07$; yes
 9. 42 even digits, accept.
 13. $\chi_0^2 = \frac{(355 - 358.5)^2}{358.5} + \frac{(123 - 119.5)^2}{119.5} = 0.137 < c = 3.84$ (1 degree of freedom, 95%)
 15. Combining the last three nonzero values, we have $K - r - 1 = 9$ ($r = 1$ since we estimated the mean, $\frac{10,094}{2608} \approx 3.87$). $\chi_0^2 = 12.8 < c = 16.92$. Accept the hypothesis.

Problem Set 25.8, page 1102

3. $(\frac{1}{2})^8 + 8 \cdot (\frac{1}{2})^8 = 3.5\%$ is the probability that 7 cases in 8 trials favor A under the hypothesis that A and B are equally good. Reject.
 5. $(\frac{1}{2})^{18}(1 + 18 + 153 + 816) = 0.0038$
 7. $\bar{x} = 9.67$, $s = 11.87$. $t_0 = 9.67/(11.87/\sqrt{15}) = 3.16 > c = 1.76$ ($\alpha = 5\%$). Hypothesis rejected.
 9. Hypothesis $\tilde{\mu} = 0$. Alternative $\tilde{\mu} > 0$, $\bar{x} = 1.58$,
 $t = \sqrt{10} \cdot 1.58/1.23 = 4.06 > c = 1.83$ ($\alpha = 5\%$). Hypothesis rejected.
 11. Consider $y_j = x_j - \tilde{\mu}_0$.
 13. $n = 8$; 4 transpositions, $P(T \leq 4) = 0.007$. Assert that fertilizing increases yield.
 15. $P(T \leq 2) = 2.8\%$. Assert that there is an increase.

Problem Set 25.9, page 1111

1. $y = 0.98 + 0.495x$
 3. $y = -11,457.9 + 43.2x$
 5. $y = -10 + 0.55x$
 7. $y = 0.5932 + 0.1138x$, $R = 1/0.1138$
 9. $y = 0.32923 + 0.00032x$, $y(66) = 0.35035$
 13. $c = 3.18$ (Table A9), $k_1 = 43.2$, $q_0 = 54,878$, $K = 1.502$,
 $\text{CONF}_{0.95}\{41.7 \leq \kappa_1 \leq 44.7\}$.
 15. $y - 1.875 = 0.067(x - 25)$, $3s_x^2 = 500$, $q_0 = 0.023$, $K = 0.021$,
 $\text{CONF}_{0.95}\{0.046 \leq \kappa_1 \leq 0.088\}$

Chapter 25 Review Questions and Problems, page 1111

15. $\hat{\mu} = 20.325$, $\hat{\sigma}^2 = (\frac{7}{8})s^2 = 3.982$
 17. $\text{CONF}_{0.99}\{27.94 \leq \mu \leq 34.81\}$
 19. $c = 14.74 > 14.5$, reject μ_0 ; $\Phi((14.74 - 14.5)/\sqrt{0.025}) = 0.9353$
 21. $2.58 \cdot \sqrt{0.00024}/\sqrt{2} = 0.028$, LCL = 2.722, UCL = 2.778
 23. $\alpha = 1 - (1 - \theta)^6 = 5.85\%$, when $\theta = 0.01$. For $\theta = 15\%$ we obtain
 $\beta = (1 - \theta)^6 = 37.7\%$. If n increases, so does α , whereas β decreases.
 25. $y = 3.4 - 1.85x$



APPENDIX 3

Auxiliary Material

A3.1 Formulas for Special Functions

For tables of numeric values, see Appendix 5.

Exponential function e^x (Fig. 545)

$$e = 2.71828\ 18284\ 59045\ 23536\ 02874\ 71353$$

$$(1) \quad e^x e^y = e^{x+y}, \quad e^x / e^y = e^{x-y}, \quad (e^x)^y = e^{xy}$$

Natural logarithm (Fig. 546)

$$(2) \quad \ln(xy) = \ln x + \ln y, \quad \ln(x/y) = \ln x - \ln y, \quad \ln(x^a) = a \ln x$$

$\ln x$ is the inverse of e^x , and $e^{\ln x} = x$, $e^{-\ln x} = e^{\ln(1/x)} = 1/x$.

Logarithm of base ten $\log_{10} x$ or simply $\log x$

$$(3) \quad \log x = M \ln x, \quad M = \log e = 0.43429\ 44819\ 03251\ 82765\ 11289\ 18917$$

$$(4) \quad \ln x = \frac{1}{M} \log x, \quad \frac{1}{M} = \ln 10 = 2.30258\ 50929\ 94045\ 68401\ 79914\ 54684$$

$\log x$ is the inverse of 10^x , and $10^{\log x} = x$, $10^{-\log x} = 1/x$.

Sine and cosine functions (Figs. 547, 548). In calculus, angles are measured in radians, so that $\sin x$ and $\cos x$ have period 2π .

$\sin x$ is odd, $\sin(-x) = -\sin x$, and $\cos x$ is even, $\cos(-x) = \cos x$.

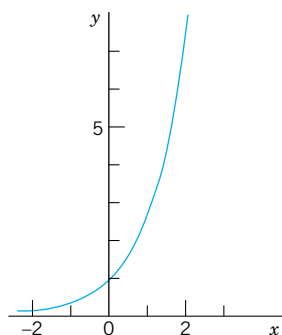


Fig. 545. Exponential function e^x

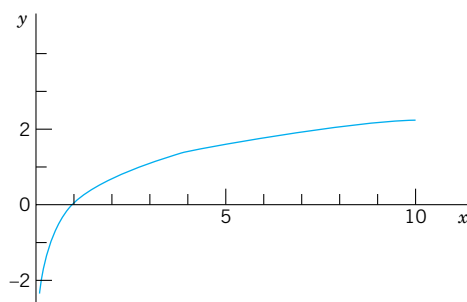
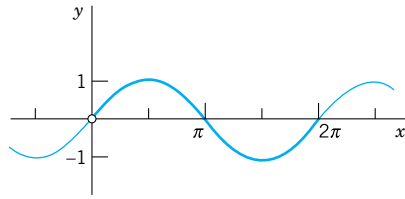
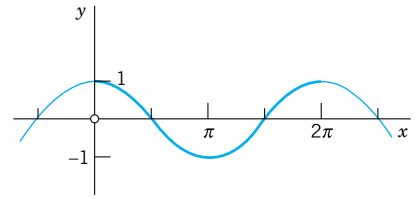


Fig. 546. Natural logarithm $\ln x$

Fig. 547. $\sin x$ Fig. 548. $\cos x$

$$1^\circ = 0.01745\ 32925\ 19943\ \text{radian}$$

$$1\ \text{radian} = 57^\circ\ 17'\ 44.80625''$$

$$= 57.29577\ 95131^\circ$$

$$(5) \quad \sin^2 x + \cos^2 x = 1$$

$$(6) \quad \begin{cases} \sin(x+y) = \sin x \cos y + \cos x \sin y \\ \sin(x-y) = \sin x \cos y - \cos x \sin y \\ \cos(x+y) = \cos x \cos y - \sin x \sin y \\ \cos(x-y) = \cos x \cos y + \sin x \sin y \end{cases}$$

$$(7) \quad \sin 2x = 2 \sin x \cos x, \quad \cos 2x = \cos^2 x - \sin^2 x$$

$$(8) \quad \begin{cases} \sin x = \cos\left(x - \frac{\pi}{2}\right) = \cos\left(\frac{\pi}{2} - x\right) \\ \cos x = \sin\left(x + \frac{\pi}{2}\right) = \sin\left(\frac{\pi}{2} - x\right) \end{cases}$$

$$(9) \quad \sin(\pi - x) = \sin x, \quad \cos(\pi - x) = -\cos x$$

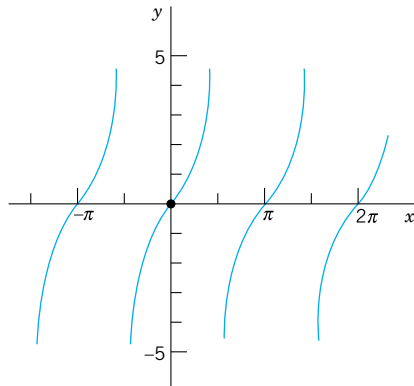
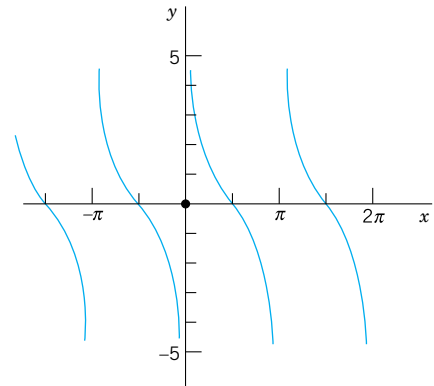
$$(10) \quad \cos^2 x = \frac{1}{2}(1 + \cos 2x), \quad \sin^2 x = \frac{1}{2}(1 - \cos 2x)$$

$$(11) \quad \begin{cases} \sin x \sin y = \frac{1}{2}[-\cos(x+y) + \cos(x-y)] \\ \cos x \cos y = \frac{1}{2}[\cos(x+y) + \cos(x-y)] \\ \sin x \cos y = \frac{1}{2}[\sin(x+y) + \sin(x-y)] \end{cases}$$

$$(12) \quad \begin{cases} \sin u + \sin v = 2 \sin \frac{u+v}{2} \cos \frac{u-v}{2} \\ \cos u + \cos v = 2 \cos \frac{u+v}{2} \cos \frac{u-v}{2} \\ \cos v - \cos u = 2 \sin \frac{u+v}{2} \sin \frac{u-v}{2} \end{cases}$$

$$(13) \quad A \cos x + B \sin x = \sqrt{A^2 + B^2} \cos(x \pm \delta), \quad \tan \delta = \frac{\sin \delta}{\cos \delta} = \mp \frac{B}{A}$$

$$(14) \quad A \cos x + B \sin x = \sqrt{A^2 + B^2} \sin(x \pm \delta), \quad \tan \delta = \frac{\sin \delta}{\cos \delta} = \pm \frac{A}{B}$$

Fig. 549. $\tan x$ Fig. 550. $\cot x$

Tangent, cotangent, secant, cosecant (Figs. 549, 550)

$$(15) \quad \tan x = \frac{\sin x}{\cos x}, \quad \cot x = \frac{\cos x}{\sin x}, \quad \sec x = \frac{1}{\cos x}, \quad \csc x = \frac{1}{\sin x}$$

$$(16) \quad \tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}, \quad \tan(x - y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$$

Hyperbolic functions (hyperbolic sine $\sinh x$, etc.; Figs. 551, 552)

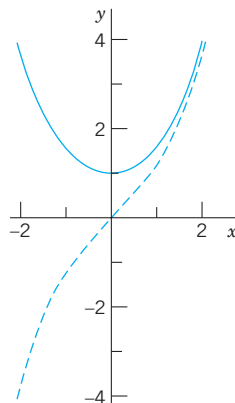
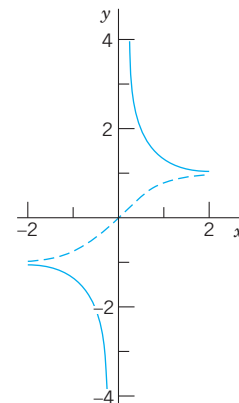
$$(17) \quad \sinh x = \frac{1}{2}(e^x - e^{-x}), \quad \cosh x = \frac{1}{2}(e^x + e^{-x})$$

$$(18) \quad \tanh x = \frac{\sinh x}{\cosh x}, \quad \coth x = \frac{\cosh x}{\sinh x}$$

$$(19) \quad \cosh x + \sinh x = e^x, \quad \cosh x - \sinh x = e^{-x}$$

$$(20) \quad \cosh^2 x - \sinh^2 x = 1$$

$$(21) \quad \sinh^2 x = \frac{1}{2}(\cosh 2x - 1), \quad \cosh^2 x = \frac{1}{2}(\cosh 2x + 1)$$

Fig. 551. $\sinh x$ (dashed) and $\cosh x$ Fig. 552. $\tanh x$ (dashed) and $\coth x$

$$(22) \quad \begin{cases} \sinh(x \pm y) = \sinh x \cosh y \pm \cosh x \sinh y \\ \cosh(x \pm y) = \cosh x \cosh y \pm \sinh x \sinh y \end{cases}$$

$$(23) \quad \tanh(x \pm y) = \frac{\tanh x \pm \tanh y}{1 \pm \tanh x \tanh y}$$

Gamma function (Fig. 553 and Table A2 in App. 5). The gamma function $\Gamma(\alpha)$ is defined by the integral

$$(24) \quad \Gamma(\alpha) = \int_0^{\infty} e^{-t} t^{\alpha-1} dt \quad (\alpha > 0),$$

which is meaningful only if $\alpha > 0$ (or, if we consider complex α , for those α whose real part is positive). Integration by parts gives the important *functional relation of the gamma function*,

$$(25) \quad \Gamma(\alpha + 1) = \alpha \Gamma(\alpha).$$

From (24) we readily have $\Gamma(1) = 1$; hence if α is a positive integer, say k , then by repeated application of (25) we obtain

$$(26) \quad \Gamma(k + 1) = k! \quad (k = 0, 1, \dots).$$

This shows that *the gamma function can be regarded as a generalization of the elementary factorial function*. [Sometimes the notation $(\alpha - 1)!$ is used for $\Gamma(\alpha)$, even for noninteger values of α , and the gamma function is also known as the **factorial function**.]

By repeated application of (25) we obtain

$$\Gamma(\alpha) = \frac{\Gamma(\alpha + 1)}{\alpha} = \frac{\Gamma(\alpha + 2)}{\alpha(\alpha + 1)} = \dots = \frac{\Gamma(\alpha + k + 1)}{\alpha(\alpha + 1)(\alpha + 2) \cdots (\alpha + k)}$$

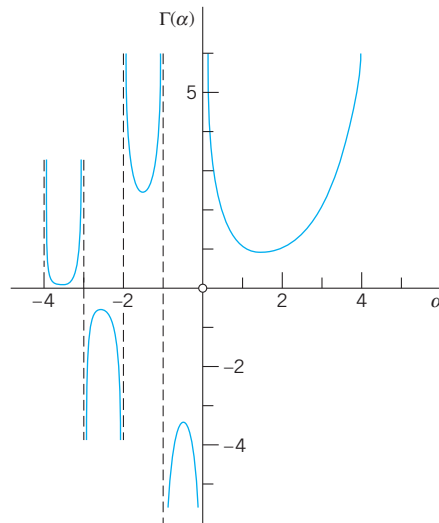


Fig. 553. Gamma function

and we may use this relation

$$(27) \quad \Gamma(\alpha) = \frac{\Gamma(\alpha + k + 1)}{\alpha(\alpha + 1) \cdots (\alpha + k)} \quad (\alpha \neq 0, -1, -2, \dots),$$

for defining the gamma function for negative α ($\neq -1, -2, \dots$), choosing for k the smallest integer such that $\alpha + k + 1 > 0$. Together with (24), this then gives a definition of $\Gamma(\alpha)$ for all α not equal to zero or a negative integer (Fig. 553).

It can be shown that the gamma function may also be represented as the limit of a product, namely, by the formula

$$(28) \quad \Gamma(\alpha) = \lim_{n \rightarrow \infty} \frac{n! n^\alpha}{\alpha(\alpha + 1)(\alpha + 2) \cdots (\alpha + n)} \quad (\alpha \neq 0, -1, \dots).$$

From (27) or (28) we see that, for complex α , the gamma function $\Gamma(\alpha)$ is a meromorphic function with simple poles at $\alpha = 0, -1, -2, \dots$.

An approximation of the gamma function for large positive α is given by the **Stirling formula**

$$(29) \quad \Gamma(\alpha + 1) \approx \sqrt{2\pi\alpha} \left(\frac{\alpha}{e}\right)^\alpha$$

where e is the base of the natural logarithm. We finally mention the special value

$$(30) \quad \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

Incomplete gamma functions

$$(31) \quad P(\alpha, x) = \int_0^x e^{-t} t^{\alpha-1} dt, \quad Q(\alpha, x) = \int_x^\infty e^{-t} t^{\alpha-1} dt \quad (\alpha > 0)$$

$$(32) \quad \Gamma(\alpha) = P(\alpha, x) + Q(\alpha, x)$$

Beta function

$$(33) \quad B(x, y) = \int_0^1 t^{x-1} (1-t)^{y-1} dt \quad (x > 0, y > 0)$$

Representation in terms of gamma functions:

$$(34) \quad B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$$

Error function (Fig. 554 and Table A4 in App. 5)

$$(35) \quad \operatorname{erf} x = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

$$(36) \quad \operatorname{erf} x = \frac{2}{\sqrt{\pi}} \left(x - \frac{x^3}{1!3} + \frac{x^5}{2!5} - \frac{x^7}{3!7} + \cdots \right)$$

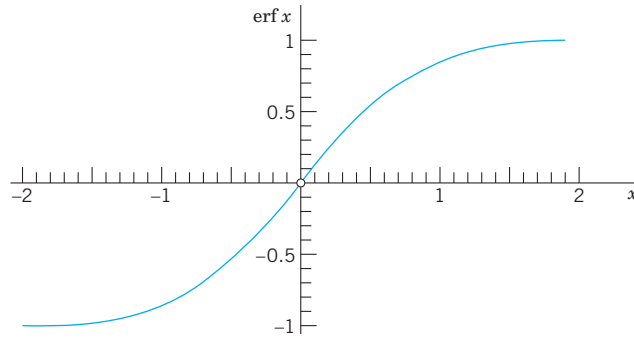


Fig. 554. Error function

$\operatorname{erf}(\infty) = 1$, *complementary error function*

$$(37) \quad \operatorname{erfc} x = 1 - \operatorname{erf} x = \frac{2}{\sqrt{\pi}} \int_x^{\infty} e^{-t^2} dt$$

Fresnel integrals¹ (Fig. 555)

$$(38) \quad C(x) = \int_0^x \cos(t^2) dt, \quad S(x) = \int_0^x \sin(t^2) dt$$

$C(\infty) = \sqrt{\pi/8}$, $S(\infty) = \sqrt{\pi/8}$, *complementary functions*

$$(39) \quad c(x) = \sqrt{\frac{\pi}{8}} - C(x) = \int_x^{\infty} \cos(t^2) dt$$

$$s(x) = \sqrt{\frac{\pi}{8}} - S(x) = \int_x^{\infty} \sin(t^2) dt$$

Sine integral (Fig. 556 and Table A4 in App. 5)

$$(40) \quad \operatorname{Si}(x) = \int_0^x \frac{\sin t}{t} dt$$

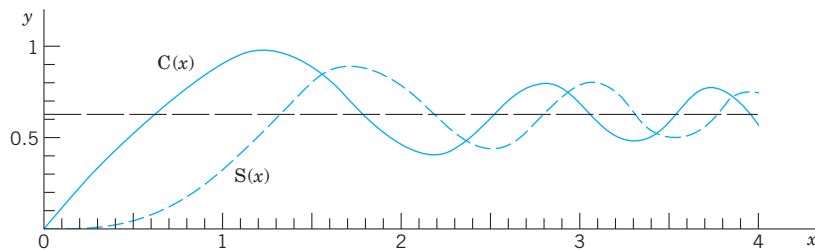


Fig. 555. Fresnel integrals

¹AUGUSTIN FRESNEL (1788–1827), French physicist and mathematician. For tables see Ref. [GenRef1].

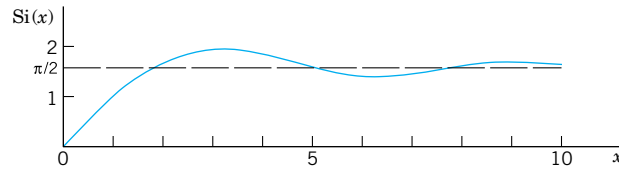


Fig. 556. Sine integral

$\text{Si}(\infty) = \pi/2$, complementary function

$$(41) \quad \text{si}(x) = \frac{\pi}{2} - \text{Si}(x) = \int_x^\infty \frac{\sin t}{t} dt$$

Cosine integral (Table A4 in App. 5)

$$(42) \quad \text{ci}(x) = \int_x^\infty \frac{\cos t}{t} dt \quad (x > 0)$$

Exponential integral

$$(43) \quad \text{Ei}(x) = \int_x^\infty \frac{e^{-t}}{t} dt \quad (x > 0)$$

Logarithmic integral

$$(44) \quad \text{li}(x) = \int_0^x \frac{dt}{\ln t}$$

A3.2 Partial Derivatives

For differentiation formulas, see inside of front cover.

Let $z = f(x, y)$ be a real function of two independent real variables, x and y . If we keep y constant, say, $y = y_1$, and think of x as a variable, then $f(x, y_1)$ depends on x alone. If the derivative of $f(x, y_1)$ with respect to x for a value $x = x_1$ exists, then the value of this derivative is called the **partial derivative** of $f(x, y)$ with respect to x at the point (x_1, y_1) and is denoted by

$$\left. \frac{\partial f}{\partial x} \right|_{(x_1, y_1)} \quad \text{or by} \quad \left. \frac{\partial z}{\partial x} \right|_{(x_1, y_1)}.$$

Other notations are

$$f_x(x_1, y_1) \quad \text{and} \quad z_x(x_1, y_1);$$

these may be used when subscripts are not used for another purpose and there is no danger of confusion.

We thus have, by the definition of the derivative,

$$(1) \quad \left. \frac{\partial f}{\partial x} \right|_{(x_1, y_1)} = \lim_{\Delta x \rightarrow 0} \frac{f(x_1 + \Delta x, y_1) - f(x_1, y_1)}{\Delta x}.$$

The partial derivative of $z = f(x, y)$ with respect to y is defined similarly; we now keep x constant, say, equal to x_1 , and differentiate $f(x_1, y)$ with respect to y . Thus

$$(2) \quad \left. \frac{\partial f}{\partial y} \right|_{(x_1, y_1)} = \left. \frac{\partial z}{\partial y} \right|_{(x_1, y_1)} = \lim_{\Delta y \rightarrow 0} \frac{f(x_1, y_1 + \Delta y) - f(x_1, y_1)}{\Delta y}.$$

Other notations are $f_y(x_1, y_1)$ and $z_y(x_1, y_1)$.

It is clear that the values of those two partial derivatives will in general depend on the point (x_1, y_1) . Hence the partial derivatives $\partial z/\partial x$ and $\partial z/\partial y$ at a variable point (x, y) are functions of x and y . The function $\partial z/\partial x$ is obtained as in ordinary calculus by differentiating $z = f(x, y)$ with respect to x , *treating y as a constant*, and $\partial z/\partial y$ is obtained by differentiating z with respect to y , *treating x as a constant*.

EXAMPLE 1

Let $z = f(x, y) = x^2y + x \sin y$. Then

$$\frac{\partial f}{\partial x} = 2xy + \sin y, \quad \frac{\partial f}{\partial y} = x^2 + x \cos y.$$

The partial derivatives $\partial z/\partial x$ and $\partial z/\partial y$ of a function $z = f(x, y)$ have a very simple **geometric interpretation**. The function $z = f(x, y)$ can be represented by a surface in space. The equation $y = y_1$ then represents a vertical plane intersecting the surface in a curve, and the partial derivative $\partial z/\partial x$ at a point (x_1, y_1) is the slope of the tangent (that is, $\tan \alpha$ where α is the angle shown in Fig. 557) to the curve. Similarly, the partial derivative $\partial z/\partial y$ at (x_1, y_1) is the slope of the tangent to the curve $x = x_1$ on the surface $z = f(x, y)$ at (x_1, y_1) .

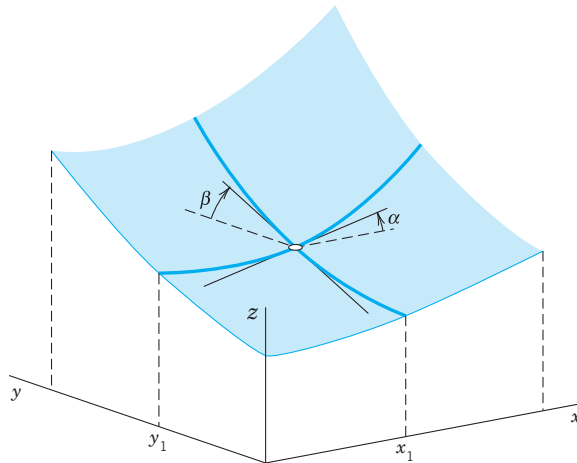


Fig. 557. Geometrical interpretation of first partial derivatives

The partial derivatives $\partial z/\partial x$ and $\partial z/\partial y$ are called *first partial derivatives* or *partial derivatives of first order*. By differentiating these derivatives once more, we obtain the four *second partial derivatives* (or *partial derivatives of second order*)²

$$\begin{aligned}
 \frac{\partial^2 f}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = f_{xx} \\
 \frac{\partial^2 f}{\partial x \partial y} &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = f_{yx} \\
 \frac{\partial^2 f}{\partial y \partial x} &= \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = f_{xy} \\
 \frac{\partial^2 f}{\partial y^2} &= \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = f_{yy}.
 \end{aligned}
 \tag{3}$$

It can be shown that if all the derivatives concerned are continuous, then the two mixed partial derivatives are equal, so that the order of differentiation does not matter (see Ref. [GenRef4] in App. 1), that is,

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}.
 \tag{4}$$

EXAMPLE 2 For the function in Example 1.

$$f_{xx} = 2y, \quad f_{xy} = 2x + \cos y = f_{yx}, \quad f_{yy} = -x \sin y.$$

By differentiating the second partial derivatives again with respect to x and y , respectively, we obtain the *third partial derivatives* or *partial derivatives of the third order* of f , etc.

If we consider a function $f(x, y, z)$ of **three independent variables**, then we have the three first partial derivatives $f_x(x, y, z)$, $f_y(x, y, z)$, and $f_z(x, y, z)$. Here f_x is obtained by differentiating f with respect to x , **treating both y and z as constants**. Thus, analogous to (1), we now have

$$\left. \frac{\partial f}{\partial x} \right|_{(x_1, y_1, z_1)} = \lim_{\Delta x \rightarrow 0} \frac{f(x_1 + \Delta x, y_1, z_1) - f(x_1, y_1, z_1)}{\Delta x},$$

etc. By differentiating f_x , f_y , f_z again in this fashion we obtain the second partial derivatives of f , etc.

EXAMPLE 3 Let $f(x, y, z) = x^2 + y^2 + z^2 + xy e^z$. Then

$$\begin{aligned}
 f_x &= 2x + y e^z, & f_y &= 2y + x e^z, & f_z &= 2z + xy e^z, \\
 f_{xx} &= 2, & f_{xy} &= f_{yx} = e^z, & f_{xz} &= f_{zx} = y e^z, \\
 f_{yy} &= 2, & f_{yz} &= f_{zy} = x e^z, & f_{zz} &= 2 + xy e^z.
 \end{aligned}$$

² **CAUTION!** In the subscript notation, the subscripts are written in the order in which we differentiate, whereas in the “ ∂ ” notation the order is opposite.

A3.3 Sequences and Series

See also Chap. 15.

Monotone Real Sequences

We call a real sequence $x_1, x_2, \dots, x_n, \dots$ a **monotone sequence** if it is either **monotone increasing**, that is,

$$x_1 \leq x_2 \leq x_3 \leq \dots$$

or **monotone decreasing**, that is,

$$x_1 \geq x_2 \geq x_3 \geq \dots$$

We call $x_1, x_2, \dots$ a **bounded sequence** if there is a positive constant K such that $|x_n| < K$ for all n .

THEOREM 1

If a real sequence is bounded and monotone, it converges.

PROOF

Let $x_1, x_2, \dots$ be a bounded monotone increasing sequence. Then its terms are smaller than some number B and, since $x_1 \leq x_n$ for all n , they lie in the interval $x_1 \leq x_n \leq B$, which will be denoted by I_0 . We bisect I_0 ; that is, we subdivide it into two parts of equal length. If the right half (together with its endpoints) contains terms of the sequence, we denote it by I_1 . If it does not contain terms of the sequence, then the left half of I_0 (together with its endpoints) is called I_1 . This is the first step.

In the second step we bisect I_1 , select one half by the same rule, and call it I_2 , and so on (see Fig. 558).

In this way we obtain shorter and shorter intervals $I_0, I_1, I_2, \dots$ with the following properties. Each I_m contains all x_n for $n > m$. No term of the sequence lies to the right of I_m , and, since the sequence is monotone increasing, all x_n with n greater than some number N lie in I_m ; of course, N will depend on m , in general. The lengths of the I_m approach zero as m approaches infinity. Hence there is precisely one number, call it L , that lies in all those intervals,³ and we may now easily prove that the sequence is convergent with the limit L .

In fact, given an $\epsilon > 0$, we choose an m such that the length of I_m is less than ϵ . Then L and all the x_n with $n > N(m)$ lie in I_m , and, therefore, $|x_n - L| < \epsilon$ for all those n . This completes the proof for an increasing sequence. For a decreasing sequence the proof is the same, except for a suitable interchange of “left” and “right” in the construction of those intervals. ■

³This statement seems to be obvious, but actually it is not; it may be regarded as an axiom of the real number system in the following form. Let $J_1, J_2, \dots$ be closed intervals such that each J_m contains all J_n with $n > m$, and the lengths of the J_m approach zero as m approaches infinity. Then there is precisely one real number that is contained in all those intervals. This is the so-called **Cantor–Dedekind axiom**, named after the German mathematicians GEORG CANTOR (1845–1918), the creator of set theory, and RICHARD DEDEKIND (1831–1916), known for his fundamental work in number theory. For further details see Ref. [GenRef2] in App. 1. (An interval I is said to be **closed** if its two endpoints are regarded as points belonging to I . It is said to be **open** if the endpoints are not regarded as points of I .)

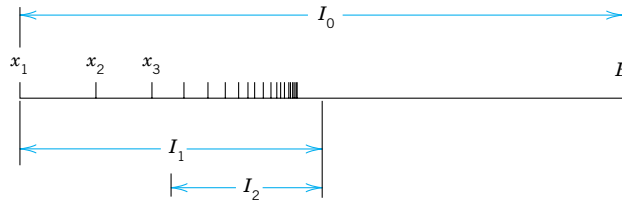


Fig. 558. Proof of Theorem 1

Real Series

THEOREM 2

Leibniz Test for Real Series

Let $x_1, x_2, \dots$ be real and monotone decreasing to zero, that is,

$$(1) \quad (a) \quad x_1 \geq x_2 \geq x_3 \geq \dots, \quad (b) \quad \lim_{m \rightarrow \infty} x_m = 0.$$

Then the series with terms of alternating signs

$$x_1 - x_2 + x_3 - x_4 + \dots$$

converges, and for the remainder R_n after the n th term we have the estimate

$$(2) \quad |R_n| \leq x_{n+1}.$$

PROOF Let s_n be the n th partial sum of the series. Then, because of (1a),

$$s_1 = x_1, \quad s_2 = x_1 - x_2 \leq s_1,$$

$$s_3 = s_2 + x_3 \geq s_2, \quad s_3 = s_1 - (x_2 - x_3) \leq s_1,$$

so that $s_2 \leq s_3 \leq s_1$. Proceeding in this fashion, we conclude that (Fig. 559)

$$(3) \quad s_1 \geq s_3 \geq s_5 \geq \dots \geq s_6 \geq s_4 \geq s_2$$

which shows that the odd partial sums form a bounded monotone sequence, and so do the even partial sums. Hence, by Theorem 1, both sequences converge, say,

$$\lim_{n \rightarrow \infty} s_{2n+1} = s, \quad \lim_{n \rightarrow \infty} s_{2n} = s^*.$$

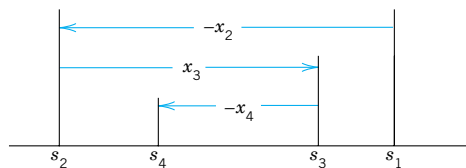


Fig. 559. Proof of the Leibniz test

Now, since $s_{2n+1} - s_{2n} = x_{2n+1}$, we readily see that (1b) implies

$$s - s^* = \lim_{n \rightarrow \infty} s_{2n+1} - \lim_{n \rightarrow \infty} s_{2n} = \lim_{n \rightarrow \infty} (s_{2n+1} - s_{2n}) = \lim_{n \rightarrow \infty} x_{2n+1} = 0.$$

Hence $s^* = s$, and the series converges with the sum s .

We prove the estimate (2) for the remainder. Since $s_n \rightarrow s$, it follows from (3) that

$$s_{2n+1} \geq s \geq s_{2n} \quad \text{and also} \quad s_{2n-1} \geq s \geq s_{2n}.$$

By subtracting s_{2n} and s_{2n-1} , respectively, we obtain

$$s_{2n+1} - s_{2n} \geq s - s_{2n} \geq 0, \quad 0 \geq s - s_{2n-1} \geq s_{2n} - s_{2n-1}.$$

In these inequalities, the first expression is equal to x_{2n+1} , the last is equal to $-x_{2n}$, and the expressions between the inequality signs are the remainders R_{2n} and R_{2n-1} . Thus the inequalities may be written

$$x_{2n+1} \geq R_{2n} \geq 0, \quad 0 \geq R_{2n-1} \geq -x_{2n}$$

and we see that they imply (2). This completes the proof. ■

A3.4 Grad, Div, Curl, ∇^2 in Curvilinear Coordinates

To simplify formulas, we write Cartesian coordinates $x = x_1, y = x_2, z = x_3$. We denote curvilinear coordinates by q_1, q_2, q_3 . Through each point P there pass three coordinate surfaces $q_1 = \text{const}, q_2 = \text{const}, q_3 = \text{const}$. They intersect along coordinate curves. We assume the three coordinate curves through P to be **orthogonal** (perpendicular to each other). We write coordinate transformations as

$$(1) \quad x_1 = x_1(q_1, q_2, q_3), \quad x_2 = x_2(q_1, q_2, q_3), \quad x_3 = x_3(q_1, q_2, q_3).$$

Corresponding transformations of grad, div, curl, and ∇^2 can all be written by using

$$(2) \quad h_j^2 = \sum_{k=1}^3 \left(\frac{\partial x_k}{\partial q_j} \right)^2.$$

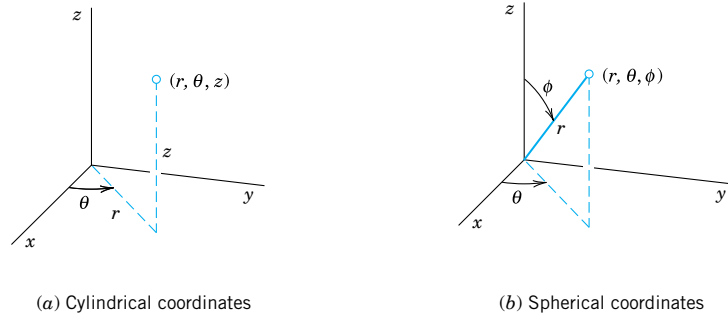
Next to Cartesian coordinates, most important are **cylindrical coordinates** $q_1 = r, q_2 = \theta, q_3 = z$ (Fig. 560a) defined by

$$(3) \quad x_1 = q_1 \cos q_2 = r \cos \theta, \quad x_2 = q_1 \sin q_2 = r \sin \theta, \quad x_3 = q_3 = z$$

and **spherical coordinates** $q_1 = r, q_2 = \theta, q_3 = \phi$ (Fig. 560b) defined by⁴

$$(4) \quad \begin{aligned} x_1 &= q_1 \cos q_2 \sin q_3 = r \cos \theta \sin \phi, & x_2 &= q_1 \sin q_2 \sin q_3 = r \sin \theta \sin \phi \\ x_3 &= q_1 \cos q_3 = r \cos \phi. \end{aligned}$$

⁴This is the notation used in calculus and in many other books. It is logical since in it, θ plays the same role as in polar coordinates. **CAUTION!** Some books interchange the roles of θ and ϕ .

**Fig. 560.** Special curvilinear coordinates

In addition to the general formulas for any orthogonal coordinates q_1, q_2, q_3 , we shall give additional formulas for these important special cases.

Linear Element ds . In Cartesian coordinates,

$$ds^2 = dx_1^2 + dx_2^2 + dx_3^2 \quad (\text{Sec. 9.5}).$$

For the q -coordinates,

$$(5) \quad ds^2 = h_1^2 dq_1^2 + h_2^2 dq_2^2 + h_3^2 dq_3^2.$$

$$(5') \quad ds^2 = dr^2 + r^2 d\theta^2 + dz^2 \quad (\text{Cylindrical coordinates}).$$

For polar coordinates set $dz^2 = 0$.

$$(5'') \quad ds^2 = dr^2 + r^2 \sin^2 \phi d\theta^2 + r^2 d\phi^2 \quad (\text{Spherical coordinates}).$$

Gradient. $\text{grad } f = \nabla f = [f_{x_1}, f_{x_2}, f_{x_3}]$ (partial derivatives; Sec. 9.7). In the q -system, with $\mathbf{u}, \mathbf{v}, \mathbf{w}$ denoting unit vectors in the positive directions of the q_1, q_2, q_3 coordinate curves, respectively,

$$(6) \quad \text{grad } f = \nabla f = \frac{1}{h_1} \frac{\partial f}{\partial q_1} \mathbf{u} + \frac{1}{h_2} \frac{\partial f}{\partial q_2} \mathbf{v} + \frac{1}{h_3} \frac{\partial f}{\partial q_3} \mathbf{w}$$

$$(6') \quad \text{grad } f = \nabla f = \frac{\partial f}{\partial r} \mathbf{u} + \frac{1}{r} \frac{\partial f}{\partial \theta} \mathbf{v} + \frac{\partial f}{\partial z} \mathbf{w} \quad (\text{Cylindrical coordinates})$$

$$(6'') \quad \text{grad } f = \nabla f = \frac{\partial f}{\partial r} \mathbf{u} + \frac{1}{r \sin \phi} \frac{\partial f}{\partial \theta} \mathbf{v} + \frac{1}{r} \frac{\partial f}{\partial \phi} \mathbf{w} \quad (\text{Spherical coordinates}).$$

Divergence $\text{div } \mathbf{F} = \nabla \cdot \mathbf{F} = (F_1)_{x_1} + (F_2)_{x_2} + (F_3)_{x_3}$ ($\mathbf{F} = [F_1, F_2, F_3]$, Sec. 9.8);

$$(7) \quad \text{div } \mathbf{F} = \nabla \cdot \mathbf{F} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} (h_2 h_3 F_1) + \frac{\partial}{\partial q_2} (h_3 h_1 F_2) + \frac{\partial}{\partial q_3} (h_1 h_2 F_3) \right]$$

$$(7') \quad \text{div } \mathbf{F} = \nabla \cdot \mathbf{F} = \frac{1}{r} \frac{\partial}{\partial r} (r F_1) + \frac{1}{r} \frac{\partial F_2}{\partial \theta} + \frac{\partial F_3}{\partial z} \quad (\text{Cylindrical coordinates})$$

$$(7'') \quad \operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 F_1) + \frac{1}{r \sin \phi} \frac{\partial F_2}{\partial \theta} + \frac{1}{r \sin \phi} \frac{\partial}{\partial \phi} (\sin \phi F_3)$$

(Spherical coordinates).

Laplacian $\nabla^2 f = \nabla \cdot \nabla f = \operatorname{div} (\operatorname{grad} f) = f_{x_1 x_1} + f_{x_2 x_2} + f_{x_3 x_3}$ (Sec. 9.8):

$$(8) \quad \nabla^2 f = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial q_1} \right) + \frac{\partial}{\partial q_2} \left(\frac{h_3 h_1}{h_2} \frac{\partial f}{\partial q_2} \right) + \frac{\partial}{\partial q_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial q_3} \right) \right]$$

$$(8') \quad \nabla^2 f = \frac{\partial^2 f}{\partial r^2} + \frac{1}{r} \frac{\partial f}{\partial r} + \frac{1}{r^2} \frac{\partial^2 f}{\partial \theta^2} + \frac{\partial^2 f}{\partial z^2} \quad (\text{Cylindrical coordinates})$$

$$(8'') \quad \nabla^2 f = \frac{\partial^2 f}{\partial r^2} + \frac{2}{r} \frac{\partial f}{\partial r} + \frac{1}{r^2 \sin^2 \phi} \frac{\partial^2 f}{\partial \theta^2} + \frac{1}{r^2} \frac{\partial^2 f}{\partial \phi^2} + \frac{\cot \phi}{r^2} \frac{\partial f}{\partial \phi}$$

(Spherical coordinates).

Curl (Sec. 9.9):

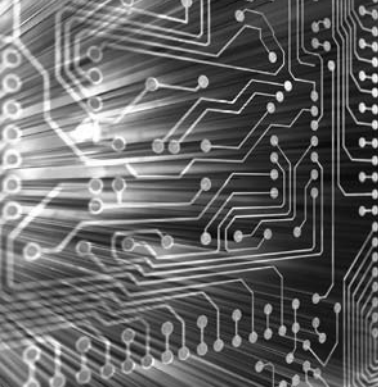
$$(9) \quad \operatorname{curl} \mathbf{F} = \nabla \times \mathbf{F} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \mathbf{u} & h_2 \mathbf{v} & h_3 \mathbf{w} \\ \frac{\partial}{\partial q_1} & \frac{\partial}{\partial q_2} & \frac{\partial}{\partial q_3} \\ h_1 F_1 & h_2 F_2 & h_3 F_3 \end{vmatrix}.$$

For cylindrical coordinates we have in (9) (as in the previous formulas)

$$h_1 = h_r = 1, \quad h_2 = h_\theta = q_1 = r, \quad h_3 = h_z = 1$$

and for spherical coordinates we have

$$h_1 = h_r = 1, \quad h_2 = h_\theta = q_1 \sin q_3 = r \sin \phi, \quad h_3 = h_\phi = q_1 = r.$$



APPENDIX 4

Additional Proofs

Section 2.6, page 74

PROOF OF THEOREM 1 Uniqueness¹

Assuming that the problem consisting of the ODE

$$(1) \quad y'' + p(x)y' + q(x)y = 0$$

and the two initial conditions

$$(2) \quad y(x_0) = K_0, \quad y'(x_0) = K_1$$

has two solutions $y_1(x)$ and $y_2(x)$ on the interval I in the theorem, we show that their difference

$$y(x) = y_1(x) - y_2(x)$$

is identically zero on I ; then $y_1 \equiv y_2$ on I , which implies uniqueness.

Since (1) is homogeneous and linear, y is a solution of that ODE on I , and since y_1 and y_2 satisfy the same initial conditions, y satisfies the conditions

$$(11) \quad y(x_0) = 0, \quad y'(x_0) = 0.$$

We consider the function

$$z(x) = y(x)^2 + y'(x)^2$$

and its derivative

$$z' = 2yy' + 2y'y''.$$

From the ODE we have

$$y'' = -py' - qy.$$

By substituting this in the expression for z' we obtain

$$(12) \quad z' = 2yy' - 2py'^2 - 2qyy'.$$

Now, since y and y' are real,

$$(y \pm y')^2 = y^2 \pm 2yy' + y'^2 \geq 0.$$

¹This proof was suggested by my colleague, Prof. A. D. Ziebur. In this proof, we use some formula numbers that have not yet been used in Sec. 2.6.

From this and the definition of z we obtain the two inequalities

$$(13) \quad (a) \quad 2yy' \leq y^2 + y'^2 = z, \quad (b) \quad -2yy' \leq y^2 + y'^2 = z.$$

From (13b) we have $2yy' \geq -z$. Together, $|2yy'| \leq z$. For the last term in (12) we now obtain

$$-2qyy' \leq |-2qyy'| = |q||2yy'| \leq |q|z.$$

Using this result as well as $-p \leq |p|$ and applying (13a) to the term $2yy'$ in (12), we find

$$z' \leq z + 2|p|y'^2 + |q|z.$$

Since $y'^2 \leq y^2 + y'^2 = z$, from this we obtain

$$z' \leq (1 + 2|p| + |q|)z$$

or, denoting the function in parentheses by h ,

$$(14a) \quad z' \leq hz \quad \text{for all } x \text{ on } I.$$

Similarly, from (12) and (13) it follows that

$$(14b) \quad \begin{aligned} -z' &= -2yy' + 2py'^2 + 2qyy' \\ &\leq z + 2|p|z + |q|z = hz. \end{aligned}$$

The inequalities (14a) and (14b) are equivalent to the inequalities

$$(15) \quad z' - hz \leq 0, \quad z' + hz \geq 0.$$

Integrating factors for the two expressions on the left are

$$F_1 = e^{-\int h(x) dx} \quad \text{and} \quad F_2 = e^{\int h(x) dx}.$$

The integrals in the exponents exist because h is continuous. Since F_1 and F_2 are positive, we thus have from (15)

$$F_1(z' - hz) = (F_1 z)' \leq 0 \quad \text{and} \quad F_2(z' + hz) = (F_2 z)' \geq 0.$$

This means that $F_1 z$ is nonincreasing and $F_2 z$ is nondecreasing on I . Since $z(x_0) = 0$ by (11), when $x \leq x_0$ we thus obtain

$$F_1 z \geq (F_1 z)_{x_0} = 0, \quad F_2 z \leq (F_2 z)_{x_0} = 0$$

and similarly, when $x \geq x_0$,

$$F_1 z \leq 0, \quad F_2 z \geq 0.$$

Dividing by F_1 and F_2 and noting that these functions are positive, we altogether have

$$z \leq 0, \quad z \geq 0 \quad \text{for all } x \text{ on } I.$$

This implies that $z = y^2 + y'^2 \equiv 0$ on I . Hence $y \equiv 0$ or $y_1 \equiv y_2$ on I . ■

Section 5.3, page 182

PROOF OF THEOREM 2 Frobenius Method. Basis of Solutions. Three Cases

The formula numbers in this proof are the same as in the text of Sec. 5.3. An additional formula not appearing in Sec. 5.3 will be called (A) (see below).

The ODE in Theorem 2 is

$$(1) \quad y'' + \frac{b(x)}{x} y' + \frac{c(x)}{x^2} y = 0,$$

where $b(x)$ and $c(x)$ are analytic functions. We can write it

$$(1') \quad x^2 y'' + x b(x) y' + c(x) y = 0.$$

The indicial equation of (1) is

$$(4) \quad r(r-1) + b_0 r + c_0 = 0.$$

The roots r_1, r_2 of this quadratic equation determine the general form of a basis of solutions of (1), and there are three possible cases as follows.

Case 1. Distinct Roots Not Differing by an Integer. A first solution of (1) is of the form

$$(5) \quad y_1(x) = x^{r_1}(a_0 + a_1 x + a_2 x^2 + \cdots)$$

and can be determined as in the power series method. For a proof that in this case, the ODE (1) has a second independent solution of the form

$$(6) \quad y_2(x) = x^{r_2}(A_0 + A_1 x + A_2 x^2 + \cdots),$$

see Ref. [A11] listed in App. 1.

Case 2. Double Root. The indicial equation (4) has a double root r if and only if $(b_0 - 1)^2 - 4c_0 = 0$, and then $r = \frac{1}{2}(1 - b_0)$. A first solution

$$(7) \quad y_1(x) = x^r(a_0 + a_1 x + a_2 x^2 + \cdots), \quad r = \frac{1}{2}(1 - b_0),$$

can be determined as in Case 1. We show that a second independent solution is of the form

$$(8) \quad y_2(x) = y_1(x) \ln x + x^r(A_1 x + A_2 x^2 + \cdots) \quad (x > 0).$$

We use the method of reduction of order (see Sec. 2.1), that is, we determine $u(x)$ such that $y_2(x) = u(x)y_1(x)$ is a solution of (1). By inserting this and the derivatives

$$y_2' = u' y_1 + u y_1', \quad y_2'' = u'' y_1 + 2u' y_1' + u y_1''$$

into the ODE (1') we obtain

$$x^2(u'' y_1 + 2u' y_1' + u y_1'') + x b(u' y_1 + u y_1') + c u y_1 = 0.$$

Since y_1 is a solution of (1'), the sum of the terms involving u is zero, and this equation reduces to

$$x^2 y_1 u'' + 2x^2 y_1' u' + x b y_1 u' = 0.$$

By dividing by $x^2 y_1$ and inserting the power series for b we obtain

$$u'' + \left(2 \frac{y_1'}{y_1} + \frac{b_0}{x} + \cdots \right) u' = 0.$$

Here, and in the following, the dots designate terms that are constant or involve positive powers of x . Now, from (7), it follows that

$$\begin{aligned} \frac{y_1'}{y_1} &= \frac{x^{r-1}[ra_0 + (r+1)a_1x + \cdots]}{x^r[a_0 + a_1x + \cdots]} \\ &= \frac{1}{x} \left(\frac{ra_0 + (r+1)a_1x + \cdots}{a_0 + a_1x + \cdots} \right) = \frac{r}{x} + \cdots. \end{aligned}$$

Hence the previous equation can be written

$$(A) \quad u'' + \left(\frac{2r + b_0}{x} + \cdots \right) u' = 0.$$

Since $r = (1 - b_0)/2$, the term $(2r + b_0)/x$ equals $1/x$, and by dividing by u' we thus have

$$\frac{u''}{u'} = -\frac{1}{x} + \cdots.$$

By integration we obtain $\ln u' = -\ln x + \cdots$, hence $u' = (1/x)e^{\cdots}$. Expanding the exponential function in powers of x and integrating once more, we see that u is of the form

$$u = \ln x + k_1x + k_2x^2 + \cdots.$$

Inserting this into $y_2 = uy_1$, we obtain for y_2 a representation of the form (8).

Case 3. Roots Differing by an Integer. We write $r_1 = r$ and $r_2 = r - p$ where p is a *positive* integer. A first solution

$$(9) \quad y_1(x) = x^{r_1}(a_0 + a_1x + a_2x^2 + \cdots)$$

can be determined as in Cases 1 and 2. We show that a second independent solution is of the form

$$(10) \quad y_2(x) = ky_1(x) \ln x + x^{r_2}(A_0 + A_1x + A_2x^2 + \cdots)$$

where we may have $k \neq 0$ or $k = 0$. As in Case 2 we set $y_2 = uy_1$. The first steps are literally as in Case 2 and give Eq. (A),

$$u'' + \left(\frac{2r + b_0}{x} + \cdots \right) u' = 0.$$

Now by elementary algebra, the coefficient $b_0 - 1$ of r in (4) equals minus the sum of the roots,

$$b_0 - 1 = -(r_1 + r_2) = -(r + r - p) = -2r + p.$$

Hence $2r + b_0 = p + 1$, and division by u' gives

$$\frac{u''}{u'} = - \left(\frac{p+1}{x} + \dots \right).$$

The further steps are as in Case 2. Integrating, we find

$$\ln u' = -(p+1) \ln x + \dots, \quad \text{thus} \quad u' = x^{-(p+1)} e^{(\dots)}$$

where dots stand for some series of nonnegative integer powers of x . By expanding the exponential function as before we obtain a series of the form

$$u' = \frac{1}{x^{p+1}} + \frac{k_1}{x^p} + \dots + \frac{k_{p-1}}{x^2} + \frac{k_p}{x} + k_{p+1} + k_{p+2}x + \dots.$$

We integrate once more. Writing the resulting logarithmic term first, we get

$$u = k_p \ln x + \left(-\frac{1}{px^p} - \dots - \frac{k_{p-1}}{x} + k_{p+1}x + \dots \right).$$

Hence, by (9) we get for $y_2 = uy_1$ the formula

$$y_2 = k_p y_1 \ln x + x^{r_1-p} \left(-\frac{1}{p} - \dots - k_{p-1}x^{p-1} + \dots \right) (a_0 + a_1x + \dots).$$

But this is of the form (10) with $k = k_p$ since $r_1 - p = r_2$ and the product of the two series involves nonnegative integer powers of x only. ■

Section 7.7, page 293

THEOREM

Determinants

The definition of a determinant

$$(7) \quad D = \det \mathbf{A} = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdot & \cdot & \cdots & \cdot \\ \cdot & \cdot & \cdots & \cdot \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}$$

as given in Sec. 7.7 is unambiguous, that is, it yields the same value of D no matter which rows or columns we choose in the development.

PROOF In this proof we shall use formula numbers not yet used in Sec. 7.7.

We shall prove first that *the same value is obtained no matter which row is chosen*.

The proof is by induction. The statement is true for a second-order determinant, for which the developments by the first row $a_{11}a_{22} + a_{12}(-a_{21})$ and by the second row $a_{21}(-a_{12}) + a_{22}a_{11}$ give the same value $a_{11}a_{22} - a_{12}a_{21}$. Assuming the statement to be true for an $(n - 1)$ st-order determinant, we prove that it is true for an n th-order determinant.

For this purpose we expand D in terms of each of two arbitrary rows, say, the i th and the j th, and compare the results. Without loss of generality let us assume $i < j$.

First Expansion. We expand D by the i th row. A typical term in this expansion is

$$(19) \quad a_{ik}C_{ik} = a_{ik} \cdot (-1)^{i+k}M_{ik}.$$

The minor M_{ik} of a_{ik} in D is an $(n - 1)$ st-order determinant. By the induction hypothesis we may expand it by any row. We expand it by the row corresponding to the j th row of D . This row contains the entries a_{jl} ($l \neq k$). It is the $(j - 1)$ st row of M_{ik} , because M_{ik} does not contain entries of the i th row of D , and $i < j$. We have to distinguish between two cases as follows.

Case I. If $l < k$, then the entry a_{jl} belongs to the l th column of M_{ik} (see Fig. 561). Hence the term involving a_{jl} in this expansion is

$$(20) \quad a_{jl} \cdot (\text{cofactor of } a_{jl} \text{ in } M_{ik}) = a_{jl} \cdot (-1)^{(j-1)+l}M_{ikjl}$$

where M_{ikjl} is the minor of a_{jl} in M_{ik} . Since this minor is obtained from M_{ik} by deleting the row and column of a_{jl} , it is obtained from D by deleting the i th and j th rows and the k th and l th columns of D . We insert the expansions of the M_{ik} into that of D . Then it follows from (19) and (20) that the terms of the resulting representation of D are of the form

$$(21a) \quad a_{ik}a_{jl} \cdot (-1)^b M_{ikjl} \quad (l < k)$$

where

$$b = i + k + j + l - 1.$$

Case II. If $l > k$, the only difference is that then a_{jl} belongs to the $(l - 1)$ st column of M_{ik} , because M_{ik} does not contain entries of the k th column of D , and $k < l$. This causes an additional minus sign in (20), and, instead of (21a), we therefore obtain

$$(21b) \quad -a_{ik}a_{jl} \cdot (-1)^b M_{ikjl} \quad (l > k)$$

where b is the same as before.

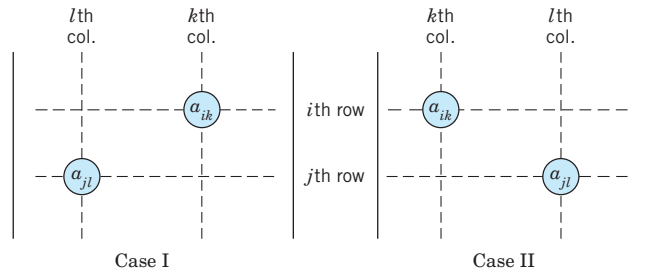


Fig. 561. Cases I and II of the two expansions of D

Second Expansion. We now expand D at first by the j th row. A typical term in this expansion is

$$(22) \quad a_{jl}C_{jl} = a_{jl} \cdot (-1)^{j+l}M_{jl}.$$

By the induction hypothesis we may expand the minor M_{jl} of a_{jl} in D by its i th row, which corresponds to the i th row of D , since $j > i$.

Case I. If $k > l$, the entry a_{ik} in that row belongs to the $(k-1)$ st column of M_{jl} , because M_{jl} does not contain entries of the l th column of D , and $l < k$ (see Fig. 561). Hence the term involving a_{ik} in this expansion is

$$(23) \quad a_{ik} \cdot (\text{cofactor of } a_{ik} \text{ in } M_{jl}) = a_{ik} \cdot (-1)^{i+(k-1)}M_{ikjl},$$

where the minor M_{ikjl} of a_{ik} in M_{jl} is obtained by deleting the i th and j th rows and the k th and l th columns of D [and is, therefore, identical with M_{ikjl} in (20), so that our notation is consistent]. We insert the expansions of the M_{jl} into that of D . It follows from (22) and (23) that this yields a representation whose terms are identical with those given by (21a) when $l < k$.

Case II. If $k < l$, then a_{ik} belongs to the k th column of M_{jl} , we obtain an additional minus sign, and the result agrees with that characterized by (21b).

We have shown that the two expansions of D consist of the same terms, and this proves our statement concerning rows.

The proof of the statement concerning **columns** is quite similar; if we expand D in terms of two arbitrary columns, say, the k th and the l th, we find that the general term involving $a_{jl}a_{ik}$ is exactly the same as before. This proves that not only all column expansions of D yield the same value, but also that their common value is equal to the common value of the row expansions of D .

This completes the proof and shows that *our definition of an n th-order determinant is unambiguous.* ■

Section 9.3, page 368

PROOF OF FORMULA (2)

We prove that in right-handed Cartesian coordinates, the vector product

$$\mathbf{v} = \mathbf{a} \times \mathbf{b} = [a_1, \ a_2, \ a_3] \times [b_1, \ b_2, \ b_3]$$

has the components

$$(2) \quad v_1 = a_2b_3 - a_3b_2, \quad v_2 = a_3b_1 - a_1b_3, \quad v_3 = a_1b_2 - a_2b_1.$$

We need only consider the case $\mathbf{v} \neq \mathbf{0}$. Since $\mathbf{v}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$, Theorem 1 in Sec. 9.2 gives $\mathbf{a} \cdot \mathbf{v} = 0$ and $\mathbf{b} \cdot \mathbf{v} = 0$; in components [see (2), Sec. 9.2],

$$(3) \quad \begin{aligned} a_1v_1 + a_2v_2 + a_3v_3 &= 0 \\ b_1v_1 + b_2v_2 + b_3v_3 &= 0. \end{aligned}$$

Multiplying the first equation by b_3 , the last by a_3 , and subtracting, we obtain

$$(a_3b_1 - a_1b_3)v_1 = (a_2b_3 - a_3b_2)v_2.$$

Multiplying the first equation by b_1 , the last by a_1 , and subtracting, we obtain

$$(a_1b_2 - a_2b_1)v_2 = (a_3b_1 - a_1b_3)v_3.$$

We can easily verify that these two equations are satisfied by

$$(4) \quad v_1 = c(a_2b_3 - a_3b_2), \quad v_2 = c(a_3b_1 - a_1b_3), \quad v_3 = c(a_1b_2 - a_2b_1)$$

where c is a constant. The reader may verify, by inserting, that (4) also satisfies (3). Now each of the equations in (3) represents a plane through the origin in $v_1v_2v_3$ -space. The vectors $\mathbf{a}$ and $\mathbf{b}$ are normal vectors of these planes (see Example 6 in Sec. 9.2). Since $\mathbf{v} \neq \mathbf{0}$, these vectors are not parallel and the two planes do not coincide. Hence their intersection is a straight line L through the origin. Since (4) is a solution of (3) and, for varying c , represents a straight line, we conclude that (4) represents L , and every solution of (3) must be of the form (4). In particular, the components of $\mathbf{v}$ must be of this form, where c is to be determined. From (4) we obtain

$$|\mathbf{v}|^2 = v_1^2 + v_2^2 + v_3^2 = c^2[(a_2b_3 - a_3b_2)^2 + (a_3b_1 - a_1b_3)^2 + (a_1b_2 - a_2b_1)^2].$$

This can be written

$$|\mathbf{v}|^2 = c^2[(a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2) - (a_1b_1 + a_2b_2 + a_3b_3)^2],$$

as can be verified by performing the indicated multiplications in both formulas and comparing. Using (2) in Sec. 9.2, we thus have

$$|\mathbf{v}|^2 = c^2[(\mathbf{a} \cdot \mathbf{a})(\mathbf{b} \cdot \mathbf{b}) - (\mathbf{a} \cdot \mathbf{b})^2].$$

By comparing this with formula (12) in Prob. 4 of Problem Set 9.3 we conclude that $c = \pm 1$.

We show that $c = +1$. This can be done as follows.

If we change the lengths and directions of $\mathbf{a}$ and $\mathbf{b}$ continuously and so that at the end $\mathbf{a} = \mathbf{i}$ and $\mathbf{b} = \mathbf{j}$ (Fig. 188a in Sec. 9.3), then $\mathbf{v}$ will change its length and direction continuously, and at the end, $\mathbf{v} = \mathbf{i} \times \mathbf{j} = \mathbf{k}$. Obviously we may effect the change so that both $\mathbf{a}$ and $\mathbf{b}$ remain different from the zero vector and are not parallel at any instant. Then $\mathbf{v}$ is never equal to the zero vector, and since the change is continuous and c can only assume the values $+1$ or -1 , it follows that at the end c must have the same value as before. Now at the end $\mathbf{a} = \mathbf{i}$, $\mathbf{b} = \mathbf{j}$, $\mathbf{v} = \mathbf{k}$ and, therefore, $a_1 = 1$, $b_2 = 1$, $v_3 = 1$, and the other components in (4) are zero. Hence from (4) we see that $v_3 = c = +1$. This proves Theorem 1.

For a left-handed coordinate system, $\mathbf{i} \times \mathbf{j} = -\mathbf{k}$ (see Fig. 188b in Sec. 9.3), resulting in $c = -1$. This proves the statement right after formula (2). ■

Section 9.9, page 408

PROOF OF THE INVARIANCE OF THE CURL

This proof will follow from two theorems (A and B), which we prove first.

THEOREM A

Transformation Law for Vector Components

For any vector $\mathbf{v}$ the components v_1, v_2, v_3 and v_1^*, v_2^*, v_3^* in any two systems of Cartesian coordinates x_1, x_2, x_3 and x_1^*, x_2^*, x_3^* , respectively, are related by

$$\begin{aligned} v_1^* &= c_{11}v_1 + c_{12}v_2 + c_{13}v_3 \\ (1) \quad v_2^* &= c_{21}v_1 + c_{22}v_2 + c_{23}v_3 \\ v_3^* &= c_{31}v_1 + c_{32}v_2 + c_{33}v_3, \end{aligned}$$

and conversely

$$\begin{aligned} v_1 &= c_{11}v_1^* + c_{21}v_2^* + c_{31}v_3^* \\ (2) \quad v_2 &= c_{12}v_1^* + c_{22}v_2^* + c_{32}v_3^* \\ v_3 &= c_{13}v_1^* + c_{23}v_2^* + c_{33}v_3^* \end{aligned}$$

with coefficients

$$\begin{aligned} (3) \quad c_{11} &= \mathbf{i}^* \cdot \mathbf{i} & c_{12} &= \mathbf{i}^* \cdot \mathbf{j} & c_{13} &= \mathbf{i}^* \cdot \mathbf{k} \\ c_{21} &= \mathbf{j}^* \cdot \mathbf{i} & c_{22} &= \mathbf{j}^* \cdot \mathbf{j} & c_{23} &= \mathbf{j}^* \cdot \mathbf{k} \\ c_{31} &= \mathbf{k}^* \cdot \mathbf{i} & c_{32} &= \mathbf{k}^* \cdot \mathbf{j} & c_{33} &= \mathbf{k}^* \cdot \mathbf{k} \end{aligned}$$

satisfying

$$(4) \quad \sum_{j=1}^3 c_{kj}c_{mj} = \delta_{km} \quad (k, m = 1, 2, 3),$$

where the **Kronecker delta**² is given by

$$\delta_{km} = \begin{cases} 0 & (k \neq m) \\ 1 & (k = m) \end{cases}$$

and $\mathbf{i}, \mathbf{j}, \mathbf{k}$ and $\mathbf{i}^*, \mathbf{j}^*, \mathbf{k}^*$ denote the unit vectors in the positive x_1 -, x_2 -, x_3 - and x_1^* -, x_2^* -, x_3^* -directions, respectively.

²LEOPOLD KRONECKER (1823–1891), German mathematician at Berlin, who made important contributions to algebra, group theory, and number theory.

We shall keep our discussion completely independent of Chap. 7, but readers familiar with matrices should recognize that we are dealing with **orthogonal transformations and matrices** and that our present theorem follows from Theorem 2 in Sec. 8.3.

PROOF The representation of $\mathbf{v}$ in the two systems are

$$(5) \quad (a) \quad \mathbf{v} = v_1 \mathbf{i} + v_2 \mathbf{j} + v_3 \mathbf{k} \quad (b) \quad \mathbf{v} = v_1^* \mathbf{i}^* + v_2^* \mathbf{j}^* + v_3^* \mathbf{k}^*.$$

Since $\mathbf{i}^* \cdot \mathbf{i}^* = 1$, $\mathbf{i}^* \cdot \mathbf{j}^* = 0$, $\mathbf{i}^* \cdot \mathbf{k}^* = 0$, we get from (5b) simply $\mathbf{i}^* \cdot \mathbf{v} = v_1^*$ and from this and (5a)

$$v_1^* = \mathbf{i}^* \cdot \mathbf{v} = \mathbf{i}^* \cdot v_1 \mathbf{i} + \mathbf{i}^* \cdot v_2 \mathbf{j} + \mathbf{i}^* \cdot v_3 \mathbf{k} = v_1 \mathbf{i}^* \cdot \mathbf{i} + v_2 \mathbf{i}^* \cdot \mathbf{j} + v_3 \mathbf{i}^* \cdot \mathbf{k}.$$

Because of (3), this is the first formula in (1), and the other two formulas are obtained similarly, by considering $\mathbf{j}^* \cdot \mathbf{v}$ and then $\mathbf{k}^* \cdot \mathbf{v}$. Formula (2) follows by the same idea, taking $\mathbf{i} \cdot \mathbf{v} = v_1$ from (5a) and then from (5b) and (3)

$$v_1 = \mathbf{i} \cdot \mathbf{v} = v_1^* \mathbf{i} \cdot \mathbf{i}^* + v_2^* \mathbf{i} \cdot \mathbf{j}^* + v_3^* \mathbf{i} \cdot \mathbf{k}^* = c_{11} v_1^* + c_{21} v_2^* + c_{31} v_3^*,$$

and similarly for the other two components.

We prove (4). We can write (1) and (2) briefly as

$$(6) \quad (a) \quad v_j = \sum_{m=1}^3 c_{mj} v_m^*, \quad (b) \quad v_k^* = \sum_{j=1}^3 c_{kj} v_j.$$

Substituting v_j into v_k^* , we get

$$v_k^* = \sum_{j=1}^3 c_{kj} \sum_{m=1}^3 c_{mj} v_m^* = \sum_{m=1}^3 v_m^* \left(\sum_{j=1}^3 c_{kj} c_{mj} \right),$$

where $k = 1, 2, 3$. Taking $k = 1$, we have

$$v_1^* = v_1^* \left(\sum_{j=1}^3 c_{1j} c_{1j} \right) + v_2^* \left(\sum_{j=1}^3 c_{1j} c_{2j} \right) + v_3^* \left(\sum_{j=1}^3 c_{1j} c_{3j} \right).$$

For this to hold for *every* vector $\mathbf{v}$, the first sum must be 1 and the other two sums 0. This proves (4) with $k = 1$ for $m = 1, 2, 3$. Taking $k = 2$ and then $k = 3$, we obtain (4) with $k = 2$ and 3, for $m = 1, 2, 3$. ■

THEOREM B

Transformation Law for Cartesian Coordinates

The transformation of any Cartesian $x_1 x_2 x_3$ -coordinate system into any other Cartesian $x_1^* x_2^* x_3^*$ -coordinate system is of the form

$$(7) \quad x_m^* = \sum_{j=1}^3 c_{mj} x_j + b_m, \quad m = 1, 2, 3,$$

with coefficients (3) and constants b_1, b_2, b_3 ; conversely,

$$(8) \quad x_k = \sum_{n=1}^3 c_{nk} x_n^* + \tilde{b}_k, \quad k = 1, 2, 3.$$

Theorem B follows from Theorem A by noting that the most general transformation of a Cartesian coordinate system into another such system may be decomposed into a transformation of the type just considered and a translation; and under a translation, corresponding coordinates differ merely by a constant.

PROOF OF THE INVARIANCE OF THE CURL

We write again x_1, x_2, x_3 instead of x, y, z , and similarly x_1^*, x_2^*, x_3^* for other Cartesian coordinates, assuming that both systems are right-handed. Let a_1, a_2, a_3 denote the components of $\text{curl } \mathbf{v}$ in the $x_1x_2x_3$ -coordinates, as given by (1), Sec. 9.9, with

$$x = x_1, \quad y = x_2, \quad z = x_3.$$

Similarly, let a_1^*, a_2^*, a_3^* denote the components of $\text{curl } \mathbf{v}$ in the $x_1^*x_2^*x_3^*$ -coordinate system. We prove that the length and direction of $\text{curl } \mathbf{v}$ are independent of the particular choice of Cartesian coordinates, as asserted. We do this by showing that the components of $\text{curl } \mathbf{v}$ satisfy the transformation law (2), which is characteristic of vector components. We consider a_1 . We use (6a), and then the chain rule for functions of several variables (Sec. 9.6). This gives

$$\begin{aligned} a_1 &= \frac{\partial v_3}{\partial x_2} - \frac{\partial v_2}{\partial x_3} = \sum_{m=1}^3 \left(c_{m3} \frac{\partial v_m^*}{\partial x_2} - c_{m2} \frac{\partial v_m^*}{\partial x_3} \right) \\ &= \sum_{m=1}^3 \sum_{j=1}^3 \left(c_{m3} \frac{\partial v_m^*}{\partial x_j^*} \frac{\partial x_j^*}{\partial x_2} - c_{m2} \frac{\partial v_m^*}{\partial x_j^*} \frac{\partial x_j^*}{\partial x_3} \right). \end{aligned}$$

From this and (7) we obtain

$$\begin{aligned} a_1 &= \sum_{m=1}^3 \sum_{j=1}^3 (c_{m3}c_{j2} - c_{m2}c_{j3}) \frac{\partial v_m^*}{\partial x_j^*} \\ &= (c_{33}c_{22} - c_{32}c_{23}) \left(\frac{\partial v_3^*}{\partial x_2^*} - \frac{\partial v_2^*}{\partial x_3^*} \right) + \cdots \\ &= (c_{33}c_{22} - c_{32}c_{23})a_1^* + (c_{13}c_{32} - c_{12}c_{33})a_2^* + (c_{23}c_{12} - c_{22}c_{13})a_3^*. \end{aligned}$$

Note what we did. The double sum had $3 \times 3 = 9$ terms, 3 of which were zero (when $m = j$), and the remaining 6 terms we combined in pairs as we needed them in getting a_1^*, a_2^*, a_3^* .

We now use (3), Lagrange's identity (see Formula (15) in Team Project 24 in Problem Set 9.3) and $\mathbf{k}^* \times \mathbf{j}^* = -\mathbf{i}^*$ and $\mathbf{k} \times \mathbf{j} = -\mathbf{i}$. Then

$$\begin{aligned} c_{33}c_{22} - c_{32}c_{23} &= (\mathbf{k}^* \cdot \mathbf{k})(\mathbf{j}^* \cdot \mathbf{j}) - (\mathbf{k}^* \cdot \mathbf{j})(\mathbf{j}^* \cdot \mathbf{k}) \\ &= (\mathbf{k}^* \times \mathbf{j}^*) \cdot (\mathbf{k} \times \mathbf{j}) = \mathbf{i}^* \cdot \mathbf{i} = c_{11}, \quad \text{etc.} \end{aligned}$$

Hence $a_1 = c_{11}a_1^* + c_{21}a_2^* + c_{31}a_3^*$. This is of the form of the first formula in (2) in Theorem A, and the other two formulas of the form (2) are obtained similarly. This proves the theorem for right-handed systems. If the $x_1x_2x_3$ -coordinates are left-handed, then $\mathbf{k} \times \mathbf{j} = +\mathbf{i}$, but then there is a minus sign in front of the determinant in (1), Sec. 9.9. ■

Section 10.2, page 420

PROOF OF THEOREM 1, PART (b) We prove that if

$$(1) \quad \int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r} = \int_C (F_1 dx + F_2 dy + F_3 dz)$$

with continuous F_1, F_2, F_3 in a domain D is independent of path in D , then $F = \text{grad } f$ in D for some f ; in components

$$(2') \quad F_1 = \frac{\partial f}{\partial x}, \quad F_2 = \frac{\partial f}{\partial y}, \quad F_3 = \frac{\partial f}{\partial z}.$$

We choose any fixed $A: (x_0, y_0, z_0)$ in D and any $B: (x, y, z)$ in D and define f by

$$(3) \quad f(x, y, z) = f_0 + \int_A^B (F_1 dx^* + F_2 dy^* + F_3 dz^*)$$

with any constant f_0 and any path from A to B in D . Since A is fixed and we have independence of path, the integral depends only on the coordinates x, y, z , so that (3) defines a function $f(x, y, z)$ in D . We show that $\mathbf{F} = \text{grad } f$ with this f , beginning with the first of the three relations (2'). Because of independence of path we may integrate from A to $B_1: (x_1, y, z)$ and then parallel to the x -axis along the segment B_1B in Fig. 562 with B_1 chosen so that the whole segment lies in D . Then

$$f(x, y, z) = f_0 + \int_A^{B_1} (F_1 dx^* + F_2 dy^* + F_3 dz^*) + \int_{B_1}^B (F_1 dx^* + F_2 dy^* + F_3 dz^*).$$

We now take the partial derivative with respect to x on both sides. On the left we get $\partial f / \partial x$. We show that on the right we get F_1 . The derivative of the first integral is zero because $A: (x_0, y_0, z_0)$ and $B_1: (x_1, y, z)$ do not depend on x . We consider the second integral. Since on the segment B_1B , both y and z are constant, the terms $F_2 dy^*$ and

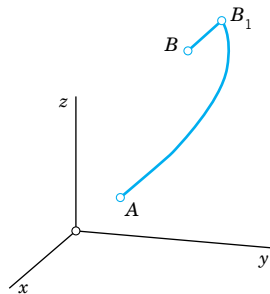


Fig. 562. Proof of Theorem 1

$F_3 dz^*$ do not contribute to the derivative of the integral. The remaining part can be written as a definite integral,

$$\int_{B_1}^B F_1 dx^* = \int_{x_1}^x F_1(x^*, y, z) dx^*.$$

Hence its partial derivative with respect to x is $F_1(x, y, z)$, and the first of the relations (2') is proved. The other two formulas in (2') follow by the same argument. ■

Section 11.5, page 500

THEOREM

Reality of Eigenvalues

If p, q, r , and p' in the Sturm–Liouville equation (1) of Sec. 11.5 are real-valued and continuous on the interval $a \leq x \leq b$ and $r(x) > 0$ throughout that interval (or $r(x) < 0$ throughout that interval), then all the eigenvalues of the Sturm–Liouville problem (1), (2), Sec. 11.5, are real.

PROOF Let $\lambda = \alpha + i\beta$ be an eigenvalue of the problem and let

$$y(x) = u(x) + iv(x)$$

be a corresponding eigenfunction; here α, β, u , and v are real. Substituting this into (1), Sec. 11.5, we have

$$(pu' + ipv')' + (q + \alpha r + i\beta r)(u + iv) = 0.$$

This complex equation is equivalent to the following pair of equations for the real and the imaginary parts:

$$(pu')' + (q + \alpha r)u - \beta r v = 0$$

$$(pv')' + (q + \alpha r)v + \beta r u = 0.$$

Multiplying the first equation by v , the second by $-u$ and adding, we get

$$\begin{aligned} -\beta(u^2 + v^2)r &= u(pv')' - v(pu')' \\ &= [(pv')u - (pu')v]'. \end{aligned}$$

The expression in brackets is continuous on $a \leq x \leq b$, for reasons similar to those in the proof of Theorem 1, Sec. 11.5. Integrating over x from a to b , we thus obtain

$$-\beta \int_a^b (u^2 + v^2)r dx = \left[p(uv' - u'v) \right]_a^b.$$

Because of the boundary conditions, the right side is zero; this is as in that proof. Since y is an eigenfunction, $u^2 + v^2 \not\equiv 0$. Since y and r are continuous and $r > 0$ (or $r < 0$) on the interval $a \leq x \leq b$, the integral on the left is not zero. Hence, $\beta = 0$, which means that $\lambda = \alpha$ is real. This completes the proof. ■

Section 13.4, page 627

PROOF OF THEOREM 2 Cauchy–Riemann Equations

We prove that Cauchy–Riemann equations

$$(1) \quad u_x = v_y, \quad u_y = -v_x$$

are sufficient for a complex function $f(z) = u(x, y) + iv(x, y)$ to be analytic; precisely, *if the real part u and the imaginary part v of $f(z)$ satisfy (1) in a domain D in the complex plane and if the partial derivatives in (1) are **continuous** in D , then $f(z)$ is analytic in D .*

In this proof we write $\Delta z = \Delta x + i\Delta y$ and $\Delta f = f(z + \Delta z) - f(z)$. The idea of proof is as follows.

(a) We express Δf in terms of first partial derivatives of u and v , by applying the mean value theorem of Sec. 9.6.

(b) We get rid of partial derivatives with respect to y by applying the Cauchy–Riemann equations.

(c) We let Δz approach zero and show that then $\Delta f/\Delta z$, as obtained, approaches a limit, which is equal to $u_x + iv_x$, the right side of (4) in Sec. 13.4, regardless of the way of approach to zero.

(a) Let $P: (x, y)$ be any fixed point in D . Since D is a domain, it contains a neighborhood of P . We can choose a point $Q: (x + \Delta x, y + \Delta y)$ in this neighborhood such that the straight-line segment PQ is in D . Because of our continuity assumptions we may apply the mean value theorem in Sec. 9.6. This yields

$$u(x + \Delta x, y + \Delta y) - u(x, y) = (\Delta x)u_x(M_1) + (\Delta y)u_y(M_1)$$

$$v(x + \Delta x, y + \Delta y) - v(x, y) = (\Delta x)v_x(M_2) + (\Delta y)v_y(M_2)$$

where M_1 and M_2 ($\neq M_1$ in general!) are suitable points on that segment. The first line is $\text{Re } \Delta f$ and the second is $\text{Im } \Delta f$, so that

$$\Delta f = (\Delta x)u_x(M_1) + (\Delta y)u_y(M_1) + i[(\Delta x)v_x(M_2) + (\Delta y)v_y(M_2)].$$

(b) $u_y = -v_x$ and $v_y = u_x$ by the Cauchy–Riemann equations, so that

$$\Delta f = (\Delta x)u_x(M_1) - (\Delta y)v_x(M_1) + i[(\Delta x)v_x(M_2) + (\Delta y)u_x(M_2)].$$

Also $\Delta z = \Delta x + i\Delta y$, so that we can write $\Delta x = \Delta z - i\Delta y$ in the first term and $\Delta y = (\Delta z - \Delta x)/i = -i(\Delta z - \Delta x)$ in the second term. This gives

$$\Delta f = (\Delta z - i\Delta y)u_x(M_1) + i(\Delta z - \Delta x)v_x(M_1) + i[(\Delta x)v_x(M_2) + (\Delta y)u_x(M_2)].$$

By performing the multiplications and reordering we obtain

$$\begin{aligned} \Delta f &= (\Delta z)u_x(M_1) - i\Delta y\{u_x(M_1) - u_x(M_2)\} \\ &\quad + i[(\Delta z)v_x(M_1) - \Delta x\{v_x(M_1) - v_x(M_2)\}]. \end{aligned}$$

Division by Δz now yields

$$(A) \quad \frac{\Delta f}{\Delta z} = u_x(M_1) + iv_x(M_1) - \frac{i\Delta y}{\Delta z} \{u_x(M_1) - u_x(M_2)\} - \frac{i\Delta x}{\Delta z} \{v_x(M_1) - v_x(M_2)\}.$$

(c) We finally let Δz approach zero and note that $|\Delta y/\Delta z| \leq 1$ and $|\Delta x/\Delta z| \leq 1$ in (A). Then $Q: (x + \Delta x, y + \Delta y)$ approaches $P: (x, y)$, so that M_1 and M_2 must approach P . Also, since the partial derivatives in (A) are assumed to be continuous, they approach their value at P . In particular, the differences in the braces $\{\cdot \cdot \cdot\}$ in (A) approach zero. Hence the limit of the right side of (A) exists and is independent of the path along which $\Delta z \rightarrow 0$. We see that this limit equals the right side of (4) in Sec. 13.4. This means that $f(z)$ is analytic at every point z in D , and the proof is complete. ■

Section 14.2, pages 653–654

GOURSAT'S PROOF OF CAUCHY'S INTEGRAL THEOREM Goursat proved Cauchy's integral theorem without assuming that $f'(z)$ is continuous, as follows.

We start with the case when C is the boundary of a triangle. We orient C counterclockwise. By joining the midpoints of the sides we subdivide the triangle into four congruent triangles (Fig. 563). Let $C_I, C_{II}, C_{III}, C_{IV}$ denote their boundaries. We claim that (see Fig. 563).

$$(1) \quad \oint_C f dz = \oint_{C_I} f dz + \oint_{C_{II}} f dz + \oint_{C_{III}} f dz + \oint_{C_{IV}} f dz.$$

Indeed, on the right we integrate along each of the three segments of subdivision in both possible directions (Fig. 563), so that the corresponding integrals cancel out in pairs, and the sum of the integrals on the right equals the integral on the left. We now pick an integral on the right that is biggest in absolute value and call its path C_1 . Then, by the triangle inequality (Sec. 13.2),

$$\left| \oint_C f dz \right| \leq \left| \oint_{C_I} f dz \right| + \left| \oint_{C_{II}} f dz \right| + \left| \oint_{C_{III}} f dz \right| + \left| \oint_{C_{IV}} f dz \right| \leq 4 \left| \oint_{C_1} f dz \right|.$$

We now subdivide the triangle bounded by C_1 as before and select a triangle of subdivision with boundary C_2 for which

$$\left| \oint_{C_1} f dz \right| \leq 4 \left| \oint_{C_2} f dz \right|. \quad \text{Then} \quad \left| \oint_C f dz \right| \leq 4^2 \left| \oint_{C_2} f dz \right|.$$

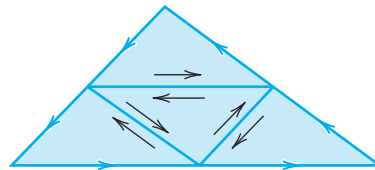


Fig. 563. Proof of Cauchy's integral theorem

Continuing in this fashion, we obtain a sequence of triangles $T_1, T_2, \dots$ with boundaries $C_1, C_2, \dots$ that are similar and such that T_n lies in T_m when $n > m$, and

$$(2) \quad \left| \oint_C f \, dz \right| \leq 4^n \left| \oint_{C_n} f \, dz \right|, \quad n = 1, 2, \dots$$

Let z_0 be the point that belongs to all these triangles. Since f is differentiable at $z = z_0$, the derivative $f'(z_0)$ exists. Let

$$(3) \quad h(z) = \frac{f(z) - f(z_0)}{z - z_0} - f'(z_0).$$

Solving this algebraically for $f(z)$ we have

$$f(z) = f(z_0) + (z - z_0)f'(z_0) + h(z)(z - z_0).$$

Integrating this over the boundary C_n of the triangle T_n gives

$$\oint_{C_n} f(z) \, dz = \oint_{C_n} f(z_0) \, dz + \oint_{C_n} (z - z_0)f'(z_0) \, dz + \oint_{C_n} h(z)(z - z_0) \, dz.$$

Since $f(z_0)$ and $f'(z_0)$ are constants and C_n is a closed path, the first two integrals on the right are zero, as follows from Cauchy's proof, which is applicable because the integrands do have continuous derivatives (0 and const, respectively). We thus have

$$\oint_{C_n} f(z) \, dz = \oint_{C_n} h(z)(z - z_0) \, dz.$$

Since $f'(z_0)$ is the limit of the difference quotient in (3), for given $\epsilon > 0$ we can find a $\delta > 0$ such that

$$(4) \quad |h(z)| < \epsilon \quad \text{when} \quad |z - z_0| < \delta.$$

We may now take n so large that the triangle T_n lies in the disk $|z - z_0| < \delta$. Let L_n be the length of C_n . Then $|z - z_0| < L_n$ for all z on C_n and z_0 in T_n . From this and (4) we have $|h(z)(z - z_0)| < \epsilon L_n$. The *ML*-inequality in Sec. 14.1 now gives

$$(5) \quad \left| \oint_{C_n} f(z) \, dz \right| = \left| \oint_{C_n} h(z)(z - z_0) \, dz \right| \leq \epsilon L_n \cdot L_n = \epsilon L_n^2.$$

Now denote the length of C by L . Then the path C_1 has the length $L_1 = L/2$, the path C_2 has the length $L_2 = L_1/2 = L/4$, etc., and C_n has the length $L_n = L/2^n$. Hence $L_n^2 = L^2/4^n$. From (2) and (5) we thus obtain

$$\left| \oint_C f \, dz \right| \leq 4^n \left| \oint_{C_n} f \, dz \right| \leq 4^n \epsilon L_n^2 = 4^n \epsilon \frac{L^2}{4^n} = \epsilon L^2.$$

By choosing $\epsilon (> 0)$ sufficiently small we can make the expression on the right as small as we please, while the expression on the left is the definite value of an integral. Consequently, this value must be zero, and the proof is complete.

The proof for the case in which C is the boundary of a polygon follows from the previous proof by subdividing the polygon into triangles (Fig. 564). The integral corresponding to each such triangle is zero. The sum of these integrals is equal to the integral over C , because we integrate along each segment of subdivision in both directions, the corresponding integrals cancel out in pairs, and we are left with the integral over C .

The case of a general simple closed path C can be reduced to the preceding one by inscribing in C a closed polygon P of chords, which approximates C “sufficiently accurately,” and it can be shown that there is a polygon P such that the integral over P differs from that over C by less than any preassigned positive real number $\tilde{\epsilon}$, no matter how small. The details of this proof are somewhat involved and can be found in Ref. [D6] listed in App. 1. ■

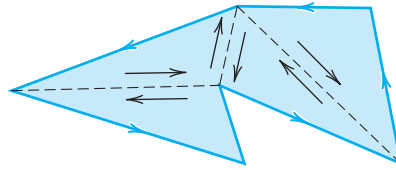


Fig. 564. Proof of Cauchy's integral theorem for a polygon

Section 15.1, page 674

PROOF OF THEOREM 4 Cauchy's Convergence Principle for Series

(a) In this proof we need two concepts and a theorem, which we list first.

1. A bounded sequence $s_1, s_2, \dots$ is a sequence whose terms all lie in a disk of (sufficiently large, finite) radius K with center at the origin; thus $|s_n| < K$ for all n .

2. A limit point a of a sequence $s_1, s_2, \dots$ is a point such that, given an $\epsilon > 0$, there are infinitely many terms satisfying $|s_n - a| < \epsilon$. (Note that this does *not* imply convergence, since there may still be infinitely many terms that do not lie within that circle of radius ϵ and center a .)

Example: $\frac{1}{4}, \frac{3}{4}, \frac{1}{8}, \frac{7}{8}, \frac{1}{16}, \frac{15}{16}, \dots$ has the limit points 0 and 1 and diverges.

3. A bounded sequence in the complex plane has at least one limit point. (Bolzano–Weierstrass theorem; proof below. Recall that “sequence” always means *infinite* sequence.)

(b) We now turn to the actual proof that $z_1 + z_2 + \dots$ converges if and only if, for every $\epsilon > 0$, we can find an N such that

$$(1) \quad |z_{n+1} + \dots + z_{n+p}| < \epsilon \quad \text{for every } n > N \text{ and } p = 1, 2, \dots$$

Here, by the definition of partial sums,

$$s_{n+p} - s_n = z_{n+1} + \dots + z_{n+p}.$$

Writing $n + p = r$, we see from this that (1) is equivalent to

$$(1^*) \quad |s_r - s_n| < \epsilon \quad \text{for all } r > N \text{ and } n > N.$$

Suppose that $s_1, s_2, \dots$ converges. Denote its limit by s . Then for a given $\epsilon > 0$ we can find an N such that

$$|s_n - s| < \frac{\epsilon}{2} \quad \text{for every } n > N.$$

Hence, if $r > N$ and $n > N$, then by the triangle inequality (Sec. 13.2),

$$|s_r - s_n| = |(s_r - s) - (s_n - s)| \leq |s_r - s| + |s_n - s| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon,$$

that is, (1*) holds.

(c) Conversely, assume that $s_1, s_2, \dots$ satisfies (1*). We first prove that then the sequence must be bounded. Indeed, choose a fixed ϵ and a fixed $n = n_0 > N$ in (1*). Then (1*) implies that all s_r with $r > N$ lie in the disk of radius ϵ and center s_{n_0} and only *finitely many terms* $s_1, \dots, s_N$ may not lie in this disk. Clearly, we can now find a circle so large that this disk and these finitely many terms all lie within this new circle. Hence the sequence is bounded. By the Bolzano–Weierstrass theorem, it has at least one limit point, call it s .

We now show that the sequence is convergent with the limit s . Let $\epsilon > 0$ be given. Then there is an N^* such that $|s_r - s_n| < \epsilon/2$ for all $r > N^*$ and $n > N^*$, by (1*). Also, by the definition of a limit point, $|s_n - s| < \epsilon/2$ for *infinitely many* n , so that we can find and fix an $n > N^*$ such that $|s_n - s| < \epsilon/2$. Together, for *every* $r > N^*$,

$$|s_r - s| = |(s_r - s_n) + (s_n - s)| \leq |s_r - s_n| + |s_n - s| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon,$$

that is, the sequence $s_1, s_2, \dots$ is convergent with the limit s . ■

THEOREM

Bolzano–Weierstrass Theorem³

A bounded infinite sequence $z_1, z_2, z_3, \dots$ in the complex plane has at least one limit point.

PROOF

It is obvious that we need both conditions: a finite sequence cannot have a limit point, and the sequence $1, 2, 3, \dots$, which is infinite but not bounded, has no limit point. To prove the theorem, consider a bounded infinite sequence $z_1, z_2, \dots$ and let K be such that $|z_n| < K$ for all n . If only finitely many values of the z_n are different, then, since the sequence is infinite, some number z must occur infinitely many times in the sequence, and, by definition, this number is a limit point of the sequence.

We may now turn to the case when the sequence contains infinitely many *different* terms. We draw a large square Q_0 that contains all z_n . We subdivide Q_0 into four congruent squares, which we number 1, 2, 3, 4. Clearly, at least one of these squares (each taken with its complete boundary) must contain infinitely many terms of the sequence. The square of this type with the lowest number (1, 2, 3, or 4) will be denoted by Q_1 . This is

³BERNARD BOLZANO (1781–1848), Austrian mathematician and professor of religious studies, was a pioneer in the study of point sets, the foundation of analysis, and mathematical logic.

For Weierstrass, see Sec. 15.5.

the first step. In the next step we subdivide Q_1 into four congruent squares and select a square Q_2 by the same rule, and so on. This yields an infinite sequence of squares $Q_0, Q_1, Q_2, \dots, Q_n, \dots$ with the property that the side of Q_n approaches zero as n approaches infinity, and Q_m contains all Q_n with $n > m$. It is not difficult to see that the number which belongs to all these squares,⁴ call it $z = a$, is a limit point of the sequence. In fact, given an $\epsilon > 0$, we can choose an N so large that the side of the square Q_N is less than ϵ and, since Q_N contains infinitely many z_n , we have $|z_n - a| < \epsilon$ for infinitely many n . This completes the proof. ■

Section 15.3, pages 688–689

PART (b) OF THE PROOF OF THEOREM 5

We have to show that

$$\begin{aligned} & \sum_{n=2}^{\infty} a_n \left[\frac{(z + \Delta z)^n - z^n}{\Delta z} - nz^{n-1} \right] \\ &= \sum_{n=2}^{\infty} a_n \Delta z [(z + \Delta z)^{n-2} + 2z(z + \Delta z)^{n-3} + \dots + (n-1)z^{n-2}], \end{aligned}$$

thus,

$$\begin{aligned} & \frac{(z + \Delta z)^n - z^n}{\Delta z} - nz^{n-1} \\ &= \Delta z [(z + \Delta z)^{n-2} + 2z(z + \Delta z)^{n-3} + \dots + (n-1)z^{n-2}]. \end{aligned}$$

If we set $z + \Delta z = b$ and $z = a$, thus $\Delta z = b - a$, this becomes simply

$$(7a) \quad \frac{b^n - a^n}{b - a} - na^{n-1} = (b - a)A_n \quad (n = 2, 3, \dots),$$

where A_n is the expression in the brackets on the right,

$$(7b) \quad A_n = b^{n-2} + 2ab^{n-3} + 3a^2b^{n-4} + \dots + (n-1)a^{n-2};$$

thus, $A_2 = 1, A_3 = b + 2a$, etc. We prove (7) by induction. When $n = 2$, then (7) holds, since then

$$\frac{b^2 - a^2}{b - a} - 2a = \frac{(b + a)(b - a)}{b - a} - 2a = b - a = (b - a)A_2.$$

Assuming that (7) holds for $n = k$, we show that it holds for $n = k + 1$. By adding and subtracting a term in the numerator and then dividing we first obtain

$$\frac{b^{k+1} - a^{k+1}}{b - a} = \frac{b^{k+1} - ba^k + ba^k - a^{k+1}}{b - a} = b \frac{b^k - a^k}{b - a} + a^k.$$

⁴The fact that such a unique number $z = a$ exists seems to be obvious, but it actually follows from an axiom of the real number system, the so-called *Cantor–Dedekind axiom*: see footnote 3 in App. A3.3.

By the induction hypothesis, the right side equals $b[(b - a)A_k + ka^{k-1}] + a^k$. Direct calculation shows that this is equal to

$$(b - a)\{bA_k + ka^{k-1}\} + aka^{k-1} + a^k.$$

From (7b) with $n = k$ we see that the expression in the braces $\{\cdot \cdot \cdot\}$ equals

$$b^{k-1} + 2ab^{k-2} + \cdots + (k - 1)ba^{k-2} + ka^{k-1} = A_{k+1}.$$

Hence our result is

$$\frac{b^{k+1} - a^{k+1}}{b - a} = (b - a)A_{k+1} + (k + 1)a^k.$$

Taking the last term to the left, we obtain (7) with $n = k + 1$. This proves (7) for any integer $n \geq 2$ and completes the proof. ■

Section 18.2, page 763

ANOTHER PROOF OF THEOREM 1 *without the use of a harmonic conjugate*

We show that if $w = u + iv = f(z)$ is analytic and maps a domain D conformally onto a domain D^* and $\Phi^*(u, v)$ is harmonic in D^* , then

$$(1) \quad \Phi(x, y) = \Phi^*(u(x, y), v(x, y))$$

is harmonic in D , that is, $\nabla^2 \Phi = 0$ in D . We make no use of a harmonic conjugate of Φ^* , but use straightforward differentiation. By the chain rule,

$$\Phi_x = \Phi_u^* u_x + \Phi_v^* v_x.$$

We apply the chain rule again, underscoring the terms that will drop out when we form $\nabla^2 \Phi$:

$$\begin{aligned} \Phi_{xx} &= \Phi_{uu}^* u_{xx} + (\Phi_{uv}^* u_x + \Phi_{vu}^* v_x) u_x \\ &\quad + \Phi_{vv}^* v_{xx} + (\Phi_{vu}^* u_x + \Phi_{vv}^* v_x) v_x. \end{aligned}$$

Φ_{yy} is the same with each x replaced by y . We form the sum $\nabla^2 \Phi$. In it, $\Phi_{vu}^* = \Phi_{uv}^*$ is multiplied by

$$u_x v_x + u_y v_y$$

which is 0 by the Cauchy–Riemann equations. Also $\nabla^2 u = 0$ and $\nabla^2 v = 0$. There remains

$$\nabla^2 \Phi = \Phi_{uu}^* (u_x^2 + u_y^2) + \Phi_{vv}^* (v_x^2 + v_y^2).$$

By the Cauchy–Riemann equations this becomes

$$\nabla^2 \Phi = (\Phi_{uu}^* + \Phi_{vv}^*) (u_x^2 + v_x^2)$$

and is 0 since Φ^* is harmonic. ■



APPENDIX 5

Tables

For Tables of Laplace Transforms see Secs. 6.8 and 6.9.

For Tables of Fourier Transforms see Sec. 11.10.

If you have a Computer Algebra System (CAS), you may not need the present tables, but you may still find them convenient from time to time.

Table A1 Bessel Functions

For more extensive tables see Ref. [GenRef1] in App. 1.

x	$J_0(x)$	$J_1(x)$	x	$J_0(x)$	$J_1(x)$	x	$J_0(x)$	$J_1(x)$
0.0	1.0000	0.0000	3.0	-0.2601	0.3391	6.0	0.1506	-0.2767
0.1	0.9975	0.0499	3.1	-0.2921	0.3009	6.1	0.1773	-0.2559
0.2	0.9900	0.0995	3.2	-0.3202	0.2613	6.2	0.2017	-0.2329
0.3	0.9776	0.1483	3.3	-0.3443	0.2207	6.3	0.2238	-0.2081
0.4	0.9604	0.1960	3.4	-0.3643	0.1792	6.4	0.2433	-0.1816
0.5	0.9385	0.2423	3.5	-0.3801	0.1374	6.5	0.2601	-0.1538
0.6	0.9120	0.2867	3.6	-0.3918	0.0955	6.6	0.2740	-0.1250
0.7	0.8812	0.3290	3.7	-0.3992	0.0538	6.7	0.2851	-0.0953
0.8	0.8463	0.3688	3.8	-0.4026	0.0128	6.8	0.2931	-0.0652
0.9	0.8075	0.4059	3.9	-0.4018	-0.0272	6.9	0.2981	-0.0349
1.0	0.7652	0.4401	4.0	-0.3971	-0.0660	7.0	0.3001	-0.0047
1.1	0.7196	0.4709	4.1	-0.3887	-0.1033	7.1	0.2991	0.0252
1.2	0.6711	0.4983	4.2	-0.3766	-0.1386	7.2	0.2951	0.0543
1.3	0.6201	0.5220	4.3	-0.3610	-0.1719	7.3	0.2882	0.0826
1.4	0.5669	0.5419	4.4	-0.3423	-0.2028	7.4	0.2786	0.1096
1.5	0.5118	0.5579	4.5	-0.3205	-0.2311	7.5	0.2663	0.1352
1.6	0.4554	0.5699	4.6	-0.2961	-0.2566	7.6	0.2516	0.1592
1.7	0.3980	0.5778	4.7	-0.2693	-0.2791	7.7	0.2346	0.1813
1.8	0.3400	0.5815	4.8	-0.2404	-0.2985	7.8	0.2154	0.2014
1.9	0.2818	0.5812	4.9	-0.2097	-0.3147	7.9	0.1944	0.2192
2.0	0.2239	0.5767	5.0	-0.1776	-0.3276	8.0	0.1717	0.2346
2.1	0.1666	0.5683	5.1	-0.1443	-0.3371	8.1	0.1475	0.2476
2.2	0.1104	0.5560	5.2	-0.1103	-0.3432	8.2	0.1222	0.2580
2.3	0.0555	0.5399	5.3	-0.0758	-0.3460	8.3	0.0960	0.2657
2.4	0.0025	0.5202	5.4	-0.0412	-0.3453	8.4	0.0692	0.2708
2.5	-0.0484	0.4971	5.5	-0.0068	-0.3414	8.5	0.0419	0.2731
2.6	-0.0968	0.4708	5.6	0.0270	-0.3343	8.6	0.0146	0.2728
2.7	-0.1424	0.4416	5.7	0.0599	-0.3241	8.7	-0.0125	0.2697
2.8	-0.1850	0.4097	5.8	0.0917	-0.3110	8.8	-0.0392	0.2641
2.9	-0.2243	0.3754	5.9	0.1220	-0.2951	8.9	-0.0653	0.2559

$J_0(x) = 0$ for $x = 2.40483, 5.52008, 8.65373, 11.7915, 14.9309, 18.0711, 21.2116, 24.3525, 27.4935, 30.6346$

$J_1(x) = 0$ for $x = 3.83171, 7.01559, 10.1735, 13.3237, 16.4706, 19.6159, 22.7601, 25.9037, 29.0468, 32.1897$

Table A1 (continued)

x	$Y_0(x)$	$Y_1(x)$	x	$Y_0(x)$	$Y_1(x)$	x	$Y_0(x)$	$Y_1(x)$
0.0	$(-\infty)$	$(-\infty)$	2.5	0.498	0.146	5.0	-0.309	0.148
0.5	-0.445	-1.471	3.0	0.377	0.325	5.5	-0.339	-0.024
1.0	0.088	-0.781	3.5	0.189	0.410	6.0	-0.288	-0.175
1.5	0.382	-0.412	4.0	-0.017	0.398	6.5	-0.173	-0.274
2.0	0.510	-0.107	4.5	-0.195	0.301	7.0	-0.026	-0.303

Table A2 Gamma Function [see (24) in App. A3.1]

α	$\Gamma(\alpha)$	α	$\Gamma(\alpha)$	α	$\Gamma(\alpha)$	α	$\Gamma(\alpha)$	α	$\Gamma(\alpha)$
1.00	1.000 000	1.20	0.918 169	1.40	0.887 264	1.60	0.893 515	1.80	0.931 384
1.02	0.988 844	1.22	0.913 106	1.42	0.886 356	1.62	0.895 924	1.82	0.936 845
1.04	0.978 438	1.24	0.908 521	1.44	0.885 805	1.64	0.898 642	1.84	0.942 612
1.06	0.968 744	1.26	0.904 397	1.46	0.885 604	1.66	0.901 668	1.86	0.948 687
1.08	0.959 725	1.28	0.900 718	1.48	0.885 747	1.68	0.905 001	1.88	0.955 071
1.10	0.951 351	1.30	0.897 471	1.50	0.886 227	1.70	0.908 639	1.90	0.961 766
1.12	0.943 590	1.32	0.894 640	1.52	0.887 039	1.72	0.912 581	1.92	0.968 774
1.14	0.936 416	1.34	0.892 216	1.54	0.888 178	1.74	0.916 826	1.94	0.976 099
1.16	0.929 803	1.36	0.890 185	1.56	0.889 639	1.76	0.921 375	1.96	0.983 743
1.18	0.923 728	1.38	0.888 537	1.58	0.891 420	1.78	0.926 227	1.98	0.991 708
1.20	0.918 169	1.40	0.887 264	1.60	0.893 515	1.80	0.931 384	2.00	1.000 000

Table A3 Factorial Function and Its Logarithm with Base 10

n	$n!$	$\log(n!)$	n	$n!$	$\log(n!)$	n	$n!$	$\log(n!)$
1	1	0.000 000	6	720	2.857 332	11	39 916 800	7.601 156
2	2	0.301 030	7	5 040	3.702 431	12	479 001 600	8.680 337
3	6	0.778 151	8	40 320	4.605 521	13	6 227 020 800	9.794 280
4	24	1.380 211	9	362 880	5.559 763	14	87 178 291 200	10.940 408
5	120	2.079 181	10	3 628 800	6.559 763	15	1 307 674 368 000	12.116 500

Table A4 Error Function, Sine and Cosine Integrals [see (35), (40), (42) in App. A3.1]

x	$\operatorname{erf} x$	$\operatorname{Si}(x)$	$\operatorname{ci}(x)$	x	$\operatorname{erf} x$	$\operatorname{Si}(x)$	$\operatorname{ci}(x)$
0.0	0.0000	0.0000	∞	2.0	0.9953	1.6054	-0.4230
0.2	0.2227	0.1996	1.0422	2.2	0.9981	1.6876	-0.3751
0.4	0.4284	0.3965	0.3788	2.4	0.9993	1.7525	-0.3173
0.6	0.6039	0.5881	0.0223	2.6	0.9998	1.8004	-0.2533
0.8	0.7421	0.7721	-0.1983	2.8	0.9999	1.8321	-0.1865
1.0	0.8427	0.9461	-0.3374	3.0	1.0000	1.8487	-0.1196
1.2	0.9103	1.1080	-0.4205	3.2	1.0000	1.8514	-0.0553
1.4	0.9523	1.2562	-0.4620	3.4	1.0000	1.8419	0.0045
1.6	0.9763	1.3892	-0.4717	3.6	1.0000	1.8219	0.0580
1.8	0.9891	1.5058	-0.4568	3.8	1.0000	1.7934	0.1038
2.0	0.9953	1.6054	-0.4230	4.0	1.0000	1.7582	0.1410

Table A5 Binomial DistributionProbability function $f(x)$ [see (2), Sec. 24.7] and distribution function $F(x)$

n	x	$p = 0.1$		$p = 0.2$		$p = 0.3$		$p = 0.4$		$p = 0.5$	
		$f(x)$	$F(x)$	$f(x)$	$F(x)$	$f(x)$	$F(x)$	$f(x)$	$F(x)$	$f(x)$	$F(x)$
1	0	0.9000	0.9000	0.8000	0.8000	0.7000	0.7000	0.6000	0.6000	0.5000	0.5000
	1	0.1000	1.0000	0.2000	1.0000	0.3000	1.0000	0.4000	1.0000	0.5000	1.0000
2	0	0.8100	0.8100	0.6400	0.6400	0.4900	0.4900	0.3600	0.3600	0.2500	0.2500
	1	0.1800	0.9900	0.3200	0.9600	0.4200	0.9100	0.4800	0.8400	0.5000	0.7500
	2	0.0100	1.0000	0.0400	1.0000	0.0900	1.0000	0.1600	1.0000	0.2500	1.0000
3	0	0.7290	0.7290	0.5120	0.5120	0.3430	0.3430	0.2160	0.2160	0.1250	0.1250
	1	0.2430	0.9720	0.3840	0.8960	0.4410	0.7840	0.4320	0.6480	0.3750	0.5000
	2	0.0270	0.9990	0.0960	0.9920	0.1890	0.9730	0.2880	0.9360	0.3750	0.8750
	3	0.0010	1.0000	0.0080	1.0000	0.0270	1.0000	0.0640	1.0000	0.1250	1.0000
4	0	0.6561	0.6561	0.4096	0.4096	0.2401	0.2401	0.1296	0.1296	0.0625	0.0625
	1	0.2916	0.9477	0.4096	0.8192	0.4116	0.6517	0.3456	0.4752	0.2500	0.3125
	2	0.0486	0.9963	0.1536	0.9728	0.2646	0.9163	0.3456	0.8208	0.3750	0.6875
	3	0.0036	0.9999	0.0256	0.9984	0.0756	0.9919	0.1536	0.9744	0.2500	0.9375
	4	0.0001	1.0000	0.0016	1.0000	0.0081	1.0000	0.0256	1.0000	0.0625	1.0000
5	0	0.5905	0.5905	0.3277	0.3277	0.1681	0.1681	0.0778	0.0778	0.0313	0.0313
	1	0.3281	0.9185	0.4096	0.7373	0.3602	0.5282	0.2592	0.3370	0.1563	0.1875
	2	0.0729	0.9914	0.2048	0.9421	0.3087	0.8369	0.3456	0.6826	0.3125	0.5000
	3	0.0081	0.9995	0.0512	0.9933	0.1323	0.9692	0.2304	0.9130	0.3125	0.8125
	4	0.0005	1.0000	0.0064	0.9997	0.0284	0.9976	0.0768	0.9898	0.1563	0.9688
	5	0.0000	1.0000	0.0003	1.0000	0.0024	1.0000	0.0102	1.0000	0.0313	1.0000
6	0	0.5314	0.5314	0.2621	0.2621	0.1176	0.1176	0.0467	0.0467	0.0156	0.0156
	1	0.3543	0.8857	0.3932	0.6554	0.3025	0.4202	0.1866	0.2333	0.0938	0.1094
	2	0.0984	0.9841	0.2458	0.9011	0.3241	0.7443	0.3110	0.5443	0.2344	0.3438
	3	0.0146	0.9987	0.0819	0.9830	0.1852	0.9295	0.2765	0.8208	0.3125	0.6563
	4	0.0012	0.9999	0.0154	0.9984	0.0595	0.9891	0.1382	0.9590	0.2344	0.8906
	5	0.0001	1.0000	0.0015	0.9999	0.0102	0.9993	0.0369	0.9959	0.0938	0.9844
	6	0.0000	1.0000	0.0001	1.0000	0.0007	1.0000	0.0041	1.0000	0.0156	1.0000
7	0	0.4783	0.4783	0.2097	0.2097	0.0824	0.0824	0.0280	0.0280	0.0078	0.0078
	1	0.3720	0.8503	0.3670	0.5767	0.2471	0.3294	0.1306	0.1586	0.0547	0.0625
	2	0.1240	0.9743	0.2753	0.8520	0.3177	0.6471	0.2613	0.4199	0.1641	0.2266
	3	0.0230	0.9973	0.1147	0.9667	0.2269	0.8740	0.2903	0.7102	0.2734	0.5000
	4	0.0026	0.9998	0.0287	0.9953	0.0972	0.9712	0.1935	0.9037	0.2734	0.7734
	5	0.0002	1.0000	0.0043	0.9996	0.0250	0.9962	0.0774	0.9812	0.1641	0.9375
	6	0.0000	1.0000	0.0004	1.0000	0.0036	0.9998	0.0172	0.9984	0.0547	0.9922
	7	0.0000	1.0000	0.0000	1.0000	0.0002	1.0000	0.0016	1.0000	0.0078	1.0000
8	0	0.4305	0.4305	0.1678	0.1678	0.0576	0.0576	0.0168	0.0168	0.0039	0.0039
	1	0.3826	0.8131	0.3355	0.5033	0.1977	0.2553	0.0896	0.1064	0.0313	0.0352
	2	0.1488	0.9619	0.2936	0.7969	0.2965	0.5518	0.2090	0.3154	0.1094	0.1445
	3	0.0331	0.9950	0.1468	0.9437	0.2541	0.8059	0.2787	0.5941	0.2188	0.3633
	4	0.0046	0.9996	0.0459	0.9896	0.1361	0.9420	0.2322	0.8263	0.2734	0.6367
	5	0.0004	1.0000	0.0092	0.9988	0.0467	0.9887	0.1239	0.9502	0.2188	0.8555
	6	0.0000	1.0000	0.0011	0.9999	0.0100	0.9987	0.0413	0.9915	0.1094	0.9648
	7	0.0000	1.0000	0.0001	1.0000	0.0012	0.9999	0.0079	0.9993	0.0313	0.9961
	8	0.0000	1.0000	0.0000	1.0000	0.0001	1.0000	0.0007	1.0000	0.0039	1.0000

Table A6 Poisson Distribution

Probability function $f(x)$ [see (5), Sec. 24.7] and distribution function $F(x)$

x	$\mu = 0.1$		$\mu = 0.2$		$\mu = 0.3$		$\mu = 0.4$		$\mu = 0.5$	
	$f(x)$	$F(x)$	$f(x)$	$F(x)$	$f(x)$	$F(x)$	$f(x)$	$F(x)$	$f(x)$	$F(x)$
0	0. 9048	0.9048	0. 8187	0.8187	0. 7408	0.7408	0. 6703	0.6703	0. 6065	0.6065
1	0905	0.9953	1637	0.9825	2222	0.9631	2681	0.9384	3033	0.9098
2	0045	0.9998	0164	0.9989	0333	0.9964	0536	0.9921	0758	0.9856
3	0002	1.0000	0011	0.9999	0033	0.9997	0072	0.9992	0126	0.9982
4	0000	1.0000	0001	1.0000	0003	1.0000	0007	0.9999	0016	0.9998
5							0001	1.0000	0002	1.0000

[illegible][illegible]

Table A7 Normal DistributionValues of the distribution function $\Phi(z)$ [see (3), Sec. 24.8]. $\Phi(-z) = 1 - \Phi(z)$

z	$\Phi(z)$	z	$\Phi(z)$	z	$\Phi(z)$	z	$\Phi(z)$	z	$\Phi(z)$	z	$\Phi(z)$
	0.		0.		0.		0.		0.		0.
0.01	5040	0.51	6950	1.01	8438	1.51	9345	2.01	9778	2.51	9940
0.02	5080	0.52	6985	1.02	8461	1.52	9357	2.02	9783	2.52	9941
0.03	5120	0.53	7019	1.03	8485	1.53	9370	2.03	9788	2.53	9943
0.04	5160	0.54	7054	1.04	8508	1.54	9382	2.04	9793	2.54	9945
0.05	5199	0.55	7088	1.05	8531	1.55	9394	2.05	9798	2.55	9946
0.06	5239	0.56	7123	1.06	8554	1.56	9406	2.06	9803	2.56	9948
0.07	5279	0.57	7157	1.07	8577	1.57	9418	2.07	9808	2.57	9949
0.08	5319	0.58	7190	1.08	8599	1.58	9429	2.08	9812	2.58	9951
0.09	5359	0.59	7224	1.09	8621	1.59	9441	2.09	9817	2.59	9952
0.10	5398	0.60	7257	1.10	8643	1.60	9452	2.10	9821	2.60	9953
0.11	5438	0.61	7291	1.11	8665	1.61	9463	2.11	9826	2.61	9955
0.12	5478	0.62	7324	1.12	8686	1.62	9474	2.12	9830	2.62	9956
0.13	5517	0.63	7357	1.13	8708	1.63	9484	2.13	9834	2.63	9957
0.14	5557	0.64	7389	1.14	8729	1.64	9495	2.14	9838	2.64	9959
0.15	5596	0.65	7422	1.15	8749	1.65	9505	2.15	9842	2.65	9960
0.16	5636	0.66	7454	1.16	8770	1.66	9515	2.16	9846	2.66	9961
0.17	5675	0.67	7486	1.17	8790	1.67	9525	2.17	9850	2.67	9962
0.18	5714	0.68	7517	1.18	8810	1.68	9535	2.18	9854	2.68	9963
0.19	5753	0.69	7549	1.19	8830	1.69	9545	2.19	9857	2.69	9964
0.20	5793	0.70	7580	1.20	8849	1.70	9554	2.20	9861	2.70	9965
0.21	5832	0.71	7611	1.21	8869	1.71	9564	2.21	9864	2.71	9966
0.22	5871	0.72	7642	1.22	8888	1.72	9573	2.22	9868	2.72	9967
0.23	5910	0.73	7673	1.23	8907	1.73	9582	2.23	9871	2.73	9968
0.24	5948	0.74	7704	1.24	8925	1.74	9591	2.24	9875	2.74	9969
0.25	5987	0.75	7734	1.25	8944	1.75	9599	2.25	9878	2.75	9970
0.26	6026	0.76	7764	1.26	8962	1.76	9608	2.26	9881	2.76	9971
0.27	6064	0.77	7794	1.27	8980	1.77	9616	2.27	9884	2.77	9972
0.28	6103	0.78	7823	1.28	8997	1.78	9625	2.28	9887	2.78	9973
0.29	6141	0.79	7852	1.29	9015	1.79	9633	2.29	9890	2.79	9974
0.30	6179	0.80	7881	1.30	9032	1.80	9641	2.30	9893	2.80	9974
0.31	6217	0.81	7910	1.31	9049	1.81	9649	2.31	9896	2.81	9975
0.32	6255	0.82	7939	1.32	9066	1.82	9656	2.32	9898	2.82	9976
0.33	6293	0.83	7967	1.33	9082	1.83	9664	2.33	9901	2.83	9977
0.34	6331	0.84	7995	1.34	9099	1.84	9671	2.34	9904	2.84	9977
0.35	6368	0.85	8023	1.35	9115	1.85	9678	2.35	9906	2.85	9978
0.36	6406	0.86	8051	1.36	9131	1.86	9686	2.36	9909	2.86	9979
0.37	6443	0.87	8078	1.37	9147	1.87	9693	2.37	9911	2.87	9979
0.38	6480	0.88	8106	1.38	9162	1.88	9699	2.38	9913	2.88	9980
0.39	6517	0.89	8133	1.39	9177	1.89	9706	2.39	9916	2.89	9981
0.40	6554	0.90	8159	1.40	9192	1.90	9713	2.40	9918	2.90	9981
0.41	6591	0.91	8186	1.41	9207	1.91	9719	2.41	9920	2.91	9982
0.42	6628	0.92	8212	1.42	9222	1.92	9726	2.42	9922	2.92	9982
0.43	6664	0.93	8238	1.43	9236	1.93	9732	2.43	9925	2.93	9983
0.44	6700	0.94	8264	1.44	9251	1.94	9738	2.44	9927	2.94	9984
0.45	6736	0.95	8289	1.45	9265	1.95	9744	2.45	9929	2.95	9984
0.46	6772	0.96	8315	1.46	9279	1.96	9750	2.46	9931	2.96	9985
0.47	6808	0.97	8340	1.47	9292	1.97	9756	2.47	9932	2.97	9985
0.48	6844	0.98	8365	1.48	9306	1.98	9761	2.48	9934	2.98	9986
0.49	6879	0.99	8389	1.49	9319	1.99	9767	2.49	9936	2.99	9986
0.50	6915	1.00	8413	1.50	9332	2.00	9772	2.50	9938	3.00	9987

Table A8 Normal Distribution

Values of z for given values of $\Phi(z)$ [see (3), Sec. 24.8] and $D(z) = \Phi(z) - \Phi(-z)$
 Example: $z = 0.279$ if $\Phi(z) = 61\%$; $z = 0.860$ if $D(z) = 61\%$.

%	$z(\Phi)$	$z(D)$	%	$z(\Phi)$	$z(D)$	%	$z(\Phi)$	$z(D)$
1	-2.326	0.013	41	-0.228	0.539	81	0.878	1.311
2	-2.054	0.025	42	-0.202	0.553	82	0.915	1.341
3	-1.881	0.038	43	-0.176	0.568	83	0.954	1.372
4	-1.751	0.050	44	-0.151	0.583	84	0.994	1.405
5	-1.645	0.063	45	-0.126	0.598	85	1.036	1.440
6	-1.555	0.075	46	-0.100	0.613	86	1.080	1.476
7	-1.476	0.088	47	-0.075	0.628	87	1.126	1.514
8	-1.405	0.100	48	-0.050	0.643	88	1.175	1.555
9	-1.341	0.113	49	-0.025	0.659	89	1.227	1.598
10	-1.282	0.126	50	0.000	0.674	90	1.282	1.645
11	-1.227	0.138	51	0.025	0.690	91	1.341	1.695
12	-1.175	0.151	52	0.050	0.706	92	1.405	1.751
13	-1.126	0.164	53	0.075	0.722	93	1.476	1.812
14	-1.080	0.176	54	0.100	0.739	94	1.555	1.881
15	-1.036	0.189	55	0.126	0.755	95	1.645	1.960
16	-0.994	0.202	56	0.151	0.772	96	1.751	2.054
17	-0.954	0.215	57	0.176	0.789	97	1.881	2.170
18	-0.915	0.228	58	0.202	0.806	97.5	1.960	2.241
19	-0.878	0.240	59	0.228	0.824	98	2.054	2.326
20	-0.842	0.253	60	0.253	0.842	99	2.326	2.576
21	-0.806	0.266	61	0.279	0.860	99.1	2.366	2.612
22	-0.772	0.279	62	0.305	0.878	99.2	2.409	2.652
23	-0.739	0.292	63	0.332	0.896	99.3	2.457	2.697
24	-0.706	0.305	64	0.358	0.915	99.4	2.512	2.748
25	-0.674	0.319	65	0.385	0.935	99.5	2.576	2.807
26	-0.643	0.332	66	0.412	0.954	99.6	2.652	2.878
27	-0.613	0.345	67	0.440	0.974	99.7	2.748	2.968
28	-0.583	0.358	68	0.468	0.994	99.8	2.878	3.090
29	-0.553	0.372	69	0.496	1.015	99.9	3.090	3.291
30	-0.524	0.385	70	0.524	1.036			
31	-0.496	0.399	71	0.553	1.058	99.91	3.121	3.320
32	-0.468	0.412	72	0.583	1.080	99.92	3.156	3.353
33	-0.440	0.426	73	0.613	1.103	99.93	3.195	3.390
34	-0.412	0.440	74	0.643	1.126	99.94	3.239	3.432
35	-0.385	0.454	75	0.674	1.150	99.95	3.291	3.481
36	-0.358	0.468	76	0.706	1.175	99.96	3.353	3.540
37	-0.332	0.482	77	0.739	1.200	99.97	3.432	3.615
38	-0.305	0.496	78	0.772	1.227	99.98	3.540	3.719
39	-0.279	0.510	79	0.806	1.254	99.99	3.719	3.891
40	-0.253	0.524	80	0.842	1.282			

Table A9 t-Distribution

Values of z for given values of the distribution function $F(z)$ (see (8) in Sec. 25.3).
 Example: For 9 degrees of freedom, $z = 1.83$ when $F(z) = 0.95$.

$F(z)$	Number of Degrees of Freedom									
	1	2	3	4	5	6	7	8	9	10
0.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.6	0.32	0.29	0.28	0.27	0.27	0.26	0.26	0.26	0.26	0.26
0.7	0.73	0.62	0.58	0.57	0.56	0.55	0.55	0.55	0.54	0.54
0.8	1.38	1.06	0.98	0.94	0.92	0.91	0.90	0.89	0.88	0.88
0.9	3.08	1.89	1.64	1.53	1.48	1.44	1.41	1.40	1.38	1.37
0.95	6.31	2.92	2.35	2.13	2.02	1.94	1.89	1.86	1.83	1.81
0.975	12.7	4.30	3.18	2.78	2.57	2.45	2.36	2.31	2.26	2.23
0.99	31.8	6.96	4.54	3.75	3.36	3.14	3.00	2.90	2.82	2.76
0.995	63.7	9.92	5.84	4.60	4.03	3.71	3.50	3.36	3.25	3.17
0.999	318.3	22.3	10.2	7.17	5.89	5.21	4.79	4.50	4.30	4.14

$F(z)$	Number of Degrees of Freedom									
	11	12	13	14	15	16	17	18	19	20
0.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.6	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.26
0.7	0.54	0.54	0.54	0.54	0.54	0.54	0.53	0.53	0.53	0.53
0.8	0.88	0.87	0.87	0.87	0.87	0.86	0.86	0.86	0.86	0.86
0.9	1.36	1.36	1.35	1.35	1.34	1.34	1.33	1.33	1.33	1.33
0.95	1.80	1.78	1.77	1.76	1.75	1.75	1.74	1.73	1.73	1.72
0.975	2.20	2.18	2.16	2.14	2.13	2.12	2.11	2.10	2.09	2.09
0.99	2.72	2.68	2.65	2.62	2.60	2.58	2.57	2.55	2.54	2.53
0.995	3.11	3.05	3.01	2.98	2.95	2.92	2.90	2.88	2.86	2.85
0.999	4.02	3.93	3.85	3.79	3.73	3.69	3.65	3.61	3.58	3.55

$F(z)$	Number of Degrees of Freedom									
	22	24	26	28	30	40	50	100	200	∞
0.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.6	0.26	0.26	0.26	0.26	0.26	0.26	0.25	0.25	0.25	0.25
0.7	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.52
0.8	0.86	0.86	0.86	0.85	0.85	0.85	0.85	0.85	0.84	0.84
0.9	1.32	1.32	1.31	1.31	1.31	1.30	1.30	1.29	1.29	1.28
0.95	1.72	1.71	1.71	1.70	1.70	1.68	1.68	1.66	1.65	1.65
0.975	2.07	2.06	2.06	2.05	2.04	2.02	2.01	1.98	1.97	1.96
0.99	2.51	2.49	2.48	2.47	2.46	2.42	2.40	2.36	2.35	2.33
0.995	2.82	2.80	2.78	2.76	2.75	2.70	2.68	2.63	2.60	2.58
0.999	3.50	3.47	3.43	3.41	3.39	3.31	3.26	3.17	3.13	3.09

Table A10 Chi-square Distribution

Values of x for given values of the distribution function $F(z)$ (see Sec. 25.3 before (17)).
 Example: For 3 degrees of freedom, $z = 11.34$ when $F(z) = 0.99$.

$F(z)$	Number of Degrees of Freedom									
	1	2	3	4	5	6	7	8	9	10
0.005	0.00	0.01	0.07	0.21	0.41	0.68	0.99	1.34	1.73	2.16
0.01	0.00	0.02	0.11	0.30	0.55	0.87	1.24	1.65	2.09	2.56
0.025	0.00	0.05	0.22	0.48	0.83	1.24	1.69	2.18	2.70	3.25
0.05	0.00	0.10	0.35	0.71	1.15	1.64	2.17	2.73	3.33	3.94
0.95	3.84	5.99	7.81	9.49	11.07	12.59	14.07	15.51	16.92	18.31
0.975	5.02	7.38	9.35	11.14	12.83	14.45	16.01	17.53	19.02	20.48
0.99	6.63	9.21	11.34	13.28	15.09	16.81	18.48	20.09	21.67	23.21
0.995	7.88	10.60	12.84	14.86	16.75	18.55	20.28	21.95	23.59	25.19

$F(z)$	Number of Degrees of Freedom									
	11	12	13	14	15	16	17	18	19	20
0.005	2.60	3.07	3.57	4.07	4.60	5.14	5.70	6.26	6.84	7.43
0.01	3.05	3.57	4.11	4.66	5.23	5.81	6.41	7.01	7.63	8.26
0.025	3.82	4.40	5.01	5.63	6.26	6.91	7.56	8.23	8.91	9.59
0.05	4.57	5.23	5.89	6.57	7.26	7.96	8.67	9.39	10.12	10.85
0.95	19.68	21.03	22.36	23.68	25.00	26.30	27.59	28.87	30.14	31.41
0.975	21.92	23.34	24.74	26.12	27.49	28.85	30.19	31.53	32.85	34.17
0.99	24.72	26.22	27.69	29.14	30.58	32.00	33.41	34.81	36.19	37.57
0.995	26.76	28.30	29.82	31.32	32.80	34.27	35.72	37.16	38.58	40.00

$F(z)$	Number of Degrees of Freedom									
	21	22	23	24	25	26	27	28	29	30
0.005	8.0	8.6	9.3	9.9	10.5	11.2	11.8	12.5	13.1	13.8
0.01	8.9	9.5	10.2	10.9	11.5	12.2	12.9	13.6	14.3	15.0
0.025	10.3	11.0	11.7	12.4	13.1	13.8	14.6	15.3	16.0	16.8
0.05	11.6	12.3	13.1	13.8	14.6	15.4	16.2	16.9	17.7	18.5
0.95	32.7	33.9	35.2	36.4	37.7	38.9	40.1	41.3	42.6	43.8
0.975	35.5	36.8	38.1	39.4	40.6	41.9	43.2	44.5	45.7	47.0
0.99	38.9	40.3	41.6	43.0	44.3	45.6	47.0	48.3	49.6	50.9
0.995	41.4	42.8	44.2	45.6	46.9	48.3	49.6	51.0	52.3	53.7

$F(z)$	Number of Degrees of Freedom							
	40	50	60	70	80	90	100	> 100 (Approximation)
0.005	20.7	28.0	35.5	43.3	51.2	59.2	67.3	$\frac{1}{2}(h - 2.58)^2$
0.01	22.2	29.7	37.5	45.4	53.5	61.8	70.1	$\frac{1}{2}(h - 2.33)^2$
0.025	24.4	32.4	40.5	48.8	57.2	65.6	74.2	$\frac{1}{2}(h - 1.96)^2$
0.05	26.5	34.8	43.2	51.7	60.4	69.1	77.9	$\frac{1}{2}(h - 1.64)^2$
0.95	55.8	67.5	79.1	90.5	101.9	113.1	124.3	$\frac{1}{2}(h + 1.64)^2$
0.975	59.3	71.4	83.3	95.0	106.6	118.1	129.6	$\frac{1}{2}(h + 1.96)^2$
0.99	63.7	76.2	88.4	100.4	112.3	124.1	135.8	$\frac{1}{2}(h + 2.33)^2$
0.995	66.8	79.5	92.0	104.2	116.3	128.3	140.2	$\frac{1}{2}(h + 2.58)^2$

In the last column, $h = \sqrt{2m - 1}$, where m is the number of degrees of freedom.

Table A11 F-Distribution with (m, n) Degrees of Freedom

Values of z for which the distribution function $F(z)$ [see (13), Sec. 25.4] has the value **0.95**
 Example: For $(7, 4)$ d.f., $z = 6.09$ if $F(z) = 0.95$.

n	$m = 1$	$m = 2$	$m = 3$	$m = 4$	$m = 5$	$m = 6$	$m = 7$	$m = 8$	$m = 9$
1	161	200	216	225	230	234	237	239	241
2	18.5	19.0	19.2	19.2	19.3	19.3	19.4	19.4	19.4
3	10.1	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21
32	4.15	3.29	2.90	2.67	2.51	2.40	2.31	2.24	2.19
34	4.13	3.28	2.88	2.65	2.49	2.38	2.29	2.23	2.17
36	4.11	3.26	2.87	2.63	2.48	2.36	2.28	2.21	2.15
38	4.10	3.24	2.85	2.62	2.46	2.35	2.26	2.19	2.14
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12
50	4.03	3.18	2.79	2.56	2.40	2.29	2.20	2.13	2.07
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04
70	3.98	3.13	2.74	2.50	2.35	2.23	2.14	2.07	2.02
80	3.96	3.11	2.72	2.49	2.33	2.21	2.13	2.06	2.00
90	3.95	3.10	2.71	2.47	2.32	2.20	2.11	2.04	1.99
100	3.94	3.09	2.70	2.46	2.31	2.19	2.10	2.03	1.97
150	3.90	3.06	2.66	2.43	2.27	2.16	2.07	2.00	1.94
200	3.89	3.04	2.65	2.42	2.26	2.14	2.06	1.98	1.93
1000	3.85	3.00	2.61	2.38	2.22	2.11	2.02	1.95	1.89
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88

Table A11 F-Distribution with (m, n) Degrees of Freedom (continued)Values of z for which the distribution function $F(z)$ [see (13), Sec. 25.4] has the value **0.95**

n	$m = 10$	$m = 15$	$m = 20$	$m = 30$	$m = 40$	$m = 50$	$m = 100$	∞
1	242	246	248	250	251	252	253	254
2	19.4	19.4	19.4	19.5	19.5	19.5	19.5	19.5
3	8.79	8.70	8.66	8.62	8.59	8.58	8.55	8.53
4	5.96	5.86	5.80	5.75	5.72	5.70	5.66	5.63
5	4.74	4.62	4.56	4.50	4.46	4.44	4.41	4.37
6	4.06	3.94	3.87	3.81	3.77	3.75	3.71	3.67
7	3.64	3.51	3.44	3.38	3.34	3.32	3.27	3.23
8	3.35	3.22	3.15	3.08	3.04	3.02	2.97	2.93
9	3.14	3.01	2.94	2.86	2.83	2.80	2.76	2.71
10	2.98	2.85	2.77	2.70	2.66	2.64	2.59	2.54
11	2.85	2.72	2.65	2.57	2.53	2.51	2.46	2.40
12	2.75	2.62	2.54	2.47	2.43	2.40	2.35	2.30
13	2.67	2.53	2.46	2.38	2.34	2.31	2.26	2.21
14	2.60	2.46	2.39	2.31	2.27	2.24	2.19	2.13
15	2.54	2.40	2.33	2.25	2.20	2.18	2.12	2.07
16	2.49	2.35	2.28	2.19	2.15	2.12	2.07	2.01
17	2.45	2.31	2.23	2.15	2.10	2.08	2.02	1.96
18	2.41	2.27	2.19	2.11	2.06	2.04	1.98	1.92
19	2.38	2.23	2.16	2.07	2.03	2.00	1.94	1.88
20	2.35	2.20	2.12	2.04	1.99	1.97	1.91	1.84
22	2.30	2.15	2.07	1.98	1.94	1.91	1.85	1.78
24	2.25	2.11	2.03	1.94	1.89	1.86	1.80	1.73
26	2.22	2.07	1.99	1.90	1.85	1.82	1.76	1.69
28	2.19	2.04	1.96	1.87	1.82	1.79	1.73	1.65
30	2.16	2.01	1.93	1.84	1.79	1.76	1.70	1.62
32	2.14	1.99	1.91	1.82	1.77	1.74	1.67	1.59
34	2.12	1.97	1.89	1.80	1.75	1.71	1.65	1.57
36	2.11	1.95	1.87	1.78	1.73	1.69	1.62	1.55
38	2.09	1.94	1.85	1.76	1.71	1.68	1.61	1.53
40	2.08	1.92	1.84	1.74	1.69	1.66	1.59	1.51
50	2.03	1.87	1.78	1.69	1.63	1.60	1.52	1.44
60	1.99	1.84	1.75	1.65	1.59	1.56	1.48	1.39
70	1.97	1.81	1.72	1.62	1.57	1.53	1.45	1.35
80	1.95	1.79	1.70	1.60	1.54	1.51	1.43	1.32
90	1.94	1.78	1.69	1.59	1.53	1.49	1.41	1.30
100	1.93	1.77	1.68	1.57	1.52	1.48	1.39	1.28
150	1.89	1.73	1.64	1.54	1.48	1.44	1.34	1.22
200	1.88	1.72	1.62	1.52	1.46	1.41	1.32	1.19
1000	1.84	1.68	1.58	1.47	1.41	1.36	1.26	1.08
∞	1.83	1.67	1.57	1.46	1.39	1.35	1.24	1.00

Table A11 F-Distribution with (m, n) Degrees of Freedom (continued)Values of z for which the distribution function $F(z)$ [see (13), Sec. 25.4] has the value **0.99**

n	$m = 1$	$m = 2$	$m = 3$	$m = 4$	$m = 5$	$m = 6$	$m = 7$	$m = 8$	$m = 9$
1	4052	4999	5403	5625	5764	5859	5928	5981	6022
2	98.5	99.0	99.2	99.2	99.3	99.3	99.4	99.4	99.4
3	34.1	30.8	29.5	28.7	28.2	27.9	27.7	27.5	27.3
4	21.2	18.0	16.7	16.0	15.5	15.2	15.0	14.8	14.7
5	16.3	13.3	12.1	11.4	11.0	10.7	10.5	10.3	10.2
6	13.7	10.9	9.78	9.15	8.75	8.47	8.26	8.10	7.98
7	12.2	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72
8	11.3	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91
9	10.6	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35
10	10.0	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94
11	9.65	7.21	6.22	5.67	5.32	5.07	4.89	4.74	4.63
12	9.33	6.93	5.95	5.41	5.06	4.82	4.64	4.50	4.39
13	9.07	6.70	5.74	5.21	4.86	4.62	4.44	4.30	4.19
14	8.86	6.51	5.56	5.04	4.69	4.46	4.28	4.14	4.03
15	8.68	6.36	5.42	4.89	4.56	4.32	4.14	4.00	3.89
16	8.53	6.23	5.29	4.77	4.44	4.20	4.03	3.89	3.78
17	8.40	6.11	5.18	4.67	4.34	4.10	3.93	3.79	3.68
18	8.29	6.01	5.09	4.58	4.25	4.01	3.84	3.71	3.60
19	8.18	5.93	5.01	4.50	4.17	3.94	3.77	3.63	3.52
20	8.10	5.85	4.94	4.43	4.10	3.87	3.70	3.56	3.46
22	7.95	5.72	4.82	4.31	3.99	3.76	3.59	3.45	3.35
24	7.82	5.61	4.72	4.22	3.90	3.67	3.50	3.36	3.26
26	7.72	5.53	4.64	4.14	3.82	3.59	3.42	3.29	3.18
28	7.64	5.45	4.57	4.07	3.75	3.53	3.36	3.23	3.12
30	7.56	5.39	4.51	4.02	3.70	3.47	3.30	3.17	3.07
32	7.50	5.34	4.46	3.97	3.65	3.43	3.26	3.13	3.02
34	7.44	5.29	4.42	3.93	3.61	3.39	3.22	3.09	2.98
36	7.40	5.25	4.38	3.89	3.57	3.35	3.18	3.05	2.95
38	7.35	5.21	4.34	3.86	3.54	3.32	3.15	3.02	2.92
40	7.31	5.18	4.31	3.83	3.51	3.29	3.12	2.99	2.89
50	7.17	5.06	4.20	3.72	3.41	3.19	3.02	2.89	2.78
60	7.08	4.98	4.13	3.65	3.34	3.12	2.95	2.82	2.72
70	7.01	4.92	4.07	3.60	3.29	3.07	2.91	2.78	2.67
80	6.96	4.88	4.04	3.56	3.26	3.04	2.87	2.74	2.64
90	6.93	4.85	4.01	3.54	3.23	3.01	2.84	2.72	2.61
100	6.90	4.82	3.98	3.51	3.21	2.99	2.82	2.69	2.59
150	6.81	4.75	3.91	3.45	3.14	2.92	2.76	2.63	2.53
200	6.76	4.71	3.88	3.41	3.11	2.89	2.73	2.60	2.50
1000	6.66	4.63	3.80	3.34	3.04	2.82	2.66	2.53	2.43
∞	6.63	4.61	3.78	3.32	3.02	2.80	2.64	2.51	2.41

Table A11 F-Distribution with (m, n) Degrees of Freedom (continued)Values of z for which the distribution function $F(z)$ [see (13), Sec. 25.4] has the value **0.99**

n	$m = 10$	$m = 15$	$m = 20$	$m = 30$	$m = 40$	$m = 50$	$m = 100$	∞
1	6056	6157	6209	6261	6287	6303	6334	6366
2	99.4	99.4	99.4	99.5	99.5	99.5	99.5	99.5
3	27.2	26.9	26.7	26.5	26.4	26.4	26.2	26.1
4	14.5	14.2	14.0	13.8	13.7	13.7	13.6	13.5
5	10.1	9.72	9.55	9.38	9.29	9.24	9.13	9.02
6	7.87	7.56	7.40	7.23	7.14	7.09	6.99	6.88
7	6.62	6.31	6.16	5.99	5.91	5.86	5.75	5.65
8	5.81	5.52	5.36	5.20	5.12	5.07	4.96	4.86
9	5.26	4.96	4.81	4.65	4.57	4.52	4.42	4.31
10	4.85	4.56	4.41	4.25	4.17	4.12	4.01	3.91
11	4.54	4.25	4.10	3.94	3.86	3.81	3.71	3.60
12	4.30	4.01	3.86	3.70	3.62	3.57	3.47	3.36
13	4.10	3.82	3.66	3.51	3.43	3.38	3.27	3.17
14	3.94	3.66	3.51	3.35	3.27	3.22	3.11	3.00
15	3.80	3.52	3.37	3.21	3.13	3.08	2.98	2.87
16	3.69	3.41	3.26	3.10	3.02	2.97	2.86	2.75
17	3.59	3.31	3.16	3.00	2.92	2.87	2.76	2.65
18	3.51	3.23	3.08	2.92	2.84	2.78	2.68	2.57
19	3.43	3.15	3.00	2.84	2.76	2.71	2.60	2.49
20	3.37	3.09	2.94	2.78	2.69	2.64	2.54	2.42
22	3.26	2.98	2.83	2.67	2.58	2.53	2.42	2.31
24	3.17	2.89	2.74	2.58	2.49	2.44	2.33	2.21
26	3.09	2.81	2.66	2.50	2.42	2.36	2.25	2.13
28	3.03	2.75	2.60	2.44	2.35	2.30	2.19	2.06
30	2.98	2.70	2.55	2.39	2.30	2.25	2.13	2.01
32	2.93	2.65	2.50	2.34	2.25	2.20	2.08	1.96
34	2.89	2.61	2.46	2.30	2.21	2.16	2.04	1.91
36	2.86	2.58	2.43	2.26	2.18	2.12	2.00	1.87
38	2.83	2.55	2.40	2.23	2.14	2.09	1.97	1.84
40	2.80	2.52	2.37	2.20	2.11	2.06	1.94	1.80
50	2.70	2.42	2.27	2.10	2.01	1.95	1.82	1.68
60	2.63	2.35	2.20	2.03	1.94	1.88	1.75	1.60
70	2.59	2.31	2.15	1.98	1.89	1.83	1.70	1.54
80	2.55	2.27	2.12	1.94	1.85	1.79	1.65	1.49
90	2.52	2.24	2.09	1.92	1.82	1.76	1.62	1.46
100	2.50	2.22	2.07	1.89	1.80	1.74	1.60	1.43
150	2.44	2.16	2.00	1.83	1.73	1.66	1.52	1.33
200	2.41	2.13	1.97	1.79	1.69	1.63	1.48	1.28
1000	2.34	2.06	1.90	1.72	1.61	1.54	1.38	1.11
∞	2.32	2.04	1.88	1.70	1.59	1.52	1.36	1.00

Table A12 Distribution Function $F(x) = P(T \leq x)$ of the Random Variable T in Section 25.8

[illegible]

- Abel, Niels Henrik, 79n.6
- Abel's formula, 79
- Absolute convergence (series):
 - defined, 674
 - and uniform convergence, 704
- Absolute frequency (probability):
 - of an event, 1019
 - cumulative, 1012
 - of a value, 1012
- Absolutely integrable nonperiodic function, 512–513
- Absolute value (complex numbers), 613
- Acceleration, 386–389
- Acceleration of gravity, 8
- Acceleration vector, 386
- Acceptable lots, 1094
- Acceptable quality level (AQL), 1094
- Acceptance:
 - of a hypothesis, 1078
 - of products, 1092
- Acceptance number, 1092
- Acceptance sampling, 1092–1096, 1113
 - errors in, 1093–1094
 - rectification, 1094–1095
- Adams, John Couch, 912n.2
- Adams–Bashforth methods, 911–914, 947
- Adams–Moulton methods, 913–914, 947
- Adaptive integration, 835–836, 843
- Addition:
 - for arbitrary events, 1021–1022
 - of complex numbers, 609, 610
 - of matrices and vectors, 126, 259–261
 - of means, 1057–1058
 - for mutually exclusive events, 1021
 - of power series, 687
 - termwise, 173, 687
 - of variances, 1058–1059
 - vector, 309, 357–359
- ADI (alternating direction implicit) method, 928–930
- Adjacency matrix:
 - of a digraph, 973
 - of a graph, 972–973
- Adjacent vertices, 971, 977
- Airy, Sir George Bidell, 556n.2, 918n.4
- Airy equation, 556
 - RK method, 917–919
 - RKN method, 919–920
- Airy function:
 - RK method, 917–919
 - RKN method, 919–920
- Algebraic equations, 798
- Algebraic multiplicity, 326, 878
- Algorithms:
 - complexity of, 978–979
 - defined, 796
 - numeric analysis, 796
 - numeric methods as, 788
 - numeric stability of, 796, 842
- ALGORITHMS:
 - BISECT, A46
 - DIJKSTRA, 982
 - EULER, 903
 - FORD–FULKERSON, 998
 - GAUSS, 849
 - GAUSS–SEIDEL, 860
 - INTERPOL, 814
 - KRUSKAL, 985
 - MATCHING, 1003
 - MOORE, 977
 - NEWTON, 802
 - PRIM, 989
 - RUNGE–KUTTA, 905
 - SIMPSON, 832
- Aliasing, 531
- Alternating direction implicit (ADI) method, 928–930
- Alternating path, 1002
- Alternative hypothesis, 1078
- Ampère, André Marie, 93n.7
- Amplification, 91
- Amplitude, 90
- Amplitude spectrum, 511
- Analytic functions, 172, 201, 641
 - complex analysis, 623–624
 - conformal mapping, 737–742
 - derivatives of, 664–668, 688–689, A95–A96
 - integration of:
 - indefinite, 647
 - by use of path, 647–650
- Analytic functions (*Cont.*)
 - Laurent series:
 - analytics at infinity, 718–719
 - zeros of, 717–718
 - maximum modulus theorem, 782–783
 - mean value property, 781–782
 - power series representation of, 688–689
 - real functions vs., 694
- Analyticity, 623
- Angle of intersection:
 - conformal mapping, 738
 - between two curves, 36
- Angular speed (rotation), 372
- Angular velocity (fluid flow), 775
- AOQ (average outgoing quality), 1095
- AOQL (average outgoing quality limit), 1095
- Apparent resistance (RLC circuits), 95
- Approximation(s):
 - errors involved in, 794
 - polynomial, 808
 - by trigonometric polynomials, 495–498
- Approximation theory, 495
- A priori estimates, 805
- AQL (acceptable quality level), 1094
- Arbitrary positive, 191
- Arc, of a curve, 383
- Archimedes, 391n.4
- Arc length (curves), 385–386
- Area:
 - of a region, 428
 - of region bounded by ellipses, 436
 - of a surface, 448–450
- Argand, Jean Robert, 611n.2
- Argand diagram, 611n.2
- Argument (complex numbers), 613
- Artificial variables, 965–968
- Assignment problems (combinatorial optimization), 1001–1006
- Associative law, 264
- Asymptotically equal, 189, 1027, 1050
- Asymptotically normal, 1076

- Asymptotically stable critical points, 149
- Augmented matrices, 258, 272, 273, 321, 845, 959
- Augmenting path, 1002–1003. *See also* Flow augmenting paths
- Autonomous ODEs, 11, 33
- Autonomous systems, 152, 165
- Auxiliary equation, 54. *See also* Characteristic equation
- Average flow, 458
- Average outgoing quality (AOQ), 1095
- Average outgoing quality limit (AOQL), 1095
- Axioms of probability, 1020

- Back substitution (linear systems), 274–276, 846
- Backward edges:
 - cut sets, 994
 - initial flow, 998
 - of a path, 992
- Backward Euler formula, 909
- Backward Euler method (BEM):
 - first-order ODEs, 909–910
 - stiff systems, 920–921
- Backward Euler scheme, 909
- Balance law, 14
- Band matrices, 928
- Bashforth, Francis, 912n.2
- Basic feasible solution:
 - normal form of linear optimization problems, 957
 - simplex method, 959
- Basic Rule (method of undetermined coefficients):
 - higher-order homogeneous linear ODEs, 115
 - second-order nonhomogeneous linear ODEs, 81, 82
- Basic variables, 960
- Basis:
 - eigenvectors, 339–340
 - of solutions:
 - higher-order linear ODEs, 106, 113, 123
 - homogeneous linear systems, 290
 - homogeneous ODEs, 50–52, 75, 104, 106, 113
 - second-order homogeneous linear ODEs, 50–52, 75, 104
 - systems of ODEs, 139
 - standard, 314
 - vector spaces, 286, 311, 314
- Beats (oscillation), 89
- Bellman, Richard, 981n.3
- Bellman equations, 981
- Bellman's principle, 980–981
- Bell-shaped curve, 13, 574
- BEM, *see* Backward Euler method
- Benoulli, Niklaus, 31n.7
- Bernoulli, Daniel, 31n.7
- Bernoulli, Jakob, 31n.7
- Bernoulli, Johann, 31n.7
- Bernoulli distribution, 1040. *See also* Binomial distributions
- Bernoulli equation, 45
 - defined, 31
 - linear ODEs, 31–33
- Bernoulli's law of large numbers, 1051
- Bessel, Friedrich Wilhelm, 187n.6
- Bessel functions, 167, 187–191, 202
 - of the first kind, 189–190
 - with half-integer ν , 193–194
 - of order 1, 189
 - of order ν , 191
 - orthogonality of, 506
 - of the second kind:
 - general solution, 196–200
 - of order ν , 198–200
 - table, A97–A98
 - of the third kind, 200
- Bessel's equation, 167, 187–196, 202
- Bessel functions, 167, 187–191, 196–200
 - circular membrane, 587
 - general solution, 194–200
- Bessel's inequality:
 - for Fourier coefficients, 497
 - orthogonal series, 508–509
- Beta function, formula for, A67
- Bezier curve, 827
- BFS algorithms, *see* Breadth First search algorithms
- Bijjective mapping, 737n.1
- Binomial coefficients:
 - Newton's forward difference formula, 816
 - probability theory, 1027–1028
- Binomial distributions, 1039–1041, 1061
 - normal approximation of, 1049–1050
 - sampling with replacement for, 1042
 - table, A99
- Binomial series, 696
- Binomial theorem, 1029
- Bipartite graphs, 1001–1006, 1008
- BISECT, ALGORITHM, A46
- Bisection method, 807–808
- Bolzano, Bernard, A94n.3
- Bolzano–Weierstrass theorem, A94–A95
- Bonnet, Ossian, 180n.3
- Bonnet's recursion, 180
- Borda, J. C., 16n.4
- Boundaries:
 - ODEs, 39
 - of regions, 426n.2
 - sets in complex plane, 620
- Boundary conditions:
 - one-dimensional heat equation, 559
 - PDEs, 541, 605
 - periodic, 501
 - two-dimensional wave equation, 577
 - vibrating string, 545–547
- Boundary points, 426n.2
- Boundary value problem (BVP), 499
 - conformal mapping for, 763–767, A96
 - first, *see* Dirichlet problem
 - mixed, *see* Mixed boundary value problem
 - second, *see* Neumann problem
 - third, *see* Mixed boundary value problem
 - two-dimensional heat equation, 564
- Bounded domains, 652
- Bounded regions, 426n.2
- Bounded sequence, A93–A95
- Boxplots, 1013
- Boyle, Robert, 19n.5
- Boyle–Mariotte's law for idea gases, 19
- Bragg, Sir William Henry, 938n.5
- Bragg, Sir William Lawrence, 938n.5
- Branch, of logarithm, 639
- Branch cut, of logarithm, 639
- Branch point (Riemann surfaces), 755
- Breadth First search (BFS)
 - algorithms, 977
 - defined, 977, 998
 - Moore's, 977–980
- BVP, *see* Boundary value problem

- CAD (computer-aided design), 820
- Cancellation laws, 306–307
- Canonical form, 344
- Cantor, Georg, A72n.3
- Cantor–Dedekind axiom, A72n.3, A95n.4
- Capacity:
 - cut sets, 994
 - networks, 991
- Cardano, Girolamo, 608n.1
- Cardioid, 391, 437

- Cartesian coordinates:
 - linear element in, A75
 - transformation law, A86–A87
 - vector product in, A83–A84
 - writing, A74
- Cartesian coordinate systems:
 - complex plane, 611
 - left-handed, 369, 370, A84
 - right-handed, 368–369, A83–A84
 - in space, 315, 356
 - transformation law for vector components, A85–A86
- Cartesius, Renatus, 356n.1
- Cauchy, Augustin-Louis, 71n.4, 625n.4, 683n.1
- Cauchy determinant, 113
- Cauchy–Goursat theorem, *see* Cauchy’s integral theorem
- Cauchy–Hadamard formula, 683
- Cauchy principal value, 727, 730
- Cauchy–Riemann equations, 38, 642
 - complex analysis, 623–629
 - proof of, A90–A91
- Cauchy–Schwarz inequality, 363, 871–872
- Cauchy’s convergence principle, 674–675, A93–A94
- Cauchy’s inequality, 666
- Cauchy’s integral formula, 660–663, 670
- Cauchy’s integral theorem, 652–660, 669
 - existence of indefinite integral, 656–658
 - Goursat’s proof of, A91–A93
 - independence of path, 655
 - for multiply connected domains, 658–659
 - principle of deformation of path, 656
- Cayley, Arthur, 748n.2
- c*-charts, 1092
- Center:
 - as critical point, 144, 165
 - of a graph, 991
 - of power series, 680
- Center control line (CL), 1088
- Center of gravity, of mass in a region, 429
- Central difference notation, 819
- Central limit theorem, 1076
- Central vertex, 991
- Centrifugal force, 388
- Centripetal acceleration, 387–388
- Chain rules, 392–394
- Characteristics, 555
- Characteristics, method of, 555
- Characteristic determinant, of a matrix, 129, 325, 326, 353, 877
- Characteristic equation:
 - matrices, 129, 325, 326, 353, 877
 - PDEs, 555
 - second-order homogeneous linear ODEs, 54
- Characteristic matrix, 326
- Characteristic polynomial, 325, 353, 877
- Characteristic values, 87, 324, 353. *See also* Eigenvalues
- Characteristic vectors, 324, 877. *See also* eigenvectors
- Chebyshev, Pafnuti, 504n.6
- Chebyshev equation, 504
- Chebyshev polynomials, 504
- Checkerboard pattern (determinants), 294
- Chi-square (χ^2) distribution, 1074–1076, A104
- Chi-square (χ^2) test, 1096–1097, 1113
- Choice of numeric method, for matrix eigenvalue problems, 879
- Cholesky, André-Louis, 855n.3
- Cholesky’s method, 855–856, 898
- Chopping, error caused by, 792
- Chromatic number, 1006
- Circle, 386
- Circle of convergence (power series), 682
- Circulation, of flow, 467, 774
- CL (center control line), 1088
- Clairaut equation, 35
- Clamped condition (spline interpolation), 823
- Class intervals, 1012
- Class marks, 1012
- Closed annulus, 619
- Closed circular disk, 619
- Closed integration formulas, 833, 838
- Closed intervals, A72n.3
- Closed Newton–Cotes formulas, 833
- Closed paths, 414, 645, 975–976
- Closed regions, 426n.2
- Closed sets, 620
- Closed trails, 975–976
- Closed walks, 975–976
- CN (Crank–Nicolson) method, 938–941
- Coefficients:
 - binomial:
 - Newton’s forward difference formula, 816
 - probability theory, 1027–1028
 - constant:
 - higher-order homogeneous linear ODEs, 111–116
 - second-order homogeneous linear ODEs, 53–60
- Coefficients: (*Cont.*)
 - second-order nonhomogeneous linear ODEs, 81
 - systems of ODEs, 140–151
 - correlation, 1108–1111, 1113
 - Fourier, 476, 484, 538, 582–583
 - of kinetic friction, 19
 - of linear systems, 272, 845
 - of ODEs, 47
 - higher-order homogeneous linear ODEs, 105
 - second-order homogeneous linear ODEs, 53–60, 73
 - second-order nonhomogeneous linear ODEs, 81–85
 - series of ODEs, 168, 174
 - variable, 167, 240–241
 - of power series, 680
 - regression, 1105, 1107–1108
 - variable:
 - Frobenius method, 180–187
 - Laplace transforms ODEs with, 240–241
 - of ODEs, 167, 240–241
 - power series method, 167–175
 - second-order homogeneous linear ODEs, 73
- Coefficient matrices, 257, 273
- Hermitian or skew-Hermitian forms, 351
- linear systems, 845
- quadratic form, 343
- Cofactor (determinants), 294
- Collatz, Lothar, 883n.9
- Collatz inclusion theorem, 883–884
- Columns:
 - determinants, 294
 - matrix, 125, 257, 320
- Column “sum” norm, 861
- Column vectors, 126
- matrices, 257, 284–285, 320
- rank in terms of, 284–285
- Combinations (probability theory), 1024, 1026–1027
 - of n things taken k at a time without repetitions, 1026
 - of n things taken k at a time with repetitions, 1026
- Combinatorial optimization, 970, 975–1008
 - assignment problems, 1001–1006
 - flow problems in networks, 991–997
 - cut sets, 994–996
 - flow augmenting paths, 992–993
 - paths, 992
- Ford–Fulkerson algorithm for maximum flow, 998–1001

Combinatorial optimization (*Cont.*)
 shortest path problems, 975–980
 Bellman's principle, 980–981
 complexity of algorithms, 978–980
 Dijkstra's algorithm, 981–983
 Moore's BFS algorithm, 977–980
 shortest spanning trees:
 Greedy algorithm, 984–988
 Prim's algorithm, 988–991
 Commutation (matrices), 271
 Complements:
 of events, 1016
 of sets in complex plane, 620
 Complementation rule, 1020–1021
 Complete bipartite graphs, 1005
 Complete graphs, 974
 Complete matching, 1002
 Completeness (orthogonal series), 508–509
 Complete orthonormal set, 508
 Complex analysis, 607
 analytic functions, 623–624
 Cauchy–Riemann equations, 623–629
 circles and disks, 619
 complex functions, 620–623
 exponential, 630–633
 general powers, 639–640
 hyperbolic, 635
 logarithm, 636–639
 trigonometric, 633–635
 complex integration, 643–670
 Cauchy's integral formula, 660–663, 670
 Cauchy's integral theorem, 652–660, 669
 derivatives of analytic functions, 664–668
 Laurent series, 708–719
 line integrals, 643–652, 669
 power series, 671–707
 residue integration, 719–733
 complex numbers, 608–619
 addition of, 609, 610
 conjugate, 612
 defined, 608
 division of, 610
 multiplication of, 609, 610
 polar form of, 613–618
 subtraction of, 610
 complex plane, 611
 conformal mapping, 736–757
 geometry of analytic functions, 737–742
 linear fractional transformations, 742–750

Complex analysis (*Cont.*)
 Riemann surfaces, 754–756
 by trigonometric and hyperbolic analytic functions, 750–754
 half-planes, 619–620
 harmonic functions, 628–629
 Laplace's equation, 628–629
 Laurent series, 708–719, 734
 analytic or singular at infinity, 718–719
 point at infinity, 718
 Riemann sphere, 718
 singularities, 715–717
 zeros of analytic functions, 717
 power series, 168, 671–707
 convergence behavior of, 680–682
 convergence tests, 674–676, A93–A94
 functions given by, 685–690
 Maclaurin series, 690
 in powers of x , 168
 radius of convergence, 682–684
 ratio test, 676–678
 root test, 678–679
 sequences, 671–673
 series, 673–674
 Taylor series, 690–697
 uniform convergence, 698–705
 residue integration, 719–733
 formulas for residues, 721–722
 of real integrals, 725–733
 several singularities inside contour, 723–725
 Taylor series, 690–697, 707
 Complex conjugate numbers, 612
 Complex conjugate roots, 72–73
 Complex Fourier integral, 523
 Complex functions, 620–623
 exponential, 630–633
 general powers, 639–640
 hyperbolic, 635
 logarithm, 636–639
 trigonometric, 633–635
 Complex heat potential, 767
 Complex integration, 643–670
 Cauchy's integral formula, 660–663, 670
 Cauchy's integral theorem, 652–660, 669
 existence of indefinite integral, 656–658
 independence of path, 655
 for multiply connected domains, 658–659
 principle of deformation of path, 656

Complex integration (*Cont.*)
 derivatives of analytic functions, 664–668
 Laurent series, 708–719
 analytic or singular at infinity, 718–719
 point at infinity, 718
 Riemann sphere, 718
 singularities, 715–717
 zeros of analytic functions, 717–718
 line integrals, 643–652, 669
 basic properties of, 645
 bounds for, 650–651
 definition of, 643–645
 existence of, 646
 indefinite integration and substitution of limits, 646–647
 representation of a path, 647–650
 power series, 671–707
 convergence behavior of, 680–682
 convergence tests, 674–676
 functions given by, 685–690
 Maclaurin series, 690
 radius of convergence of, 682–684
 ratio test, 676–678
 root test, 678–679
 sequences, 671–673
 series, 673–674
 Taylor series, 690–697
 uniform convergence, 698–705
 residue integration, 719–733
 formulas for residues, 721–722
 of real integrals, 725–733
 several singularities inside contour, 723–725
 Complexity, of algorithms, 978–979
 Complex line integrals, *see* Line integrals
 Complex matrices and forms, 346–352
 Complex numbers, 608–619, 641
 addition of, 609, 610
 conjugate, 612
 defined, 608
 division of, 610
 multiplication of, 609, 610
 polar form of, 613–618
 subtraction of, 610
 Complex plane, 611
 extended, 718, 744–745
 sets in, 620
 Complex potential, 786
 electrostatic fields, 760–761
 of fluid flow, 771, 773–774

- Complex roots:
 - higher-order homogeneous linear ODEs:
 - multiple, 115
 - simple, 113–114
 - second-order homogeneous linear ODEs, 57–59
- Complex trigonometric polynomials, 529
- Complex variables, 620–621
- Complex vector space, 309, 310, 349
- Components (vectors), 126, 356, 365
- Composition, of linear transformations, 316–317
- Computer-aided design (CAD), 820
- Condition:
 - of incompressibility, 405
 - spline interpolation, 823
- Conditionally convergent series, 675
- Conditional probability, 1022–1023, 1061
- Condition number, 868–870, 899
- Confidence intervals, 1063, 1068–1077, 1113
 - interval estimates, 1065
 - for mean of normal distribution:
 - with known variance, 1069–1071
 - with unknown variance, 1071–1073
 - for parameters of distributions
 - other than normal, 1076
 - in regression analysis, 1107–1108
 - for variance of a normal distribution, 1073–1076
- Confidence level, 1068
- Conformality, 738
- Conformal mapping, 736–757
 - boundary value problems, 763–767, A96
 - defined, 738
 - geometry of analytic functions, 737–742
 - linear fractional transformations, 742–750
 - extended complex plane, 744–745
 - mapping standard domains, 747–750
 - Riemann surfaces, 754–756
 - by trigonometric and hyperbolic analytic functions, 750–754
- Connected graphs, 977, 981, 984
- Connected set, in complex plane, 620
- Conservative physical systems, 422
- Conservative vector fields, 400, 408
- Consistent linear systems, 277
- Constant coefficients:
 - higher-order homogeneous linear ODEs, 111–116
 - distinct real roots, 112–113
 - multiple real roots, 114–115
 - simple complex roots, 113–114
 - second-order homogeneous linear ODEs, 53–60
 - complex roots, 57–59
 - real double root, 55–56
 - two distinct real roots, 54–55
 - second-order nonhomogeneous linear ODEs, 81
 - systems of ODEs, 140–151
 - critical points, 142–146, 148–151
 - graphing solutions in phase plane, 141–142
- Constant of gravity, at the Earth's surface, 63
- Constant of integration, 18
- Constant revenue, lines of, 954
- Constrained (linear) optimization, 951, 954–958, 969
 - normal form of problems, 955–957
 - simplex method, 958–968
 - degenerate feasible solution, 962–965
 - difficulties in starting, 965–968
- Constraints, 951
- Consumers, 1092
- Consumer's risk, 1094
- Consumption matrix, 334
- Continuity equation (compressible fluid flow), 405
- Continuous complex functions, 621
- Continuous distributions, 1029, 1032–1034
 - marginal distribution of, 1055
 - two-dimensional, 1053
- Continuous random variables, 1029, 1032–1034, 1061
- Continuous vector functions, 378–379
- Contour integral, 653
- Contour lines, 21, 36
- Control charts, 1088
 - for mean, 1088–1089
 - for range, 1090–1091
 - for standard deviation, 1090
 - for variance, 1089–1090
- Controlled variables, in regression analysis, 1103
- Control limits, 1088, 1089
- Control variables, 951
- Convergence:
 - absolute:
 - defined, 674
 - and uniform convergence, 704
 - of approximate and exact solutions, 936
- Convergence: (*Cont.*)
 - circle of, 682
 - defined, 861
 - Gauss–Seidel iteration, 861–862
 - mean square (orthogonal series), 507–508
 - in the norm, 507
 - power series, 680–682
 - convergence tests, 674–676, A93–A94
 - radius of convergence of, 682–684, 706
 - uniform convergence, 698–705
 - radius of, 172
 - defined, 172
 - power series, 682–684, 706
 - sequence of vectors, 378
 - speed of (numeric analysis), 804–805
 - superlinear, 806
 - uniform:
 - and absolute convergence, 704
 - power series, 698–705
- Convergence interval, 171, 683
- Convergence tests, 674–676
 - power series, 674–676, A93–A94
 - uniform convergence, 698–705
- Convergent iteration processes, 800
- Convergent sequence of functions, 507–508, 672
- Convergent series, 171, 673
- Convolution:
 - defined, 232
 - Fourier transforms, 527–528
 - Laplace transforms, 232–237
- Convolution theorem, 232–233
- Coriolis, Gustave Gaspard, 389n.3
- Coriolis acceleration, 388–389
- Corrector (improved Euler method), 903
- Correlation analysis, 1063, 1108–1111, 1113
 - defined, 1103
 - test for correlation coefficient, 1110–1111
- Correlation coefficient, 1108–1111, 1113
- Cosecant, formula for, A65
- Cosine function:
 - conformal mapping by, 752
 - formula for, A63–A65
- Cosine integral:
 - formula for, A69
 - table, A98
- Cosine series, 781
- Cotangent, formula for, A65
- Coulomb, Charles Augustin de, 19n.6, 93n.7, 401n.6
- Coulomb's law, 19, 401

- Covariance:
 in correlation analysis, 1109
 defined, 1058
- Cramer, Gabriel, 31n.7, 298n.2
- Cramer's rule, 292, 298–300, 321
 for three equations, 293
 for two equations, 292
- Cramer's Theorem, 298
- Crank, John, 938n.5
- Crank–Nicolson (CN) method,
 938–941
- Critical damping, 65, 66
- Critical points, 33, 165
 asymptotically stable, 149
 and conformal mapping, 738, 757
 constant-coefficient systems of
 ODEs, 142–146
 center, 144
 criteria for, 148–151
 degenerate node, 145–146
 improper node, 142
 proper node, 143
 saddle point, 143
 spiral point, 144–145
 stability of, 149–151
 isolated, 152
 nonlinear systems, 152
 stable, 140, 149
 stable and attractive, 140, 149
 unstable, 140, 149
- Critical region, 1079
- Cross product, 368, 410. *See also*
 Vector product
- Crout, Prescott Durand, 853n.2
- Crout's method, 853, 898
- Cubic spline, 821
- Cumulative absolute frequencies (of
 values), 1012
- Cumulative distribution functions,
 1029
- Cumulative relative frequencies (of
 values), 1012
- Curl, A76
 invariance of, A85–A88
 of vector fields, 406–409, 412
- Curvature, of a curve, 389–390
- Curves: (Cont.)
 regression, 1103
 simple, 383
 simple closed, 646
 smooth, 414, 644
 solution, 4–6
 twisted, 383
 vector differential calculus,
 381–392, 411
 arc length of, 385–386
 length of, 385
 in mechanics, 386–389
 tangents to, 384–385
 and torsion, 389–390
- Curve fitting, 872–876
 method of least squares, 872–874
 by polynomials of degree m ,
 874–875
- Curvilinear coordinates, 354, 412, A74
- Cut sets, 994–996, 1008
- Cycle (paths), 976, 984
- Cylindrical coordinates, 593–594,
 A74–A76
- D'Alembert, Jean le Rond, 554n.1
- D'Alembert's solution, 553–556
- Damped oscillations, 67
- Damping constant, 65
- Dantzig, George Bernard, 959
- Data processing:
 frequency distributions,
 1011–1012
 and randomness, 1064
- Data representation:
 frequency distributions,
 1011–1015
 Empirical Rule, 1014
 graphic, 1012
 mean, 1013–1014
 standard deviation, 1014
 variation, 1014
 and randomness, 1064
- Decisions:
 false, risks of making, 1080
 statistics for, 1077–1078
- Dedekind, Richard, A72n.3
- Defect (eigenvalue), 328
- Defectives, 1092
- Definite integrals, complex, *see* Line
 integrals
- Deflection curve, 120
- Deformation of path, principle of,
 656
- Degenerate feasible solution (simplex
 method), 962–965
- Degenerate node, 145–146
- Degrees of freedom (d.f.), number of,
 1071, 1074
- Degree of incidence, 971
- Degree of precision (DP), 833
- Deleted neighborhood, 720
- Demand vector, 334
- De Moivre, Abraham, 616n.3
- De Moivre–Laplace limit theorem,
 1050
- De Moivre's formula, 616
- De Morgan's laws, 1018
- Density, 1061
 continuous two-dimensional
 distributions, 1053
 of a distribution, 1033
- Dependent random variables, 1055,
 1056
- Dependent variables, 393, 1055, 1056
- Depth First Search (DFS) algorithms,
 977
- Derivatives:
 of analytic functions, 664–668,
 688–689, A95–A96
 of complex functions, 622, 641
 Laplace transforms of, 211–212
 of matrices or vectors, 127
 of vector functions, 379–380
- Derived series, 687
- Descartes, René, 356n.1, 391n.4
- Determinants, 293–301, 321
 Cauchy, 113
 Cramer's rule, 298–300
 defined, A81
 general properties of, 295–298
 of a matrix, 128
 of matrix products, 307–308
 of order n , 293
 proof of, A81–A83
 second-order, 291–292
 second-order homogeneous linear
 ODEs, 76
 third-order, 292–293
 Vandermonde, 113
 Wronski:
 second-order homogeneous
 linear ODEs, 75–78
 systems of ODEs, 139
- Developed, in a power series, 683
- D.f. (degrees of freedom), number of,
 1071, 1074
- DFS (Depth First Search) algorithms,
 977
- DFTs (discrete Fourier transforms),
 528–531
- Diagonalization of matrices, 341–342
- Diagonally dominant matrices, 881
- Diagonal matrices, 268
 inverse of, 305–306
 scalar, 268
- Diameter (graphs), 991
- Difference:
 complex numbers, 610
 scalar multiplication, 260

- Difference equations (elliptic PDEs), 923–925
- Difference quotients, 923
- Difference table, 814
- Differentiable complex functions, 622–623
- Differentiable vector functions, 379
- Differential (total differential), 20, 45
- Differential equations:
 - applications of, 3
 - defined, 2
- Differential form, 422
 - exact, 21, 470
 - first fundamental form, of S , 451
 - floating-point, of numbers, 791–792
 - path independence and exactness of, 422, 470
- Differential geometry, 381
- Differential operators:
 - second-order, 60
 - for second-order homogeneous linear ODEs, 60–62
- Differentiation:
 - of Laplace transforms, 238–240
 - matrices or vectors, 127
 - numeric, 838–839
 - of power series, 687–688, 703
 - termwise, 173, 687–688, 703
- Diffusion equation, 459–460, 558.
See also Heat equation
- Digraphs (directed graphs), 971–972, 1007
 - computer representation of, 972–974
 - defined, 972
 - incidence matrix of, 975
 - subgraphs, 972
- Dijkstra, Edsger Wybe, 981n.4
- Dijkstra's algorithm, 981–983, 1008
- DIJKSTRA, ALGORITHM, 982
- Dimension of vector spaces, 286, 311, 359
- Diocles, 391n.4
- Dirac, Paul, 226n.2
- Dirac delta function, 226–228, 237
- Directed graphs, *see* Digraphs (directed graphs)
- Directed path, 1000
- Directional derivatives (scalar functions), 396–397, 411
- Direction field (slope field), 9–10, 44
- Direct methods (linear system solutions), 858, 898. *See also* iteration
- Dirichlet, Peter Gustav LeJeune, 462n.8
- Dirichlet boundary condition, 564
- Dirichlet problem, 605, 923
 - ADI method, 929
 - heat equation, 564–566
 - Laplace equation, 593–596, 925–928, 934–935
 - Poisson equation, 925–928
 - two-dimensional heat equation, 564–565
 - uniqueness theorem for, 462, 784
- Dirichlet's discontinuous factor, 514
- Discharge (flow modeling), 776
- Discrete distributions, 1029–1032
 - marginal distributions of, 1053–1054
 - two-dimensional, 1052–1053
- Discrete Fourier transforms (DFTs), 528–531
- Discrete random variables, 1029, 1030–1032, 1061
 - defined, 1030
 - marginal distributions of, 1054
- Discrete spectrum, 525
- Disjoint events, 1016
- Disks:
 - circular, open and closed, 619
 - mapping, 748–750
 - Poisson's integral formula, 779–780
- Dissipative physical systems, 422
- Distance:
 - graphs, 991
 - vector norms, 866
- Distinct real roots:
 - higher-order homogeneous linear ODEs, 112–113
 - second-order homogeneous linear ODEs, 54–55
- Distinct roots (Frobenius method), 182
- Distributions, 226n.2. *See also* Frequency distributions; Probability distributions
- Distribution-free tests, 1100
- Distribution function, 1029–1032
 - cumulative, 1029
 - normal distributions, 1046–1047
 - of random variables, 1056, A109
 - sample, 1096
 - two-dimensional probability distributions, 1051–1052
- Distributive laws, 264
- Distributivity, 363
- Divergence, A75
 - fluid flow, 775
 - of vector fields, 402–406
 - of vector functions, 411, 453
- Divergence theorem of Gauss, 405, 470
 - applications, 458–463
 - vector integral calculus, 453–457
- Divergent sequence, 672
- Divergent series, 171, 673
- Division, of complex numbers, 610, 615–616
- Domain(s), 393
 - bounded, 652
 - doubly connected, 658, 659
 - of f , 620
 - holes of, 653
 - mapping, 737, 747–750
 - multiply connected:
 - Cauchy's integral formula, 662–663
 - Cauchy's integral theorem, 658–659
 - p -fold connected, 652–653
 - sets in complex plane, 620
 - simply connected, 423, 646, 652, 653
 - triply connected, 653, 658, 659
- Dominant eigenvalue, 883
- Doolittle, Myrick H., 853n.1
- Doolittle's method, 853–855, 898
- Dot product, 312, 410. *See also* Inner product
- Double Fourier series:
 - defined, 582
 - rectangular membrane, 577–585
- Double integrals (vector integral calculus), 426–432, 470
 - applications of, 428–429
 - change of variables in, 429–431
 - evaluation of, by two successive integrations, 427–428
- Double precision, floating-point standard for, 792
- Double root (Frobenius method), 183
- Double subscript notation, 125
- Doubly connected domains, 658, 659
- DP (degree of precision), 833
- Driving force, *see* Input (driving force)
- Duffing equation, 160
- Duhamel, Jean-Marie Constant, 603n.4
- Duhamel's formula, 603
- Eccentricity, of vertices, 991
- Edges:
 - backward:
 - cut sets, 994
 - initial flow, 998
 - of a path, 992
 - forward:
 - cut sets, 994
 - initial flow, 998
 - of a path, 992
 - graphs, 971, 1007
 - incident, 971
- Edge chromatic number, 1006

- Edge condition, 991
- Edge incidence list (graphs), 973
- Efficient algorithms, 979
- Eigenbases, 339–341
- Eigenfunctions, 605
 - circular membrane, 588
 - one-dimensional heat equation, 560
 - Sturm–Liouville Problems, 499–500
 - two-dimensional heat equation, 565
 - two-dimensional wave equation, 578, 580
 - vibrating string, 547
- Eigenfunction expansion, 504
- Eigenspaces, 326, 878
- Eigenvalues, 129–130, 166, 353, 605, 877, 899. *See also* Matrix eigenvalue problems
 - circular membrane, 588
 - complex matrices, 347–351
 - and critical points, 149
 - defined, 324
 - determining, 323–329
 - dominant, 883
 - finding, 324–328
 - one-dimensional heat equation, 560
 - Sturm–Liouville Problems, 499–500, A89
 - two-dimensional wave equation, 580
 - vibrating string, 547
- Eigenvalues of A, 322
- Eigenvalue problem, 140
- Eigenvectors, 129–130, 166, 353, 877, 899
 - basis of, 339–340
 - convergent sequence of, 886
 - defined, 324
 - determining, 323–329
 - finding, 324–328
- Eigenvectors of A, 322
- EISPACK, 789
- Elastic curve, 120
- Electric circuits:
 - analogy of electrical and mechanical quantities, 97–98
 - second-order nonhomogeneous linear ODEs, 93–99
- Electrostatic fields (potential theory), 759–763
 - complex potential, 760–761
 - superposition, 761–762
- Electrostatic potential, 759
- Electrostatics (Laplace’s equation), 593
- Elementary matrix, 281
- Elementary row operations (linear systems), 277
- Ellipses, area of region bounded by, 436
- Elliptic PDEs:
 - defined, 923
 - numeric analysis, 922–936
 - ADI method, 928–930
 - difference equations, 923–925
 - Dirichlet problem, 925–928
 - irregular boundary, 933–935
 - mixed boundary value problems, 931–933
 - Neumann problem, 931
- Empirical Rule, 1014
- Energies, 157
- Entire function, 630, 642, 707, 718
- Entries:
 - determinants, 294
 - matrix, 125, 257
- Equal complex numbers, 609
- Equality:
 - of matrices, 126, 259
 - of vectors, 355
- Equally likely events, 1018
- Equal spacing (interpolation):
 - Newton’s backward difference formula, 818–819
 - Newton’s forward difference formula, 815–818
- Equilibrium harvest, 36
- Equilibrium solutions (equilibrium points), 33–34
- Equipotential curves, 36, 759, 761
- Equipotential lines, 38
 - electrostatic fields, 759, 761
 - fluid flow, 771
- Equipotential surfaces, 759
- Equivalent vector norms, 871
- Error(s):
 - in acceptance sampling, 1093–1094
 - of approximations, 495
 - in numeric analysis, 842
 - basic error principle, 796
 - error propagation, 795
 - errors of numeric results, 794–795
 - roundoff, 792
 - in statistical tests, 1080–1081
 - and step size control, 906–907
 - trapezoidal rule, 830
 - vector norms, 866
- Error bounds, 795
- Error estimate, 908
- Error function, 828, A67–A68, A98
- Essential singularity, 715–716
- Estimation of parameters, 1063
- EULER, ALGORITHM, 903
- Euler, Leonhard, 31n.7, 71n.4
- Euler–Cauchy equations, 71–74, 104
 - higher-order nonhomogeneous linear ODEs, 119–120
 - Laplace’s equation, 595
 - third-order, IVP for, 108
- Euler–Cauchy method, 901
- Euler constant, 198
- Euler formulas, 58
 - complex Fourier integral, 523
 - derivation of, 479–480
 - exponential function, 631
 - Fourier coefficients given by, 476, 484
 - generalized, 582
 - Taylor series, 694
 - trigonometric function, 634
- Euler graph, 980
- Euler’s method:
 - defined, 10
 - error of, 901–902, 906, 908
 - first-order ODEs, 10–11, 901–902
 - backward method, 909–910
 - improved method, 902–904
 - higher order ODEs, 916–917
- Euler trail, 980
- Even functions, 486–488
- Even periodic extension, 488–490
- Events (probability theory), 1016–1017, 1060
 - addition rule for, 1021–1022
 - arbitrary, 1021–1022
 - complements of, 1016
 - defined, 1015
 - disjoint, 1016
 - equally likely, 1018
 - independent, 1022–1023
 - intersection, 1016, 1017
 - mutually exclusive, 1016, 1021
 - simple, 1015
 - union, 1016–1017
- Exact differential equation, 21
- Exact differential form, 422, 470
- Exact ODEs, 20–27, 45
 - defined, 21
 - integrating factors, 23–26
- Existence, problem of, 39
- Existence theorems:
 - cubic splines, 822
 - first-order ODEs, 39–42
 - homogeneous linear ODEs:
 - higher-order, 108
 - second-order, 74
 - of the inverse, 301–302
 - Laplace transforms, 209–210
 - linear systems, 138
 - power series solutions, 172
 - systems of ODEs, 137
- Expectation, 1035, 1037–1038, 1057

- Experiments:
 - defined, 1015, 1060
 - in probability theory, 1015–1016
 - random, 1011, 1015–1016, 1060
- Experimental error, 794
- Explicit formulas, 913
- Explicit method:
 - heat equation, 937, 940–941
 - wave equation, 943
- Explicit solution, 21
- Exponential decay, 5, 7
- Exponential function, 630–633, 642
 - formula for, A63
 - Taylor series, 694
- Exponential growth, 5
- Exponential integral, formula for, A69
- Exposed vertices, 1001, 1003
- Extended complex plane:
 - conformal mapping, 744–745
 - defined, 718
- Extended method (separable ODEs), 17–18
- Extended problems, 966
- Extrapolation, 808
- Extrema (unconstrained optimization), 951

- Factorial function, 1027, A66, A98.
 - See also* Gamma functions
- Failing to reject a hypothesis, 1081
- Fair die, 1018, 1019
- False decisions, risks of making, 1080
- False position, method of, 807–808
- Family of curves, one-parameter, 36–37
- Family of solutions, 5
- Faraday, Michael, 93n.7
- Fast Fourier transforms (FFTs), 531–532
- F -distribution, 1086, A105–A108
- Feasibility region, 954
- Feasible solutions, 954–955
 - basic, 957, 959
 - degenerate, 962–965
 - normal form of linear optimization problems, 957
- Fehlberg, E., 907
- Fehlberg's fifth-order RK method, 907–908
- Fehlberg's fourth-order RK method, 907–908
- FFTs (fast Fourier transforms), 531–532
- Fibonacci (Leonardo of Pisa), 690n.2
- Fibonacci numbers, 690
- Fibonacci's rabbit problem, 690
- Finite complex plane, 718. *See also* Complex plane
- Finite jumps, 209
- First boundary value problem, *see* Dirichlet problem
- First fundamental form, of S , 451
- First-order method, Euler method as, 902
- First-order ODEs, 2–45, 44
 - defined, 4
 - direction fields, 9–10
 - Euler's method, 10–11
 - exact, 20–27, 45
 - defined, 21
 - integrating factors, 23–26
 - explicit form, 4
 - geometric meanings of, 9–12
 - implicit form, 4
 - initial value problem, 38–43
 - linear, 27–36
 - Bernoulli equation, 31–33
 - homogeneous, 28
 - nonhomogeneous, 28–29
 - population dynamics, 33–34
 - modeling, 2–8
 - numeric analysis, 901–915
 - Adams–Bashforth methods, 911–914
 - Adams–Moulton methods, 913–914
 - backward Euler method, 909–910
 - Euler's method, 901–902
 - improved Euler's method, 902–904
 - multistep methods, 911–915
 - Runge–Kutta–Fehlberg method, 906–908
 - Runge–Kutta methods, 904–906
 - orthogonal trajectories, 36–38
 - separable, 12–20, 44
 - extended method, 17–18
 - modeling, 13–17
 - systems of, 165
 - transformation of systems to, 157–159
- First (first order) partial derivatives, A71
- First shifting theorem (s -shifting), 208–209
- First transmission line equation, 599
- Fisher, Sir Ronald Aylmer, 1086
- Fixed points:
 - defined, 799
 - of a mapping, 745
- Fixed-point iteration (numeric analysis), 798–801, 842
- Fixed-point systems, numbers in, 791
- Floating, 793
- Floating-point form of numbers, 791–792
- Flow augmenting paths, 992–993, 998, 1008
- Flow problems in networks (combinatorial optimization), 991–997
 - cut sets, 994–996
 - flow augmenting paths, 992–993
 - paths, 992
- Fluid flow:
 - Laplace's equation, 593
 - potential theory, 771–777
- Fluid state, 404
- Flux (motion of a fluid), 404
- Flux integral, 444, 450
- Forced motions, 68, 86
- Forced oscillations:
 - Fourier analysis, 492–495
 - second-order nonhomogeneous
 - linear ODEs, 85–92
 - damped, 89–90
 - resonance, 88–91
 - undamped, 87–89
- Forcing function, 86
- Ford, Lester Randolph, Jr., 998n.7
- FORD–FULKERSON, ALGORITHM, 998
- Ford–Fulkerson algorithm for maximum flow, 998–1001, 1008
- Forest (graph), 987
- Form(s):
 - canonical, 344
 - complex, 351
 - differential, 422
 - exact, 21, 470
 - path independence and exactness of, 422
 - Hesse's normal, 366
 - Lagrange's, 812
 - normal (linear optimization problems), 955–957, 959, 969
 - Pfaffian, 422
 - polar, of complex numbers, 613–618, 631
 - quadratic, 343–344, 346
 - reduced echelon, 279
 - row echelon, 279–280
 - skew-Hermitian and Hermitian, 351
 - standard:
 - first-order ODEs, 27
 - higher-order homogeneous
 - linear ODEs, 105
 - higher-order linear ODEs, 123
 - power series method, 172
 - second-order linear ODEs, 46, 103
 - triangular (Gauss elimination), 846

- Forward edge:
 cut sets, 994
 initial flow, 998
 of a path, 992
- Four-color theorem, 1006
- Fourier, Jean-Baptiste Joseph, 473n.1
- Fourier analysis, 473–539
 approximation by trigonometric polynomials, 495–498
 forced oscillations, 492–495
 Fourier integral, 510–517
 applications, 513–515
 complex form of, 522–523
 sine and cosine, 515–516
 Fourier series, 474–483
 convergence and sum of, 480–481
 derivation of Euler formulas, 479–480
 even and odd functions, 486–488
 half-range expansions, 488–490
 from period 2π to $2L$, 483–486
 Fourier transforms, 522–536
 complex form of Fourier integral, 522–523
 convolution, 527–528
 cosine, 518–522, 534
 discrete, 528–531
 fast, 531–532
 and its inverse, 523–524
 linearity, 526–527
 sine, 518–522, 535
 spectrum representation, 525
 orthogonal series (generalized Fourier series), 504–510
 completeness, 508–509
 mean square convergence, 507–508
 Sturm–Liouville Problems, 498–504
 eigenvalues, eigenfunctions, 499–500
 orthogonal functions, 500–503
 Fourier–Bessel series, 506–507, 589
 Fourier coefficients, 476, 484, 538, 582–583
 Fourier constants, 504–505
 Fourier cosine integral, 515–516
 Fourier cosine series, 484, 486, 538
 Fourier cosine transforms, 518–522, 534
 Fourier cosine transform method, 518
 Fourier integrals, 510–517, 539
 applications, 513–515
 complex form of, 522–523
 heat equation, 568–571
 residue integration, 729–730
 sine and cosine, 515–516
 Fourier–Legendre series, 505–506, 596–598
 Fourier matrix, 530
 Fourier series, 473–483, 538
 convergence and sum of, 480–481
 derivation of Euler formulas, 479–480
 double, 577–585
 even and odd functions, 486–488
 half-range expansions, 488–490
 heat equation, 558–563
 from period 2π to $2L$, 483–486
 Fourier sine integral, 515–516
 Fourier sine series, 477, 486, 538
 one-dimensional heat equation, 561
 vibrating string, 548
 Fourier sine transforms, 518–522, 535
 Fourier transforms, 522–536, 539
 complex form of Fourier integral, 522–523
 convolution, 527–528
 cosine, 518–522, 534, 539
 defined, 522, 523
 discrete, 528–531
 fast, 531–532
 heat equation, 571–574
 and its inverse, 523–524
 linearity of, 526–527
 sine, 518–522, 535, 539
 spectrum representation, 525
 Fourier transform method, 524
 Four-point formulas, 841
 Fraction defective chars, 1091–1092
 Francis, J. G. F., 892
 Fredholm, Erik Ivar, 198n.7, 263n.3
 Free condition (spline interpolation), 823
 Free oscillations of mass–spring system (second-order ODEs), 62–70
 critical damping, 65, 66
 damped system, 64–65
 overdamping, 65–66
 undamped system, 63–64
 underdamping, 65, 67
 Frenet, Jean-Frédéric, 392
 Frenet formulas, 392
 Frequency (in statistics):
 absolute, 1012, 1019
 cumulative absolute, 1012
 cumulative relative, 1012
 relative class, 1012
 Frequency (of vibrating string), 547
 Frequency distributions, mean and variance of:
 expectation, 1037–1038
 moments, 1038
 transformation of, 1036–1037
 Fresnel, Augustin, 697n.4, A68n.1
 Fresnel integrals, 697, A68
 Frobenius, Georg, 180n.4
 Frobenius method, 167, 180–187, 201
 indicial equation, 181–183
 proof of, A77–A81
 typical applications, 183–185
 Frobenius norm, 861
 Fulkerson, Delbert Ray, 998n.7
 Function, of complex variable, 620–621
 Function spaces, 313
 Fundamental matrix, 139
 Fundamental period, 475
 Fundamental region (exponential function), 632
 Fundamental system, 50, 104. *See also* Basis, of solutions
 Fundamental Theorem:
 higher-order homogeneous linear ODEs, 106
 for linear systems, 288
 PDEs, 541–542
 second-order homogeneous linear ODEs, 48
 Galilei, Galileo, 16n.4
 Gamma functions, 190–191, 208
 formula for, A66–A67
 incomplete, A67
 table, A98
 GAMS (Guide to Available Mathematical Software), 789
 GAUSS, ALGORITHM, 849
 Gauss, Carl Friedrich, 186n.5, 608n.1, 1103
 Gauss distribution, 1045. *See also* Normal distributions
 Gauss “Double Ring,” 451
 Gauss elimination, 320, 849
 linear systems, 274–280, 844–852, 898
 back substitution, 274–276, 846
 elementary row operations, 277
 if infinitely many solutions exist, 278
 if no solution exists, 278–279
 operation count, 850–851
 row echelon form, 279–280
 operation count, 850–851
 Gauss integration formulas, 807, 836–838, 843
 Gauss–Jordan elimination, 302–304, 856–857
 GAUSS–SEIDEL, ALGORITHM, 860

- Gauss–Seidel iteration, 858–863, 898
- Gauss’s hypergeometric ODE, 186, 202
- Geiger, H., 1044, 1100
- Generalized Euler formula, 582
- Generalized Fourier series, *see* Orthogonal series
- Generalized solution (vibrating string), 550
- Generalized triangle inequality, 615
- General powers, 639–640, 642
- General solution:
- Bessel’s equation, 194–200
 - first-order ODEs, 6, 44
 - higher-order linear ODEs, 106, 110–111, 123
 - nonhomogeneous linear systems, 160
 - second-order linear ODEs:
 - homogeneous, 49–51, 77–78, 104
 - nonhomogeneous, 80–81
 - systems of ODEs, 131–132, 139
- Generating functions, 179, 241
- Geometric interpretation:
- partial derivatives, A70
 - scalar triple product, 373, 374
- Geometric multiplicity, 326, 878
- Geometric series, 168, 675
- Taylor series, 694
 - uniformly convergent, 698
- Gerschgorin, Semyon Aranovich, 879n.6
- Gerschgorin’s theorem, 879–881, 899
- Gibbs phenomenon, 515
- Global error, 902
- Golden Rule, 15, 24
- Gompertz model, 19
- Goodness of fit, 1096–1100
- Gosset, William Sealy, 1086n.4
- Goursat, Édouard, 654n.1
- Goursat’s proof, 654
- Gradient, A75
- fluid flow, 771
 - of a scalar field, 395–402
 - directional derivatives, 396–397
 - maximum increase, 398
 - as surface normal vector, 398–399
 - vector fields that are, 400–401
 - of a scalar function, 396, 411
 - unconstrained optimization, 952
- Gradient method, 952. *See also* Method of steepest descent
- Graphs, 970–971, 1007
- bipartite, 1001–1006, 1008
 - center of, 991
 - complete, 974
- Graphs (*Cont.*)
- complete bipartite, 1005
 - computer representation of, 972–974
 - connected, 977, 981, 984
 - diameter of, 991
 - digraphs (directed graphs), 971–974, 1007
 - computer representation of, 972–974
 - defined, 972
 - incidence matrix of, 975
 - subgraphs, 972
- Euler, 980
- forest, 987
- incidence matrix of, 975
- planar, 1005
- radius of, 991
- sparse, 974
- subgraphs, 972
- trees, 984
- vertices, 971, 977, 1007
- adjacent, 971, 977
 - central, 991
 - coloring, 1005–1006
 - double labeling of, 986
 - eccentricity of, 991
 - exposed, 1001, 1003
 - four-color theorem, 1006
 - scanning, 998
 - weighted, 976
- Graphic data representation, 1012
- Gravitation (Laplace’s equation), 593
- Gravity, acceleration of, 8
- Gravity constant, at the Earth’s surface, 63
- Greedy algorithm, 984–988
- Green, George, 433n.4
- Green’s first formula, 461, 470
- Green’s second formula, 461, 470
- Green’s theorem:
- first and second forms of, 461
 - in the plane, 433–438, 470
- Gregory, James, 816n.2
- Gregory–Newton’s (Newton’s)
- backward difference interpolation formula, 818–819
- Gregory–Newton’s (Newton’s)
- forward difference interpolation formula, 815–818
- Growth restriction, 209
- Guideposts, 827
- Guide to Available Mathematical Software (GAMS), 789
- Guldin, Habakuk, 452n.7
- Guldin’s theorem, 452n.7
- Hadamard, Jacques, 683n.1
- Half-planes:
- complex analysis, 619–620
 - mapping, 747–749
- Half-range expansions (Fourier series), 488–490, 538
- Hamilton, William Rowan, 976n.1
- Hamiltonian cycle, 976
- Hankel, Hermann, 200n.8
- Hankel functions, 200
- Harmonic conjugate function (Laplace’s equation), 629
- Harmonic functions, 460, 462, 758
- complex analysis, 628–629
 - under conformal mapping, 763
 - defined, 758
 - Laplace’s equation, 593, 628–629
 - maximum modulus theorem, 783–784
 - potential theory, 781–784, 786
- Harmonic oscillation, 63–64
- Heat equation, 459–460, 557–558
- Dirichlet problem, 564–566
 - Laplace’s equation, 564
 - numeric analysis, 936–941, 948
 - Crank–Nicolson method, 938–941
 - explicit method, 937, 940–941
 - one-dimensional, 559
 - solution:
 - by Fourier integrals, 568–571
 - by Fourier series, 558–563
 - by Fourier transforms, 571–574
 - steady two-dimensional heat problems, 546–566
 - two-dimensional, 564–566
 - unifying power of methods, 566
- Heat flow:
- Laplace’s equation, 593
 - potential theory, 767–770
- Heat flow lines, 767
- Heaviside, Oliver, 204n.1
- Heaviside calculus, 204n.1
- Heaviside expansions, 228
- Heaviside function, 217–219
- Helix, 386
- Henry, Joseph, 93n.7
- Hermite, Charles, 510n.8
- Hermite interpolation, 826
- Hermitian form, 351
- Hermitian matrices, 347, 348, 350, 353
- Hertz, Heinrich, 63n.3
- Hesse, Ludwig Otto, 366n.2
- Hesse’s normal form, 366
- Heun, Karl, 905n.1
- Heun’s method, 903. *See also* Improved Euler’s method
- Higher functions, 167. *See also* Special functions

- Higher-order linear ODEs, 105–123
 - homogeneous, 105–116, 123
 - nonhomogeneous, 116–123
 - systems of, *see* Systems of ODEs
- Higher order ODEs (numeric analysis), 915–922
 - Euler method, 916–917
 - Runge–Kutta methods, 917–919
 - Runge–Kutta–Nyström methods, 919–921
- Higher transcendental functions, 920
- High-frequency line equations, 600
- Hilbert, David, 198n.7, 312n.4
- Hilbert spaces, 363
- Histograms, 1012
- Holes, of domains, 653
- Homogeneous first-order linear ODEs, 28
- Homogeneous higher-order linear ODEs, 105–111
- Homogeneous linear systems, 138, 165, 272, 290–291, 845
 - constant-coefficient systems, 140–151
 - matrices and vectors, 124–130, 321
 - trivial solution, 290
- Homogeneous PDEs, 541
- Homogeneous second-order linear ODEs, 46–48
 - basis, 50–52
 - with constant coefficients, 53–60
 - complex roots, 57–59
 - real double root, 55–56
 - two distinct real-roots, 54–55
 - differential operators, 60–62
 - Euler–Cauchy equations, 71–74
 - existence and uniqueness of solutions, 74–79
 - general solution, 49–51, 77–78
 - initial value problem, 49–50
 - modeling free oscillations of mass–spring system, 62–70
 - particular solution, 49–51
 - reduction of order, 51–52
 - Wronskian, 75–78
- Hooke, Robert, 62
- Hooke’s law, 62
- Householder, Alston Scott, 888n.11
- Householder’s tridiagonalization method, 888–892
- Hyperbolic analytic functions (conformal mapping), 750–754
- Hyperbolic cosine, 635, 752
- Hyperbolic functions, 635, 642
 - formula for, A65–A66
 - inverse, 640
 - Taylor series, 695
- Hyperbolic PDEs:
 - defined, 923
 - numeric analysis, 942–945
- Hyperbolic sine, 635, 752
- Hypergeometric distributions, 1042–1044, 1061
- Hypergeometric equations, 167, 185–187
- Hypergeometric functions, 167, 186
- Hypergeometric series, 186
- Hypothesis, 1077
- Hypothesis testing (in statistics), 1063, 1077–1087
 - comparison of means, 1084–1085
 - comparison of variances, 1086
 - errors in tests, 1080–1081
 - for mean of normal distribution
 - with known variance, 1081–1083
 - for mean of normal distribution
 - with unknown variance, 1083–1084
 - one- and two-sided alternatives, 1079–1080
- Idempotent matrices, 270
- Identity mapping, 745
- Identity matrices, 268
- Identity operator (second-order homogeneous linear ODEs), 60
- Ill-conditioned equations, 805
- Ill-conditioned problems, 864
- Ill-conditioned systems, 864, 865, 899
- Ill-conditioning (linear systems), 864–872
 - condition number of a matrix, 868–870
 - matrix norms, 866–868
 - vector norms, 866
- Image:
 - conformal mapping, 737
 - linear transformations, 313
- Imaginary axis (complex plane), 611
- Imaginary part (complex numbers), 609
- Imaginary unit, 609
- Impedance (RLC circuits), 95
- Implicit formulas, 913
- Implicit method:
 - backward Euler scheme as, 909
 - for hyperbolic PDEs, 943
- Implicit solution, 21
- Improper integrals:
 - defined, 205
 - residue integration, 726–732
- Improper node, 142
- Improved Euler’s method:
 - error of, 904, 906, 908
 - first-order ODEs, 902–904
- Impulse, of a force, 225
 - short impulses, 225–226
 - unit impulse function, 226
- Incidence matrices (graphs and digraphs), 975
- Incident edges, 971
- Inclusion theorems:
 - defined, 882
 - matrix eigenvalue problems, 879–884
- Incomplete gamma functions, formula for, A67
- Inconsistent linear systems, 277
- Indefinite (quadratic form), 346
- Indefinite integrals:
 - defined, 643
 - existence of, 656–658
- Indefinite integration (complex line integral), 646–647
- Independence:
 - of path, 669
 - of path in domain (integrals), 470, 655
 - of random variables, 1055–1056
- Independent events, 1022–1023, 1061
- Independent sample values, 1064
- Independent variables:
 - in calculus, 393
 - in regression analysis, 1103
- Indicial equation, 181–183, 188, 202
- Indirect methods (solving linear systems), 858, 898
- Inference, statistical, 1059, 1063
- Infinite dimensional vector space, 311
- Infinite populations, 1044
- Infinite sequences:
 - bounded, A93–A95
 - monotone real, A72–A73
 - power series, 671–673
- Infinite series, 673–674
- Infinity:
 - analytic of singular at, 718–719
 - point at, 718
- Initial conditions:
 - first-order ODEs, 6, 7, 44
 - heat equation, 559, 568, 569
 - higher-order linear ODEs:
 - homogeneous, 107
 - nonhomogeneous, 117
 - one-dimensional heat equation, 559
 - PDEs, 541, 605
 - second-order homogeneous linear ODEs, 49–50, 104
 - systems of ODEs, 137
 - two-dimensional wave equation, 577
 - vibrating string, 545
- Initial point (vectors), 355
- Initial value problem (IVP):
 - defined, 6
 - first-order ODEs, 6, 39, 44, 901

- Initial value problem (IVP): (*Cont.*)
 bell-shaped curve, 13
 existence and uniqueness of
 solutions for, 38–43
 higher-order linear ODEs, 123
 homogeneous, 107–108
 nonhomogeneous, 117
 Laplace transforms, 213–216
 for RLC circuit, 99
 second-order homogeneous linear
 ODEs, 49, 74–75, 104
 systems of ODEs, 137
- Injective mapping, 737n.1
- Inner product (dot product), 312
 for complex vectors, 349
 invariance of, 336
 vector differential calculus,
 361–367, 410
 applications, 364–366
 orthogonality, 361–363
- Inner product spaces, 311–313
- Input (driving force), 27, 86, 214
- Instability, numeric vs. mathematical,
 796
- Integrals, *see* Line integrals
- Integral equations:
 defined, 236
 Laplace transforms, 236–237
- Integral of a function, Laplace
 transforms of, 212–213
- Integral transforms, 205, 518
- Integrand, 414, 644
- Integrating factors, 23–26, 45
 defined, 24
 finding, 24–26
- Integration. *See also* Complex
 integration
 constant of, 18
 of Laplace transforms, 238–240
 numeric, 827–838
 adaptive, 835–836
 Gauss integration formulas,
 836–838
 rectangular rule, 828
 Simpson's rule, 831–835
 trapezoidal rule, 828–831
 termwise, of power series, 687,
 688
- Intermediate value theorem, 807–808
- Intermediate variables, 393
- Intermittent harvesting, 36
- INTERPOL, ALGORITHM, 814
- Interpolation, 529
 defined, 808
 numeric analysis, 808–820, 842
 equal spacing, 815–819
 Lagrange, 809–812
 Newton's backward difference
 formula, 818–819
 Newton's divided difference,
 812–815
- Interpolation (*Cont.*)
 Newton's forward difference
 formula, 815–818
 spline, 820–827
- Interpolation polynomial, 808, 842
- Interquartile range, 1013
- Intersection, of events, 1016, 1017
- Intervals. *See also* Confidence
 intervals
 class, 1012
 closed, A72n.3
 convergence, 171, 683
 open, 4, A72n.3
- Interval estimates, 1065
- Invariance, of curl, A85–A88
- Invariant rank, 283
- Invariant subspace, 878
- Inverse cosine, 640
- Inverse cotangent, 640
- Inverse Fourier cosine transform, 518
- Inverse Fourier sine transform, 519
- Inverse Fourier sine transform
 method, 519
- Inverse Fourier transform, 524
- Inverse hyperbolic function, 640
- Inverse hyperbolic sine, 640
- Inverse mapping, 741, 745
- Inverse of a matrix, 128, 301–309,
 321
 cancellation laws, 306–307
 determinants of matrix
 products, 307–308
 formulas for, 304–306
 Gauss–Jordan method,
 302–304, 856–857
- Inverse sine, 640
- Inverse tangent, 640
- Inverse transform, 205, 253
- Inverse transformation, 315
- Inverse trigonometric function, 640
- Irreducible, 883
- Irregular boundary (elliptic PDEs),
 933–935
- Irrotational flow, 774
- Isocline, 10
- Isolated critical point, 152
- Isolated essential singularity, 715
- Isolated singularity, 715
- Isotherms, 36, 38, 402, 767
- Iteration (iterative) methods:
 numeric analysis, 798–808
 fixed-point iteration, 798–801
 Newton's (Newton–Raphson)
 method, 801–805
 secant method, 805–806
 speed of convergence, 804–805
 numeric linear algebra, 858–864,
 898
 Gauss–Seidel iteration, 858–862
 Jacobi iteration, 862–863
- IVP, *see* Initial value problem
- Jacobi, Carl Gustav Jacob, 430n.3
- Jacobians, 430, 741
- Jacobi iteration, 862–863
- Jordan, Wilhelm, 302n.3
- Joukowski airfoil, 739–740
- Kantorovich, Leonid Vitaliyevich,
 959n.1
- KCL (Kirchhoff's Current Law),
 93n.7, 274
- Kernel, 205
- Kinetic friction, coefficient of, 19
- Kirchhoff, Gustav Robert, 93n.7
- Kirchhoff's Current Law (KCL),
 93n.7, 274
- Kirchhoff's law, 991
- Kirchhoff's Voltage Law (KVL), 29,
 93, 274
- Koopmans, Tjalling Charles, 959n.1
- Kreyszig, Erwin, 855n.3
- Kronecker, Leopold, 500n.5
- Kronecker delta, A85
- Kronecker symbol, 500
- Kruskal, Joseph Bernard, 985n.5
- KRUSKAL, ALGORITHM, 985
- Kruskal's Greedy algorithm,
 984–988, 1008
- k th backward difference, 818
- k th central moment, 1038
- k th divided difference, 813
- k th forward difference, 815–816
- k th moment, 1038, 1065
- Kublanovskaya, V. N., 892
- Kutta, Wilhelm, 905n.1
- Kutta's third-order method, 911
- KVL, *see* Kirchhoff's Voltage Law
- Lagrange, Joseph Louis, 51n.1
- Lagrange interpolation, 809–812
- Lagrange's form, 812, 842
- Laguerre, Edmond, 504n.7
- Laguerre polynomials, 241, 504
- Laguerre's equation, 240–241
- LAPACK, 789
- Laplace, Pierre Simon Marquis de,
 204n.1
- Laplace equation, 400, 564, 593–600,
 642, 923
 boundary value problem in
 spherical coordinates,
 594–596
 complex analysis, 628–629
 in cylindrical coordinates,
 593–594
 Fourier–Legendre series, 596–598
 heat equation, 564
 numeric analysis, 922–936, 948
 ADI method, 928–930
 difference equations, 923–925

- Laplace equation (*Cont.*)
 Dirichlet problem, 925–928, 934–935
 Liebmann's method, 926–928
 in spherical coordinates, 594
 theory of solutions of, 460, 786.
See also Potential theory
 two-dimensional heat equation, 564
 two-dimensional problems, 759
 uniqueness theorem for, 462
- Laplace integrals, 516
- Laplace operator, 401. *See also* Laplacian
- Laplace transforms, 203–253
 convolution, 232–237
 defined, 204, 205
 of derivatives, 211–212
 differentiation of, 238–240
 Dirac delta function, 226–228
 existence, 209–210
 first shifting theorem (*s*-shifting), 208–209
 general formulas, 248
 initial value problems, 213–216
 integral equations, 236–237
 of integral of a function, 212–213
 integration of, 238–240
 linearity of, 206–208
 notation, 205
 ODEs with variable coefficients, 240–241
 partial differential equations, 600–603
 partial fractions, 228–230
 second shifting theorem (*t*-shifting), 219–223
 short impulses, 225–226
 systems of ODEs, 242–247
 table of, 249–251
 uniqueness, 210
 unit step function (Heaviside function), 217–219
- Laplacian, 400, 463, 605, A76
 in cylindrical coordinates, 593–594
 heat equation, 557
 Laplace's equation, 593
 in polar coordinates, 585–592
 in spherical coordinates, 594
 of *u* in polar coordinates, 586
- Lattice points, 925–926
- Laurent, Pierre Alphonse, 708n.1
- Laurent series, 708–719, 734
 analytic or singular at infinity, 718–719
 point at infinity, 718
 Riemann sphere, 718
 singularities, 715–717
 zeros of analytic functions, 717
- Laurent's theorem, 709
- LCL (lower control limit), 1088
- Least squares approximation, of a function, 875–876
- Least squares method, 872–876, 899
- Least squares principle, 1103
- Lebesgue, Henri, 876n.5
- Left-handed Cartesian coordinate system, 369, 370, A84
- Left-hand limit (Fourier series), 480
- Left-sided tests, 1079, 1082
- Legendre, Adrien-Marie, 175n.1, 1103
- Legendre function, 175
- Legendre polynomials, 167, 177–179, 202
- Legendre's equation, 167, 175–177, 201, 202
 Laplace's equation, 595–596
 special, 169–170
- Leibniz, Gottfried Wilhelm, 15n.3
- Leibniz test for real series, A73–A74
- Length:
 curves, 385
 vectors, 355, 356, 410
- Leonardo of Pisa, 690n.2
- Leontief, Wassily, 334n.1
- Leontief input–output model, 334
- Leslie model, 331
- Level surfaces, 380, 398
- LFTs, *see* Linear fractional transformations
- Libby, Willard Frank, 13n.2
- Liebmann's method, 926–928
- Likelihood function, 1066
- Limit (sequences), 672
- Limit cycle, 158–159, 621
- Limit *l*, 378
- Limit point, A93
- Limit vector, 378
- Linear algebra, 255. *See also*
 Numeric linear algebra
 determinants, 293–301
 Cramer's rule, 298–300
 general properties of, 295–298
 of matrix products, 307–308
 second-order, 291–292
 third-order, 292–293
 inverse of a matrix, 301–309
 cancellation laws, 306–307
 determinants of matrix products, 307–308
 formulas for, 304–306
 Gauss–Jordan method, 302–304
 linear systems, 272–274
 back substitution, 274–276
 elementary row operations, 277
 Gauss elimination, 274–280
 homogeneous, 290–291
- Linear algebra (*Cont.*)
 nonhomogeneous, 291
 solutions of, 288–291
 matrices and vectors, 257–262
 addition and scalar multiplication of, 259–261
 diagonal matrices, 268
 linear independence and dependence of vectors, 282–283
 matrix multiplication, 263–266, 269–279
 notation, 258
 rank of, 283–285
 symmetric and skew-symmetric matrices, 267–268
 transposition of, 266–267
 triangular matrices, 268
 matrix eigenvalue problems, 322–353
 applications, 329–334
 complex matrices and forms, 346–352
 determining eigenvalues and eigenvectors, 323–329
 diagonalization of matrices, 341–342
 eigenbases, 339–341
 orthogonal matrices, 337–338
 orthogonal transformations, 336
 quadratic forms, 343–344
 symmetric and skew-symmetric matrices, 334–336
 transformation to principal axes, 344
 vector spaces:
 inner product spaces, 311–313
 linear transformations, 313–317
 real, 309–311
 special, 285–287
- Linear combination:
 homogeneous linear ODEs:
 higher-order, 107
 second-order, 48
 of matrices, 129, 271
 of vectors, 129, 282
 of vectors in vector space, 311
- Linear dependence, of vectors, 282–283
- Linear element, 386
- Linear equations, systems of, *see* Linear systems
- Linear fractional transformations (LFTs), 742–750, 757
 extended complex plane, 744–745
 mapping standard domains, 747–750

- Linear independence:
 - scalar triple product, 373
 - of vectors, 282–283
- Linear inequalities, 954
- Linear interpolation, 809–810
- Linearity:
 - Fourier transforms, 526–527
 - Laplace transforms, 206–208
 - line integrals, 645
- Linearity principle, *see* Superposition principle
- Linearization, 152–155
- Linearized system, 153
- Linearly dependent functions:
 - higher-order homogeneous linear ODEs, 106, 109
 - second-order homogeneous linear ODEs, 50, 75
- Linearly dependent sets, 129, 311
- Linearly dependent vectors, 282–283, 285
- Linearly independent functions:
 - higher-order homogeneous linear ODEs, 106, 109, 113
 - second-order homogeneous linear ODEs, 50, 75
- Linearly independent sets, 128–129, 311
- Linearly independent vectors, 282–283
- Linearly related variables, 1109
- Linear mapping, 314. *See also* Linear transformations
- Linear ODEs, 45, 46
 - first order, 27–36
 - Bernoulli equation, 31–33
 - homogeneous, 28
 - nonhomogeneous, 28–29
 - population dynamics, 33–34
 - higher-order, 105–123
 - homogeneous, 105–116
 - nonhomogeneous, 116–122
 - higher-order homogeneous, 105
 - second-order, 46–104
 - homogeneous, 46–78, 103
 - nonhomogeneous, 79–102, 103
- Linear operations:
 - Fourier cosine and sine transforms as, 520
 - integration as, 645
- Linear operators (second-order homogeneous linear ODEs), 61
- Linear optimization, *see* Constrained (linear) optimization
- Linear PDEs, 541
- Linear programming problems, 954–958
 - normal form of problems, 955–957
 - simplex method, 958–968
 - degenerate feasible solution, 962–965
 - difficulties in starting, 965–968
- Linear systems, 138–139, 165, 272–274, 320, 845
 - back substitution, 274–276
 - defined, 267, 845
 - elementary row operations, 277
 - Gauss elimination, 274–280, 844–852
 - applications, 277–180
 - back substitution, 274–276
 - elementary row operations, 277
 - operation count, 850–851
 - row echelon form, 279–280
 - Gauss–Jordan elimination, 856–857
 - homogeneous, 138, 165, 272, 290–291
 - constant-coefficient systems, 140–151
 - matrices and vectors, 124–130
 - ill-conditioning, 864–872
 - condition number of a matrix, 868–870
 - matrix norms, 866–868
 - vector norms, 866
 - iterative methods, 858–864
 - Gauss–Seidel iteration, 858–882
 - Jacobi iteration, 862–863
 - LU-factorization, 852–855
 - Cholesky’s method, 855–856
 - of m equations in n unknowns, 272
 - nonhomogeneous, 138, 160–163, 272, 290, 291
 - solutions of, 288–291, 898
- Linear transformations, 320
 - motivation of multiplication by, 265–266
 - vector spaces, 313–317
- Line integrals, 643–652, 669
 - basic properties of, 645
 - bounds for, 650–651
 - definition of, 414, 643–645
 - existence of, 646
 - indefinite integration and substitution of limits, 646–647
 - path dependence of, and integration around closed curves, 421–425
 - representation of a path, 647–650
 - vector integral calculus, 413–419
 - definition and evaluation of, 414–416
 - path dependence of, 418–426
 - work done by a force, 416–417
- Lines of constant revenue, 954
- Lines of force, 760–762
- LINPACK, 789
- Liouville, Joseph, 499n.4
- Liouville’s theorem, 666–667
- Lipschitz, Rudolf, 42n.9
- Lipschitz condition, 42
- Ljapunov, Alexander Michailovich, 149n.2
- Local error, 830
- Local maximum (unconstrained optimization), 952
- Local minimum (unconstrained optimization), 951
- Local truncation error, 902
- Logarithm, 636–639
 - natural, 636–638, 642, A63
 - Taylor series, 695
- Logarithmic decrement, 70
- Logarithmic integral, formula for, A69
- Logarithm of base ten, formula for, A63
- Logistic equation, 32–33
- Longest path, 976
- Loss of significant digits (numeric analysis), 793–794
- Lotka, Alfred J., 155n.3
- Lotka–Volterra population model, 155–156
- Lot tolerance percent defective (LTPD), 1094
- Lower confidence limits, 1068
- Lower control limit (LCL), 1088
- Lower triangular matrices, 268
- LTPD (lot tolerance percent defective), 1094
- LU-factorization (linear systems), 852–855
- Machine numbers, 792
- Maclaurin, Colin, 690n.2, 712
- Maclaurin series, 690, 694–696
- Main diagonal:
 - determinants, 294
 - matrix, 125, 258
- Malthus, Thomas Robert, 5n.1
- Malthus’ law, 5, 33
- Maple, 789
- Maple Computer Guide, 789
- Mapping, 313, 736, 737, 757
 - bijjective, 737n.1
 - conformal, 736–757
 - boundary value problems, 763–767, A96
 - defined, 738
 - geometry of analytic functions, 737–742
 - linear fractional transformations, 742–750
- Riemann surfaces, 754–756
- by trigonometric and hyperbolic analytic functions, 750–754

- Mapping (*Cont.*)
 of disks, 748–750
 fixed points of, 745
 of half-planes onto half-planes, 748
 identity, 745
 injective, 737n.1
 inverse, 741, 745
 linear, 314. *See also* Linear transformations
 one-to-one, 737n.1
 spectral mapping theorem, 878
 surjective, 737n.1
- Marconi, Guglielmo, 63n.3
- Marginal distributions, 1053–1055, 1062
 of continuous distributions, 1055
 of discrete distributions, 1053–1054
- Mariotte, Edme, 19n.5
- Markov, Andrei Andrejevitch, 270n.1
- Markov process, 270, 331
- MATCHING, ALGORITHM, 1003
- Matching, 1008
 assignment problems, 1001
 complete, 1002
 maximum cardinality, 1001, 1008
- Mathcad, 789
- Mathematica, 789
- Mathematica Computer Guide, 789
- Mathematical models, *see* Models
- Mathematical modeling, *see* Modeling
- Mathematical statistics, 1009, 1063–1113
 acceptance sampling, 1092–1096
 errors in, 1093–1094
 rectification, 1094–1095
 confidence intervals, 1068–1077
 for mean of normal distribution with known variance, 1069–1071
 for mean of normal distribution with unknown variance, 1071–1073
 for parameters of distributions other than normal, 1076
 for variance of a normal distribution, 1073–1076
 correlation analysis, 1108–1111
 defined, 1103
 test for correlation coefficient, 1110–1111
 defined, 1063
 goodness of fit, 1096–1100
 hypothesis testing, 1077–1087
 comparison of means, 1084–1085
 comparison of variances, 1086
 errors in tests, 1080–1081
 for mean of normal distribution with known variance, 1081–1083
 for mean of normal distribution with unknown variance, 1083–1084
 one- and two-sided alternatives, 1079–1080
 main purpose of, 1015
 nonparametric tests, 1100–1102
 point estimation of parameters, 1065–1068
 quality control, 1087–1092
 for mean, 1088–1089
 for range, 1090–1091
 for standard deviation, 1090
 for variance, 1089–1090
 random sampling, 1063–1065
 regression analysis, 1103–1108
 confidence intervals in, 1107–1108
 defined, 1103
- Matlab, 789
- Matrices, 124–130, 256–262, 320
 addition and scalar multiplication of, 259–261
 calculations with, 126–127
 condition number of, 868–870
 definitions and terms, 125–126, 128, 257
 diagonal, 268
 diagonalization of, 341–342
 eigenvalues, 129–130
 equality of, 126, 259
 fundamental, 139
 inverse of, 128, 301–309, 321
 cancellation laws, 306–307
 determinants of matrix products, 307–308
 formulas for, 304–306
 Gauss–Jordan method, 302–304, 856–857
 matrix multiplication, 127, 263–266, 269–279
 applications of, 269–279
 cancellation laws, 306–307
 determinants of matrix products, 307–308
 scalar, 259–261
 normal, 352, 882
 notation, 258
 orthogonal, 337–338
 rank of, 283–285
 square, 126
 symmetric and skew-symmetric, 267–268
 transposition of, 266–267
 triangular, 268
 unitary, 347–350, 353
- Matrix eigenvalue problems, 322–353, 876–896
 applications, 329–334
 choice of numeric method for, 879
 complex matrices and forms, 346–352
 determining eigenvalues and eigenvectors, 323–329
 diagonalization of matrices, 341–342
 eigenbases, 339–341
 inclusion theorems, 879–884
 orthogonal matrices, 337–338
 orthogonal transformations, 336
 power method, 885–888
 QR-factorization, 892–896
 quadratic forms, 343–344
 symmetric and skew-symmetric matrices, 334–336
 transformation to principal axes, 344
 tridiagonalization, 888–892
- Matrix multiplication, 127, 263–266, 269–279
 applications of, 269–279
 cancellation laws, 306–307
 determinants of matrix products, 307–308
 scalar, 259–261
- Matrix norms, 861, 866–868
- Maximum cardinality matching, 1001, 1003–1004, 1008
- Maximum flow:
 Ford–Fulkerson algorithm, 998–1000
 and minimum cut set, 996
- Maximum increase:
 gradient of a scalar field, 398
 unconstrained optimization, 951
- Maximum likelihood estimates (MLEs), 1066–1067
- Maximum likelihood method, 1066–1067, 1113
- Maximum modulus theorem, 782–784
- Maximum principle, 783
- Mean(s), 1013–1014, 1061
 comparison of, 1084–1085
 control chart for, 1088–1089
 of normal distributions:
 confidence intervals for, 1069–1073
 hypothesis testing for, 1081–1084
 probability distributions, 1035–1039
 addition of, 1057–1058
 transformation of, 1036–1037
 sample, 1064

- Mean square convergence (orthogonal series), 507–508
- Mean value (fluid flow), 774n.1
- Mean value property:
analytic functions, 781–782
harmonic functions, 782
- Mean value theorem, 395
for double integrals, 427
for surface integrals, 448
for triple integrals, 456–457
- Median, 1013, 1100–1101
- Mendel, Gregor, 1100
- Meromorphic function, 719
- Mesh incidence matrix, 262
- Mesh points (lattice points, nodes), 925–926
- Mesh size, 924
- Method of characteristics (PDEs), 555
- Method of least squares, 872–876, 899
- Method of moments, 1065
- Method of separating variables, 12–13
circular membrane, 587
partial differential equations, 545–553, 605
Fourier series, 548–551
satisfying boundary conditions, 546–548
two ODEs from wave equation, 545–546
vibrating string, 545–546
- Method of steepest descent, 952–954
- Method of undetermined coefficients:
higher-order homogeneous linear ODEs, 115, 123
nonhomogeneous linear systems of ODEs, 161
second-order nonhomogeneous linear ODEs, 81–85, 104
- Method of variation of parameters:
higher-order nonhomogeneous linear ODEs, 118–120, 123
nonhomogeneous linear systems of ODEs, 162–163
second-order nonhomogeneous linear ODEs, 99–102, 104
- Minimization (normal form of linear optimization problems), 957
- Minimum (unconstrained optimization), 951
- Minimum cut set, 996
- Minors, of determinants, 294
- Mixed boundary condition (two-dimensional heat equation), 564
- Mixed boundary value problem, 605, 923. *See also* Robin problem
elliptic PDEs, 931–933
heat conduction, 768–769
- Mixed type PDEs, 555
- Mixing problems, 14
- MLEs (maximum likelihood estimates), 1066–1067
- ML*-inequality, 650–651
- Möbius, August Ferdinand, 447n.5
- Möbius strip, 447
- Möbius transformations, 743. *See also* Linear fractional transformations (LFTs)
- Models, 2
- Modeling, 1, 2–8, 44
and concept of solution, 4–6
defined, 2
first-order ODEs, 2–8
initial value problem, 6
separable ODEs, 13–17
typical steps of, 6–7
and unifying power of mathematics, 766
- Modification Rule (method of undetermined coefficients):
higher-order homogeneous linear ODEs, 115–116
second-order nonhomogeneous linear ODEs, 81, 83
- Modulus (complex numbers), 613
- Moments, method of, 1065
- Moments of inertia, of a region, 429
- Moment vector (vector moment), 371
- Monotone real sequences, A72–A73
- Moore, Edward Forrest, 977n.2
- MOORE, ALGORITHM, 977
- Moore's BFS algorithm, 977–980, 1008
- Morera's theorem, 667
- Moulton, Forest Ray, 913n.3
- Multinomial distribution, 1045
- Multiple complex roots, 115
- Multiple points, curves with, 383
- Multiplication:
of complex numbers, 609, 610, 615
in conditional probability, 1022–1023
matrix, 127, 263–266
applications of, 269–279
cancellation laws, 306–307
determinants of matrix products, 307–308
scalar, 259–261
of means, 1057–1058
of power series, 687
scalar, 126–127, 259–261, 310
termwise, 173, 687
of transforms, 232. *See also* Convolution
- Multiplicity, algebraic, 326, 878
- Multiply connected domains, 652, 653
Cauchy's integral formula, 662–663
Cauchy's integral theorem, 658–659
- Multistep methods, 911–915, 947
Adams–Bashforth methods, 911–914
Adams–Moulton methods, 913–914
defined, 908
first-order ODEs, 911
- Mutually exclusive events, 1016, 1021
- $m \times n$ matrix, 258
- Nabla, 396
- NAG (Numerical Algorithms Group, Inc.), 789
- National Institute of Standards and Technology (NIST), 789
- Natural condition (spline interpolation), 823
- Natural frequency, 63
- Natural logarithm, 636–638, 642, A63
- Natural spline, 823
- n -dimensional vector spaces, 311
- Negative (scalar multiplication), 260
- Negative definite (quadratic form), 346
- Neighborhood, 619, 720
- Net flow, through cut set, 994–995
- NETLIB, 789
- Networks:
defined, 991
flow problems in, 991–997
cut sets, 994–996
flow augmenting paths, 992–993
paths, 992
- Neumann, Carl, 198n.7
- Neumann, John von, 959n.1
- Neumann boundary condition, 564
- Neumann problem, 605, 923
elliptic PDEs, 931
Laplace's equation, 593
two-dimensional heat equation, 564
- Neumann's function, 198
- NEWTON, ALGORITHM, 802
- Newton, Sir Isaac, 15n.3
- Newton–Cotes formulas, 833, 843
- Newton's (Gregory–Newton's)
backward difference
interpolation formula, 818–819
- Newton's divided difference
interpolation, 812–815, 842

- Newton's divided difference interpolation formula, 814–815
- Newton's (Gregory–Newton's) forward difference interpolation formula, 815–818, 842
- Newton's law of cooling, 15–16
- Newton's law of gravitation, 377
- Newton's (Newton–Raphson) method, 801–805, 842
- Newton's second law, 11, 63, 245, 544, 576
- Neyman, Jerzy, 1068n.1, 1077n.2
- Nicolson, Phyllis, 938n.5
- Nicomedes, 391n.4
- Nilpotent matrices, 270
- NIST (National Institute of Standards and Technology), 789
- Nodal incidence matrix, 262
- Nodal lines, 580–581, 588
- Nodes, 165, 925–926
 - degenerate, 145–146
 - improper, 142
 - interpolation, 808
 - proper, 143
 - spline interpolation, 820
 - trapezoidal rule, 829
 - vibrating string, 547
- Nonbasic variables, 960
- Nonconservative physical systems, 422
- Nonhomogeneous linear ODEs:
 - convolution, 235–236
 - first-order, 28–29
 - higher-order, 106, 116–122
 - second-order, 79–102
 - defined, 47
 - method of undetermined coefficients, 81–85
 - modeling electric circuits, 93–99
 - modeling forced oscillations, 85–92
 - particular solution, 80
 - solution by variation of parameters, 99–102
- Nonhomogeneous linear systems, 138, 160–163, 166, 272, 290, 291, 845
 - method of undetermined coefficients, 161
 - method of variation of parameters, 162–163
- Nonhomogeneous PDEs, 541
- Nonlinear ODEs, 46
 - first-order, 27
 - higher-order homogeneous, 105
 - second-order, 46
- Nonlinear PDEs, 541
- Nonlinear systems, qualitative
 - methods for, 152–160
 - linearization, 152–155
 - Lotka–Volterra population model, 155–156
 - transformation to first-order equation in phase plane, 157–159
- Nonparametric tests (statistics), 1100–1102, 1113
- Nonsingular matrices, 128, 301
- Norm(s):
 - matrix, 861, 866–868
 - orthogonal functions, 500
 - vector, 312, 355, 410, 866
- Normal accelerations, 391
- Normal acceleration vector, 387
- Normal derivative, 437
 - defined, 437
 - mixed problems, 768, 931
 - Neumann problems, 931
 - solutions of Laplace's equation, 460
- Normal distributions, 1045–1051, 1062
 - as approximation of binomial distribution, 1049–1050
 - confidence intervals:
 - for means of, 1069–1073
 - for variances of, 1073–1076
 - distribution function, 1046–1047
 - means of:
 - confidence intervals for, 1069–1073
 - hypothesis testing for, 1081–1084
 - numeric values, 1047–1048
 - tables, A101–A102
 - two-dimensional, 1110
 - working with normal tables, 1048–1049
- Normal equations, 873, 1105–1106
- Normal form (linear optimization problems), 955–957, 959, 969
- Normalizing, eigenvectors, 326
- Normal matrices, 352, 882
- Normal mode:
 - circular membrane, 588
 - vibrating string, 547–548
- Normal plane, 390
- Normal random variables, 1045
- Normal vectors, 366, 441
- Not rejecting a hypothesis, 1081
- No trend hypothesis, 1101
- n th order linear ODEs, 105, 123
- n th-order ODEs, 134–135
- n th partial sum, 170
 - Fourier series, 495
 - of series, 673
- n th roots, 616
- n th roots of unity, 617
- Null hypothesis, 1078
- Nullity, 287, 291
- Null space, 287, 291
- Numbers:
 - acceptance, 1092
 - Bernoulli's law of large numbers, 1051
 - chromatic, 1006
 - complex, 608–619, 641
 - addition of, 609, 610
 - conjugate, 612
 - defined, 608
 - division of, 610
 - multiplication of, 609, 610
 - polar form of, 613–618
 - subtraction of, 610
 - condition, 868–870, 899
 - Fibonacci, 690
 - floating-point form of, 791–792
 - machine, 792
 - random, 1064
- Number of degrees of freedom, 1071, 1074
- Numerics, *see* Numeric analysis
- Numerical Algorithms Group, Inc. (NAG), 789
- Numerically stable algorithms, 796, 842
- Numerical Recipes, 789
- Numeric analysis (numerics), 787–843
 - algorithms, 796
 - basic error principle, 796
 - error propagation, 795
 - errors of numeric results, 794–795
 - floating-point form of numbers, 791–792
 - interpolation, 808–820
 - equal spacing, 815–819
 - Lagrange, 809–812
 - Newton's backward difference formula, 818–819
 - Newton's divided difference, 812–815
 - Newton's forward difference formula, 815–818
 - spline, 820–827
 - loss of significant digits, 793–794
 - numeric differentiation, 838–839
 - numeric integration, 827–838
 - adaptive, 835–836
 - Gauss integration formulas, 836–838
 - rectangular rule, 828
 - Simpson's rule, 831–835
 - trapezoidal rule, 828–831
 - for ODEs, 901–922
 - first-order, 901–915
 - higher order, 915–922

- numeric integration (*Cont.*)
 - for PDEs, 922–945
 - elliptic, 922–936
 - hyperbolic, 942–945
 - parabolic, 936–942
 - roundoff, 792–793
 - software for, 788–789
 - solution of equations by iteration, 798–808
 - fixed-point iteration, 798–801
 - Newton's (Newton–Raphson) method, 801–805
 - secant method, 805–806
 - speed of convergence, 804–805
 - spline interpolation, 820–827
- Numeric differentiation, 838–839
- Numeric integration, 827–838
 - adaptive, 835–836
 - Gauss integration formulas, 836–838
 - rectangular rule, 828
 - Simpson's rule, 831–835
 - trapezoidal rule, 828–831
- Numeric linear algebra, 844–899
 - curve fitting, 872–876
 - least squares method, 872–876
 - linear systems, 845
 - Gauss elimination, 844–852
 - Gauss–Jordan elimination, 856–857
 - ill-conditioning norms, 864–872
 - iterative methods, 858–864
 - LU-factorization, 852–855
 - matrix eigenvalue problems, 876–896
 - inclusion theorems, 879–884
 - power method, 885–888
 - QR-factorization, 892–896
 - tridiagonalization, 888–892
- Numeric methods:
 - choice of, 791, 879
 - defined, 791
- $n \times n$ matrix, 125
- Nyström, E. J., 919
- Objective function, 951, 969
- OCs (operating characteristics), 1081
- OC curve, *see* Operating characteristic curve
- Odd functions, 486–488
- Odd periodic extension, 488–490
- ODEs, *see* Ordinary differential equations
- Ohm, Georg Simon, 93n.7
- Ohm's law, 29
- One-dimensional heat equation, 559
- One-dimensional wave equation, 544–545
- One-parameter family of curves, 36–37
- One-sided alternative (hypothesis testing), 1079–1080
- One-sided tests, 1079
- One-step methods, 908, 911, 947
- One-to-one mapping, 737n.1
- Open annulus, 619
- Open circular disk, 619
- Open integration formula, 838
- Open intervals, 4, A72n.3
- Open Leontief input–output model, 334
- Open set, in complex plane, 620
- Operating characteristic curve (OC curve), 1081, 1092, 1095
- Operating characteristics (OCs), 1081
- Operational calculus, 60, 203
- Operation count (Gauss elimination), 850
- Operators, 60–61, 313
- Optimal solutions (normal form of linear optimization problems), 957
- Optimization:
 - combinatorial, 970, 975–1008
 - assignment problems, 1001–1006
 - flow problems in networks, 991–997
 - Ford–Fulkerson algorithm for maximum flow, 998–1001
 - shortest path problems, 975–980
 - constrained (linear), 951, 954–968
 - normal form of problems, 955–957
 - simplex method, 958–968
 - unconstrained:
 - basic concepts, 951–952
 - method of steepest descent, 952–954
- Optimization methods, 949
- Optimization problems, 949, 954–958
 - normal form of problems, 955–957
 - objective, 951
 - simplex method, 958–968
 - degenerate feasible solution, 962–965
 - difficulties in starting, 965–968
- Order:
 - and complexity of algorithms, 978
 - Gauss elimination, 850
 - of iteration process, 804
 - of PDE, 540
 - singularities, 714
- Ordering (Greedy algorithm), 987
- Order statistics, 1100
- Ordinary differential equations (ODEs), 44
 - autonomous, 11, 33
 - defined, 1, 3–4
 - first-order, 2–45
 - direction fields, 9–10
 - Euler's method, 10–11
 - exact, 20–27
 - geometric meanings of, 9–12
 - initial value problem, 38–43
 - linear, 27–36
 - modeling, 2–8
 - numeric analysis, 901–915
 - orthogonal trajectories, 36–38
 - separable, 12–20
 - higher-order linear, 105–123
 - homogeneous, 105–116, 123
 - nonhomogeneous, 116–123
 - systems of, *see* Systems of ODEs
- Laplace transforms, 203–253
 - convolution, 232–237
 - defined, 204, 205
 - of derivatives, 211–212
 - differentiation of, 238–240
 - Dirac delta function, 226–228
 - existence, 209–210
 - first shifting theorem (*s*-shifting), 208–209
 - general formulas, 248
 - initial value problems, 213–216
 - integral equations, 236–237
 - of integral of a function, 212–213
 - integration of, 238–240
 - linearity of, 206–208
 - notation, 205
 - ODEs with variable coefficients, 240–241
 - partial differential equations, 600–603
 - partial fractions, 228–230
 - second shifting theorem (*t*-shifting), 219–223
 - short impulses, 225–226
 - systems of ODEs, 242–247
 - table of, 249–251
 - uniqueness, 210
 - unit step function (Heaviside function), 217–219
- linear, 46
- nonlinear, 46
- numeric analysis, 901–922
 - first-order ODEs, 901–915
 - higher order ODEs, 915–922
- second-order linear, 46–104
 - homogeneous, 46–79
 - nonhomogeneous, 79–102
- second-order nonlinear, 46

- Ordinary differential equations (*Cont.*)
 series solutions of ODEs, 167–202
 Bessel functions, 187–194, 196–200
 Bessel's equation, 187–200
 Frobenius method, 180–187
 Legendre polynomials, 177–179
 Legendre's equation, 175–179
 power series method, 167–175
 systems of, 124–166
 basic theory, 137–139
 constant-coefficient, 140–151
 conversion of n th-order ODEs to, 134–135
 homogeneous, 138
 Laplace transforms, 242–247
 linear, 124–130, 138–151, 160–163
 matrices and vectors, 124–130
 as models of applications, 130–134
 nonhomogeneous, 138, 160–163
 nonlinear, 152–160
 in phase plane, 124, 141–146, 157–159
 qualitative methods for nonlinear systems, 152–160
- Orientable surfaces, 446–447
- Oriented curve, 644
- Oriented surfaces, integrals over, 446–447
- Origin (vertex), 980
- Orthogonal, to a vector, 362
- Orthogonal coordinate curves, A74
- Orthogonal expansion, 504
- Orthogonal functions:
 defined, 500
 Sturm–Liouville Problems, 500–503
- Orthogonality:
 trigonometric system, 479–480, 538
 vector differential calculus, 361–363
- Orthogonal matrices, 335, 337–338, 353, A85n.2
- Orthogonal polynomials, 179
- Orthogonal series (generalized Fourier series), 504–510
 completeness, 508–509
 mean square convergence, 507–508
- Orthogonal trajectories:
 defined, 36
 first-order ODEs, 36–38
- Orthogonal transformations, 336, A85n.2
- Orthogonal vectors, 312, 362, 410
- Orthonormal functions, 500, 501, 508
- Orthonormal system, 337
- Oscillations:
 forced, 85–92
 free, 62–70
 harmonic, 63–64
 second-order linear ODEs:
 homogeneous, 62–70
 nonhomogeneous, 85–92
- Osculating plane, 389, 390
- Outcomes:
 of experiments, 1015, 1060
 probability theory, 1015
- Outer normal derivative, 460, 931
- Outliers, 1013–1015
- Output (response to input), 27, 86, 214
- Overdamping, 65–66
- Overdetermined linear systems, 277
- Overflow (floating-point numbers), 792
- Overrelaxation factor, 863
- Paired comparison, 1084, 1113
- Pappus, theorem of, 452
- Pappus of Alexandria, 452n.7
- Parabolic PDEs:
 defined, 923
 numeric analysis, 936–942
- Parallelogram law, 357
- Parallel processing of products (on computer), 265
- Parameters, 175, 381, 1112
 estimation of, 1063
 point estimation of, 1065–1068
 probability distributions, 1035
 of a sample, 1065
- Parameter curves, 442
- Parametric representations, 381, 439–441
- Parseval, Marc Antoine, 497n.3
- Parseval equality, 509
- Parseval's identity, 497
- Parseval's theorem, 497
- Partial derivatives, A69–A71
 defined, A69
 first (first order), A71
 second (second order), A71
 third (third order), A71
 of vector functions, 380
- Partial differential equations (PDEs), 473, 540–605
 basic concepts of, 540–543
 d'Alembert's solution, 553–556
 defined, 540
 double Fourier series solution, 577–585
 heat equation, 557–558
 Dirichlet problem, 564–566
- Partial differential equations (*Cont.*)
 Laplace's equation, 564
 solution by Fourier integrals, 568–571
 solution by Fourier series, 558–563
 solution by Fourier transforms, 571–574
 steady two-dimensional heat problems, 546–566
 unifying power of methods, 566
- homogeneous, 541
- Laplace's equation, 593–600
 boundary value problem in spherical coordinates, 594–596
 in cylindrical coordinates, 593–594
 Fourier–Legendre series, 596–598
 in spherical coordinates, 594
- Laplace transforms, solution by, 600–603
- Laplacian in polar coordinates, 585–592
- linear, 541
- method of separating variables, 545–553
 Fourier series, 548–551
 satisfying boundary conditions, 546–548
 two ODEs from wave equation, 545–546
- nonhomogeneous, 541
- nonlinear, 541
- numeric analysis, 922–945
 elliptic, 922–936
 hyperbolic, 942–945
 parabolic, 936–942
- ODEs vs., 4
- wave equation, 544–545
 d'Alembert's solution, 553–556
 solution by separating variables, 545–553
 two-dimensional, 575–584
- Partial fractions (Laplace transforms), 228–230
- Partial pivoting, 276, 846–848, 898
- Partial sums, of series, 477, 478, 495
- Particular solution(s):
 first-order ODEs, 6, 44
 higher-order homogeneous linear ODEs, 106
 nonhomogeneous linear systems, 160
 second-order linear ODEs:
 homogeneous, 49–51, 104
 nonhomogeneous, 80

- Partitioning, of a path, 645
Pascal, Blaise, 391n.4
Pascal, Étienne, 391n.4
Paths:
 alternating, 1002
 augmenting, 1002–1003
 closed, 414, 645, 975–976
 deformation of, 656
 directed, 1000
 flow augmenting, 992–993, 998, 1008
 flow problems in networks, 992
 integration by use of, 647–650
 longest, 976
 partitioning of, 645
 principle of deformation of, 656
 shortest, 976
 shortest path problems, 975–976
 simple closed, 652
Path dependence (line integrals), 418–426, 470, 649–650
 defined, 418
 and integration around closed curves, 421–425
Path independence, 669
 Cauchy's integral theorem, 655
 in a domain D in space, 419
 proof of, A88–A89
 Stokes's Theorem applied to, 468
Path of integration, 414, 644
Pauli spin matrices, 351

-charts, 1091–1092
PDEs, *see* Partial differential equations
Pearson, Egon Sharpe, 1077n.2
Pearson, Karl, 1077, 1086n.4
Period, 475
Periodic boundary conditions, 501
Periodic extensions, 488–490
Periodic function, 474–475, 538
Periodic Sturm–Liouville problem, 501
Permutations:
 of n things taken k at a time, 1025
 of n things taken k at a time with repetitions, 1025–1026
 probability theory, 1024–1026
Perron, Oskar, 882n.8
Perron–Frobenius Theorem, 883
Perron's theorem, 334, 882–883
Pfaff, Johann Friedrich, 422n.1
Pfaffian form, 422

-fold connected domains, 652–653
Phase angle, 90
Phase lag, 90
Phase plane, 134, 165
 linear systems, 141, 148
 nonlinear systems, 152
Phase plane method, 124
 linear systems:
 critical points, 142–146
 graphing solutions, 141–142
 nonlinear systems, 152
 linearization, 152–155
 Lotka–Volterra population model, 155–156
 transformation to first-order equation in, 157–159
Phase plane representations, 134
Phase portrait, 165
 linear systems, 141–142, 148
 nonlinear systems, 152
Picard, Emile, 42n.10
Picard's Iteration Method, 42
Picard's theorem, 716
Piecewise continuous functions, 209
Piecewise smooth path of integration, 414, 645
Piecewise smooth surfaces, 442, 447
Pivot, 276, 898, 960
Pivot equation, 276, 846, 898, 960
Planar graphs, 1005
Plane:
 complex, 611
 extended, 718, 744–745
 finite, 718
 sets in, 620
 normal, 390
 osculating, 389, 390
 phase, 134, 165
 linear systems, 141, 148
 nonlinear systems, 152
 rectifying, 390
 tangent, 398, 441–442
 vectors in, 309
Plane curves, 383
Planimeters, 436
Poincaré, Henri, 141n.1, 510n.8
Points:
 boundary, 426n.2, 620
 branch, 755
 center, 144, 165
 critical, 33, 144, 165
 asymptotically stable, 149
 and conformal mapping, 738, 757
 constant-coefficient systems of ODEs, 142–151
 isolated, 152
 nonlinear systems, 152
 stable, 140, 149
 stable and attractive, 140, 149
 unstable, 140, 149
 equilibrium, 33–34
 fixed, 745, 799
 guidepoints, 827
 at infinity, 718
 initial (vectors), 355
Points: (*Cont.*)
 lattice, 925–926
 limit, A93
 mesh, 925–926
 regular, 181
 regular singular, 180n.4
 saddle, 143, 165
 sample, 1015
 singular, 181, 201
 analytic functions, 693
 regular, 180n.4
 spiral, 144–145, 165
 stagnation, 773
 stationary, 952
 terminal (vectors), 355
Point estimation of parameters (statistics), 1065–1068, 1113
 defined, 1065
 maximum likelihood method, 1066–1067
Point set, in complex plane, 620
Point source (flow modeling), 776
Point spectrum, 525
Poisson, Siméon Denis, 779n.2
Poisson distributions, 1041–1042, 1061, A100
Poisson equation:
 defined, 923
 numeric analysis, 922–936
 ADI method, 928–930
 difference equations, 923–925
 Dirichlet problem, 925–928
 mixed boundary value problem, 931–933
Poisson's integral formula:
 derivation of, 778–778
 potential theory, 777–781
 series for potentials in disks, 779–780
Polar coordinates, 431
 Laplacian in, 585–592
 notation for, 594
 two-dimensional wave equation in, 586
Polar form, of complex numbers, 613–618, 631
Polar moment of inertia, of a region, 429
Poles (singularities), 714–715
 of order m , 735
 and zeros, 717
Polynomials, 624
 characteristic, 325, 353, 877
 Chebyshev, 504
 interpolation, 808, 842
 Laguerre, 241, 504
 Legendre, 167, 177–179, 202
 orthogonal, 179
 trigonometric:
 approximation by, 495–498

- Polynomials (*Cont.*)
 complex, 529
 of the same degree N , 495
 Polynomial approximations, 808
 Polynomial interpolation, 808, 842
 Polynomially bounded, 979
 Polynomial matrix, 334, 878–879
 Populations:
 infinite, 1044
 for statistical sampling, 1063
 Population dynamics:
 defined, 33
 logistic equation, 33–34
 Position vector, 356
 Positive correlation, 1111
 Positive definite (quadratic form), 346
 Positive sense, on curve, 644
 Possible values (random variables), 1030
 Postman problem, 980
 Potential (potential function), 400
 complex, 760–761
 Laplace's equation, 593
 Poisson's integral formula for, 777–781
 Potential theory, 179, 420, 460, 758–786
 conformal mapping for boundary value problems, 763–767
 defined, 758
 electrostatic fields, 759–763
 complex potential, 760–761
 superposition, 761–762
 fluid flow, 771–777
 harmonic functions, 781–784
 heat problems, 767–770
 Laplace's equation, 593, 628
 Poisson's integral formula, 777–781
 Power function, of a test, 1081, 1113
 Power method (matrix eigenvalue problems), 885–888, 899
 Power series, 168, 671–707
 convergence behavior of, 680–682
 convergence tests, 674–676, A93–A94
 functions given by, 685–690
 Maclaurin series, 690
 in powers of x , 168
 radius of convergence, 682–684
 ratio test, 676–678
 root test, 678–679
 sequences, 671–673
 series, 673–674
 Taylor series, 690–697
 uniform convergence, 698–705
 and absolute convergence, 704
 properties of, 700–701
 termwise integration, 701–703
 test for, 703–704
 Power series method, 167–175, 201
 extension of, *see* Frobenius method
 idea and technique of, 168–170
 operations on, 173–174
 theory of, 170–174
 Practical resonance, 90
 Predator–prey population model, 155–156
 Predictor–corrector method, 913
 PRIM, ALGORITHM, 989
 Prim, Robert Clay, 988n.6
 Prim's algorithm, 988–991, 1008
 Principal axes, transformation to, 344
 Principal branch, of logarithm, 639
 Principal directions, 330
 Principal minors, 346
 Principal part, 735
 of isolated singularities, 715
 of singularities, 708, 709
 Principal value (complex numbers), 614, 617, 642
 complex logarithm, 637
 general powers, 639
 Principle of deformation of path, 656
 Prior estimates, 805
 Probability, 1060
 axioms of, 1020
 basic theorems of, 1020–1022
 conditional, 1022–1023
 definitions of, 1018–1020
 independent events, 1023
 Probability distributions, 1029, 1061
 binomial, 1039–1042
 continuous, 1032–1034
 discrete, 1030–1032
 hypergeometric, 1042–1044
 mean and variance of, 1035–1039
 multinomial, 1045
 normal, 1045–1051
 Poisson, 1041–1042
 of several random variables, 1051–1060
 addition of means, 1057–1058
 addition of variances, 1058–1059
 continuous two-dimensional distributions, 1053
 discrete two-dimensional distributions, 1052–1053
 function of random variables, 1056
 independence of random variables, 1055–1056
 marginal distributions, 1053–1055
 symmetric, 1036
 two-dimensional, 1051
 continuous, 1053
 discrete, 1052–1053
 uniform, 1035–1036
 Probability function, 1030–1032, 1052, 1061
 Probability theory, 1009, 1015–1062
 binomial coefficients, 1027–1028
 combinations, 1024, 1026–1027
 distributions (probability distributions), 1029
 binomial, 1039–1042
 continuous, 1032–1034
 discrete, 1030–1032
 hypergeometric, 1042–1044
 mean and variance of, 1035–1039
 normal, 1045–1051
 Poisson, 1041–1042
 of several random variables, 1051–1060
 events, 1016–1017
 experiments, 1015–1016
 factorial function, 1027
 outcomes, 1015
 permutations, 1024–1026
 probability:
 basic theorems of, 1020–1022
 conditional, 1022–1023
 definition of, 1018–1020
 independent events, 1023
 random variables, 1029–1030
 continuous, 1032–1034
 discrete, 1030–1032
 Problem of existence, 39
 Problem of uniqueness, 39
 Producers, 1092
 Producer's risk, 1094
 Product:
 inner (dot), 312
 for complex vectors, 349
 invariance of, 336
 vector differential calculus, 361–367, 410
 of matrix, 260
 determinants of, 307–308
 inverting, 306
 matrix multiplication, 263, 320
 parallel processing of (on computer), 265
 scalar multiplication, 260
 scalar triple, 373–374, 411
 vector (cross):
 in Cartesian coordinates, A83–A84
 vector differential calculus, 368–375, 410
 Product method, 605. *See also* Method of separating variables
 Projection (vectors), 365
 Proper node, 143
 Pseudocode, 796
 Pure imaginary complex numbers, 609

- QR-factorization, 892–896
- Quadrant, of a circle, 604
- Quadratic forms (matrix eigenvalue problems), 343–344
- Quadratic interpolation, 810–811
- Qualitative methods, 124, 141n.1
 - defined, 152
 - for nonlinear systems, 152–160
 - linearization, 152–155
 - Lotka–Volterra population model, 155–156
 - transformation to first-order equation in phase plane, 157–159
- Quality control (statistics), 1087–1092, 1113
 - for mean, 1088–1089
 - for range, 1090–1091
 - for standard deviation, 1090
 - for variance, 1089–1090
- Quantitative methods, 124
- Quasilinear equations, 555, 923
- Quotient:
 - complex numbers, 610
 - difference, 923
 - Rayleigh, 885, 899
- Radius:
 - of convergence, 172
 - defined, 172
 - power series, 682–684, 706
 - of a graph, 991
- Random experiments, 1011, 1015–1016, 1060
- Randomly selected samples, 1064
- Randomness, 1015, 1064. *See also* Random variables
- Random numbers, 1064
- Random number generators, 1064
- Random sampling (statistics), 1063–1065
- Random selections, 1064
- Random variables, 1011, 1029–1030, 1061
 - continuous, 1029, 1032–1034, 1055
 - defined, 1030
 - dependent, 1055
 - discrete, 1029–1032, 1054
 - function of, 1056
 - independence of, 1055–1056
 - marginal distribution of, 1054, 1055
 - normal, 1045
 - occurrence of, 1063
 - probability distributions of, 1051–1060
- addition of means, 1057–1058
 - addition of variances, 1058–1059
 - continuous two-dimensional distributions, 1053
 - discrete two-dimensional distributions, 1052–1053
 - function of random variables, 1056
 - independence of random variables, 1055–1056
 - marginal distributions, 1053–1055
 - skewness of, 1039
 - standardized, 1037
 - two-dimensional, 1051, 1062
- Random variation, 1063
- Range, 1013
 - control chart for, 1090–1091
 - defined, 1090
 - of f , 620
- Rank:
 - of A , 279
 - of a matrix, 279, 283, 321
 - in terms of column vectors, 284–285
 - in terms of determinants, 297
 - of R , 279
- Raphson, Joseph, 801n.1
- Rational functions, 624, 725–729
- Ratio test (power series), 676–678
- Rayleigh, Lord (John William Strutt), 160n.5, 885n.10
- Rayleigh equation, 160
- Rayleigh quotient, 885, 899
- Reactance (RLC circuits), 94
- Real axis (complex plane), 611
- Real different roots, 71
- Real double root, 55–56, 72
- Real functions, complex analytic functions vs., 694
- Real inner product space, 312
- Real integrals, residue integration of, 725–733
 - Fourier integrals, 729–730
 - improper integrals, 730–732
 - of rational functions of $\cos \theta$ $\sin \theta$, 725–729
- Real part (complex numbers), 609
- Real pre-Hilbert space, 312
- Real roots:
 - different, 71
 - double, 55–56
 - higher-order homogeneous linear ODEs:
 - distinct, 112–113
 - multiple, 114–115
 - second-order homogeneous linear ODEs:
 - distinct, 54–55
 - double, 55–56
- Real sequence, 671
- Real series, A73–A74
- Real vector spaces, 309–311, 359, 410
- Recording, of sample values, 1011–1012
- Rectangular cross-section, 120
- Rectangular matrix, 258
- Rectangular membrane R , 577–584
- Rectangular rule (numeric integration), 828
- Rectifiable (curves), 385
- Rectification (acceptance sampling), 1094–1095
- Rectifying plane, 390
- Recurrence formula, 201
- Recurrence relation, 176
- Recursion formula, 176
- Reduced echelon form, 279
- Reduction of order (second-order homogeneous linear ODEs), 51–52
- Regions, 426n.2
 - bounded, 426n.2
 - center of gravity of mass in, 429
 - closed, 426n.2
 - critical, 1079
 - feasibility, 954
 - fundamental (exponential function), 632
 - moments of inertia of, 429
 - polar moment of inertia of, 429
 - rejection, 1079
 - sets in complex plane, 620
 - total mass of, 429
 - volume of, 428
- Regression analysis, 1063, 1103–1108, 1113
 - confidence intervals in, 1107–1108
 - defined, 1103
- Regression coefficient, 1105, 1107–1108
- Regression curve, 1103
- Regression line, 1103, 1104, 1106
- Regular point, 181
- Regular singular point, 180n.4
- Regular Sturm–Liouville problem, 501
- Rejectable quality level (RQL), 1094
- Rejection:
 - of a hypothesis, 1078
 - of products, 1092
- Rejection region, 1079
- Relative class frequency, 1012
- Relative error, 794
- Relative frequency (probability):
 - of an event, 1019
 - class, 1012
 - cumulative, 1012

- Relaxation methods, 862
- Remainder, 170
 - of a series, 673
 - of Taylor series, 691
- Remarkable parallelogram, 375
- Removable singularities, 717
- Repeated factors, 220, 221
- Representation, 315
 - by Fourier series, 476
 - by power series, 683
 - spectral, 525
- Residual, 805, 862, 899
- Residues, 708, 720, 735
 - at m th-order pole, 722
 - at simple poles, 721–722
- Residue integration, 719–733
 - formulas for residues, 721–722
 - of real integrals, 725–733
 - Fourier integrals, 729–730
 - improper integrals, 730–732
 - of rational functions of $\cos \theta$
 - $\sin \theta$, 725–729
 - several singularities inside
 - contour, 723–725
- Residue theorem, 723–724
- Resistance, apparent, 95
- Resonance:
 - practical, 90
 - undamped forced oscillations, 88–89
- Resonance factor, 88
- Response to input, *see* Output
 - (response to input)
- Resultant, of forces, 357
- Riccati equation, 35
- Riemann, Bernhard, 625n.4
- Riemannian geometry, 625n.4
- Riemann sphere, 718
- Riemann surfaces (conformal mapping), 754–757
- Right-hand derivatives (Fourier series), 480
- Right-handed Cartesian coordinate system, 368–369, A83–A84
- Right-handed triple, 369
- Right-hand limit (Fourier series), 480
- Right-sided tests, 1079, 1082
- Risks of making false decisions, 1080
- RKF method, *see* Runge–Kutta–Fehlberg method
- RK methods, *see* Runge–Kutta methods
- RKN methods, *see* Runge–Kutta–Nyström methods
- Robin problem:
 - Laplace’s equation, 593
 - two-dimensional heat equation, 564
- Rodrigues, Olinde, 179n.2
- Rodrigues’s formula, 179, 241
- Romberg integration, 840, 843
- Roots:
 - complex:
 - higher-order homogeneous linear ODEs, 113–115
 - second-order homogeneous linear ODEs, 57–59
 - complex conjugate, 72–73
 - differing by an integer, 183
 - Frobenius method, 183
 - distinct (Frobenius method), 182
 - double (Frobenius method), 183
 - of equations, 798
 - multiple complex, 115
 - n th, 616
 - n th roots of unity, 617
 - simple complex, 113–114
- Root test (power series), 678–679
- Rotation (vorticity of flow), 774
- Rounding, 792
- Rounding unit, 793
- Roundoff (numeric analysis), 792–793
- Roundoff errors, 792, 794, 902
- Roundoff rule, 793
- Rows:
 - determinants, 294
 - matrix, 125, 257, 320
- Row echelon form, 279–280
- Row-equivalent matrices, 283–284
- Row-equivalent systems, 277
- Row operations (linear systems), 276, 277
- Row scaling (Gauss elimination), 850
- Row “sum” norm, 861
- Row vectors, 126, 257, 320
- RQL (rejectable quality level), 1094
- Runge, Carl, 820n.3
- Runge, Karl, 905n.1
- RUNGE–KUTTA, ALGORITHM, 905
- Runge–Kutta–Fehlberg (RKF)
 - method, 947
 - error of, 908
 - first-order ODEs, 906–908
- Runge–Kutta (RK) methods, 915, 947
 - error of, 908
 - first-order ODEs, 904–906
 - higher order ODEs, 917–919
- Runge–Kutta–Nyström (RKN)
 - methods, 919–921, 947
- Rutherford, E., 1044, 1100
- Rutherford–Geiger experiments, 1044, 1100
- Rutishauser, Heinz, 892n.12
- Saddle point, 143, 165
- Samples:
 - for experiments, 1015
 - in mathematical statistics, 1063–1064
 - selection of, 1063–1064
- Sample covariance, 1105
- Sampled function, 529
- Sample distribution function, 1096
- Sample mean, 1064, 1113
- Sample points, 1015
- Sample regression line, 1104
- Sample size, 1015, 1064
- Sample space, 1015, 1016, 1060
- Sample standard deviation, 1065
- Sample variance, 1015, 1113
- Sampling:
 - from a population, 1023
 - random, 1063–1065
 - with replacement, 1023
 - binomial distribution, 1042
 - hypergeometric distribution, 1043–1044
 - in statistics, 1063
 - without replacement, 1018, 1023
 - binomial distribution, 1042–1043
 - hypergeometric distribution, 1043–1044
- Sampling plan, 1092–1093
- Scalar(s), 260, 310, 354
- Scalar fields, vector fields that are
 - gradients of, 400–401
- Scalar functions:
 - defined, 376
 - vector differential calculus, 376
- Scalar matrices, 268
- Scalar multiplication, 126–127, 310
 - of matrices and vectors, 259–261
 - vectors in 2-space and 3-space, 358–359
- Scalar triple product, 373–374, 411
- Scale (vectors), 886–887
- Scanning labeled vertices, 998
- Schrödinger, Erwin, 226n.2
- Schur, Issai, 882n.7
- Schur’s inequality, 882
- Schur’s theorem, 882
- Schwartz, Laurent, 226n.2
- Secant, formula for, A65
- Secant method (numeric analysis), 805–806, 842
- Second boundary value problem, *see* Neumann problem
- Second-order determinants, 291–292
- Second-order differential operator, 60
- Second-order linear ODEs, 46–104
 - homogeneous, 46–79
 - basis, 50–52
 - with constant coefficients, 53–60
 - differential operators, 60–62
 - Euler–Cauchy equations, 71–74
 - existence and uniqueness of solutions, 74–79

- Second-order linear ODEs (*Cont.*)
 general solution, 49–51, 77–78
 initial value problem, 49–50
 modeling free oscillations of
 mass–spring system,
 62–70
 reduction of order, 51–52
 superposition principle, 47–48
 Wronskian, 75–78
 nonhomogeneous, 79–102
 defined, 47
 general solution, 80–81
 method of undetermined
 coefficients, 81–85
 modeling electric circuits, 93–99
 modeling forced oscillations,
 85–92
 solution by variation of
 parameters, 99–102
- Second-order method, improved
 Euler method as, 904
- Second-order nonlinear ODEs, 46
- Second-order PDEs, 540–541
- Second (second order) partial
 derivatives, A71
- Second shifting theorem (t -shifting),
 219–223
- Second transmission line equation,
 599
- Seidel, Philipp Ludwig von, 858n.4
- Self-starting methods, 911
- Sense reversal (complex line
 integrals), 645
- Separable equations, 12–13
- Separable ODEs, 44
 first-order, 12–20
 extended method, 17–18
 modeling, 13–17
 reduction of nonseparable ODEs
 to, 17–18
- Separating variables, method of,
 12–13
 circular membrane, 587
 partial differential equations,
 545–553, 605
 Fourier series, 548–551
 satisfying boundary conditions,
 546–548
 two ODEs from wave
 equation, 545–546
 vibrating string, 545–546
- Separation constant, 546
- Sequences (infinite sequences):
 bounded, A93–A95
 convergent, 507–508, 672
 divergent, 672
 limit point of, A93
 monotone real, A72–A73
 power series, 671–673
 real, 671
- Series, A73–A74
 binomial, 696
 conditionally convergent, 675
 convergent, 171, 673
 cosine, 781
 derived, 687
 divergent, 171, 673
 double Fourier:
 defined, 582
 rectangular membrane,
 577–585
 Fourier, 473–483, 538
 convergence and sum or,
 480–481
 derivation of Euler formulas,
 479–480
 double, 577–585
 even and odd functions,
 486–488
 half-range expansions, 488–490
 heat equation, 558–563
 from period 2π to $2L$,
 483–486
 Fourier–Bessel, 506–507, 589
 Fourier cosine, 484, 486, 538
 Fourier–Legendre, 505–506,
 596–598
 Fourier sine, 477, 486, 538
 one-dimensional heat equation,
 561
 vibrating string, 548
 geometric, 168, 675
 Taylor series, 694
 uniformly convergent, 698
 hypergeometric, 186
 infinite, 673–674
 Laurent, 708–719, 734
 analytic or singular at infinity,
 718–719
 point at infinity, 718
 Riemann sphere, 718
 singularities, 715–717
 zeros of analytic functions, 717
 Maclaurin, 690, 694–696
 orthogonal, 504–510
 completeness, 508–509
 mean square convergence,
 507–508
 power, 168, 671–707
 convergence behavior of,
 680–682
 convergence tests, 674–676,
 A93–A94
 functions given by, 685–690
 Maclaurin series, 690
 in powers of x , 168
 radius of convergence,
 682–684
 ratio test, 676–678
 root test, 678–679
- Series (*Cont.*)
 sequences, 671–673
 series, 673–674
 Taylor series, 690–697
 uniform convergence, 698–705
 real, A73–A74
 Taylor, 690–697, 707
 trigonometric, 476, 484
 value (sum) of, 171, 673
- Series solutions of ODEs, 167–202
 Bessel functions, 187–188
 of the first kind, 189–194
 of the second kind, 196–200
 Bessel's equation, 187–196
 Bessel functions, 187–188,
 196–200
 general solution, 194–200
 Frobenius method, 180–187
 indicial equation, 181–183
 typical applications, 183–185
 Legendre polynomials, 177–179
 Legendre's equation, 175–179
 power series method, 167–175
 idea and technique of,
 168–170
 operations on, 173–174
 theory of, 170–174
- Sets:
 complete orthonormal, 508
 in the complex plane, 620
 cut, 994–996, 1008
 linearly dependent, 129, 311
 linearly independent, 128–129,
 311
- Shewhart, W. A., 1088
- Shifted function, 219
- Shortest path, 976
- Shortest path problems
 (combinatorial optimization),
 975–980, 1008
 Bellman's principle, 980–981
 complexity of algorithms,
 978–980
 Dijkstra's algorithm, 981–983
 Moore's BFS algorithm, 977–980
- Shortest spanning trees:
 combinatorial optimization, 1008
 Greedy algorithm, 984–988
 Prim's algorithm, 988–991
 defined, 984
- Short impulses (Laplace transforms),
 225–226
- Sifting property, 226
- Significance (in statistics), 1078
- Significance level, 1078, 1080, 1113
- Significance tests, 1078
- Significant digits, 791–792
- Similarity transformation, 340
- Similar matrices, 340–341, 878
- Simple closed curves, 646

- Simple closed path, 652
- Simple complex roots, 113–114
- Simple curves, 383
- Simple events, 1015
- Simple general properties of the line integral, 415–416
- Simple poles, 714
- Simplex method, 958–968
 - degenerate feasible solution, 962–965
 - difficulties in starting, 965–968
- Simplex table, 960
- Simplex tableau, 960
- Simple zero, 717
- Simply connected domains, 423, 646, 652, 653
- SIMPSON, ALGORITHM, 832
- Simpson, Thomas, 832n.4
- Simpson's rule, 832, 843
 - adaptive integration with, 835–836
 - numeric integration, 831–835
- Simultaneous corrections, 862
- Sine function:
 - conformal mapping by, 750–751
 - formula for, A63–A65
- Sine integral, 514, 697, A68–A69, A98
- Single precision, floating-point standard for, 792
- Singularities (singular, having a singularity), 693, 707, 715
 - analytic functions, 693
 - essential, 715–716
 - inside a contour, 723–725
 - isolated, 715
 - isolated essential, 715
 - Laurent series, 715–719
 - principal part of, 708
 - removable, 717
- Singular matrices, 301
- Singular point, 181, 201
 - analytic functions, 693
 - regular, 180n.4
- Singular solutions:
 - first-order ODEs, 8, 35
 - higher-order homogeneous linear ODEs, 110
 - second-order homogeneous linear ODEs, 50, 78
- Singular Sturm–Liouville problem, 501, 503
- Sink(s):
 - motion of a fluid, 404, 458, 775, 776
 - networks, 991
- Size:
 - of matrices, 258
 - sample, 1015, 1064
- Skew-Hermitian form, 351
- Skew-Hermitian matrices, 347, 348, 350, 353
- Skewness, of a random variables, 1039
- Skew-symmetric matrices, 268, 320, 334–336, 353
- Slack variables, 956, 969
- Slope field (direction field), 9–10
- Smooth curves, 414, 644
- Smooth surfaces, 442
- Sobolev, Sergei L'Vovich, 226n.2
- Software:
 - for data representation in statistics, 1011
 - numeric analysis, 788–789
 - variable step size selection in, 902
- Solenoid, 405
- Solutions. *See also specific methods*
 - defined, 4, 798
 - first-order ODEs:
 - concept of, 4–6
 - equilibrium solutions, 33–34
 - explicit solutions, 21
 - family of solutions, 5
 - general solution, 6, 44
 - implicit solutions, 21
 - particular solution, 6, 44
 - singular solution, 8, 35
 - solution by calculus, 5
 - trivial solution, 28, 35
 - graphing in phase plane, 141–142
 - higher-order homogeneous linear ODEs, 106
 - general solution, 106, 110–111
 - particular solution, 106
 - singular solution, 110
 - linear systems, 273, 745
 - nonhomogeneous linear systems:
 - general solution, 160
 - particular solution, 160
 - PDEs, 541
 - second-order homogeneous linear ODEs:
 - general solution, 49–51, 77–78
 - linear dependence and independence of, 75
 - particular solution, 49–51
 - singular solution, 50, 78
 - second-order linear ODEs, 47
 - second-order nonhomogeneous linear ODEs:
 - general solution, 80–81
 - particular solution, 80
 - systems of ODEs, 137, 139
- Solution curves, 4–6
- Solution space, 290
- Solution vector, 273, 745
- SOR (successive overrelaxation), 863
- SOR formula for Gauss–Seidel, 863
- Sorting, of sample values, 1011–1012
- Source(s):
 - motion of a fluid, 404, 458, 775
 - networks, 991
- Source intensity, 458
- Source line (flow modeling), 776
- Span, of vectors, 286
- Spanning trees, 984, 988
- Sparse graphs, 974
- Sparse matrices, 823, 925
- Sparse systems, 858
- Special functions, 167, 202
 - formulas for, A63–A69
 - theory of, 175
- Special vector spaces, 285–287
- Specific circulation, of flow, 467
- Spectral density, 525
- Spectral mapping theorem, 878
- Spectral radius, 324, 861
- Spectral representation, 525
- Spectral shift, 896
- Spectrum, 877
 - of matrix, 324
 - vibrating string, 547
- Speed, 386, 391
 - angular (rotation), 372
 - of convergence, 804–805
- Spherical coordinates, A74–A76
 - boundary value problem in, 594–596
 - defined, 594
 - Laplacian in, 594
- Spiral point, 144–145, 165
- Spline, 821, 843
- Spline interpolation, 820–827
- Spring constant, 62
- Square error, 496–497, 539
- Square matrices, 126, 257, 258, 301–309, 320
- s*-shifting, 208–209
- Stability:
 - of critical points, 165
 - of solutions, 33–34, 124, 936
 - of systems, 84, 124
- Stability chart, 149
- Stable algorithms, 796, 842
- Stable and attractive critical points, 140, 149
- Stable critical points, 140, 149
- Stable equilibrium solution, 33–34
- Stable systems, 84
- Stagnation points, 773
- Standard basis, 314, 359, 365
- Standard deviation, 1014, 1035, 1090
- Standard form:
 - first-order ODEs, 27
 - higher-order homogeneous linear ODEs, 105
 - higher-order linear ODEs, 123
 - power series method, 172
 - second-order linear ODEs, 46, 103
- Standardized normal distribution, 1046

- Standardized random variables, 1037
 Standard trick (confidence intervals), 1068
 Stationary point (unconstrained optimization), 952
 Statistics, 1015, 1063. *See also* Mathematical statistics
 Statistical inference, 1059, 1063
 Steady flow, 405, 458
 Steady heat flow, 767
 Steady-state case (heat problems), 591
 Steady-state current, 98
 Steady-state heat flow, 460
 Steady-state solution, 31, 84, 89–91
 Steady two-dimensional heat problems, 546–566, 605
 Steepest descent, method of, 952–954
 Steiner, Jacob, 451n.6
 Stem-and-leaf plots, 1012
 Stencil (pattern, molecule, star), 925
 Step-by-step methods, 901
 Step function, 828, 1031
 Step size, 901, 902
 Stereographic projection, 718
 Stiff ODEs, 909–910
 Stiff systems, 920–921
 Stirling, James, 1027n.2
 Stirling formula, 1027, A67
 Stochastic matrices, 270
 Stochastic variables, 1029. *See also* Random variables
 Stokes, Sir George Gabriel, 464n.9, 703n.5
 Stokes's Theorem, 463–470
 Stream function, 771
 Streamline, 771
 Strength (flow modeling), 776
 Strictly diagonally dominant matrices, 881
 Sturm, Jacques Charles François, 499n.4
 Sturm–Liouville equation, 499
 Sturm–Liouville expansions, 474
 Sturm–Liouville Problems, 498–504
 eigenvalues, eigenfunctions, 499–500
 orthogonal functions, 500–503
 Subgraphs, 972
 Submarine cable equations, 599
 Submatrices, 288
 Subsidiary equation, 203, 253
 Subspace, of vector space, 286
 Subtraction:
 of complex numbers, 610
 termwise, of power series, 687
 Success corrections, 862
 Successive overrelaxation (SOR), 863
 Sufficient convergence condition, 861
 Sum:
 of matrices, 320
 partial, of series, 477, 478, 495
 of a series, 171, 673
 of vectors, 357
 Sum Rule (method of undetermined coefficients):
 higher-order homogeneous linear ODEs, 115
 second-order nonhomogeneous linear ODEs, 81, 83–84
 Superlinear convergence, 806
 Superposition (electrostatic fields), 761–762
 Superposition (linearity) principle:
 higher-order homogeneous linear ODEs, 106
 higher-order linear ODEs, 123
 homogeneous linear systems, 138
 PDEs, 541–542
 second-order homogeneous linear ODEs, 47–48, 104
 undamped forced oscillations, 87
 Surfaces, for surface integrals, 439–443
 orientation of, 446–447
 representation of surfaces, 439–441
 tangent plane and surface normal, 441–442
 Surface integrals, 470
 defined, 443
 surfaces for, 439–443
 orientation of, 446–447
 representation of surfaces, 439–441
 tangent plane and surface normal, 441–442
 vector integral calculus, 443–452
 orientation of surfaces, 446–447
 without regard to orientation, 448–450
 Surface normal, 398–399, 442
 Surface normal vector, 398–399
 Surjective mapping, 737n.1
 Sustainable yield, 36
 Symbol O , 979
 Symmetric coefficient matrix, 343
 Symmetric distributions, 1036
 Symmetric matrices, 267–268, 320, 334–336, 353
 Systems of ODEs, 124–166
 basic theory of, 137–139
 constant-coefficient, 140–151
 critical points, 142–146, 148–151
 graphing solutions in phase plane, 141–142
 Systems of ODEs (*Cont.*)
 conversion of n th-order ODEs to, 134–135
 homogeneous, 138
 Laplace transforms, 242–247
 linear, 138–139. *See also* Linear systems
 constant-coefficient systems, 140–151
 matrices and vectors, 124–130
 nonhomogeneous, 138, 160–163
 calculations with, 125–127
 definitions and terms, 125–126, 128–129
 eigenvalues and eigenvectors, 129–130
 systems of ODEs as vector equations, 127–128
 as models of applications:
 electrical network, 132–134
 mixing problem involving two tanks, 130–132
 nonhomogeneous, 138, 160–163
 method of undetermined coefficients, 161
 method of variation of parameters, 162–163
 nonlinear systems:
 qualitative methods for, 152–160
 transformation to first-order equation in phase plane, 157–159
 in phase plane, 124
 critical points, 142–146
 graphing solutions in, 141–142
 transformation to first-order equation in, 157–159
 qualitative methods for nonlinear systems, 152–160
 linearization, 152–155
 Lotka–Volterra population model, 155–156
 Tangent:
 to a curve, 384
 formula for, A65
 Tangent function, conformal mapping by, 752–753
 Tangential accelerations, 391
 Tangential acceleration vector, 387
 Tangent plane, 398, 441–442
 Tangent vector, 384, 411
 Target (networks), 991
 Taylor, Brook, 690n.2
 Taylor series, 690–697, 707
 Taylor's formula, 691
 Taylor's theorem, 691

- t -distribution, 1071–1073, 1078, A103
 Telegraph equations, 599
 Term(s):
 of a sequence, 671
 of a series, 673
 Terminal point (vectors), 355
 Termination criterion, 802–803
 Termwise addition, 173, 687
 Termwise differentiation, 173, 687–688, 703
 Termwise integration, 687, 688, 701–703
 Termwise multiplication, 173, 687
 Termwise subtraction, 687
 Tests, statistical, 1077, 1113
 Theory of special functions, 175
 Thermal diffusivity, 460
 Third boundary value problem, *see* Robin problem
 Third-order determinants, 292–293
 Third (third order) partial derivatives, A71
 3-space, vectors in, 309, 354
 components of a vector, 356–357
 scalar multiplication, 358–359
 vector addition, 357–359
 Three-sigma limits, 1047
 Time (curves in mechanics), 386
 TI-Nspire, 789
 Todd, John, 855n.3
 Tolerance (adaptive integration), 835
 Torricelli, Evangelista, 16n.4
 Torricelli's law, 16–17
 Torsion, curvature and, 389–390
 Total differential, 20, 45
 Total energy, of physical system, 525
 Total error, 902
 Total mass, of a region, 429
 Total orthonormal set, 508
 Total pivoting, 846
 Trace, 345
 Trail (shortest path problems), 975
 closed trails, 975–976
 Euler trail, 980
 Trajectories, 134, 165
 linear systems, 141–142, 148
 nonlinear systems, 152
 Transcendental equations, 798
 Transducers, 98
 Transfer function, 214
 Transformation(s), 313
 orthogonal, 336
 to principal axes, 344
 Transient solution, 84, 89
 Transient-state solution, 31
 Translation (vectors), 355
 Transposition(s):
 of matrices or vectors, 128, 320
 in samples, 1101
 Trapezoidal rule, 828, 843
 error bounds and estimate for, 829–831
 numeric integration, 828–831
 Trees (graphs), 984, 988. *See also* Shortest spanning trees
 Trials (experiments), 1011, 1015
 Triangle inequality, 363, 614–615
 Triangular form (Gauss elimination), 846
 Triangular matrices, 268
 Tricomi, Francesco, 556n.2
 Tricomi equation, 555, 556
 Tridiagonalization (matrix eigenvalue problems), 888–892
 Tridiagonal matrices, 823, 888, 928
 Trigonometric analytic functions (conformal mapping), 750–754
 Trigonometric function, 633–635, 642
 inverse, 640
 Taylor series, 695
 Trigonometric polynomials:
 approximation by, 495–498
 complex, 529
 of the same degree N , 495
 Trigonometric series, 476, 484
 Trigonometric system, 475, 479–480, 538
 Trihedron, 390
 Triple integrals, 470
 defined, 452
 mean value theorem for, 456–457
 vector integral calculus, 452–458
 Triply connected domains, 653, 658, 659
 Trivial solution, 28, 35
 homogeneous linear systems, 290
 linear systems, 273
 Sturm–Liouville problem, 499
 Truncating, 794
 t -shifting, 219–223
 Tuning (vibrating string), 548
 Twisted curves, 383
 2-space (plane), vectors in, 354
 components of a vector, 356–357
 scalar multiplication, 358–359
 vector addition, 357–359
 2×2 matrix, 125
 Two-dimensional heat equation, 564–566
 Two-dimensional normal distribution, 1110
 Two-dimensional probability distributions:
 continuous, 1053
 discrete, 1052–1053
 Two-dimensional problems (potential theory), 759, 771
 Two-dimensional random variables, 1051, 1062
 Two-dimensional wave equation, 575–584, 586
 Two-sided alternative (hypothesis testing), 1079–1080
 Two-sided tests, 1079, 1082–1083
 Type I errors, 1080, 1081
 Type II errors, 1080–1081
 UCL (upper control limit), 1088
 Unacceptable lots, 1094
 Unconstrained optimization, 969
 basic concepts, 951–952
 method of steepest descent, 952–954
 Uncorrelated related variables, 1109
 Underdamping, 65, 67
 Underdetermined linear systems, 277
 Underflow (floating-point numbers), 792
 Undetermined coefficients, method of:
 higher-order homogeneous linear ODEs, 115
 higher-order linear ODEs, 123
 nonhomogeneous linear systems of ODEs, 161
 second-order linear ODEs:
 homogeneous, 104
 nonhomogeneous, 81–85
 Uniform convergence:
 and absolute convergence, 704
 power series, 698–705
 properties of uniform convergence, 700–701
 termwise integration, 701–703
 test for, 703–704
 Uniform distributions, 1035–1036, 1053
 Unifying power of mathematics, 97
 Union, of events, 1016–1017
 Uniqueness:
 of Laplace transforms, 210
 of Laurent series, 712
 of power series representation, 685–686
 problem of, 39
 Uniqueness theorems:
 cubic splines, 822
 Dirichlet problem, 462, 784
 first-order ODEs, 39–42
 higher-order homogeneous linear ODEs, 108
 Laplace's equation, 462
 linear systems, 138
 proof of, A77–A79
 second-order homogeneous linear ODEs, 74
 systems of ODEs, 137

- Unitary matrices, 347–350, 353
- Unitary systems, 349
- Unitary transformation, 349
- Unit binormal vector, 389
- Unit circle, 617, 619
- Unit impulse function, 226. *See also* Dirac delta function
- Unit matrices, 128, 268
- Unit normal vectors, 366, 441
- Unit principal normal vector, 389
- Unit step function (Heaviside function), 217–219
- Unit tangent vector, 384
- Unit vectors, 312, 355
- Universal gravitational constant, 63
- Unknowns, 257
- Unrepeated factors, 220–221
- Unstable algorithms, 796
- Unstable critical points, 140, 149
- Unstable equilibrium solution, 33–34
- Unstable systems, 84
- Upper bound, for flows, 995
- Upper confidence limits, 1068
- Upper control limit (UCL), 1088
- Upper triangular matrices, 268

- Value (sum) of series, 171, 673
- Vandermonde, Alexandre Théophile, 113n.1
- Vandermonde determinant, 113
- Van der Pol, Balthasar, 158n.4
- Van der Pol equation, 158–160
- Variables:
 - artificial, 965–968
 - basic, 960
 - complex, 620–621
 - control, 951
 - controlled, 1103
 - dependent, 393, 1055, 1056
 - independent, 393, 1103
 - intermediate, 393
 - linearly, 1109
 - nonbasic, 960
 - random, 1011, 1029–1030, 1061
 - continuous, 1029, 1032–1034, 1055
 - defined, 1030
 - dependent, 1055
 - discrete, 1029–1032, 1054
 - function of, 1056
 - independence of, 1055–1056
 - marginal distribution of, 1054, 1055
 - normal, 1045
 - occurrence of, 1063
 - probability distributions of, 1051–1060
 - skewness of, 1039
- Variables: (*Cont.*)
 - standardized, 1037
 - two-dimensional, 1051, 1062
 - slack, 956, 969
 - stochastic, 1029
 - uncorrelated related, 1109
- Variable coefficients:
 - Frobenius method, 180–187
 - indicial equation, 181–183
 - typical applications, 183–185
 - Laplace transforms ODEs with, 240–241
 - power series method, 167–175
 - idea and technique of, 168–170
 - operations on, 173–174
 - theory of, 170–174
 - second-order homogeneous linear ODEs, 73
- Variance(s), 1014, 1061
 - comparison of, 1086
 - control chart for, 1089–1090
 - equality of, 1084n.3
 - of normal distributions,
 - confidence intervals for, 1073–1076
 - of probability distributions, 1035–1039
 - addition of, 1058–1059
 - transformation of, 1036–1037
 - sample, 1015
- Variation, random, 1063
- Variation of parameters, method of:
 - higher-order linear ODEs, 123
 - high-order nonhomogeneous linear ODEs, 118–120
 - nonhomogeneous linear systems of ODEs, 162–163
 - second-order linear ODEs:
 - homogeneous, 104
 - nonhomogeneous, 99–102
- Vectors, 256, 259
 - addition and scalar multiplication of, 259–261
 - calculations with, 126–127
 - definitions and terms, 126, 128–129, 257, 259, 309
 - eigenvalues, 129–130
 - eigenvectors, 129–130
 - linear independence and dependence of, 282–283
 - multiplying matrices by, 263–265
 - in the plane, 309, 355
 - systems of ODEs as vector equations, 127–128
 - in 3-space, 309
 - transposition of, 266–267
- Vector addition, 309, 357–359
- Vector calculus, 354, 378–380
 - differential, *see* Vector differential calculus
 - integral, *see* Vector integral calculus
- Vector differential calculus, 354–412
 - curves, 381–392
 - arc length of, 385–386
 - length of, 385
 - in mechanics, 386–389
 - tangents to, 384–385
 - and torsion, 389–390
 - gradient of a scalar field, 395–402
 - directional derivatives, 396–397
 - maximum increase, 398
 - as surface normal vector, 398–399
 - vector fields that are, 400–401
 - inner product (dot product), 361–367
 - applications, 364–366
 - orthogonality, 361–363
 - scalar functions, 376
 - and vector calculus, 378–380
 - vector fields, 377–378
 - curl of, 406–409
 - divergence of, 402–406
 - that are gradients of scalar fields, 400–401
 - vector functions, 375–376
 - partial derivatives of, 380
 - of several variables, 392–395
 - vector product (cross product), 368–375
 - applications, 371–372
 - scalar triple product, 373–374
 - vectors in 2-space and 3-space:
 - components of a vector, 356–357
 - scalar multiplication, 358–359
 - vector addition, 357–359
- Vector fields:
 - defined, 376
 - vector differential calculus, 377–378
 - curl of, 406–409, 412
 - divergence of, 402–406
 - that are gradients of scalar fields, 400–401
- Vector functions:
 - continuous, 378–379
 - defined, 375–376
 - differentiable, 379
 - divergence theorem of Gauss, 453–457
 - of several variables, 392–395
 - chain rules, 392–394
 - mean value theorem, 395

- Vector functions: (*Cont.*)
 vector differential calculus,
 375–376, 411
 partial derivatives of, 380
 of several variables, 392–395
- Vectors in 2-space and 3-space:
 components of a vector,
 356–357
 scalar multiplication, 358–359
 vector addition, 357–359
- Vector integral calculus, 413–471
 divergence theorem of Gauss,
 453–463
 double integrals, 426–432
 applications of, 428–429
 change of variables in,
 429–431
 evaluation of, by two
 successive integrations,
 427–428
- Green's theorem in the plane,
 433–438
- line integrals, 413–419
 definition and evaluation of,
 414–416
 path dependence of, 418–426
 work done by a force, 416–417
- path dependence of line integrals,
 418–426
 defined, 418
 and integration around closed
 curves, 421–425
- Stokes's Theorem, 463–469
- surface integrals, 443–452
 orientation of surfaces,
 446–447
 without regard to orientation,
 448–450
- surfaces for surface integrals,
 439–443
 representation of surfaces,
 439–441
- Vector integral calculus (*Cont.*)
 tangent plane and surface
 normal, 441–442
 triple integrals, 452–458
- Vector moment, 371
- Vector norms, 866
- Vector product (cross product):
 in Cartesian coordinates,
 A83–A84
 vector differential calculus,
 368–375, 410
 applications, 371–372
 scalar triple product, 373–374
- Vector spaces, 482
 complex, 309–310, 349
 inner product spaces, 311–313
 linear transformations, 313–317
 real, 309–311
 special, 285–287
- Velocity, 391, 411, 771
- Velocity potential, 771
- Velocity vector, 386, 771
- Venn, John, 1017n.1
- Venn diagrams, 1017
- Verhulst, Pierre-François, 32n.8
- Verhulst equation, 32–33
- Vertices (graphs), 971, 977, 1007
 adjacent, 971, 977
 central, 991
 coloring, 1005–1006
 double labeling of, 986
 eccentricity of, 991
 exposed, 1001, 1003
 four-color theorem, 1006
 scanning, 998
- Vertex condition, 991
- Vertex incidence list (graphs), 973
- Volta, Alessandro, 93n.7
- Voltage drop, 29
- Volterra, Vito, 155n.3, 198n.7, 236n.3
- Volterra integral equations, of the
 second kind, 236–237
- Volume, of a region, 428
- Vortex (fluid flow), 777
- Vorticity, 774
- Walk (shortest path problems), 975
- Wave equation, 544–545, 942
 d'Alembert's solution, 553–556
 numeric analysis, 942–944, 948
 one-dimensional, 544–545
 solution by separating variables,
 545–553
 two-dimensional, 575–584
- Weber's equation, 510
- Weber's functions, 198n.7
- Weierstrass, Karl, 625n.4, 703n.5
- Weierstrass approximation theorem,
 809
- Weierstrass M -test for uniform
 convergence, 703–704
- Weighted graphs, 976
- Weight function, 500
- Well-conditioned problems, 864
- Well-conditioning (linear systems),
 865
- Wessel, Caspar, 611n.2
- Work done by a force, 416–417
- Work integral, 415
- Wronski, Josef Maria Höne, 76n.5
- Wronskian (Wronski determinant):
 second-order homogeneous linear
 ODEs, 75–78
 systems of ODEs, 139
- Zeros, of analytic functions, 717
- Zero matrix, 260
- Zero surfaces, 598
- Zero vector, 129, 260, 357
- z -score, 1014

PHOTO CREDITS

Part A Opener: © Denis Jr. Tangney/iStockphoto
Part B Opener: © Jill Fromer/iStockphoto
Part C Opener: © Science Photo Library/Photo Researchers, Inc
Part D Opener: © Rafa Irusta/iStockphoto
Part E Opener: © Alberto Pomares/iStockphoto
Chapter 19, Figure 437: © Eddie Gerald/Alamy
Part F Opener: © Rainer Plendl/iStockphoto
Part G Opener: © Sean Locke/iStockphoto
Appendix 1 Opener: © Ricardo De Mattos/iStockphoto
Appendix 2 Opener: © joel-t/iStockphoto
Appendix 3 Opener: © Luke Daniek/iStockphoto
Appendix 4 Opener: © Andrey Prokhorov/iStockphoto
Appendix 5 Opener: © Pedro Castellano/iStockphoto

Some Constants

$$e = 2.71828\ 18284\ 59045\ 23536$$

$$\sqrt{e} = 1.64872\ 12707\ 00128\ 14685$$

$$e^2 = 7.38905\ 60989\ 30650\ 22723$$

$$\pi = 3.14159\ 26535\ 89793\ 23846$$

$$\pi^2 = 9.86960\ 44010\ 89358\ 61883$$

$$\sqrt{\pi} = 1.77245\ 38509\ 05516\ 02730$$

$$\log_{10} \pi = 0.49714\ 98726\ 94133\ 85435$$

$$\ln \pi = 1.14472\ 98858\ 49400\ 17414$$

$$\log_{10} e = 0.43429\ 44819\ 03251\ 82765$$

$$\ln 10 = 2.30258\ 50929\ 94045\ 68402$$

$$\sqrt{2} = 1.41421\ 35623\ 73095\ 04880$$

$$\sqrt[3]{2} = 1.25992\ 10498\ 94873\ 16477$$

$$\sqrt{3} = 1.73205\ 08075\ 68877\ 29353$$

$$\sqrt[3]{3} = 1.44224\ 95703\ 07408\ 38232$$

$$\ln 2 = 0.69314\ 71805\ 59945\ 30942$$

$$\ln 3 = 1.09861\ 22886\ 68109\ 69140$$

$$\gamma = 0.57721\ 56649\ 01532\ 86061$$

$$\ln \gamma = -0.54953\ 93129\ 81644\ 82234$$

(see Sec. 5.6)

$$1^\circ = 0.01745\ 32925\ 19943\ 29577\ \text{rad}$$

$$1\ \text{rad} = 57.29577\ 95130\ 82320\ 87680^\circ$$

$$= 57^\circ 17' 44.806''$$

Polar Coordinates

$$x = r \cos \theta \qquad y = r \sin \theta$$

$$r = \sqrt{x^2 + y^2} \qquad \tan \theta = \frac{y}{x}$$

$$dx\ dy = r\ dr\ d\theta$$

Series

$$\frac{1}{1-x} = \sum_{m=0}^{\infty} x^m \quad (|x| < 1)$$

$$e^x = \sum_{m=0}^{\infty} \frac{x^m}{m!}$$

$$\sin x = \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m+1}}{(2m+1)!}$$

$$\cos x = \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m}}{(2m)!}$$

$$\ln(1-x) = -\sum_{m=1}^{\infty} \frac{x^m}{m} \quad (|x| < 1)$$

$$\arctan x = \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m+1}}{2m+1} \quad (|x| < 1)$$

Greek Alphabet

α	Alpha	ν	Nu
β	Beta	ξ	Xi
γ, Γ	Gamma	$\omicron$	Omicron
δ, Δ	Delta	π	Pi
ϵ, ε	Epsilon	ρ	Rho
ζ	Zeta	σ, Σ	Sigma
η	Eta	τ	Tau
$\theta, \vartheta, \Theta$	Theta	υ, Υ	Upsilon
ι	Iota	ϕ, φ, Φ	Phi
κ	Kappa	χ	Chi
λ, Λ	Lambda	ψ, Ψ	Psi
μ	Mu	ω, Ω	Omega

Vectors

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$\text{grad } f = \nabla f = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k}$$

$$\text{div } \mathbf{v} = \nabla \cdot \mathbf{v} = \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z}$$

$$\text{curl } \mathbf{v} = \nabla \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_1 & v_2 & v_3 \end{vmatrix}$$